

ANVIL MINING LIMITED NI 43-101 TECHNICAL REPORT KINSEVERE COPPER PROJECT

KATANGA PROVINCE,
DEMOCRATIC REPUBLIC OF CONGO



Principal Authors:

Geoffrey Booth (formerly of Anvil Mining Ltd) (QP)
Anthony Cameron (A & J Cameron & Associates) (QP)
Gerry Fahey (CSA Global Pty Ltd) (QP)
Michael Lawlor (Anvil Mining Ltd) (QP)

Date: 31st March 2010

TABLE OF CONTENTS

ITEM 3	SUMMARY	12
3.1	Property Location, Ownership and Approvals	12
3.2	Geology and Mineralisation	13
3.3	Exploration Concept and Status	13
3.4	Development and Operations	14
3.5	Mineral Resource Estimate	14
3.6	Mineral Reserve Estimate	16
3.7	Production Schedule	17
3.8	Operating and Capital Costs	18
3.9	Economic Evaluation	19
3.10	Conclusions and Recommendations	21
ITEM 4	INTRODUCTION	26
ITEM 5	RELIANCE ON OTHER EXPERTS	30
ITEM 6	PROPERTY DESCRIPTION AND LOCATION	32
ITEM 7	ACCESSIBILITY, CLIMATE, LOCAL RESOURCES, INFRASTRUCTURE & PHYSIOGRAPHY	36
ITEM 8	HISTORY	39
8.1	Exploration	39
8.2	Previous Anvil Mineral Resource and Mineral Reserve Estimates	40
8.3	Stage I Development and Production	43
8.4	Current Status	51
ITEM 9	GEOLOGICAL SETTING	54
9.1	Regional Geology	54
9.2	Local and Property Geology	57
ITEM 10	DEPOSIT TYPE	60
ITEM 11	MINERALISATION	62
ITEM 12	EXPLORATION	66
ITEM 13	DRILLING	67
13.1	Drill Rigs and Operators	68
13.2	Collar Locations	69
13.3	Down- Hole Surveys	69
ITEM 14	SAMPLING METHOD AND APPROACH	70
14.1	Geological Logging	70
14.2	Sampling Protocols	70
14.3	Relationship Between Sample Length and Mineralised Thickness	70
14.4	Sample Representivity and Relevant Samples	70

14.5	Potential Sample Bias	73
14.6	Density Measurements.....	76
14.7	Drill Data for Mineral Resource Modelling.....	76
ITEM 15	SAMPLE PREPARATION, ANALYSES AND SECURITY.....	78
18.1	Sample Preparation.....	78
15.2	Analysis	78
15.3	Quality Assurance and Control Measures	78
15.4	Quality Control Issues	79
15.5	Security	80
ITEM 16	DATA VERIFICATION	81
ITEM 17	ADJACENT PROPERTIES.....	83
ITEM 18	MINERAL PROCESSING AND METALLURGICAL TESTING.....	84
18.1	Metallurgical Samples	84
18.2	Summary of Metallurgical Testwork Results	87
ITEM 19	MINERAL RESOURCE AND MINERAL RESERVE ESTIMATES.....	88
19.1	Mineral Resource Estimate	89
19.2	Mineral Reserve Estimate	132
ITEM 20	OTHER RELEVANT DATA AND INFORMATION	147
20.1	Stage I Mining and Processing.....	147
20.2	DRC Government Review	148
20.3	Potential Pit Expansions for Stage III	148
20.4	Metallurgical Testwork for Kinsevere Stage III	153
20.5	Viability of Cobalt Recovery	156
20.6	Project Enhancements	157
ITEM 21	INTERPRETATION AND CONCLUSIONS.....	159
21.1	Mineral Resource Modelling and Estimation	159
21.2	Mineral Reserve Estimation	161
21.3	Uncertainty and Risk	162
ITEM 22	RECOMMENDATIONS.....	166
22.1	Oxide Resampling and Assaying	166
22.2	Sulphide Assaying.....	166
22.3	Density Measurements.....	167
22.4	Grade Control Drilling.....	167
22.5	Data Management.....	168
22.6	Grade Control Modelling	169
22.7	Photogrammetry	169
22.8	Production Tracking and Reconciliation	169
22.9	Updated Mineral Resource Modelling	170

22.10	Geotechnical Engineering	170
22.11	Updated LOM Plan	170
22.12	Short-Medium Term Planning	170
22.13	Groundwater Bore Drilling, Aquifer Testing and Equipping	171
22.14	Groundwater Discharge	171
22.15	Groundwater Drawdown and Quality Monitoring	171
22.16	Geotechnical Investigations for the TSF	172
22.17	Surface Water Management Planning	172
22.18	ARD Testwork and Waste Dump Planning	172
22.19	Stage II Mining Contract	172
22.20	Environmental Compliance	173
22.21	Social Development	173
22.22	Closure Plan	173
22.23	Project Enhancement Studies	173
22.24	Summary of Cost Estimates for Recommended Work Programme	173
ITEM 23	REFERENCES	175
ITEM 24	DATE AND SIGNATURE PAGE	177
ITEM 25	ADDITIONAL REQUIREMENTS FOR TECHNICAL REPORTS ON DEVELOPMENT PROPERTIES AND PRODUCTION PROPERTIES..	182
25.1	Mining Operations	182
25.2	Infrastructure	204
25.3	Processing Operations	209
25.4	Recoverability	218
25.5	Tailings Management	242
25.6	Stage II Implementation Plan	245
25.7	Markets	248
25.8	Contracts	248
25.9	Environmental Considerations	248
25.10	Operations	251
25.11	Cost Estimates	261
25.12	Economic Analysis	268
25.13	Sensitivity Analysis	274
APPENDIX A	CRM BATCH CONTROL CHARTS	277

LIST OF FIGURES

Figure 3.1	Production Schedule Profile	17
Figure 6.1	Location of the Kinsevere Project.....	32
Figure 6.2	Location of the Kinsevere Mining Permit	33
Figure 6.3	Exploitation Permit PE 528.....	34
Figure 7.1	Satellite Image of Kinsevere Project Area (Before Stage I Mining)	37
Figure 7.2	Aerial View, Looking SE, Across Tshifufia Central and South, Towards Kinsevere Hill (Before Stage I Mining)	38
Figure 7.3	Aerial View, Looking NW, of the Tshifufia Central Open Pit Area (Before Stage I Mining)	38
Figure 8.1	Monthly Ore Production – Stage I Mining.....	44
Figure 8.2	Monthly Ore and Waste Movement – Stage I Mining	44
Figure 8.3	Monthly Crusher Feed – Stage I Processing.....	45
Figure 8.4	Monthly Plant Recovery – Stage I Processing	45
Figure 8.5	Monthly Metal Production – Stage I Processing.....	46
Figure 8.6	Stage I Process Plant – Schematic Flowsheet.....	47
Figure 9.1	Regional Geological Setting	54
Figure 9.2	Geology of the Kinsevere Project Area	56
Figure 9.3	Revised Interpretation (2008) of Surface Geology at Tshifufia & Tshifufiamashi.....	58
Figure 10.1	Schematic Model of Tshifufia Deposit	60
Figure 10.2	Diagram of Significant Developmental Stages in the Formation of Kinsevere Cu Mineralisation and its Geological Setting.....	61
Figure 11.1	Brittle Vein Network of Supergene Cu Mineralisation in GRAT, Pit Wall	63
Figure 11.2	Mineralised Intercepts from Phase II Drilling Programme	63
Figure 13.1	Plan Illustrating the Distribution of Drillholes Across Kinsevere	67
Figure 13.2	In pit RC Drilling in the Tshifufia Pit in 2008	68
Figure 14.1	QQ Plot Comparing AC Cu Assay Results With RC Cu Results For Tshifufia Oxidised Material.....	74
Figure 14.2	QQ Plot Comparing RC Cu Assay Results With Diamond Cu Results for Tshifufia Oxidised Material.....	75
Figure 14.3	QQ Plot Comparing AC Cu Assay Results With Diamond Cu Results for Tshifufia Oxidised Material.....	75
Figure 19.1	West to East Cross Section at Tshifufia, Typical of Kinsevere Mineralisation	90
Figure 19.2	Drillhole Plan of the Kinsevere Hill and Kilongo Project Areas.....	91
Figure 20.1	Conceptual Stage III Pit Limits (Green Perimeter)	152
Figure 20.2	Flotation Testwork Results – Kinsevere Sulphide Samples	156

Figure 20.3	Estimated Production Cost for Cobalt Production at Kinsevere	157
Figure 25.1	Kinsevere Mine-Mill Metal Flow	184
Figure 25.2	Proposed Production Tracking System	185
Figure 25.3	Proposed Grade Control Drilling Campaigns	186
Figure 25.4	Location of Existing Groundwater Abstraction and Monitoring Bores....	188
Figure 25.5	Groundwater Monitoring Trends.....	189
Figure 25.6	Location Plan of Proposed New Boreholes	190
Figure 25.7	Arrangement Concept, Site Water Management Plan	193
Figure 25.8	Scenario E Production Profile (Superseded).....	196
Figure 25.9	Scenario H Production Profile (Current).....	197
Figure 25.10	Life of Mine Mining Sequence (Total Ore + Waste).....	201
Figure 25.11	Base Case Material Movement Profile	201
Figure 25.12	Base Case Stockpile Reclaim and Balance Profile	202
Figure 25.13	Stage II Plant Layout Plan	215
Figure 25.14	Stage II Schematic Flowsheet	216
Figure 25.15	Variation of GAC and Leach pH	229
Figure 25.16	Pre-leach Rheogram	240
Figure 25.17	Variation of Effluent pH with Lime Addition	241
Figure 25.18	Revised TSF Design.....	244
Figure 25.19	Stage II Project Schedule	247
Figure 25.20	Stage II Operations, Organisation Structure – Administration.....	255
Figure 25.21	Stage II Operations, Organisation Structure – Technical Services.....	256
Figure 25.22	Stage II Operations, Organisation Structure – Processing.....	257
Figure 25.23	Stage II Operations, Organisation Structure – Maintenance	258
Figure 25.24	Sensitivity to Copper Price (\$/lb) - Based on Project NPV Including Sunk Costs to Date	274

LIST OF TABLES

Table 3.1	Kinsevere Project Measured and Indicated Mineral Resource Statement, at 31 st December 2009	15
Table 3.2	Kinsevere Project Inferred Mineral Resource Statement, at 31 st December 2009	15
Table 3.3	Kinsevere Project Mineral Reserve Statement, at 31 st December 2009	16
Table 3.4	Kinsevere Project Stockpile Reserve Statement, at 31 st December 2009	16
Table 3.5	Schedule Changes Since January 2010 News Release	18
Table 3.6	Stage II Unit Operating Cost Summary Averaged Over the Life of Project	18
Table 3.7	Stage II Capital Costs.....	19
Table 3.8	Economic Evaluation Parameters	20
Table 3.9	Key Economic Results.....	21
Table 3.10	Mineral Reserve and Associated Mine Waste Comparisons.....	22
Table 3.11	Summary of Cost Estimates for Recommended Work Programme	25
Table 6.1	Coordinates of the Extended PE 528	35
Table 8.1	Summary of Previous Exploration Work by EXACO & Gécamines.....	39
Table 8.2	Gécamines Historical Estimates for the Kinsevere Deposits	40
Table 8.3	Kinsevere Project Mineral Resource Estimate, September 2005.....	41
Table 8.4	Kinsevere Project Mineral Resource Estimate, April 2007	41
Table 8.5	Kinsevere Project Mineral Reserve Estimate, April 2007	42
Table 8.6	Kinsevere Project Mineral Resource Estimate, December 2007.....	42
Table 8.7	Kinsevere Project Mineral Reserve Estimate, December 2007.....	43
Table 9.1	Stratigraphy of the Katangan System.....	55
Table 13.1	Summary of Resource Drilling at Kinsevere.....	67
Table 13.2	Drill Rigs and Operators	68
Table 13.3	Down-hole Surveying and Operators	69
Table 14.1	Summary of Best Drill Intercepts (relevant samples)	71
Table 14.2	In-pit Bulk Density Sample Results	76
Table 15.1	Mean Values (in ppm) of Certified Reference Materials.....	79
Table 16.1	Summary Statistics – 20 m to 40 m Depth	81
Table 16.2	Summary Statistics – Greater than 40 m Depth	82
Table 18.1	Drill Core Samples Used for Metallurgical Testwork	84
Table 18.2	Master Composite Samples	86
Table 18.3	Variability Composite Samples.....	86
Table 19.1	Boreholes and Sample Metres for Kinsevere Hill (KV) and Kilongo (KP).....	90
Table 19.2	Domain Codes for Tshifufia and Tshifufiamashi.....	97

Table 19.3	Domain Codes for Kinsevere Hill and Kilongo.....	97
Table 19.4	Stratigraphic Flagging Codes	97
Table 19.5	Weathering Flagging Codes	97
Table 19.6	Area Flagging Codes.....	97
Table 19.7	Mineralised Envelope Flagging Codes	98
Table 19.8	Univariate Statistics for TCu by Domain Code	101
Table 19.9	Orezone codes subdivide mineralised zone by stratigraphy, grade of mineralisation and area.....	103
Table 19.10	Summary of TCu Statistics, Mineralised Zone	104
Table 19.11	Summary of ASCu Statistics, Mineralised Zone	105
Table 19.12	Summary of TCu Statistics, Unmineralised and Weakly Mineralised Zone	106
Table 19.13	Summary of ASCu Statistics, Unmineralised and Weakly Mineralised Zone	107
Table 19.14	Axes of Anisotropy for Tshifufia Variogram Models.....	110
Table 19.15	Axes of Anisotropy for Tshifufiamashi Variogram Models	111
Table 19.16	Variogram Parameters for Tshifufia using Spherical Models.....	111
Table 19.17	Variogram Parameters for Tshifufiamashi using Spherical Models..	112
Table 19.18	Axes of Anisotropy for Kinsevere Hill Variogram Models	112
Table 19.19	Variogram Parameters for Kinsevere Hill using Spherical Models ...	112
Table 19.20	Block Model Dimensions and Cell Sizes	118
Table 19.21	Average Density Values for Diamond Core Samples per Domain and Zone of Weathering.....	119
Table 19.22	In-pit Bulk Density Sample Results	119
Table 19.23	Density Values Used in Kinsevere Hill/Kilongo Model.....	121
Table 19.24	Table of Top Cuts per Domain, Tshifufia and Tshifufiamashi.....	122
Table 19.25	Top Cuts Used in Modelling, Kinsevere Hill and Kilongo	122
Table 19.26	Grade Estimation Search Parameters – Cu and ASCu	124
Table 19.27	Variogram Models for Grade Estimation – Cu and ASCu	124
Table 19.28	Table of Mean Statistics per Domain Comparing Model Estimates with Data Values	126
Table 19.29	Kinsevere Project Measured and Indicated Mineral Resource Statement, at 31 st December 2009	131
Table 19.30	Kinsevere Project Inferred Mineral Resource Statement, at 31 st December 2009.....	132
Table 19.31	Fixed Costs – Stage II	133
Table 19.32	Variable Mining Costs – Stage II	134
Table 19.33	Variable Processing Costs – Stage II	135
Table 19.34	Metal Costs at \$1.43/lb – Stage II	135
Table 19.35	Pit Optimisation Slope Angles	136

Table 19.36	Marginal Cut Off Grades for Stage II	137
Table 19.37	Pit Optimisation Results Tshifufia & Tshifufiamashi	137
Table 19.38	Pit Optimisation Results Kinsevere Hill & Kilongo	138
Table 19.39	Kinsevere Project Mineral Reserve Statement, at 31 st December 2009.	144
Table 19.40	Kinsevere Project Stockpile Reserve Statement, at 31 st December 2009	144
Table 19.41	Kinsevere Project Mineral Reserve Summary, Checklist of Assessment and Reporting Criteria	145
Table 20.1	Key Amendments to the Kinsevere Lease Agreement.....	149
Table 20.2	Marginal Cut-Off Grades at \$1.43/lb.....	150
Table 20.3	Stage III Pit Optimisation Results - Tshifufia Central & Tshifufiamashi	151
Table 20.4	Stage III Pit Optimisation Results – Kinsevere Hill/Kilongo	151
Table 20.5	Details of Drill Core Intervals Selected for Metallurgical Testwork ...	154
Table 20.6	Summary of Master Composite Analysis (TCDH 032)	155
Table 20.7	Summary of SMC Test Results	155
Table 21.1	The 2007 Oxide and Sulphide Mineral Resource Estimates	160
Table 21.2	Mineral Reserve and Associated Mine Waste Comparisons.....	161
Table 21.3	SRK Cash Flow Model Sensitivity Scenarios	165
Table 22.1	Proposed Number of Drillholes and Total Drilling Metres for each Stage of the 2010 Grade Control Programme	168
Table 22.2	Summary of Cost Estimates for Recommended Work Programme	174
Table 25.1	Revised Pit Slope Design Parameters, September 2007	187
Table 25.2	Design Storm Criteria	192
Table 25.3	ECD Design Summary	192
Table 25.4	Diversion Channel Design Summary.....	194
Table 25.5	Culvert Design Summary.....	195
Table 25.6	Life of Mine Plan Production Schedule Scenarios.....	196
Table 25.7	Scheduled Stockpile Grade Ranges.....	197
Table 25.8	Production Ramp-Up Limits.....	198
Table 25.9	Life of Mine Mining and Processing Summary	199
Table 25.10	Life of Mine Stockpile Balances.....	200
Table 25.11	Metal Balance, 27 th January 2010 News Release	203
Table 25.12	Metal Balance, This Technical Report.....	203
Table 25.13	Revised TSF Construction Quantities, June 2009.....	204
Table 25.14	Summary of SEM Analysis	219
Table 25.15	Major Constituents of the Kinsevere Deposits (Wt%).....	220
Table 25.16	Summary of Master Composite Analyses	220
Table 25.17	Sequential Leach Analysis of Master Composite Samples	221

Table 25.18	Dry Solids SG of Master Composite Samples	221
Table 25.19	Source of SMC Test Samples	222
Table 25.20	Summary of SMC Test Results	223
Table 25.21	Summary of SMC Test Ratings	223
Table 25.22	Summary of Impact Crushing Work Index and UCS Results	223
Table 25.23	Summary of Rod Mill Work Indices (kWh/t)	224
Table 25.24	Summary of Ball Mill Work Indices (kWh/t)	224
Table 25.25	Summary of Levin Tests	225
Table 25.26	Summary of Abrasion Index Results	225
Table 25.27	Comparison of Site Water and Perth Tap Water	225
Table 25.28	Summary of Acid Soluble Recoveries (% AS Cu)	228
Table 25.29	Comparison of Initial and Confirmatory Leach Tests Performed at a Grind P ₈₀ of 150 microns, pH 1.5 for 8 Hours	229
Table 25.30	Results for Variability Samples Leached at Selected Optimum Conditions (Grind P ₈₀ of 150 microns, pH 1.5 for 8 Hours)	230
Table 25.31	Summary of Leach Solution Analyses	231
Table 25.32	Tailings Leach Test Results (Sequential Copper assays)	232
Table 25.33	Overall Leach Results – Conventional (12 hours) & Tailings Leach (16 weeks)	232
Table 25.34	Summary of Pre-leach Hi-Rate Thickening Testwork	234
Table 25.35	Summary of Post-leach Hi-Rate Thickening Testwork	234
Table 25.36	Summary of Clarification Testwork	235
Table 25.37	Comparison of Gécamines Limestone and Quicklime Quality with Ndola Lime Product Specifications	236
Table 25.38	Comparison of Neutralisation Performance of Gécamines Limestone at Different Grind Sizes	236
Table 25.39	Effect of Combined Limestone and Lime for Neutralisation	237
Table 25.40	Cognis Phase Disengagement Testwork Results	238
Table 25.41	Summary of Copper Equilibrium Extraction Data and ISOCALC™ Modelling	238
Table 25.42	Range of Rheology Testwork	239
Table 25.43	Acid Consumption for Effluent pH Adjustment	241
Table 25.44	Effect of pH Adjustment on Effluent Quality	241
Table 25.45	Flocculant Effectiveness	242
Table 25.46	Production Ramp-Up Limits	242
Table 25.47	Boreholes Drilled for Hydrogeological Investigations	245
Table 25.48	Manning Numbers	254
Table 25.49	Stage I Operating Cost Inputs (Real)	261
Table 25.50	Stage II Variable Operating Cost Inputs (Real)	262
Table 25.51	Stage II Fixed Operating Cost Inputs (Real)	263

Table 25.52	Stage II Unit Operating Cost Summary Averaged Over the Life of Project	264
Table 25.53	Stage II LSTK Costs	265
Table 25.54	Stage II Capital Cost Summary	266
Table 25.55	Capital Expenditure Schedule (\$'000) (Real)	267
Table 25.56	Corporate Cost Provisions (Real)	268
Table 25.57	Production Assumptions for Stage II	269
Table 25.58	Copper Pricing Scenarios (Nominal)	269
Table 25.59	Copper Pricing Scenarios (Real)	269

ITEM 3 Summary

Anvil Mining Limited (Anvil or the Company) has prepared this Technical Report on its Kinsevere Project to **document the estimated Mineral Resources and Mineral Reserves, economic evaluations and supporting ancillary documentation** associated with the Stage II Development Project. This report complies with the reporting requirements of the Canadian National Instrument 43-101 guidelines: “Standards of Disclosure for Mineral Properties” and the JORC Code.

This report supersedes the Feasibility Study dated December 2007 (AMCK, December 2007) and the NI 43-101 Technical Report dated March 2008 (Anvil, March 2008). Updates to the previous Technical Report are noted throughout¹.

Since these previous reports there have been a number of material events and changes to the Project, most notably:

- suspension of Stage I mining and processing operations at Kinsevere in November 2008;
- suspension of Stage II construction works at the same time;
- changes to the future Stage I processing flowsheet, involving the shut down of the electric arc furnace (EAF);
- resumption of stockpile reclaim and Stage I production of copper concentrate in April 2009, followed by continued plant feed from resumed mining in August 2009;
- in the period since the Feasibility Study, a significant discrepancy in oxide rock density had become apparent, resulting in a reduction of the average Mineral Resource model rock density;
- an updated Mineral Reserve estimate was prepared and reported as at 31st December 2009, reflecting density corrections, Mineral Resource model updates, mining depletion since the Feasibility Study, available stockpile inventories, revised optimisation parameters, revised staged pit designs and updates to the Stage I and Stage II mining and processing production schedules;
- the design of the Stage II tailings facility was changed to incorporate downstream raised walls on all four sides;
- in August 2009, the Company reached an agreement with Trafigura Beheer B.V. (Trafigura) on a combined debt and equity financing arrangement amounting to \$200 million;
- negotiations resumed with Ausenco Engineering Ltd (Ausenco) for the resumption of Stage II construction works in late 2009, with a Lump Sum Turn Key (LSTK) contract awarded in January 2010;
- the commissioning of the Stage II plant is now expected in January 2011, as opposed to the first half of 2009; and
- the estimated Stage II capital costs have risen from \$277 million to \$395 million, whilst the total capital costs (ie, LSTK and owner controlled) for the Project have risen from \$302 million to \$515 million (ie, excluding corporate charges and import duties).

3.1 Property Location, Ownership and Approvals

The Kinsevere Project property is located approximately 27 km north of Lubumbashi, the provincial capital of the Katanga Province, in the southeast of the Democratic Republic of the Congo (DRC). The property portfolio consists of two separate Exploitation Permits (Kinsevere, PE 528 and Nambulwa, PE 539). The Nambulwa

¹ Reference to dollars in this Technical Report means United States dollars, unless otherwise noted.

permit is to be relinquished, following an exploration and evaluation programme completed in 2007.

The mineral rights of PE 528 are held by La Générale des Carrierés et des Mines (Gécamines), the DRC state-owned copper mining company. However, Anvil, via its subsidiary AMCK Mining s.p.r.l. (AMCK, a joint venture between Anvil (Anvil 95%) and Mining Company of Katanga s.p.r.l. (MCK) (5%)), has a *Contrat d'Amodiation* (Lease Agreement) with Gécamines to mine and process ore from PE 528 until 2024, followed by a 15 year extension.

Approval for the Lease Agreement was issued by the DRC Minister of Portfolio on the 6th December 2005 with the provision that the AMCK joint venture make royalty payments to Gécamines on each tonne of commercially viable copper metal extracted from future mining and processing operations. As a consequence of a Government commission in 2008, the terms of the Lease Agreement were amended in January 2009 to reflect revised royalty payments, whereby royalty payments are now based on 2.5% of gross revenue and the imposition of a further *pas de porte* (entry premium) of \$15 million.

There are no encumbrances that apply to this permit and the property is not subject to any environmental liabilities. Following submission of an Environmental Impact Assessment report, Government approval of the Kinsevere Copper Project, Stages I and II, was issued on 15 October 2007.

Approval of a variance to the design and operation of the Stage II tailings storage facility was issued by the Directorate for the Protection of the Mining Environment (DPEM) on 28 October 2008 (DPEM, October 2008).

3.2 Geology and Mineralisation

The Kinsevere Project is host to three known copper deposits: Tshifufia, Tshifufiamashi and Kinsevere Hill. Mineralisation extending northwest from Kinsevere Hill is sometimes referred to separately as Kilongo. All of these deposits are hosted by interbedded calcareous siltstones and silty dolomites of the Mines Group sediments. This sequence is overturned and dips steeply eastwards.

The main deposit, Tshifufia is an example of a structurally modified, stratiform copper deposit.

Deep weathering extending to approximately 100 m below surface has led to the development of a large oxide Mineral Resource. This mineralisation is dominated by malachite and lesser pseudomalachite and is hosted by highly altered dolomitic and argillaceous sediments. Underlying sulphide mineralisation at Tshifufia comprises chalcopyrite in crosscutting veins and stringers, produced during the Lufilian Orogeny (approximately 550 Ma).

3.3 Exploration Concept and Status

Following preliminary exploration and shallow mining conducted by other parties in the 1990's, Anvil has progressively reviewed its Kinsevere Project with the aim of defining a robust and economically viable Mineral Resource. Anvil's exploration programme has had three principal objectives:

1. to verify the results of previous explorers;
2. to improve the understanding of the Mineral Resources for the Stage II Project; and

3. to identify extensional oxide mineralisation, and also sulphide mineralisation, which may ultimately warrant the development of a Stage III Project.

Following an initial trenching programme in 2004, four phases of drilling were carried out between 2005 and 2008. Drilling programmes comprised a combination of air-core, diamond and reverse circulation (RC) drilling. Approximately 56% of the metres drilled were RC, followed by diamond drilling at 28%, and the remainder were by air-coring.

The sulphide drilling component of the 2008 Phase IV programme was only partially completed before the suspension of operations at Kinsevere in late 2008. RC infill and grade control drilling is planned for Tshifufia and Tshifufiamashi during 2010, and will continue during the Stage II operations phase. Once Stage II production is underway from 2011, it is expected that the Phase IV sulphide drilling programme will be resumed and extended.

3.4 Development and Operations

Mining commenced at Tshifufiamashi in December 2006 and at Tshifufia in February 2007. Mining continued successfully at Tshifufia until operations were suspended and placed under care and maintenance in November 2008. The Stage I heavy media separation (HMS) plant, which was commissioned in March 2007, was also placed under care and maintenance at this time.

Due to the sharp fall in copper prices in late 2008, together with uncertainty in the capital markets, and as part of the care and maintenance programme, the Company also suspended the fabrication and construction works associated with the Stage II Project. The production of blister copper from the Stage I EAF was subsequently suspended in Q1 2009 due to technical difficulties and better cashflows being able to be realised from the resumed sale of HMS concentrate.

Against this background, the Stage I HMS plant was restarted in March 2009, drawing feed from the run of mine (ROM) stockpiles. This feed was only sufficient to last until Q3 2009, at which time (ie, August 2009) mining was restarted in the Tshifufia Pit from readily available ore faces and broken stocks. Mining was resumed on a reduced scale relative to previous production, with the intention of just providing sufficient feed for the HMS plant (ie, operating at up to 2,000 tonnes per day) and with only minimal long-term stockpile building.

Construction of the Stage II Project was resumed in January 2010 following the signing of a Lump Sum Turn Key (LSTK) contract with Ausenco. A "Client Representative" was appointed in February 2010 to act as the Company's site representative and interface with Ausenco during the construction phase.

The 2010 production plan provides just sufficient cash flow to sustain Stage I operations and corporate overheads throughout the resumed 2010 Stage II construction programme. HMS plant recovery is projected at 60% and the concentrate sales (at mine gate) are at a discounted price of 40% to 50% of the London Metal Exchange (LME) price. Hence, high grade ores are being conserved for Stage II production, from which cathode sales will be at 100% of LME pricing.

3.5 Mineral Resource Estimate

Preliminary evaluation of the Tshifufia, Tshifufiamashi and Kinsevere Hill deposits was originally reported by Anvil in 2005. Recent work involving the third and fourth

phases of infill and resource extension drilling and an up-dated Mineral Resource estimate for Kinsevere was reported by Anvil in a News Release dated 19th February, 2009 (Anvil, February 2009). This News Release noted a reduction in the oxide Mineral Resource inventory relative to that reported in the Feasibility Study.

The February 2009 estimate has now been further updated to reflect mining depletion to the end of December 2009 (Table 3.1 and Table 3.2). The Mineral Resource has been classified and reported using the guidelines of the JORC Code (JORC, 2004), which comply with the CIM Standards on Mineral Resources and Reserves (CIM, 2000) of the Canadian Institute of Mining, Metallurgy and Petroleum (CIM).

Table 3.1 Kinsevere Project Measured and Indicated Mineral Resource Statement, at 31st December 2009

Kinsevere oxide Mineral Resource statement using a 0.7% TCu cutoff grade as at 31 st December 2009						
Area	Category	Volume (M m ³)	Density (t/m ³)	Tonnes (M t)	TCu (%)	ASCu (%)
Tshifufia	Measured	4.19	1.76	7.37	4.30	3.64
	Indicated	3.10	1.88	5.81	5.00	3.95
Tshifufiamashi	Measured	1.15	1.71	1.97	3.45	3.06
	Indicated	2.28	1.84	4.18	2.70	2.08
Kinsevere Hill	Measured	-	-	-	-	-
	Indicated	3.79	1.69	6.41	2.88	2.53
Total	Measured & Indicated	14.51	1.77	25.74	3.78	3.14
Kinsevere sulphide Mineral Resource statement using a 0.7% TCu cutoff grade as at 31 st December 2009						
Area	Category	Volume (M m ³)	Density (t/m ³)	Tonnes (M t)	TCu %	ASCu %
Tshifufia	Measured	0.03	2.32	0.06	2.16	0.97
	Indicated	1.33	2.32	3.09	3.99	2.60
Tshifufiamashi	Measured	-	-	-	-	-
	Indicated	0.26	2.32	0.61	2.40	1.59
Total	Measured & Indicated	1.62	2.32	3.76	3.70	2.41

Table 3.2 Kinsevere Project Inferred Mineral Resource Statement, at 31st December 2009

Kinsevere oxide Mineral Resource statement using a 0.7% TCu cutoff grade as at 31 st December 2009						
Area	Category	Volume (M m ³)	Density (t/m ³)	Tonnes (M t)	TCu%	ASCu %
Tshifufiamashi	Inferred	0.33	1.86	0.62	2.55	1.91
Tshifufia	Inferred	0.49	1.90	0.92	4.73	3.42
Kinsevere Hill	Inferred	-	-	-	-	-
Total	Inferred	0.82	1.88	1.54	3.85	2.81
Kinsevere sulphide Mineral Resource statement using a 0.7% TCu cutoff grade as at 31 st December 2009						
Area	Category	Volume (M m ³)	Density (t/m ³)	Tonnes (M t)	TCu%	ASCu %
Tshifufiamashi	Inferred	0.45	2.32	1.05	2.37	1.49
Tshifufia	Inferred	4.99	2.32	11.59	3.65	1.85
Total	Inferred	5.44	2.32	12.64	3.54	1.82

3.6 Mineral Reserve Estimate

The updated Mineral Reserve estimate is based on the February 2009 Mineral Resource model, as subsequently adjusted for mining depletion to the end of December 2009. The Mineral Reserve estimate relates specifically to the Measured and Indicated Resource categories and oxide mineralisation only. The oxide Measured and Indicated Mineral Resource estimate (Table 3.1) is inclusive of the Mineral Reserves listed in Table 3.3. There are additional stockpile Mineral Reserves as listed in Table 3.4.

The Mineral Reserve estimate was derived from conventional mine planning processes including Whittle optimisation, detailed pit designs according to geotechnical design parameters and production scheduling. Hence, in the conversion of Mineral Resources to Mineral Reserves, modifying factors such as mining/metallurgical parameters, economic, product marketing, legal, environmental (in respect of waste dump design and site rehabilitation allowances), social and government considerations have been taken into account.

The Proven and Probable Mineral Reserve estimate has been reported using the guidelines of the JORC Code (JORC, 2004) and is consistent with Mineral Reserves reported using the guidelines for the CIM (CIM, 2000).

Table 3.3 Kinsevere Project Mineral Reserve Statement, at 31st December 2009

Kinsevere oxide Mineral Reserve statement as at December 31 2009 (based on \$1.43/lb Cu)					
Deposit	Classification	Tonnes (Mt)	TCu (%)	ASCu (%)	Contained ASCu (k t)
Tshifufia	Proven	7.14	4.12	3.51	250.5
	Probable	4.40	5.09	4.05	178.3
Tshifufiamashi	Proven	1.65	3.56	3.21	52.8
	Probable	1.20	4.02	3.68	44.3
Kinsevere Hill	Proven	-	-	-	-
	Probable	4.02	3.42	3.07	123.5
Total	Proven & Probable	18.41	4.14	3.53	649.5

Table 3.4 Kinsevere Project Stockpile Reserve Statement, at 31st December 2009

Kinsevere oxide Stockpiles as at December 31 2009 (based on \$1.43/lb Cu)					
Stockpile	Classification	Tonnes (Mt)	TCu (%)	ASCu (%)	Contained ASCu (k t)
High Grade (+4% ASCu)	Proven	-	-	-	-
	Probable	0.03	8.56	7.18	2.1
Medium Grade (2% to 4% ASCu)	Proven	-	-	-	-
	Probable	1.11	2.90	2.43	27.0
Low Grade (<2% ASCu)	Proven	-	-	-	-
	Probable	1.26	1.04	0.87	11.0
Total	Proven & Probable	2.40	1.99	1.67	40.1

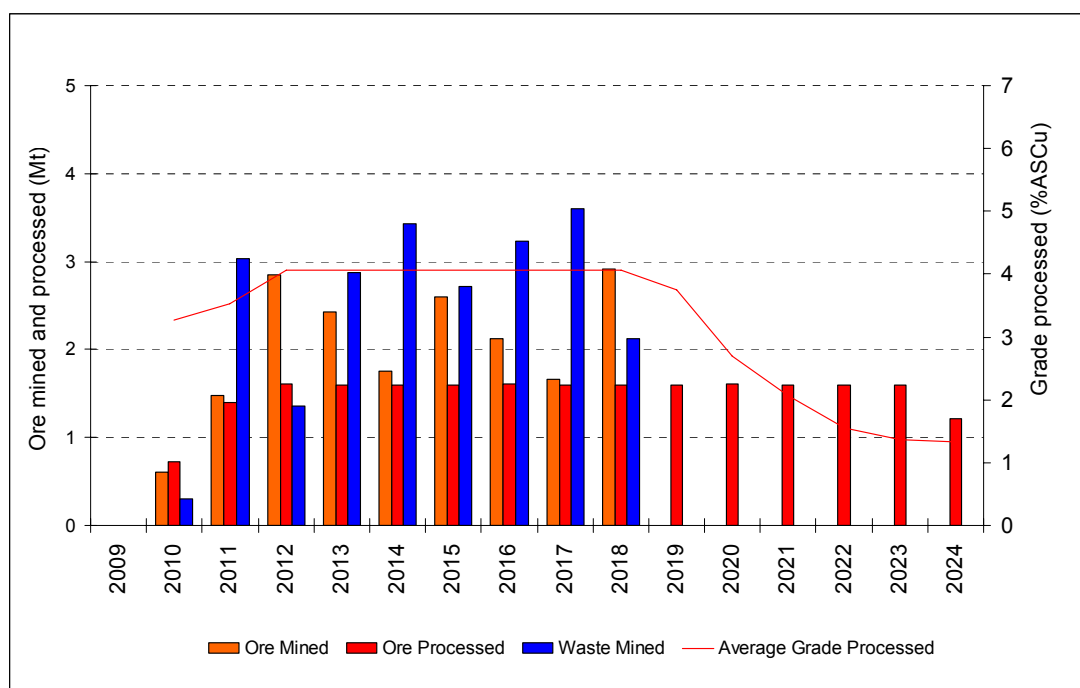
3.7 Production Schedule

A Life of Mine production schedule was prepared based on optimal, staged pit designs and the Mineral Reserve inventories listed in Tables 3.3 and 3.4.

Figure 3.1 shows the production schedule profile, from which the following can be noted:

- Stage I mining and HMS processing is shown in 2010, followed by an increased production scale for Stage II commencing in 2011.
- Stage II mining is completed in 2018, whilst SX-EW processing continues until 2023, drawing down on medium grade and low grade stockpiles.
- The mining sequence was forced to target the highest grade ore whilst honouring the staged pit development, and limiting the vertical advance rate to no more than 25 m per year.
- There is a six month Stage II production ramp up period in the first half of 2011.
- Thereafter, an output Stage II production constraint of 60,000 tonnes of copper metal per annum was imposed.
- At maximum production, the design plant throughput and recovery was set at 1.6 million tonnes per annum (with a +/-5% tolerance) and 92.14%, respectively.
- Stockpiled HMS floats and effluent can be re-processed to extend the Stage II operation to 2024.

Figure 3.1 Production Schedule Profile



Between the January 2010 News Release (Anvil, January 2010) and this Technical Report, changes were made to the Stage I and Stage II production schedules resulting in a minor reduction to the total copper metal recovered in Stage II. Table 3.5 shows the difference.

Table 3.5 Schedule Changes Since January 2010 News Release

Description	News Release	This Technical Report
Stage I mining/reclaim (million tonnes)	0.73	0.73
Average grade (TCu%)	3.71	3.84
Stage II mining/reclaim (million tonnes)	20.08	20.08
Average grade (ASCu%)	3.31	3.30
Stage II processing (million tonnes) ¹	21.86	21.82
Average processed grade (%ASCu)	3.23	3.23
Cu metal produced (tonnes)	648,525	647,075

1. Total ore processed exceeds the total ore mined and reclaimed due to additional reclamation from stockpiled Stage I HMS floats and effluent.

3.8 Operating and Capital Costs

Table 3.6 lists the Stage II unit cash operating costs, in real terms averaged over the life of the Project.

Table 3.6 Stage II Unit Operating Cost Summary Averaged Over the Life of Project

Average SX-EW Operating Costs	Units	LOM Real Cash Costs
Mining Costs	\$/lb	0.14
Fixed Technical Services Costs	\$/lb	0.01
Processing Costs	\$/lb	0.34
Maintenance Costs	\$/lb	0.07
Administrative Costs	\$/lb	0.11
Cash Cost at Mine Gate	\$/lb	0.67
Transport and Marketing Costs	\$/lb	0.20
Export Duties	\$/lb	0.02
Total C1 Cash Costs	\$/lb	0.89
Royalties Paid	\$/lb	0.11
Depreciation and Amortisation	\$/lb	0.33
Total C3 Operating Costs	\$/lb	1.33

As at Q4 2009, approximately \$199 million (\$179.5 million spent, \$19.5 million committed) of the revised budgeted capital cost of \$395 million had been invested. The increase in capital cost from the estimate reported in the March 2008 Technical Report is a result of several factors including:

- revised Ausenco estimates;
- redundancy payments to Kinsevere Stage II personnel;
- payment of demobilisation costs to contractors;
- charges associated with the cancellation of the engineering and design works;
- penalties in connection with the cancellation and suspension of orders;
- provision for re-establishment of contracts and remobilisation; and
- some escalation primarily due to the contract now being a Lump Sum Turn Key (LSTK) contract rather than an engineering, procurement and construction management (EPCM) contract.

The breakdown of Stage II capital costs is provided in Table 3.7 for the LSTK plus other costs. Before payment allowances to Gecamines and Stage II creditors, the

subtotal capital cost (ie, remaining cost) is \$195.86 million. When added to the \$199 million of funds already spent, the total cost in the Project cash flow model is approximately \$395 million.

Table 3.7 notes the “Owners” costs as being approximately \$52 million and the Owners contingency as being \$15 million. These figures were revised for the January 2010 News Release (Anvil, January 2010) and reported as \$58 million and \$12 million, respectively. Table 3.7 shows the LSTK cost as \$128.9 million, whereas in the January 2010 News Release the value of the LSTK contract was noted as \$130 million, bringing the reported subtotal capital cost (ie, remaining cost) to \$200 million.

Additional capital expenditure is included in the cash flow model for construction of the tailings dam embankments, sustaining and closure cost provisions.

Table 3.7 Stage II Capital Costs

CAPITAL COSTS	USD
LSTK COSTS	\$128,870,046
OWNERS COSTS	
Capital purchases including transport and duties	\$21,791,812
Insurance & commissioning spares	\$13,408,377
Commissioning	\$10,060,328
During construction expenses	\$3,983,004
Subcontractors	\$207,686
Services contract - complete design	\$2,534,143
	\$51,985,350
OWNERS CONTINGENCY	
Contingency on Owners Cost	\$5,000,000
Contingency on LSTK Contract	\$10,000,000
	\$15,000,000
ALLOWANCES	
payment to Gecamines	\$5,000,000
Stage II creditor payments	\$6,400,000
	\$11,400,000
TOTAL IN CASH FLOW MODEL	\$207,255,396

3.9 Economic Evaluation

An economic model has been applied to Kinsevere Stage II on the basis of the outlined Proven and Probable Mineral Reserve estimates and the production schedule summarised in Item 3.7. A list of the key inputs and parameters is provided in Table 3.8. The 2007 Feasibility Study parameters are provided for comparison. A summary of the key economic results for the Project is detailed in Table 3.9.

The January 2010 News Release (Anvil, January 2010) advised the following economic results, based on the exclusion of sunk costs incurred before 1 January 2010:

- Stage II NPV at 95% equity = \$683 million;
- Stage II Internal Rate of Return = 79%; and
- Payback period of less than 2.5 years.

These results differ slightly from the key outcomes listed in Table 3.9 due to the production schedule changes made between the date of the January 2010 News Release and this Technical Report, as follows:

- Stage II NPV at 95% equity = \$668 million;
- Stage II Internal Rate of Return = 73%; and
- Payback period of less than 2.5 years.

The economic results when sunk costs incurred before 1 January 2010 are included are also shown below and in Table 3.9:

- Stage II NPV at 95% equity = \$457 million;
- Stage II Internal Rate of Return = 34%; and
- Payback period of less than 3.6 years.

Table 3.8 Economic Evaluation Parameters

Description	Unit	2007 Feasibility Study Base Case	2010 Base Case
Operations Life			
Mining duration: Stage I, Stage II	Years	2.25, 10.75	1.0, 8.0
SX-EW duration (incl floats/effluent processing)	Years	16.5	14.00
Mining			
Ore mined	million t	25.74	18.41
Waste mined	million t	43.68	22.68
Total material mined	million t	69.42	41.10
Processing			
Stage I HMS - ore processed	million t	1.00	0.73
Stage I HMS – Cu recovered	Tonne	41,937	16,799
Stage II SX-EW - ore processed (inc floats/effluent)	million t	26.23	21.82
Stage II SX-EW – Cu recovered	tonne	842,487	647,075
Total Tonnes of Cu metal produced	tonne	882,484	663,874
Operating Costs			
Mining + Technical Services	\$/lb	\$0.14	\$0.19
Processing - Stage I	\$/lb	\$0.015	-
Maintenance - Stage I	\$/lb	\$0.002	-
Transport and Marketing - Stage I	\$/lb	\$0.01	-
Processing - Stage II	\$/lb	\$0.26	\$0.26
Maintenance - Stage II	\$/lb	\$0.06	\$0.06
Transport and Marketing - Stage II	\$/lb	\$0.18	\$0.22
Administration	\$/lb	\$0.08	\$0.08
Depreciation and Royalties	\$/lb	\$0.24	\$0.37
Total Operating Cost, Stage I and Stage II	\$/lb	\$0.98	\$1.19
Capital Costs			
Capital cost - Stage I	\$ million	\$4.22	-
Capital cost - Stage II – Expended (Sunk)	\$ million	-	\$199.09
Capital cost - Stage II - Remaining	\$ million	-	\$195.86 ¹
Capital cost - Stage II - Total	\$ million	\$241.50	\$394.95
Tailings Storage Facility	\$ million	\$39.83	\$59.49
Drilling and sampling of stockpiles	\$ million	\$1.70	\$0.80
Sustaining capital	\$ million	\$6.93	\$36.49
Closure costs	\$ million	\$8.19	\$23.45
Total Capital Cost	\$ million	\$302.37	\$515.27
Other Costs			
Corporate charges	\$ million	\$12.74	\$15.00
Revenue			
Long term Copper Price, Average Copper Price	\$/lb	\$1.43, \$1.53	\$2.24, \$2.44
Total copper revenue	\$ million	\$2,984.58	\$3,475.26
Charges			
Royalty - Gecamines	\$ million	\$55.49	\$86.88
Royalty - DRC	\$ million	\$54.79	\$63.83

1. Excludes final payment to Gecamines and Stage II creditor payments

Table 3.9 Key Economic Results

Description	Unit	2007 Feasibility Study Base Case	2010 Base Case
Economic Results, Excluding Sunk Costs			
Discount Rate	%	10%	10%
Stage I+II NPV	\$ million	\$281.22	\$720.47
Stage I NPV	\$ million	\$83.32	\$17.49
Stage II NPV	\$ million	\$197.91	\$702.97
Stage II IRR	%	25%	73%
Stage I+II NPV , 95% Equity	\$ million	\$267.16	\$684.44
Stage I NPV, 95% Equity	\$ million	\$79.15	\$16.62
Stage II NPV, 95% Equity	\$ million	\$188.01	\$667.82
Economic Results, Including Sunk Costs			
Discount Rate	%	10%	10%
Stage I+II NPV	\$ million	\$281.22	\$498.72
Stage I NPV	\$ million	\$83.32	\$17.49
Stage II NPV	\$ million	\$197.91	\$481.23
Stage II IRR	%	25%	34%
Stage I+II NPV , 95% Equity	\$ million	\$267.16	\$473.79
Stage I NPV, 95% Equity	\$ million	\$79.15	\$16.62
Stage II NPV, 95% Equity	\$ million	\$188.01	\$457.17

Due to the relatively high average oxide Mineral Resource grade at Kinsevere, the key economic outcomes listed in Table 3.9 are highly sensitive to variations in copper price, with every 10% change in the copper price resulting in a change of \$120 million in the NPV.

Movements in other key variables have a less significant impact. For example, a 10% change in capital expenditure results in only a 2% change in the NPV, whilst a 10% change in the operating costs results in a 7% change in the NPV.

Excluding the floats and effluent (as opportunistic feed) from the Stage II production profile reduces the Base Case NPV by only 4%.

3.10 Conclusions and Recommendations

3.10.1 Sampling Data and Measurements

Investigations in 2008 revealed that density applied in the Mineral Resource models up to 2007 had been based on density measurements made from core during earlier drilling programmes. Unfortunately these density measurements were made on core that was selectively sampled (hard core as opposed to decomposed core), thereby introducing a bias in the density measurements.

As a result of these investigations, the procedures for continuing density measurements with drill core were changed. For subsequent Mineral Resource modelling and estimation for this Technical Report, newly measured density from 19 drillholes (about 1,700 samples) plus four in-pit bulk sample estimates were used to assign density in the models.

Furthermore, assay results for the final Phase IV (2008) drilling samples were received during April and May 2009. QAQC results from these sample batches were analysed for sub standard sampling and assaying practice. Of the 45 sample batches submitted, nine batches failed due to poor accuracy (standards) and contamination (blanks) and twelve due to poor precision (duplicates). Analysis revealed that these QAQC issues were likely due to deficient Anvil preparation and dispatch procedures at the time of operations shutdown and staff retrenchments.

The defective data is **not included in the Mineral Resource** estimate compiled for this Technical Report and hence is not material to the Mineral Reserve inventory, production schedules, nor to the cash flow modelling outcomes.

A resampling programme which is currently in progress will enable this data to be included in the final oxide Mineral Resource estimate before commencement of Stage II production.

3.10.2 Mineral Resource Modelling and Estimation

The results of the exploration drilling up to 2008 (Phase IV programme) have contributed to the geological model, and the extent of mineralisation. The current geological model is considered robust as a basis for Mineral Resource estimation, and is supported by sample grid spacings which range from 25 m to approximately 100 m.

Continuing structural mapping is required to improve domain definition in the block models. Additional data generated from planned grade control drilling, together with the results of other planned geological research (eg, SiroVision structural analysis), will lead to a future refinement or modification to the geological and block models.

Since the 2007 estimate, the latest Mineral Resource estimate has been reduced by 9%, a total of 150,000 tonnes of Cu. Reductions are associated with mining depletion, density corrections and to a lesser extent grade. The reductions are offset by increases in the Indicated and Inferred Mineral Resources for the sulphide mineralisation.

3.10.3 Mineral Reserve Estimation

The updated Mineral Reserve estimate is based on the February 2009 Mineral Resource model, and as subsequently adjusted for mining depletion to the end of December 2009. The Mineral Reserve estimate relates specifically to the Measured and Indicated Mineral Resource categories and oxide mineralisation only.

The Mineral Reserve estimate summarised in Item 3.6 differs significantly from that previously reported in the Feasibility Study (AMCK, December 2007) and the 2008 Technical Report (Anvil, March 2008). A comparison between the two estimates is provided in Table 3.10.

Table 3.10 Mineral Reserve and Associated Mine Waste Comparisons

Estimates	Ore Tonnes (Mt)	TCu (%)	Contained TCu (kt)	Waste Tonnes (Mt)
Feasibility Study (2007)	25.75	4.03	1,037.5	43.68
Mineral Reserve Update (2009)	18.41	4.14	762.2	22.68
Difference	-7.34 -29%	3.75	-275.3 -27%	-21.00 -48%

The reasons for the difference can be listed as follows:

- changes to the Mineral Resource model as described in Item 3.5; and
- a different approach to the selection of the ultimate pit from the suite of optimal pit shells.

In relation to the second point listed above, it is worth noting that several years of mining have been subtracted from the Stage II mine life considered in the Feasibility Study. The ore mined in these years in fact contributed little extra value to the Project, considering the extra waste that needed to be mined.

3.10.4 Stage II Implementation

One of the more significant risks to the Project Implementation Plan includes not achieving the LSTK schedule and being able to commence Stage II production from January 2011. Economic sensitivity analysis indicates that a three month delay in commissioning with associated increased capital expenditure could reduce the Project NPV by around 13%.

In order to manage this risk, the Company has appointed an on-site “Client Representative” to monitor budget and timeframes and also implement remedial actions to ensure that the Project is completed on time.

3.10.5 Groundwater Abstraction

The base of the current Tshifufia and Tshifufiamashi Pits are coincident with the regional groundwater level. Success of the groundwater bore drilling and equipping programme, which is currently in progress, is critical to enabling the Stage II mining programme to proceed below the current groundwater levels.

Furthermore, efforts in 2010 need to focus on improving the availability of the existing dewatering pumps and re-engineering the groundwater disposal and discharge system.

3.10.6 Environmental Compliance

A number of deficiencies are apparent in the Company’s Environmental and Social Impact Assessment (ESIA), and further work is required in 2010 to proceed towards full Equator Principles (EP) and International Finance Corporation (IFC) Performance Standards.

3.10.7 Recommendations for Further Work

The following work programmes are recommended for the Kinsevere Project during 2010, in respect of aspects relating to the Mineral Resource and Mineral Reserve:

- Phase IV oxide resampling and assaying to enable a further Mineral Resource estimate update in 2010;
- completion of Phase IV sulphide sampling and assaying, for the same reason;
- further density measurements;
- grade control drilling programmes, based on inclined RC drilling and sampling, rather than short-length blast hole sampling;
- Mineral Resource update;
- development of routine grade control modelling in support of short-medium term mine planning;
- implementation of a production tracking system to provide more reliable data for reconciliation;
- continuing mine geotechnical engineering and a Life of Mine plan, Mineral Reserve update;
- groundwater bore drilling, aquifer testing and equipping (critical item);
- re-engineering of the groundwater discharge system;
- groundwater drawdown and quality monitoring programmes;
- geotechnical investigations for the Stage II TSF;
- acid rock drainage (ARD) testwork in relation to waste rock associated with oxide and sulphide copper mineralisation;
- surface water management planning;
- waste dump optimisation and planning;

- review of equipment requirements for Stage II mining and subsequently, tendering and award of a Stage II mining contract;
- advancement of environmental compliance studies;
- updates to the Project closure plan; and
- a number of potential Project enhancement studies.

Table 3.11 summarises the existing allowances and further cost provisions that are recommended for these work programmes.

Table 3.11 Summary of Cost Estimates for Recommended Work Programme

Item	2010 Budget Allowance		Required Additional		Total Provision	
	Operating	Corporate	Operating	Capital	Operating	Corp+Cap
Oxide Resampling and Assaying	\$105,800				\$105,800	
Sulphide Assaying	\$106,000				\$106,000	
Density Measurements						
Grade Control Drilling		\$640,000		\$437,950		\$1,077,950
Data Management						
Grade Control Modelling	\$50,000				\$50,000	
Photogrammetry	\$40,000				\$40,000	
Production Tracking and Reconciliation	\$170,000					\$170,000
Updated Mineral Resource Modelling	\$35,000		\$55,000		\$90,000	
Mine Geotechnical Engineering	\$20,000				\$20,000	
Updated LOM Plan	\$75,000				\$75,000	
Short-Medium Term Planning	\$25,000		\$44,000		\$69,000	
Groundwater Bore Drilling, Aquifer Testing and Equipping		\$591,646		\$558,956		\$1,150,602
Groundwater Drilling Supervision and Modelling		\$129,254				\$129,254
Groundwater Discharge Engineering and EIA				\$319,456		\$319,456
Groundwater Drawdown Monitoring	\$90,000				\$90,000	
Groundwater Quality Monitoring			\$132,260		\$132,260	
Geotechnical Investigations for the TSF				\$100,000		\$100,000
Surface Water Management Planning						
ARD Testwork	\$60,000				\$60,000	
Waste Dump Planning and Optimisation	\$25,000				\$25,000	
Stage II Mining Contract	\$13,600				\$13,600	
Environmental Compliance	\$102,000		\$10,000		\$112,000	
Social Development	\$18,000				\$18,000	
Closure Plan	\$16,000				\$16,000	
Project Enhancement Studies						
Equipment Review	\$13,600				\$13,600	
ROM pad earthworks				\$250,000		\$250,000
Leaching Studies				\$1,100,000		\$1,100,000
	\$965,000	\$1,360,900	\$241,260	\$2,766,362	\$1,036,260	\$4,297,262

ITEM 4 Introduction

This Technical Report is prepared for the shareholders of Anvil Mining Ltd.

The purpose of this Technical Report is to document the updated Mineral **Resource and Mineral Reserve** estimated for the property, and also to provide an **updated commentary on production planning, economic evaluations and further ancillary studies** associated with the planned Stage II development of a solvent extraction electrowinning (SX-EW) processing operation. The updates relate specifically to further and additional technical work completed since the Kinsevere Copper Project Stage II Feasibility Study (the Feasibility Study) (AMCK, December 2007) and a corresponding NI 43-101 Technical Report filed in March 2008 (the 2008 Technical Report) (Anvil, March 2008).

This Technical Report follows a News Release dated 27th January 2010 (Anvil, January 2010).

Whilst the Feasibility Study remains conceptually relevant, it is now superseded in detail due to technical work completed in 2008 and 2009 on the Mineral Resource, Mineral Reserve, Life of Mine production schedules and Cash Flow Model. Continuing technical work is also required in 2010 before Stage II commissioning on the now planned date of 1 January 2011.

Development and technical aspects which have not changed since the Feasibility Study and 2008 Technical Report are:

1. Stage II SX- EW processing is at 1.6 million (oxide and transitional) ore tpa, 92.14% overall recovery and yielding up to 60,000 tpa of copper cathode.
2. Feed into the SX-EW plant includes stockpiled heavy media separation (HMS) floats and effluent (towards the end of the Project life).
3. Although the deposits contain cobalt mineralisation, the metal is not recovered and no value is assigned to it.

Development and technical aspects which **have changed since the Feasibility Study** and 2008 Technical Report are:

1. The Feasibility Study was premised on the basis that when the Stage II Project comes on line, the Stage I HMS plant would shut down (still the case) and the electric arc furnace (EAF) would be converted to treat Dikulushi sulphide concentrate (since proven to be unfeasible).
2. Stage II production was originally envisaged to commence in the first half of 2009, but is now scheduled for January 2011.
3. Construction of the Stage I EAF was delayed and commissioning of one of the two proposed furnace trains was not completed until August 2008.
4. In November 2008, the Company placed the Stage II construction works on hold pending improvements in the global financial and commodity markets, and also until Project funding was in place. Construction resumed in December 2009.
5. Stage I HMS processing and open pit mining was also halted in November 2008, but resumed respectively in April 2009 and August 2009 (ie, with stockpiled ROM ore providing the plant feed between April and August).
6. HMS concentrate will now be the saleable product (at 40% to 50% of London Metal Exchange (LME) price payable, depending on the copper price) of the Stage I plant until December 2010.

7. The single furnace of the EAF was shut down and placed on permanent care and maintenance in March 2009. The second furnace was never completed.
8. The timing for Project funding was adversely impacted by the DRC Government's Mining Review, which in the case of the Kinsevere Project, was not concluded until January 2009.
9. The Company successfully completed its negotiations with the DRC Government, resulting in the requirement to pay an additional *pas de porte* (entry premium) fee of \$15 million in two instalments.

The status of technical work which **had been completed for the Feasibility Study** and 2008 Technical Report was as follows:

1. A Phase III infill (oxide) Mineral Resource drilling programme had been completed on the Tshifufia and Tshifufiamashi deposits resulting in a Mineral Resource estimate completed in September 2007. The drilling programme for the Kinsevere Hill extension (ie, Kilongo), however, was not completed in time for the Feasibility Study and hence, the Feasibility Study was reliant on a Kinsevere Hill model dated December 2005.
2. An oxide Measured and Indicated Mineral Resource of 33.5 Mt at 3.7%TCu was reported.
3. A Mineral Reserve estimate was prepared according to conventional methodology comprising pit optimisations, geotechnical engineering, detailed pit designs and Life of Mine Production scheduling. The reported oxide Proven and Probable Mineral Reserve was 25.7 Mt at 4.0%TCu.
4. Metallurgical testwork for Stage II had been completed and the processing flowsheet finalised, inclusive of milling in raffinate (MIR) and direct tailings disposal (DTD) enhancements.
5. Initial groundwater investigations had been completed in September 2007, with a number of pit dewatering bores located around the pit perimeter.
6. The Stage II high density polyethylene (HDPE) lined tailings storage facility (TSF) design had been completed, based on four perimeter walls covering an area of 110 ha. The northern and eastern walls of the TSF were designed as downstream raised walls built from compacted waste hauled from the Tshifufia Pit over the life of the operation. The western and southern walls were designed to be upstream raised.
7. The location and extent of the Stage II waste rock dump had been optimised and designed, but had not taken account of the need for waste to be used in TSF embankment construction.
8. The Environmental Impact Assessment (EIA) (African Mining Consultants, 2007) had been completed and approved by the Government.
9. An interim reimbursable agreement had been put in place with Ausenco Engineering Ltd (Ausenco) to provide Engineering, Procurement, Construction and Management (EPCM) services for the development of the Stage II process plant and associated utilities and services.
10. The Stage II plant site had been cleared and foundation excavations had commenced.
11. The total capital costs over the life of the mine were at the time estimated to be \$315 million, incorporating EPCM controlled costs, owner controlled costs, plus sustaining costs, closure, corporate and import charges.

Technical work **completed subsequent to the Feasibility Study** and 2008 Technical Report was as follows:

1. The drilling programme for the Kinsevere Hill extension (ie, Kilongo) was completed and an updated Mineral Resource model and estimate was available in June 2008.
2. Phase IV infill drilling in the oxide horizons at Tshifufia and Tshifufiamashi was completed with final assay results received in April and May 2009. Phase IV sulphide drilling was partially completed before suspension of operations at Kinsevere in late 2008.
3. QAQC results from the final oxide sample batches were analysed and showed that of the last 45 batches submitted, nine batches failed due to poor accuracy and contamination, and twelve due to poor precision.
4. Accordingly, the Tshifufia and Tshifufiamashi model and Mineral Resource estimate update could only be completed on reliable data received up to December 2008.
5. In the period since the Feasibility Study, a significant discrepancy in oxide rock density had become apparent, resulting in a reduction of the average Mineral Resource model rock density from around 2.3 t/m³ to 1.8 t/m³.
6. A total oxide Measured and Indicated Mineral Resource of 26 Mt at 3.8%TCu (3.2%ASCU) was reported in a News Release dated 19th February 2009, inclusive of density corrections, mining depletion since the Feasibility Study estimate and additional Mineral Resource delineated from Phase IV drilling.
7. An updated Proven and Probable Mineral Reserve estimate was prepared and reported in a News Release dated 27th January 2010, as 20.8 Mt at 3.9%TCu (3.3%ASCU), reflecting resource model updates, mining depletion since the Feasibility Study, available stockpile inventories, revised optimisation parameters, revised staged pit designs and updates to the Stage I and Stage II mining and processing plan.
8. Detailed hydrogeological investigations were completed in late 2008, resulting in a recommendation for the drilling and equipping of five new deep groundwater abstraction bores. Further modelling indicated that as a worst case scenario, the abstraction rate may have to increase almost threefold. Due to the suspension of operations in late 2008, however, the drilling of these holes was postponed. Continued pumping from the pit sumps was maintained.
9. The design of the Stage II TSF was changed to incorporate downstream raised walls on all four sides. The implication of this design change is that essentially all of the available oxide waste rock mined will be required for TSF embankment construction.
10. A technical report and EIA addendum relating to the TSF design change, and justification for the proposed liner design, was submitted to the DRC Government's *Direction de Protection de l'Environnement Minier* (DPEM), (Anvil, July 2008).
11. The EIA addendum was approved (ie, Agreement in Principle) in August 2008 with recommendations for further analysis of hydraulic conductivity before proceeding with construction (DPEM, August 2008).
12. Further analysis was completed and reported in a subsequent letter to the DPEM (Anvil, September 2008), with approval granted subject to implementation of a tailings management plan (DPEM, October 2008).
13. As a requirement for Project funding, Independent Engineer SRK Consulting (UK) Ltd (SRK) was appointed by Standard Bank Plc, culminating in the provision of an Independent Review report for the Project in September 2009 (SRK, 2009).
14. As a requirement for International Finance Corporation (IFC) compliant Project funding, a comprehensive Environmental and Social Impact Assessment (ESIA) was completed in October 2009 (Knight Piésold, 2009). An Environmental and Social Action Plan (ESAP) was also prepared (Anvil, November 2009).

15. In August 2009, the Company reached an agreement with Trafigura Beheer B.V. (Trafigura) on a combined debt and equity financing arrangement amounting to \$200 million.
16. Whilst Stage II plant construction activities were put on hold in late 2008, negotiations resumed with Ausenco for works resumption in late 2009, but according to a Lump Sum Turn Key (LSTK) contract. The contract was awarded in January 2010.
17. An Implementation Plan has been prepared to comprehensively outline the work scopes and timelines required of Ausenco and Anvil in order to now commission the Stage II plant in January 2011 (Anvil, February 2010).
18. The estimated Stage II capital costs have risen from \$277 million to \$395 million, whilst the total capital costs (ie, LSTK and owner controlled) for the Project have risen from \$302 million to \$515 million (ie, excluding corporate charges and import duties).

The remaining technical work to be **completed before Stage II commissioning** can be summarised as follows:

1. The Phase IV oxide resampling programme (ie, on QAQC failed sample batches) will be completed.
2. A grade control drilling programme, involving reverse circulation (RC) drilling over multiple benches, will be carried out and a grade control model will be prepared.
3. The Phase IV resampling and grade control drilling data will be used to provide a final updated Mineral Resource model and estimate.
4. The waste dump and TSF waste requirements will be re-optimised, bearing in mind that oxide waste may be required to encapsulate potentially acid forming rock during future Stage III (sulphide) mining and processing. As part of this programme, acid rock drainage (ARD) testwork will be required.
5. Primary mining fleet requirements are to be reviewed and following a tender period, a contract let for Stage II mining.
6. A production tracking system will be put in place, enabling reconciliations to be carried out.
7. The Mineral Reserve estimate will be updated, based on the final Mineral Resource model, waste dump/TSF optimisations, revised mining concepts and costs, and updated Life of Mine production schedule.
8. Work on the groundwater drilling programme will be reinstated.
9. A site water management plan will be completed.
10. Various ESAP action items will be continued.

In respect of the new technical information listed above, this updated Technical Report complies with the reporting requirements of the Canadian National Instrument (NI) 43-101 guidelines: "Standards of Disclosure for Mineral Properties", and with the "Joint Ore Reserves Committee" (JORC) Code (JORC, 2004).

ITEM 5 Reliance on Other Experts

This Technical Report is based on **information updated since the Feasibility Study** (AMCK, December 2007) **and the subsequent 2008 Technical Report** (Anvil, March 2008).

The Technical Report draws on advice and information from appropriately qualified consultants as listed below, in support of updates to that Study and Technical Report. Anvil has relied on such advice and information for this Technical Report, under the assumption that the estimates, concepts, designs and conclusions supplied by these consultants have been prepared and/or reviewed by Qualified Persons.

Expert consulting advice and information was provided by:

Turner Mining and Geotechnical Pty Ltd (TMG):

- geotechnical engineering to support open pit mine design.

CSA Global Pty Ltd (CSA):

- Mineral Resource modelling and estimation for the Kinsevere Hill and Kilongo deposits.

DumpSolver Pty Ltd (DumpSolver):

- open pit optimisations.

A & J Cameron and Associates Pty Ltd (AJC):

- mine design, production scheduling and Mineral Reserve estimation

Azure Capital (Azure):

- cash flow modelling and financial analysis.

Knight Piésold (Pty) Ltd (KP):

- site geotechnical assessment and engineering analysis for the proposed tailings impoundment area
- hydrogeological studies in relation to required pumping rates for drained pit slope conditions and consequent groundwater drawdown impacts
- preparation of the Project Environmental and Social Impact Assessment (ESIA), as a condition for Project debt financing.

The remaining portions of this updated Technical Report were prepared by Anvil, including the review and revision of Project operating and capital cost estimates.

The principal report authors and **Qualified Persons** are:

Dr Geoffrey Booth: Dr Booth has a BSc (Hons) and MSc in Geology from the University of Toronto (1979,1983) and a Ph.D in Applied Geology and Geochemistry from the University of New South Wales (1988). He is a Fellow of the Australasian Institute of Mining and Metallurgy and a Certified Practising Geologist. He has visited Kinsevere on multiple occasions in 2008 to undertake audits and review mining operations. Dr Booth was formerly employed by Anvil as Feasibility Study Manager and takes responsibility for the drilling, sampling method/approach, sample preparation, analyses and security aspects of this Technical Report.

Mr Anthony Cameron: Mr Cameron of A & J Cameron and Associates has a BEng (Mining Engineering) degree from the University of Queensland (1988) and is a Fellow of the Australasian Institute of Mining and Metallurgy. He visited the Kinsevere site in March 2008 and takes responsibility for the Mineral Reserve reporting and production scheduling aspects of this Technical Report.

Mr Gerry Fahey: Mr Fahey of CSA Global Pty Ltd, has a BSc (Hons) in Geology from the National University of Ireland (1975), is a Member of the Australasian Institute of Mining and Metallurgy and a Member of the Australian Institute of Geoscientists. He visited the Kinsevere deposits in February 2005 and January 2008 and supervised the Mineral Resource estimation process. Mr Fahey takes responsibility for the Mineral Resource reporting aspects of this Technical Report.

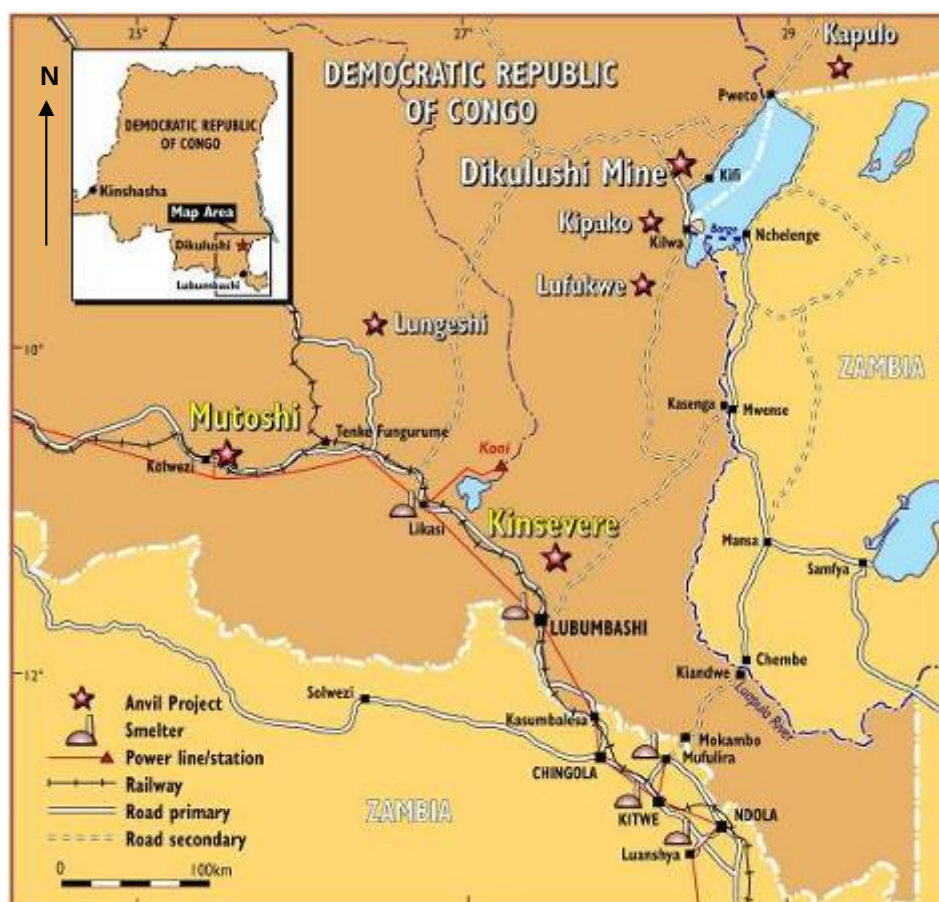
Mr Michael Lawlor: Mr Lawlor has a BEng (Hons) degree in mining engineering from the Western Australian School of Mines (1985), an MEngSc degree in mining geomechanics from the James Cook University of North Queensland (1993) and other post-graduate qualifications in mineral economics and project management. Mr Lawlor is a Fellow of the Australasian Institute of Mining and Metallurgy, and a Member and Chartered Engineer of The Institution of Engineers Australia. He visited the Kinsevere site in 2007, 2008 and 2009 to review mining operations and project development progress. Mr Lawlor is employed by Anvil as the Manager, Group Technical Services and takes responsibility for the overall technical content of this Technical Report, and specifically for those items not addressed by the other Qualified Persons (QPs).

A principal report author who is ineligible to act as a Qualified Person, but who nevertheless completed estimations and commentary relating to the Tshifufia and Tshifufiamashi Mineral Resource is Mr David Gray, who is Anvil's former Principal Resource Geologist. Mr. Gerry Fahey supervised David Gray in his estimates and commentary on the Tshifufia and Tshifufiamashi Mineral Resources.

ITEM 6 Property Description and Location

The **Kinsevere Project** is located in the Katanga Province, in the southeast of the Democratic Republic of Congo (DRC). It is **situated approximately 27 kilometres north of the provincial capital, Lubumbashi** (Figure 6.1), at latitude S 11° 21' 30" and longitude E 27° 34' 00".

Figure 6.1 Location of the Kinsevere Project

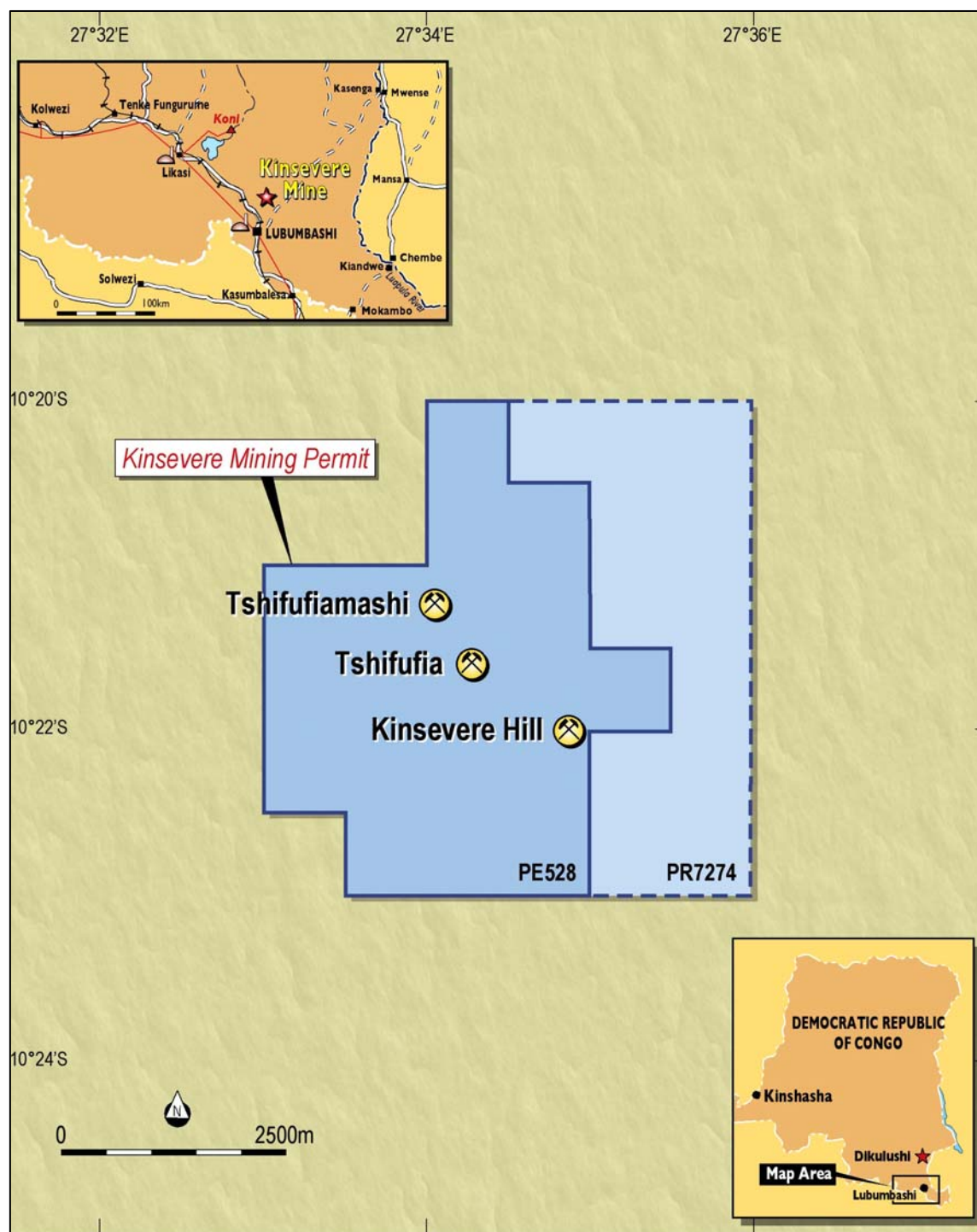


The mineral rights for the Project tenement, PE 528, covering an area of approximately 5.94 km² (Figure 6.2), are held by *La Générale des Carrières et des Mines* (Gécamines), the DRC state owned copper mining company.

Anvil, through its 95% owned subsidiary AMCK Mining s.p.r.l. (AMCK), has a *Contrat d'Amodiation*² (Lease Agreement) with Gécamines to mine and process ore from the Kinsevere Project until 2024, followed by an automatic 15 year extension. The PE528 permit covers the three major deposits (Figure 6.2) of Tshifufiamashi, Tshifufia and Kinsevere Hill/Kilongo. The Tshifufiamashi deposit is sometimes referred to as "Mashi". The Tshifufia deposit comprises Tshifufia North, Central and South, and is sometimes referred to collectively as "Central". The Kilongo deposit extends north westwards from the Kinsevere Hill deposit and is sometimes referred to as "Kinsevere Hill Extended".

² a *Contrat d'Amodiation* is provided for under the DRC Mining Code, enacted by law No 007/2002 of July 11, 2002.

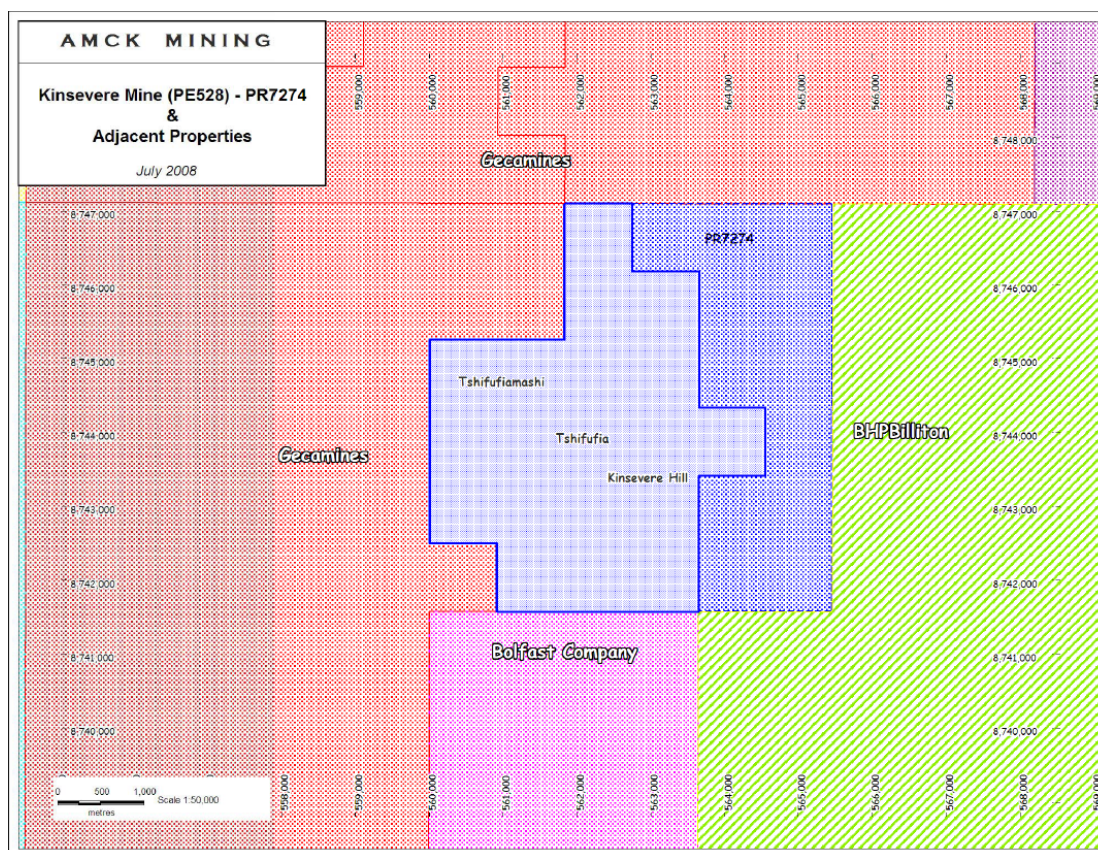
Figure 6.2 Location of the Kinsevere Mining Permit



Pursuant to an Option Agreement, AMCK has earned the right to acquire PR 7274 which is located where shown on Figure 6.2. AMCK has exercised this right, however the process of transfer of title to AMCK has not yet been completed.

PE 528 is defined by graticular blocks (Figure 6.3), and encloses the area for the planned mines, process plant, tailings storage facility and other unmovable infrastructure. In January 2007, Gécamines made an application to the *Cadastre Minière* (CAMI) to have PE 528 extended to cover the then recently defined extensions to mineralisation and provide space for the mine infrastructure (tailings dam, Stage I electric arc furnace, accommodation camps, etc).

Figure 6.3 Exploitation Permit PE 528



This application was approved and the coordinates for the extended PE 528 are presented in Table 6.1. In order to qualify for this extension, AMCK provided Gecamines with a copy of the Stage I Development Feasibility Study (AMCK, May 2006) and the Environmental Impact Assessment (African Mining Consultants, July 2007), both of which were submitted to CAMI.

The property is not subject to any environmental liabilities. Following submission of the Environmental Impact Assessment report prepared by African Mining Consultants (AMC) (July 2007), Government approval of the Kinsevere Copper Project, Stages I and II, was issued by CAMI on 15 October 2007. Approval of a variance to the design and operation of the Stage II tailings storage facility was issued by the Directorate for the Protection of the Mining Environment (DPEM) on 28 October 2008 (DPEM, October 2008).

In connection with the debt financing of the Stage II tailings storage facility, the Company anticipates AMCK will be required to grant security over its interest in the Kinsevere Copper Project.

An Environmental and Social Impact Assessment (ESIA) was prepared by KP (October, 2009) as a condition of Project debt financing. Under debt financing circumstances, the lending institution must ensure that the Company complies with the internationally recognised Equator Principles (EP) and the International Finance Corporation (IFC) Principal Standards (PS).

The ESIA document is intended to compliment the assessment information presented in the 2007 EIA. It does not overwrite any government approval or conditions of approval in the EIA of 2007 or any other regulatory requirements of the DRC Mining Code.

Table 6.1 Coordinates of the Extended PE 528

Point	Longitude	Latitude
A	27°33'00"	11°22'30"
B	27°33'00"	11°21'00"
C	27°34'00"	11°21'00"
D	27°34'00"	11°20'00"
E	27°34'30"	11°20'00"
F	27°34'30"	11°20'30"
G	27°35'00"	11°20'30"
H	27°35'00"	11°21'30"
I	27°35'30"	11°21'30"
J	27°35'30"	11°22'00"
K	27°35'00"	11°22'00"
L	27°35'00"	11°23'00"
M	27°33'30"	11°23'00"
N	27°33'30"	11°22'30"

ITEM 7 Accessibility, Climate, Local Resources, Infrastructure & Physiography

Kinsevere is situated on the Central African Plateau at an **elevation of 1,200 metres**. The surrounding area gently slopes to the north towards the Kifumashi River, though more resistant parts of the Lower Roan stratigraphy (Item 9.1) form southeast-northwest trending, low, steep-sided ridges.

Original vegetation in the Project area generally consisted of Riparian and Mungu (Acacia) vegetation (Figure 7.1 to Figure 7.3), while areas close to the Kifumashi River have been cleared by slash and burn technique for kasava, maize and vegetable cultivation.

Kinsevere has a **distinct dry and wet season**, with the wet season commencing in October and generally finishing by April. The **average rainfall of the area is approximately 1,100 mm**, although this can range from 650 mm to 1,500 mm. Approximately 90% of the rainfall occurs during the wet season. Future mining activities are not expected to be significantly affected during the wet season.

Temperatures are generally mild and **vary between 17°C and 26°C**, but it can drop to as low as 5°C during the night in July and August.

The nearest major population centre is Lubumbashi, the capital of the Katanga Province, which is situated approximately 27 km south of the property. Several small villages are located close to the property, for example Kinsevere village (on the Tshifufiamashi River, 3 km to the north of the property), and the Muombe-Bushinda village (adjacent to the Muombe River, 13 km west of the property). Local inhabitants are dependant on subsistence agriculture or production of charcoal for revenue. Several hundred artisanal miners were previously active at Kinsevere before the Project commenced. Currently no artisanal miners are active in the area.

The Kinsevere property was previously accessed via a 22 km, unsealed road, which branches off the sealed Lubumbashi–Likasi national highway at the village of Kawama (11 km northwest of the Lubumbashi International Airport turnoff). This unsealed road was refurbished as part of the Anvil Stage I Development Project. **A new unsealed and wider access road has been constructed alongside the site power line route.**

There is **good access to the national power grid** with two high tension power lines (120 and 220 kVA), operated by *Société Nationale D'Électricité* (the DRC national supply authority), running adjacent to the Lubumbashi–Likasi national highway. Nonetheless, a new 120 kVA power line has been built from Lubumbashi to Kinsevere, to connect the mine site to the 220 kVA national grid. There is also **good water availability from the mine surrounds and/or the nearby Kifumashi River** (Figure 7.1). It is expected that de-watering will provide much of the process water for Stage II.

The size of the exploitation permit and favourable topography allows Anvil significant flexibility in its choice of sites for tailings storage facilities, stockpiles, processing plant site and other infrastructure.

Figure 7.1 Satellite Image of Kinsevere Project Area (Before Stage I Mining)

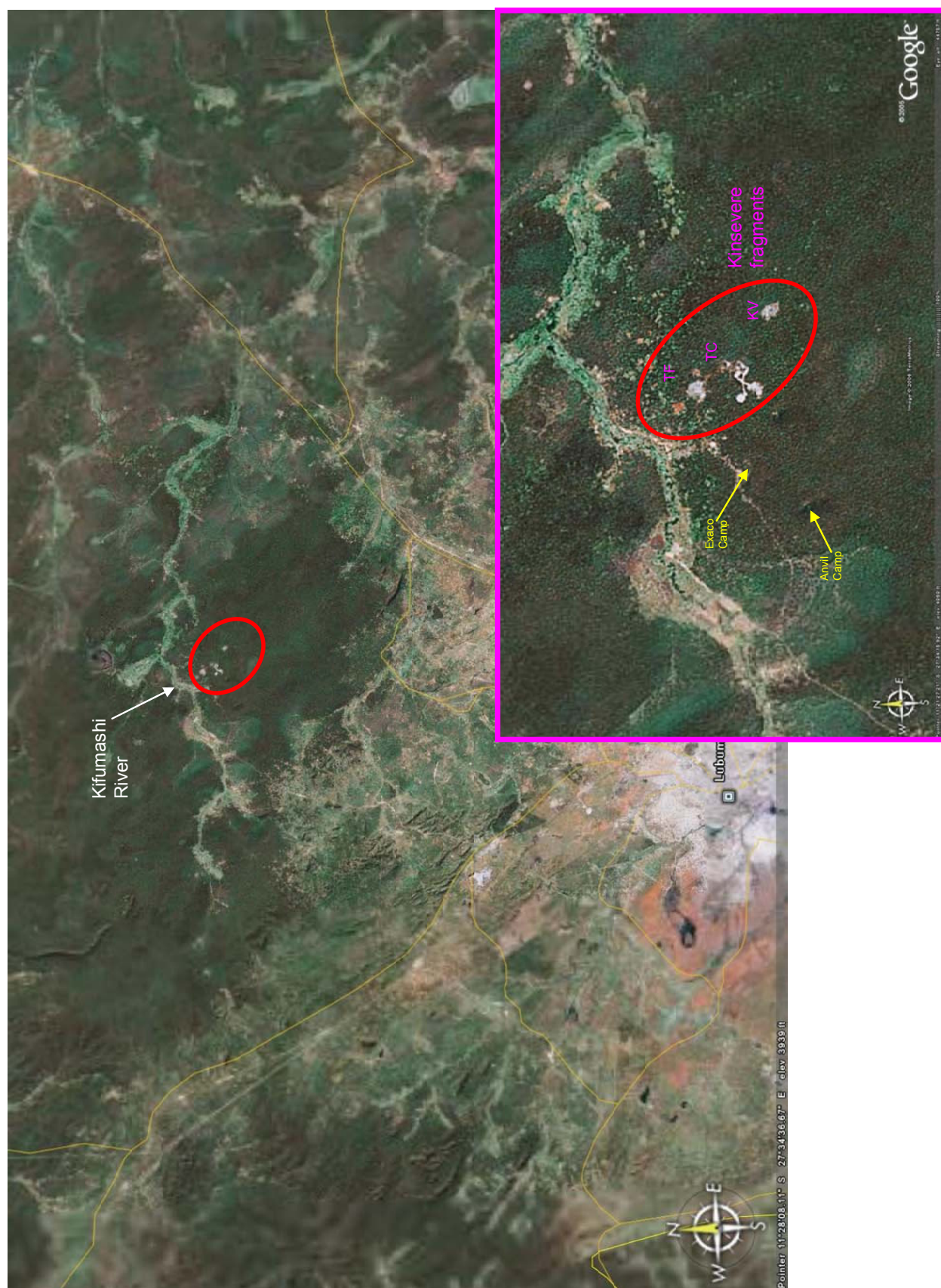


Figure 7.2 Aerial View, Looking SE, Across Tshifufia Central and South, Towards Kinsevere Hill (Before Stage I Mining)



Figure 7.3 Aerial View, Looking NW, of the Tshifufia Central Open Pit Area (Before Stage I Mining)



ITEM 8 History

Updated information is provided under this Item in respect of **historical Mineral Resource and Mineral Reserve estimates**, and **events leading up to the current status of production**.

8.1 Exploration

During the 1990's, the Tshifufia and Tshifufiamashi prospects were the subject of a joint venture between Gécamines and EXACO (a local Congolese company, interested primarily in exploiting near-surface, high-grade, oxide cobalt mineralisation). Prior exploration activity is not known. The joint venture was restricted to cobalt mineralisation within 30 m of surface. An exploration programme, involving trenching, pitting and the drilling of 67 holes, was carried out by EXACO to test the distribution and grade of oxide cobalt mineralization only (Table 8.1).

Gécamines also carried out a preliminary investigation for copper, which included limited drilling at the three Kinsevere deposits: Kinsevere Hill, Tshifufia and Tshifufiamashi (Table 8.1). However, most of the holes were drilled vertically, down the dip of the orebodies. Subsequently, an open pit was developed at Tshifufia South, to a nominal depth of 30 m, although no formal exploitation of the other deposits was undertaken.

Table 8.1 Summary of Previous Exploration Work by EXACO & Gécamines

Deposit	Pitting	Trenching		Drilling	
	No (m depth)	No. (metres)	Significant Grades	No. holes (metres)	Significant Grades
Kinsevere Hill	7 (44 max)	11 (625 m)	6.6% Cu 0.2% Co over 20 m	10 (1,021 m)	3.99% Cu 0.22% Co over 14.6 m
Tshifufia	-	17 (1,106 m)	7.6% Cu 0.3% Co over 15 m	19 (950 m)	6.3% Cu 0.6% Co over 23 m
Tshifufia South	-	39 (278 m)	7.2% Cu 0.3% Co over 40 m	11 (497 m)	
Tshifufiamashi	11	16 (1,304 m)	5.8% Cu 0.2% Co over 50 m	37 (846 m)	10.5% Cu 0.72% Co over 22.2 m

In mid 2004, Anvil obtained a series of plans and drill sections (with collar positions and some geological logs), some production records and a set of historical estimates from Gécamines (Table 8.2). Although a complete record of the work undertaken by Gécamines was not available, the data review provided sufficient information to warrant Anvil carrying out confirmatory and infill drilling, and ultimately an initial estimate of the Mineral Resource to NI 43-101 and JORC reporting standards.

Table 8.2 Gécamines Historical Estimates for the Kinsevere Deposits

Deposit	Depth (m)	Ore (tonne)	Cu (%)	Copper metal (tonne)	Co (%)	Cobalt metal (tonne)
Kinsevere Hill	0 - 50	1,140,000	5.4	61,560	0.18	2,052
	50 - 100	1,150,000	4.0	46,000	0.10	1,140
Tshifufia Central		1,017,770	6.19	63,000	0.28	2,850
Tshifufia South		56,000*	3.14	1,758	0.76	426
Tshifufiamashi	0 - 50	678,900	6.1	41,350	0.23	1,547
	50 - 100	Unknown		70,000		

Note: Gécamines divided Resources into “Proven”, “Probable” and “Possible”. At Kinsevere, Resources from 0 – 50 m depth were classified as “Probable” and those from 50 to 100 m depth were “Possible”. None of these resource estimates conformed to NI 43-101 standards.

* Mined by EXACO

In June 2004, Anvil entered into a Joint Venture (the AMCK Joint Venture) with the Mining Company of Katanga (MCK) to explore and develop the Kinsevere Project in which MCK had a preliminary agreement with Gécamines, the owner of the mineral rights. Anvil would fund all of the exploration and initial development of the Kinsevere Project including the completion of a bankable feasibility study. By carrying out these activities, Anvil acquired 70% of the Joint Venture.

In July 2004, AMCK finalised its exploration agreement with Gécamines to explore the polygons of Kinsevere and Nambulwa (*A la Prospection des Gisment du Polygon de Kinsevere and Nambulwa No 648/6743/SG/GL/2004*). This agreement allowed AMCK to carryout exploration and limited mining activities on the properties for a period of two years. At the completion of this agreement, AMCK had the exclusive right to negotiate with Gécamines to enter into either a Joint Venture or Lease Agreement for the properties subject to the completion of a feasibility study, based on exploration results.

In December 2005, AMCK concluded negotiations and signed an agreement with Gécamines on a *Contrat d'Amodiation* to mine and process ore from the two exploitation permits (Kinsevere (PE 528) and Nambulwa (PE 539) for a period to 2024. Approval for the *Contrat d'Amodiation* was issued by the DRC Minister of Portfolio (reference number 3213/MINPF/MM/CMU/CVK/05) on December 6, 2005. Following an exploration and evaluation programme completed in 2007, the Nambulwa permit will be relinquished to Gécamines.

In September 2006, Anvil increased it's equity holdings in AMCK to 80% through the purchase of 10% of the company from MCK for \$14 million. In April 2007, Anvil further increased is shareholding in AMCK to 95% for a further payment of \$43 million.

8.2 Previous Anvil Mineral Resource and Mineral Reserve Estimates

Anvil completed an initial Kinsevere Project Evaluation Study (Anvil, September 2005) in 2005, in which a Mineral Resource estimate was reported and a tabulation presented by consultants Fin Ore Ltd (now CSA Global Pty Ltd). This estimate was based on an exploration programme carried out by Anvil Mining Services (AMS) from October 2004 to June 2005, which included the collection and analysis of 532 chip

samples from surface trenches, the drilling of 5,675 m of reverse circulation and 2,213 m of diamond boreholes. The estimate was quoted at a 0.5% Cu cutoff grade as listed in Table 8.3.

Table 8.3 Kinsevere Project Mineral Resource Estimate, September 2005

Deposit	Classification	Tonnes	TCu (%)	Total TCu metal (t)
Tshifufia	Inferred	3,656,000	4.54	165,900
Tshifufiamashi	Inferred	2,816,000	4.50	125,800
Kinsevere Hill	Inferred	1,394,000	3.00	41,400
TOTAL	Inferred	7,866,000	4.24	333,200

In the absence of Measured and Indicated Resource classifications, an associated 'Mining Inventory' totalling 6.9 Mt at a grade of 4.6% Cu was reported (Anvil, September, 2005).

A significantly updated Mineral Resource estimate was reported in 2007 in the Project Feasibility Study Preliminary Report (AMCK, April 2007). This estimate was based on completion of a Phase I and Phase II exploration programme and was again prepared by Fin Ore Pty Ltd. The estimate was quoted at a 0.5% Cu cutoff grade as listed in Table 8.4.

Table 8.4 Kinsevere Project Mineral Resource Estimate, April 2007

Deposit	Classification	Tonnes	TCu (%)	Total TCu metal (t)
Tshifufia	Measured	6,010,000	4.40	266,700
	Indicated	10,950,000	4.40	485,000
	Inferred	20,120,000	3.50	683,500
Tshifufiamashi	Measured	-	-	-
	Indicated	2,245,000	4.00	90,330
	Inferred	451,000	4.90	22,100
Kinsevere Hill	Measured	-	-	-
	Indicated	875,000	2.60	22,800
	Inferred	374,000	3.10	10,900
TOTAL	Measured	6,010,000	4.44	266,700
	Indicated	14,070,000	4.25	598,130
	Inferred	20,945,000	3.42	716,500

A Mineral Reserve estimate based on the Resource listed in Table 8.4 was also reported in the Project Feasibility Study Preliminary Report (AMCK, April 2007). This estimate was produced by A & J Cameron and Associates and was based on an incremental cut-off grade relative to a copper price of \$1.20/lb. The estimate is listed in Table 8.5.

Table 8.5 Kinsevere Project Mineral Reserve Estimate, April 2007

Deposit	Classification	Tonnes	TCu (%)	Total Cu metal (t)
Tshifufia	Proved	6,041,000	4.41	266,400
	Probable	9,056,000	4.56	413,000
Tshifufiamashi	Proved	-	-	-
	Probable	1,281,000	3.65	46,800
Kinsevere Hill	Proved	-	-	-
	Probable	562,000	3.43	19,300
TOTAL	Proved	6,041,000	4.41	266,400
	Probable	10,899,000	4.39	479,100

A Phase III exploration programme was completed in 2007 to enable a further update to the Mineral Resource estimate. The estimate was reported in the Project Feasibility Study (AMCK, December 2007). This estimate was completed by CSA Global Pty Ltd, and reported using a revised cut-off grade of 0.7% Cu, as listed in Table 8.6.

Table 8.6 Kinsevere Project Mineral Resource Estimate, December 2007

Deposit	Classification	Tonnes	TCu (%)	Total Cu metal (t)
Tshifufia	Measured	14,723,999	4.05	596,300
	Indicated	6,349,000	4.65	295,200
	Inferred	1,486,000	3.70	55,000
Tshifufiamashi	Measured	3,817,000	3.61	137,800
	Indicated	7,690,000	2.46	189,200
	Inferred	769,000	2.50	19,200
Kinsevere Hill	Measured	-	-	-
	Indicated	880,000	2.60	22,900
	Inferred	350,000	3.10	10,900
TOTAL	Measured	18,540,999	3.96	734,100
	Indicated	14,919,000	3.40	507,300
	Inferred	2,605,000	3.27	85,100

An additional Inferred sulphide Mineral Resource was reported as 15,340,000 tonnes at a grade of 2.9% Cu.

A Mineral Reserve estimate based on the Mineral Resource listed in Table 8.6 was also reported in the Project Feasibility Study (AMCK, December 2007). This estimate was again produced by A & J Cameron and Associates and was based on an incremental cut-off grade relative to a copper price of \$1.43/lb. The estimate is listed in Table 8.7.

An additional Proved Mineral Reserve of 531,000 tonnes at a grade of 3.2% was reported in relation to ore stockpiled during Stage I mining to date.

Table 8.7 Kinsevere Project Mineral Reserve Estimate, December 2007

Deposit	Classification	Tonnes	TCu (%)	Total Cu metal (t)
Tshifufia	Proved	12,747,000	4.17	531,600
	Probable	5,016,000	4.92	246,800
Tshifufiamashi	Proved	3,642,000	3.55	129,300
	Probable	3,806,000	2.94	111,900
Kinsevere Hill	Proved	-	-	-
	Probable	534,000	3.29	17,600
TOTAL	Proved	16,389,000	4.03	660,800
	Probable	9,356,000	4.02	376,300

Phase III drilling continued during late 2007 and early 2008, focussing on extensional drilling across the Kilongo deposit. In 2008 Anvil undertook a further intensive drilling programme (Phase IV), completing infill investigations within the Tshifufia and Tshifufiamashi Project areas. The objective of this programme was to enhance the tonnages and classifications in these Project areas and drill deeper into the sulphide horizon. A total of eight rigs were required to complete over 26 km of air core, reverse circulation and diamond drilling (described in Item 13). Due to prevailing economic circumstances at the end of 2008, the Phase IV programme was terminated before completion.

The Mineral Resource updates for Tshifufia, Tshifufiamashi, Kinsevere Hill and Kilongo completed in 2008 and 2009 are described in Item 19.

8.3 Stage I Development and Production

8.3.1 Mining Production to Q4 2008

A preliminary mine production plan for the Kinsevere Project was produced in November 2006. The objective of this plan was to identify the most suitable location for a six-month starter pit which would provide sufficient initial high grade ore to feed the Stage I HMS plant and ultimately produce refined copper from a proposed electric arc furnace (EAF) at a rate of 23,500 tpa.

On this basis, mining commenced at Tshifufiamashi in December 2006, but due to the presence of clays and the potential effect on HMS processing, production from this source was scaled back. Following topsoil pre-stripping, mining from Tshifufia was commenced in February 2007. Up until the commencement of HMS processing in June 2007, all mined ore was stockpiled.

Mining continued successfully at Tshifufia through to the imposition of a care and maintenance programme in Q4 2008 (refer to Item 8.3.4). By that time, the Central_1 starter pit at Tshifufia had reached 1165 mRL (about 45 m below surface), and the Central_2 cutback had already commenced. The cutback provided waste materials for the construction of the Stage II TSF starter embankment. Mining in Central_1 of the Tshifufiamashi Pit was resumed and had reached 1170 mRL (about 40 m below surface) by Q4 2008.

Mining has not commenced at Kinsevere Hill nor at the adjoining Kilongo deposit. The insitu grades are such that these deposits are scheduled to be mined towards the end of the mine life. The contracted mining fleet capacity up to the end of 2008 was in excess of that required to provide feed to the HMS plant. The extra capacity was used to develop long term stockpiles for reclaim during Stage II processing.

The Tshifufia and Tshifufiamashi Pits had both reached the groundwater table by the end of 2008 and continuing efforts were required to pump water from in-pit sumps. Drained pit slope conditions were being maintained by a network of perimeter dewatering bores. No significant pit slope instability was experienced up to the end of 2008.

Figure 8.1 and Figure 8.2 show the progress of Stage I mining relative to budget.

Figure 8.1 Monthly Ore Production – Stage I Mining

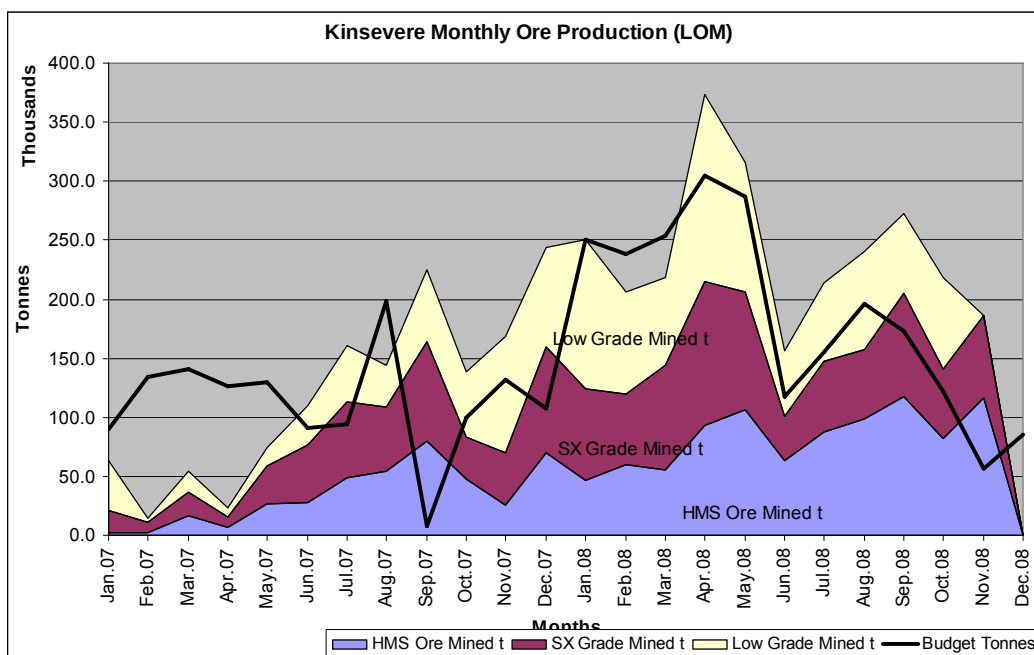
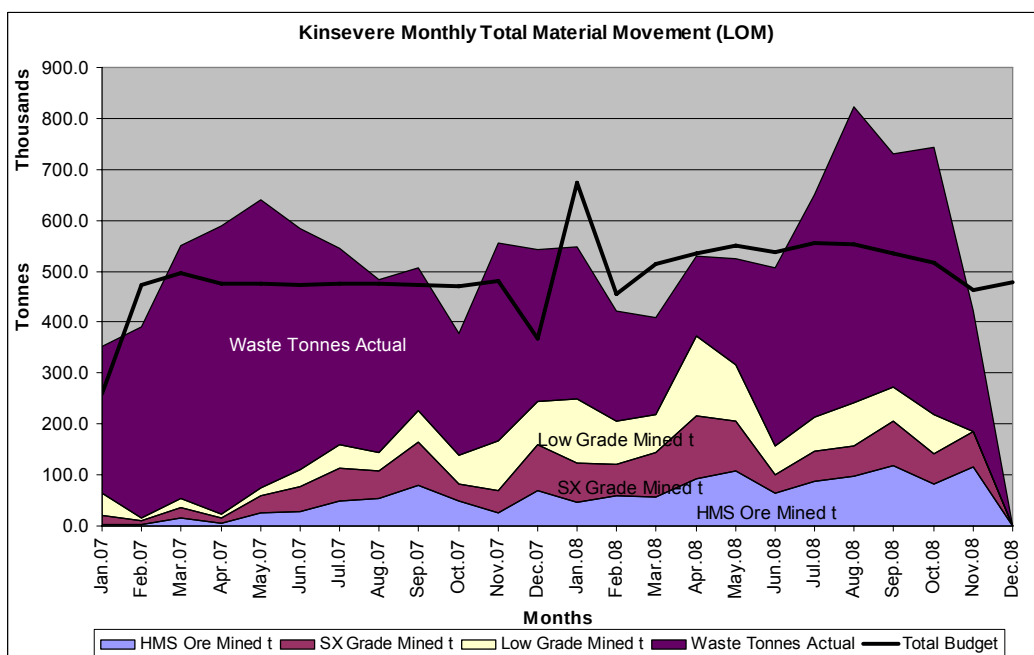


Figure 8.2 Monthly Ore and Waste Movement – Stage I Mining



8.3.2 HMS Production to Q4 2008

Following commissioning in March 2007, Figures 8.3 to 8.5 show that HMS production performance was typically in excess of that budgeted over the period to the end of 2008.

Figure 8.3 Monthly Crusher Feed – Stage I Processing

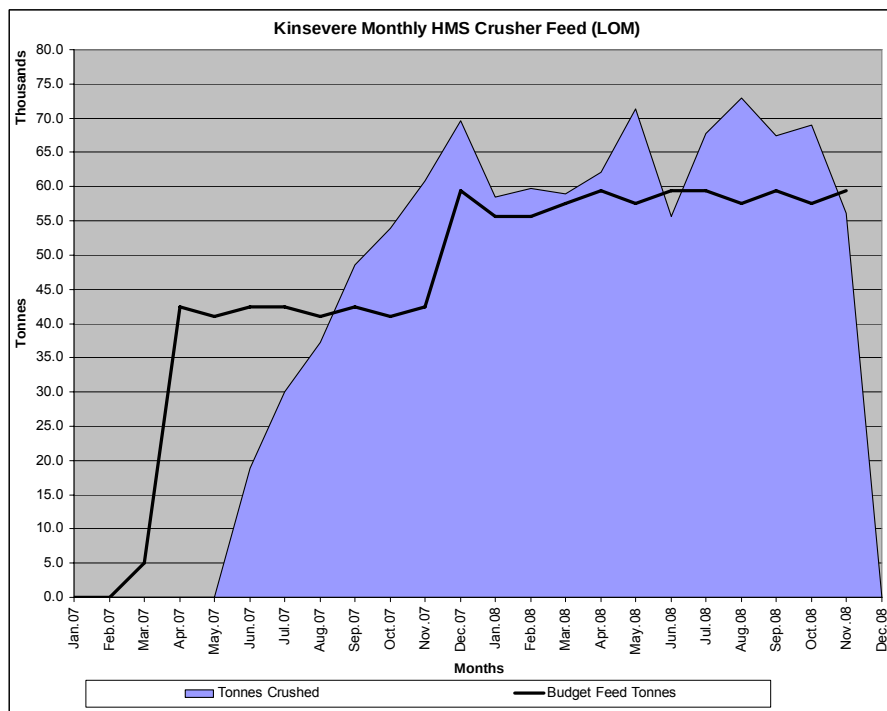


Figure 8.4 Monthly Plant Recovery – Stage I Processing

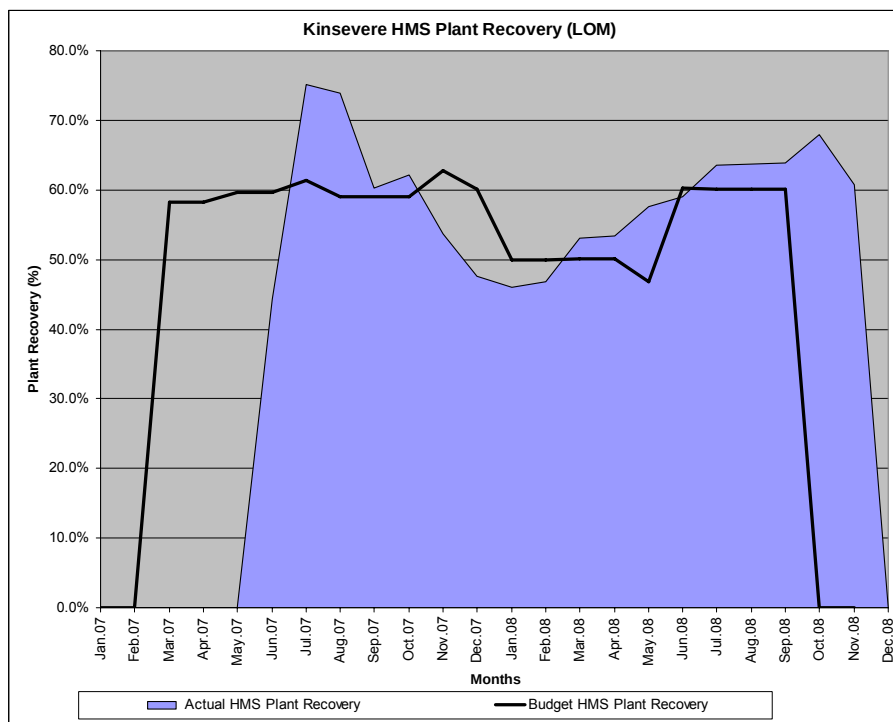
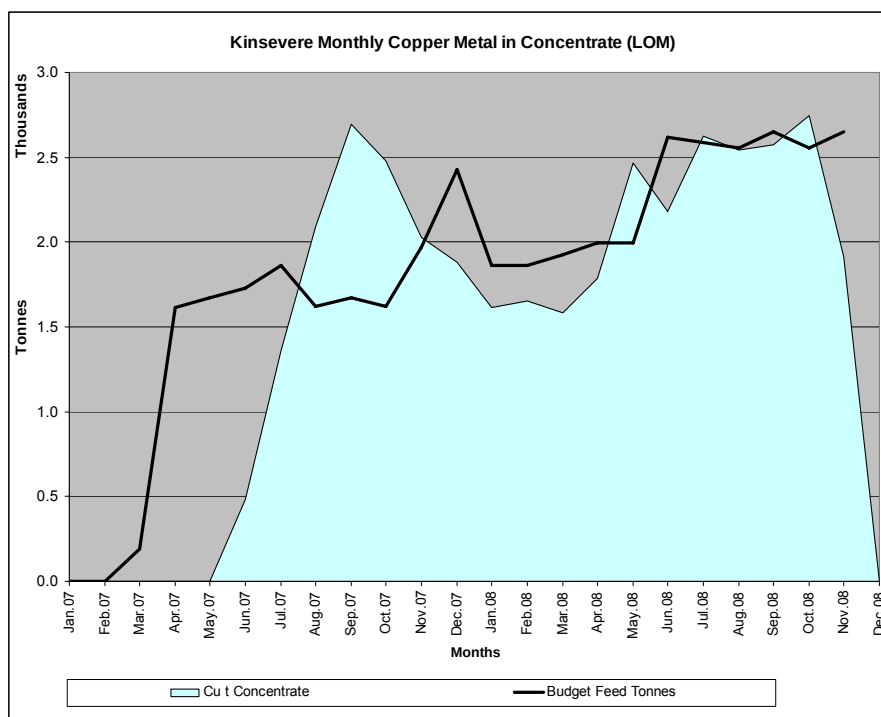


Figure 8.5 Monthly Metal Production – Stage I Processing



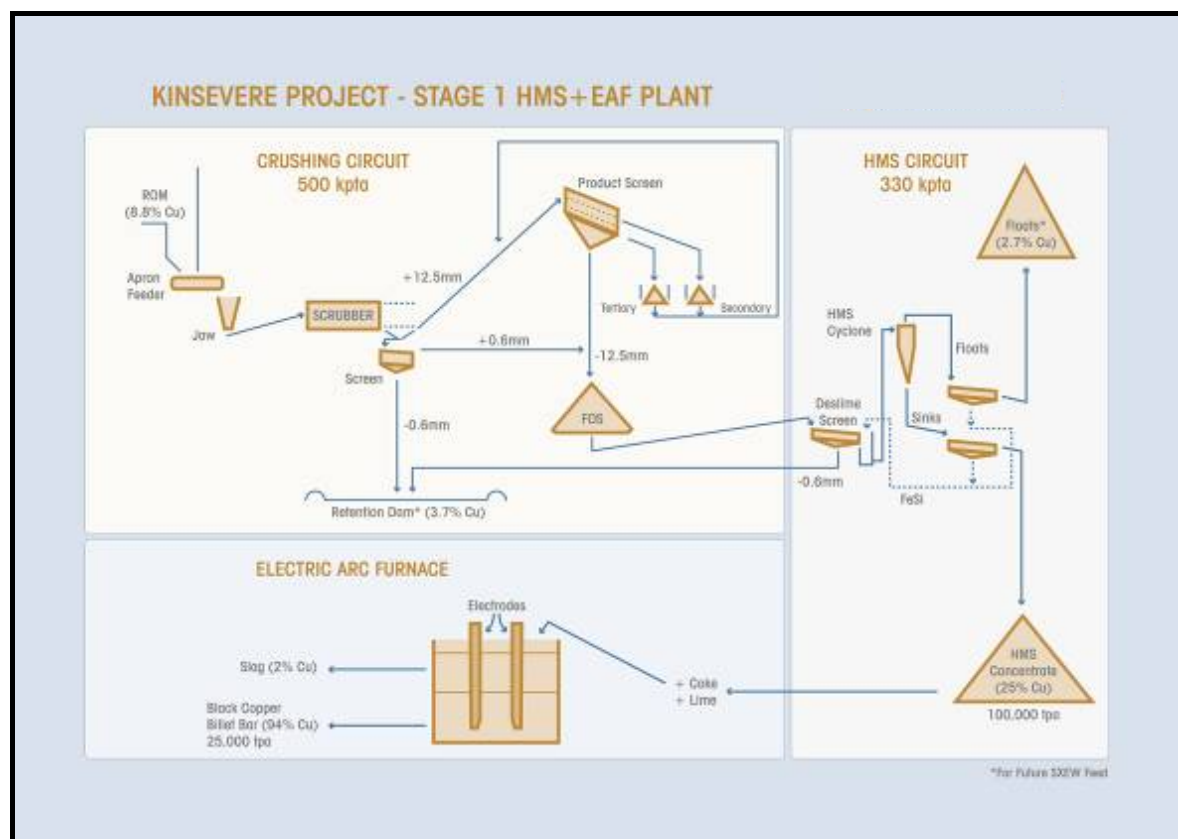
8.3.3 EAF Production

The justification strategy for the EAF was the Company's desire to maximise revenue from the sale of smelted copper during the Stage I Project and whilst construction of the Stage II SX-EW plant was in progress. Ultimately, it was envisaged that following the commencement of Stage II, the EAF could be converted to process sulphide flotation concentrate from Dikulushi. In anticipation of a future conversion, two furnaces were designed to facilitate a series configuration, with one furnace positioned higher than the other. At the time, there was an expectation that the government would eventually penalise companies for exporting concentrate out of the DRC. Against this background, the Kinsevere EAF design contract was awarded to Keech Furnace Technologies (KFT) in August 2006.

The EAF layout was designed by KFT to comprise two furnaces fed from a single feed mixing facility, with discharge to a single ingot casting station and single slag granulation pit. The reductive smelting of copper oxides is a relatively simple pyrometallurgical process involving the mixing of dry HMS concentrate with slaked lime (flux) and coke (reductant), and feeding into a furnace to produce 'black copper' (ie +94% Cu) and a slag. The matte is cast into ingots and sold, whereas the slag is granulated by quenching with water, then disposed of. Off-gases are vented to the atmosphere after any dust generated from the operation is removed via a bag house.

The sinks concentrate fraction from the Stage I HMS plant was expected to be processed in the two furnaces, both of which were planned to be commissioned in early 2008 to produce 23,500 tpa of copper. A schematic flowsheet of the Stage I plant incorporating the EAF is shown in Figure 8.6.

Figure 8.6 Stage I Process Plant – Schematic Flowsheet



A number of detailed design and engineering deficiencies became apparent in 2008, the combined effect of which was to delay EAF commissioning.

Following lengthy construction delays, commissioning of the first of the two furnaces was carried out in August 2008, and then followed a period of initial operational difficulties during the ramp-up phase. Relative to the design production capacity of 1,000 tonnes of copper metal per month per furnace, the actual production figures showed:

- September = 174 tonnes
- October = 327 tonnes
- November = 286 tonnes
- December = 75 tonnes
- January = 330 tonnes

Although there had been numerous challenges overcome during the EAF commissioning and ramp-up phase, it was apparent that there were two critical processes preventing the design production capacity (ie, from the single commissioned furnace) from being achieved:

- materials handling constraints, in respect of furnace feed arrangements and copper metal handling; and
- furnace feed blending and related control of the furnace operating temperature.

In mid March 2009, with planned furnace reline activities still in progress and with no certainty that EAF productivity would improve further in the short to medium term (despite a programme of test work and technical analysis), a decision was made to

cease furnace operations. The cashflows from recommissioning the HMS plant and selling concentrate were assessed to be more favourable than persisting with the EAF and producing black copper under then present metal price and operating cost circumstances.

8.3.4 Suspension of Operations in Q4 2008

As a result of events during Q4 of 2008, most notably a sharp fall in the copper price, highly illiquid capital markets and a deterioration in the outlook for global economic growth, Anvil put in place a strategy plan, as follows:

1. Placing the fabrication and construction works associated with the Stage II Project on hold until additional finance was available and there was greater certainty in global financial and commodity markets.
2. Suspension of full scale Stage I HMS operations in Q4 2008, followed by recommencement of processing in Q2 2009 using minimal operational staff to produce a concentrate only from existing Run of Mine (ROM) stockpiles.
3. Suspension of blister copper production from the Stage I EAF in Q1 2009 due to technical difficulties and better cashflows being able to be realised from the sale of HMS concentrate.

As at Q4 2008, the Company did not have sufficient cash or debt facilities to fund the completion of the Stage II plant fabrication and construction. The Company required additional funding of approximately \$200 million to enable Stage II to proceed.

8.3.5 Status of Stage II Construction at Q4 2008

Stage II fabrication and construction was placed on hold in November 2008. The contractor, Ausenco Engineering Ltd (Ausenco), subsequently prepared a close-out report in December 2008 (Ausenco, December 2008). As at May 2009, approximately \$190 million (\$168 million spent and \$22 million committed) of the then projected construction cost of \$380 million had been invested. The then status of the completed work was as follows:

- engineering design work was almost 80% complete;
- civils construction was approximately 90% complete;
- concrete works were 70% complete;
- construction of key infrastructure including camps, roads, offices, power and water facilities was 90% complete;
- construction of stainless steel tanks was 30% complete;
- earthworks for the process and event ponds were 40% complete;
- construction of the TSF starter embankment was 65% complete;
- procurement of major capital items was 80% complete, including the ball mill, crusher, apron feeder, tailings dam liner and half of the electrowinning cells, all of which are on site; and
- the cathode stripping machine, pin bed clarifiers, transformers and rectifiers plus half of the cathode plates and anodes were ready for transport to site.

All completed construction works, including erected tankage units and the electrowinning cells had been checked, braced and made safe as required. All contractor staff were demobilised from site, whilst Anvil retained appropriate staff to supervise off loading and secure storage of capital equipment as it continued to arrive at site.

8.3.6 Independent Engineer Review

Since finalising documentation relating to the DRC Mining Review negotiations in January 2009, work recommenced on an Independent Engineer Review (by SRK Consulting (UK) Ltd) on behalf of potential Project financiers. The review included site visits by technical specialists and focussed on technical work completed since the Feasibility Study and 2008 Technical Report.

A final report was received in September 2009 (SRK, 2009) in which SRK:

Mineral Resource

- endorsed the plan for a grade control model, between the Mineral Resource model and the mine plan
- suggested the need to consider reducing the block size in the grade control model
- endorsed the implementation of a production tracking/reconciliation system
- suggested the need to do reconciliations on a monthly, rolling quarterly and annual basis

Mineral Reserve

- endorsed the cost inputs to the pit optimisation
- endorsed the selection of optimal pit shell, and the resulting reduced waste strip ratio
- concluded that the production schedules have been appropriately developed from the pit optimisations

Mining Engineering

- concluded that mining engineering aspects have been adequately addressed
- endorsed the contingency mine plan for Stage I
- recommended redesign of the waste rock dump and TSF to account for reduced volumes of oxide waste available

Geotechnical

- concluded that data collection and analysis techniques are appropriate, although not at the required level of detail for a Bankable Feasibility Study
- noted however, that data uncertainties have been addressed by the adoption of conservative slope design parameters
- was satisfied that there are no fatal flaws in geotechnical engineering
- suggested the need to enhance the design database with more information, as gathered
- noted that dewatering is critical

Hydrogeology

- noted that latest modelling indicates the dewatering rate has increased from 250L/s in the Feasibility Study to over 500L/s
- suggested that this could result in more discharge to Kifumashi River
- endorsed completion of the proposed 2009/2010 groundwater drilling/testing/equipping programme, as a critical action item
- suggested the need to reassess capital and operating cost provisions
- suggested the need to reassess the resulting environmental impacts and documentation

Mineral Processing

- noted the technical risk of milling-in-raftinate and direct tailings discharge, given limited applications
- suggested allowing \$20 million for retro-fit and four to six months to install and commission
- suggested allowing a three month delay in the Ausenco LSTK schedule
- suggested a 10% to 15% understatement of the LSTK budget
- recommended stress testing the Base Case cash flow model for 10% higher process operating costs (influenced by adopted acid price of \$280/t)
- recommended continuous monitoring of Stage I process head grade variability and impact on recovery and mass pull
- overall, considered that metallurgical aspects have been appropriately designed and engineered for a Bankable Feasibility Study

Tailings Storage Facility (TSF)

- considered that technical investigations, design and remaining capital requirements are at a suitable level for a Bankable Feasibility Study
- noted that some design assumptions are appropriately conservative and hence safety factors are within acceptable limits
- recommended that geotechnical boreholes be drilled inside the TSF footprint, and resulting information used to confirm conservative design assumptions

Infrastructure, Capital Expenditure, Construction Schedule, Overheads and Product Costs

- noted that the LSTK contract was not sighted (it was in preparation at the time of their review)
- noted that an off-take agreement was not sighted (not available at the time of their review)
- noted that the expected cost to completion was subject to the LSTK contract and savings assumed by the Company
- noted that, whilst typical for a LSTK contract, limited supporting data had been provided for Contractor and Owner estimates
- commended the appointment of a Client Representative
- considered that the construction schedule is aggressive and there should be an allowance in the cash flow model of three months for potential delays
- noted that there was no capital spend curve in the Financial Model, and hence peak funding requirements are linked only to notional equal monthly costs (since revised)
- recommended that the ultimate Financial Model should:
 - o reflect a three month delay to the construction schedule
 - o include a 10% to 15% increase in costs to complete
 - o include allowances in 2009 and 2010 for dewatering bores
 - o include allowances for corporate costs, if required
 - o be stress tested for 10% higher process operating costs
 - o include a \$20 million allowance for plant retrofitting, in conjunction with a six month production delay

Human Resources

- considered that this aspect had been addressed to a suitable level for a Bankable Feasibility Study
- suggested, however, that further consideration be given to:
 - o the appropriateness of manning reductions
 - o terminal benefit allowances

Health and Safety

- concluded that this aspect was adequately addressed
- noted that reporting of health and safety statistics is poor, ie there was no detail on frequency rates, nor benchmarking
- noted that the social impact of HIV/AIDS lacked focus
- noted a lack of detail on clinic attendances and understanding of contributing factors
- recommended the need for an additional action plan in this respect

Environmental

- noted that a final version of the ESIA (Knight Piésold, October 2009), complete with Chapter 12 (Community Consultation) was not sighted (not available at the time of their review)
- noted that the Company has demonstrated substantial progress towards full Equator Principles (EP) and International Finance Corporation (IFC) compliance
- concluded that the Company does not comply fully with five of the nine Equator Principles, with further work required on biodiversity studies, the access road corridor, community management and social management plans, and stakeholder engagement plans
- concluded that in relation to the eight IFC Performance Standards, further work is required for full compliance on each of these, and principally on aspects of construction to operations transition, labour and working conditions, surface and groundwater impacts to the Kifumashi River, greenhouse gas emissions, community health and safety, traffic management, and compensation for land acquisition
- suggested that deficiencies could be addressed as Action Items in the Environmental and Social Action Plan (ESAP) (Anvil, November 2009) for the Project

8.4 Current Status

8.4.1 Recommencement of Stage I Operations

The Stage I HMS plant was restarted on 27 March 2009, drawing feed from the ROM stockpiles which at the time amounted to approximately 259,000 tonnes at a grade of 5.7% TCu. This feed was sufficient only to last until Q3 2009.

Mining subsequently resumed in the Tshifufia (Central) Pit in early August 2009, from readily available ore faces and broken stocks and at a reduced scale during the Stage II construction phase. HMS metal recovery of Central Pit ore is projected at 60% and the concentrate sales (at mine gate) are at an agreed discounted price of 40% to 50% of the LME price.

In proceeding with this strategy and continuing to produce concentrate in the short-term, potential revenue will be lost from Stage II under circumstances where the copper metal would otherwise be recovered at 92% in the SX-EW plant and sold as cathode at approximately 100% of the LME pricing.

Following a brief tender period in June and July 2009, a short-term mining contract was awarded to former mining contractor and JV partner, MCK. In view of the above, the contract involves a reduced scale of mining until the end of December 2010, sufficient to provide enough HMS plant feed to cover the cost of operations with only a modest buffer.

8.4.2 Resumed Stage II Construction

Following the receipt in September 2009 of Tranche 1 funds of \$32 million relating to the equity financing agreed with Trafigura, an Early Works Programme was initiated on the Stage II Project, with progress since being made in a number of areas, including (as at December 2009):

- Steel erection – 213 tonnes of pre-fabricated pipe rack steel had been erected, representing approximately 9% of the total requirement.
- Contracts – security, catering and earthworks contracts negotiated by Ausenco were in an advanced stage of documentation and expected to be finalised in early 2009.
- Critical path issues – steel for the sulphuric acid tanks had been ordered and fabrication of a variety of other tanks was near completion.
- Stockpiles of cement, sand and aggregate – orders and initial payments for the total stock requirement of cement and additives for the remainder of the project had been made and the first two of four deliveries of cement were received in November 2009.
- New orders – purchase orders had been prioritised according to schedule requirements and were being prepared and committed.
- Transportation of goods in storage – equipment items held in storage since Kinsevere Stage II was placed on hold had been reviewed and categorised according to priority. Delivery of these goods to site was underway and the Company expected to take delivery of the majority of this equipment before March 31, 2010.
- Work permit applications – Ausenco had assumed full responsibility for obtaining work permits and the process was underway for identified staff.
- Camp refurbishment – significant progress had been made in relation to the junior and senior accommodation, laundry and messes.

The LSTK contract with Ausenco was signed on 22 January 2010 for a total value of \$130 million. Owner's Costs, which include purchase and transportation of the remaining capital equipment, spares, first fills, initial training and pre-commissioning costs and associated taxes and duties amount to \$58 million, which together with contingency of \$12 million, brings the total remaining cost to \$200 million.

Group Five Projects (Group Five) is being engaged by Ausenco to undertake the Kinsevere Stage II construction and fabrication works. Prior to November 2008, when the development of Stage II was placed on hold, Group Five was contracted to perform this role and as a result has a sound understanding of the Stage II development. In addition, Group Five has an established presence in the DRC and is well equipped to manage logistical, regulatory and other issues associated with operating in the DRC.

Group Five and Ausenco have begun mobilising to site and it is expected that by March 2010, more than 100 Group Five personnel and 15 Ausenco personnel will be on site.

A "Client Representative", Green Team International (GTI), was appointed in February 2010 to act as the Company's site representative and interface with Ausenco during the construction phase.

The engineering and design works which recommenced during the third quarter of 2009 are approximately 95% complete, with 2,057 drawings of the total of 2,160 now completed. In addition, a substantial amount of vendor data has been received.

Allowing for wet season delays at the end of 2009 and into 2010, the anticipated construction period extends to the end of 2010. A six month ramp-up period for Stage II processing is then planned from 1st January 2011.

ITEM 9 Geological Setting

Updated information is provided under this Item in respect of **revisions to the local geological interpretation**.

9.1 Regional Geology

The Kinsevere Project area is located in the north-eastern section of the Central African Copperbelt. Together with the Zambian Copperbelt to the south, this celebrated metallogenic province contains some of the world's richest copper and cobalt deposits.

Both Congolese and Zambian portions of the belt are located within a continuous fold zone known as the Lufilian Arc, one of several major Pan-African structures bordering the Congo and Kalahari cratons. Each portion exhibits early Neo-Proterozoic intra-cratonic rift development, coincident with the break-up of a Meso-Proterozoic supercontinent (approximately 800 to 600 Ma). Late Neo-Proterozoic collisional deformation and metamorphism is also documented regionally, linked to the formation of central Gondwana (approximately 600 to 500 Ma).

The Lufilian Orogeny focussed various thrust events, resulting in arcuate north-northwest lineaments in Zambia and east-west lineaments at Kolwezi (Figure 9.1). These structures are best recorded in Neo-Proterozoic Roan Supergroup sediments, which outcrop in a series of faulted and brecciated anticlines or so-called “*écailles*” (mega-fragments or rafts of sediments) which can measure up to 10 km in length. This Supergroup comprises an approximately 7,000 m thick succession of predominantly shallow-marine and terrestrial meta-sediments (Table 9.1), which accumulated within intra-cratonic rift basins. Major copper-cobalt deposits of the Katangan Copperbelt, including those at Kinsevere are confined to basal sections of the Lower Roan Mines Group (

Figure 9.2).

Figure 9.1 Regional Geological Setting

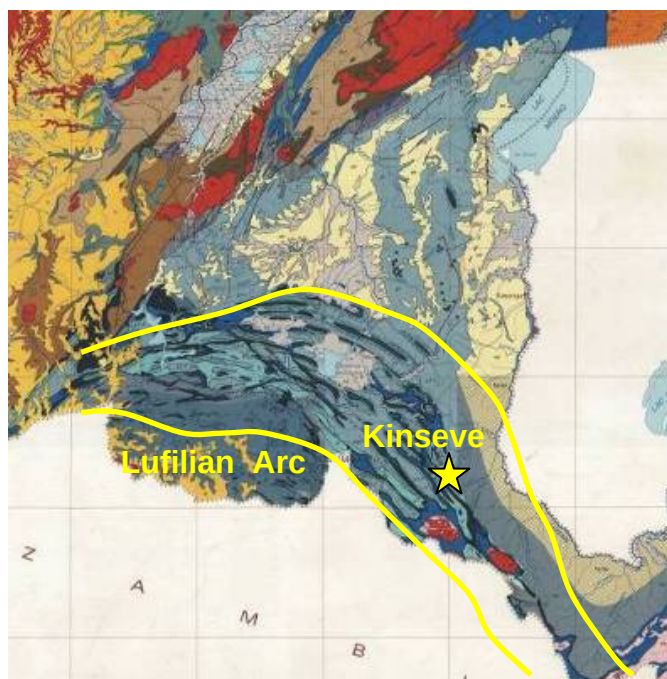
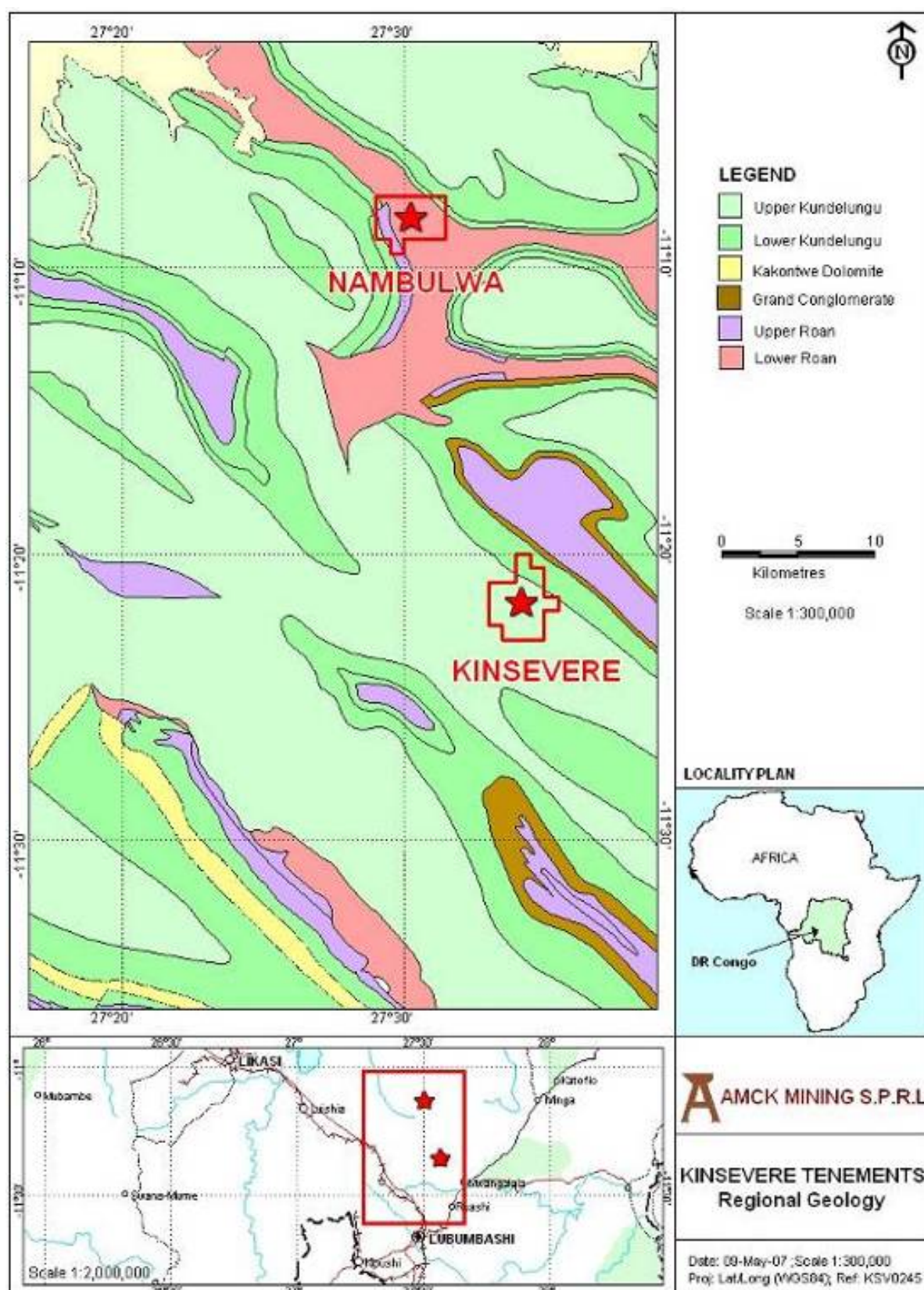


Table 9.1 Stratigraphy of the Katangan System

Super-Group	Group	Formation	Member	Lithology	
Upper Kundelungu	Plateaux (Ks-3)			arkoses with sandstones and shales	
	Kiubo (Ks-2)	Ks-2.2 / Ks-2.1		carbonated shales, sandy and argillaceous shales, sandstones	
	Kalule (Ks-1)	Ks-1.3 / Ks-1.2		carbonated siltstones, sandy and argillaceous shales, pink limestones	
		Ks-1.1		diamictite ('petit conglomerat')	
Lower Kundelungu	Monwezi (Ki-2)			arkosic sandstones with carbonated shales and siltstones	
	Likasi (Ki-1)	Ki-1.3		carbonated shales and siltstones	
		Ki-1.2		dolomites and limestones with shales	
		Ki-1.1		diamictite ('grand conglomerat')	
Roan (R)	Mwashya (R-4)	R-4.2		dolomitic shales, with sandstones or carbonaceous shales at top, 'conglomerat de Mwashya' at the base	
		R-4.1		dolostones, dolomitic shales, jasper, oolites, pyroclastic units, iron	
	Dipeta (R-3)	R-3.4 / R-3.3		dolostones, limestones, shales, sandstones and arkoses	
		R-3.2 / R-3.1		dolomitic and argillaceous siltstones (R.G.S.) dolostones	
	Mines Group (R-2)	Kambove Dolomite (formerly C.M.N.) (R-2.3)	Upper	pink brown to white dolomite, talcous dolomite, evaporitic breccia, red siltstone	Third Orebody
				dolomite, stromatolites, talcous dolomite, evaporitic breccia, grey-green siltstone	
				pink brown to white massive dolomite	
			Lower	massive stromatolitic dolomites, crypto-algal & laminated talcous dolomite	
				laminated algal dolomite	
				massive dolomite, stromatolites, dolomitic shale	
		Dolomitic Shales (S.D. or R-2.2)	S.D.3b	carbonaceous dolomitic shale	Upper Orebody
			S.D.3a	dolomitic shale, shaley dolomite	
			S.D.2d	carbonaceous dolomitic shale	
			S.D.2b+2c	dolomitic shale, shaley dolomite	
			S.D.2a	carbonaceous dolomitic shale	
			B.O.M.Z.	Black Ore Mineralized Zone	
		Kamoto Dolomite (R-2.1)	S.D.B.	Basal dolomitic shale	Lower Orebody
			R.S.C.	"Roche Siliceuse Cellulaire"	
			R.S.F.	"Roche Siliceuse Feuilletée"	
			D.Strat.	stratified dolomite	
			Grey R.A.T.	grey-green siltstone	
	R.A.T. (R-1)		Red R.A.T	"Roche Argilleuse Talcose"	
Base of R.A.T. sequence – Unknown					
Basal Conglomerate					

Figure 9.2 Geology of the Kinsevere Project Area



9.2 Local and Property Geology

Principal Copperbelt deposits are restricted to two stratigraphic horizons:

- a) the so-called “Lower Orebody” (LOB) is hosted by the D.Strat and RSF units of the Kamoto Dolomite Formation and,
- b) the so-called “Upper Orebody” (UOB) which is hosted by the SD Dolomitic Shales Formation.

A 15 m to 20 m thick barren zone of RSC (Table 9.1) frequently occurs between these orebodies. At some deposits, a “Third Orebody” (TOB) is observed, hosted by the lower part of the CMN Kambove Dolomite Formation (Table 9.1). At Kinsevere, all three ore horizons are well developed and mineralised, but, unusually, there is no barren RSC unit developed between the Lower and Upper Orebodies. Exposure in the immediate vicinity of the Kinsevere group of deposits - Tshifufiamashi, Tshifufia and Kinsevere Hill - is very limited, with less than 1% observed outcrop.

An updated structural interpretation (Figure 9.3), derived from 2008 in-pit mapping, provides the following setting for the combined Tshifufia North (TN), Tshifufia Central (TC), Tshifufia South (TS) and Tshifufiamashi (TM) areas:

- The deposits occur within an internally folded but continuous fragment of R1 (Red RAT) and R2 (Mines Group) rocks, which is interpreted to be segmented along east-west striking rotational faults.
- The folds occur as open, low-amplitude structures with moderately steeply east-southeast dipping axis.
- Faults are interpreted from stratigraphic offsets and sudden dip reversals.
- Strain partitioning is observed between the SD and the CMN. This is expressed as apparently randomly orientated cm-scale folding in the CMN and very open tens of metres scale folds in the SD.

The stratigraphy at Tshifufia North and Tshifufia Central strikes north-south, but is rotated westwards along a fault offset, immediately north of Tshifufia South. At Tshifufia North, bedding is upward facing and dips steeply to the west, while at Tshifufia Central it is overturned and dips at a moderate angle to the east. The bedding in the Tshifufia South pit is sub-vertical, with a near vertical dip, but the stratigraphy appears to wrap around a central “core” of RAT.

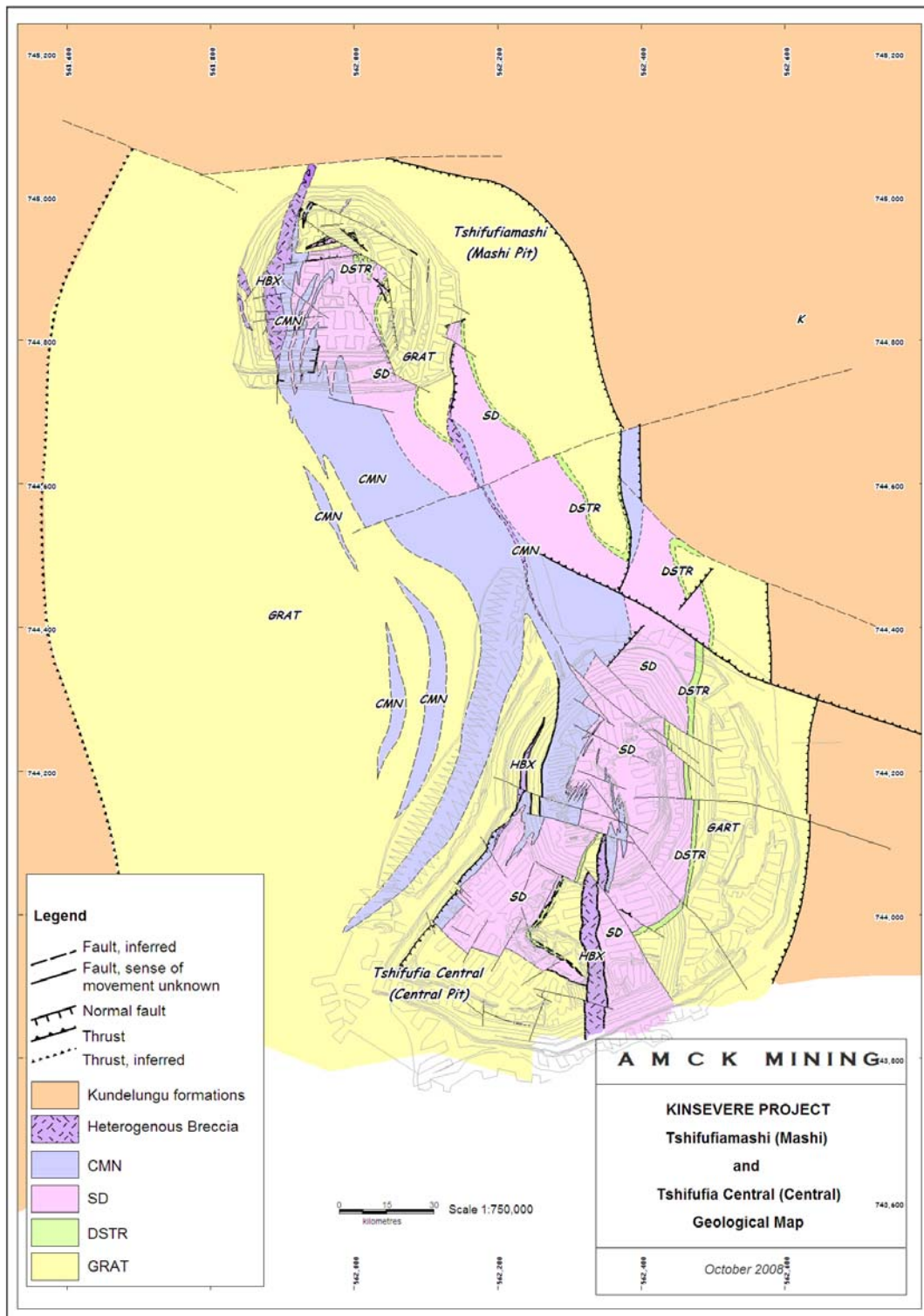
Throughout the Tshifufia deposit area, east-west-trending steeply dipping structures have been interpreted to offset stratigraphy by several tens of metres. Cross-section interpretations also suggest that compressional and extensional north-northwest-striking normal and reverse faults probably occur.

North-northeast-directed compressional deformation during the Lufilian orogeny is interpreted to have transformed nominal flat lying sediments into km-scale subvertical and overturned folds which now define the Lufilian Arc. Continued deformation is considered to have led to the dismemberment of fold segments along north-northeast directed steeply dipping faults (lateral ramps) and bedding-parallel thrusts. The deformation process was probably assisted by evaporite extrusion from the Lower Roan Group sediments along thrusts, shear zones and along the cores of fold structures. Such deformation can explain how the *écailles* of Lower Roan Group sediments became surrounded by breccias of the underlying RAT.

A second phase of deformation, or shift in the regional stress field, resulted in the development of low-amplitude east-west-trending folds. These were superimposed on the north-northeast-trending structures to generate an interference fold pattern.

These folds appear to be the loci for remobilised high-grade ore. Cross faults, associated with these folds, also focus the effects of near surface leaching / oxidation.

Figure 9.3 Revised Interpretation (2008) of Surface Geology at Tshifufia & Tshifufiamashi



The principal characteristics of the stratigraphic units observed in weathered rocks at Kinsevere are listed below:

RAT (Red) – Roche Argilo-Talqueuse:

- Light red, massive, calcareous, fine to medium grained to silty homogeneous sandstone.
- General absence of deformation fabric, however in some places this unit can be strongly fractured.

RAT (Grey) - host to LOB:

- Grey green, massive, felspathic, calcareous, medium grained, homogeneous sandstone and siltstones.
- Generally in conformable contact with the overlying dolomitic rocks of the SD and can be locally strongly deformed.

DStrat – Dolomie Stratifiee host to LOB:

- Brown-yellow (oxidised), fine-laminated, stratified, silty, dolomite with a distinct layer-parallel, possibly algal or bacterial vughiness.
- The unit contains near its footwall contact with the grey RAT very distinctive silicified and ferruginous nodular concretions.
- The unit measures between 2 m to 3 m in true thickness.

RSF (Grey) - Roche Siliceuse Feuilletée host to LOB:

- Grey brown, very fine laminated, silty dolomite with a distinct, leafy foliation.
- The unit rarely exceeds 3 m in thickness.

SD – Shale Dolomitique host to UOB:

- Brown-yellow to light red-white (oxidised), laminated, stratified, slaty and silty, dolomite with a distinct layer-parallel foliation.
- Sometimes appears massive to poorly bedded and very fine grained.
- When un-oxidised this unit is strongly carbonaceous.

CMN – Calcaire a Mineraux Noirs host to TOB:

- Brown-yellow to light red-white (oxidised), well laminated, cryptalgal-stratified, recrystallised and coarse grained dolomite.
- It appears as medium to thick stratified bedded and interlayered, containing intervals of more carbonate-dominated versus intervals with silt contamination.
- When un-oxidised this unit is moderately carbonaceous.

BH – Breche Heterogene:

- Light red-white mottled fragmental and brecciated, talcy, kaolinitic, soft matrix supported heterogeneous breccia.
- Appears as dyke-like bodies of very variable width.
- The appearance of the rock unit is often similar in the weathered and the unweathered zone.
- In the pit walls it can be observed that this unit has structural contacts with the adjacent rocks and commonly contains allochthonous fragments of the host rocks.

ITEM 10 Deposit Type

The following information is **unchanged since the March 2008 Technical Report** (Anvil, March 2008).

All three mineralised stratigraphic positions of the Katangan Copperbelt (namely, the LOB (RSF/DStrat/grey RAT), the UOB (SD) and the TOB (CMN) exist at the Kinsevere Project. Whilst these structurally modified, stratiform deposits have yet to undergo formal academic review, a three stage model has been proposed for their genesis (Figure 10.1 and Figure 10.2):

- A Late Diagenetic Mineralisation - Primary (late-diagenetic) stratiform mineralisation is well preserved at Tshifufia, with some of the sulphides occurring in veins, stringers and fault zones as a product of deformation-induced remobilisation.
- B Remobilised Mineralisation - Principal area deformation was north-directed with subsequent east-west compression resulting in the Tshifufia *écaillé*'s synformal structure. In some cases, these folds appear to have been "breached" by local faults. Some of the highest-grade (remobilised) mineralisation appears proximal to hinge zones of these cross-folds, especially where faulting is present. Remobilisation is thought to have resulted in an upgrading of primary mineralisation to $\pm 3\%$ copper.
- C Supergene Mineralisation - The oxide mineralisation blankets the sulphide deposit with no apparent stratigraphic control on grade. Superior grades are spatially associated with the base of weathering. Patchy barren zones are linked to strata permeability, the intensity of near-surface leaching and the presence of local cross-structures.

Figure 10.1 Schematic Model of Tshifufia Deposit

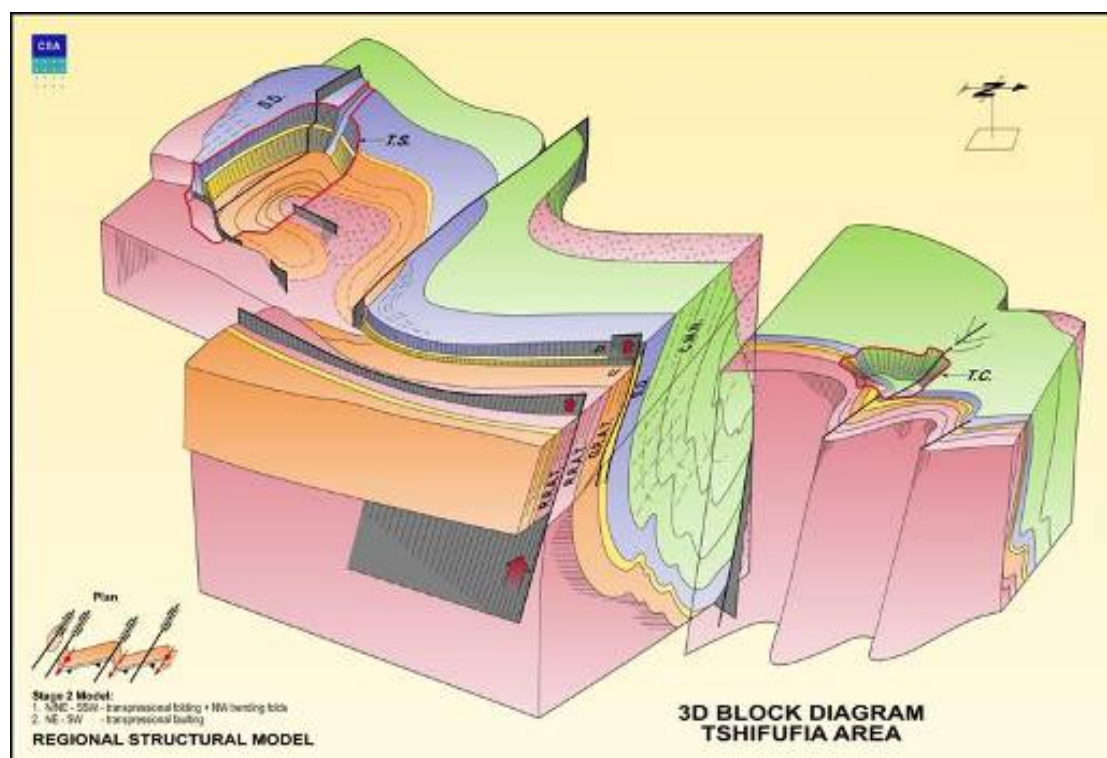
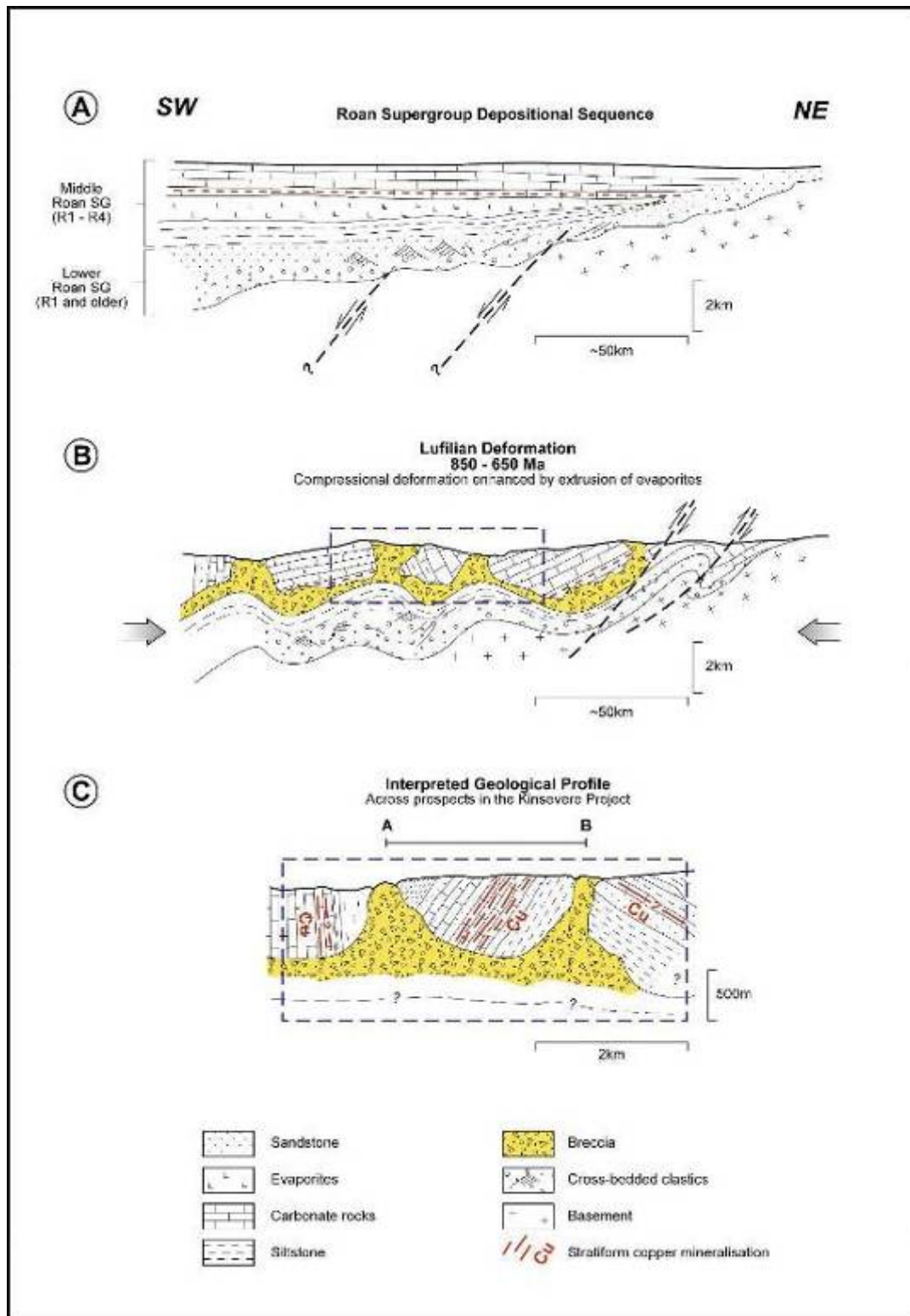


Figure 10.2 Diagram of Significant Developmental Stages in the Formation of Kinsevere Cu Mineralisation and its Geological Setting



ITEM 11 Mineralisation

The following information is **unchanged since the March 2008 Technical Report** (Anvil, March 2008).

The copper and cobalt mineralisation of the Kinsevere Project area occurs within Mines Group sediments of the Lower Roan Supergroup (R-2). To date, no mineralisation has been identified within the overlying Kundelungu Formations.

The Mines Group unit is a moderate, siliclastic carbonate (dolomitic) unit, deposited under reducing anoxic conditions in a restricted, shallow-marine or lacustrine environment. The host rocks are typical of Congolese deposits, underpinning the principal difference between DRC and Zambian portions of the belt. Parent sedimentary lithologies in Zambia are predominantly arenaceous and argillaceous in composition.

These two mineralisation styles may be present at Kinsevere: the LOB is hosted by grey RAT, DStrat and RSF, while the UOB is confined to the overlying SD unit. The unit overlying the SD, the CMN can exhibit significant mineralisation (ie, the TOB). The CMN is well developed in the central portion of Tshifufia Central and Tshifufiamashi, possibly cogenetic with higher levels of local deformation and/or remobilisation.

In general, regional metallogenesis assumes laterally stable depositional conditions, lithologic continuity and large scale ore fluid migration / remobilisation. While this straightforward interpretation remains highly practical for exploratory purposes, Kinsevere orebodies may represent local variants of a much broader, archetypal stratiform Congolese copper deposit.

The regolith profile includes a near-surface veneer of bleached clays, underlain by a zone of decreasingly oxidised country rocks. Whereas the LOB, UOB and TOB stratigraphic positions are all mineralised at Tshifufia, oxide mineralisation lacks stratigraphic control, although grade generally improves towards the base of weathering at approximately 110 m. Bonanza style oxide mineralisation is observed in central sections of the Tshifufia deposit. In general, oxide Cu grades remain significantly higher than those intersected in primary zones.

At Tshifufia Central, a large portion of oxide mineralisation is hosted in the RAT immediately below the DStrat contact in fractures and veins. High-angle faults and bedding-parallel fault breccias in the SD form another important setting for copper oxides, with mapping in the SD unit identifying widespread breccia controlled, bedding-parallel mineralisation. Mined surfaces record minimal cross-strata dispersion of supergene ore minerals, together with small scale interleaving of mineralised and unmineralised beds.

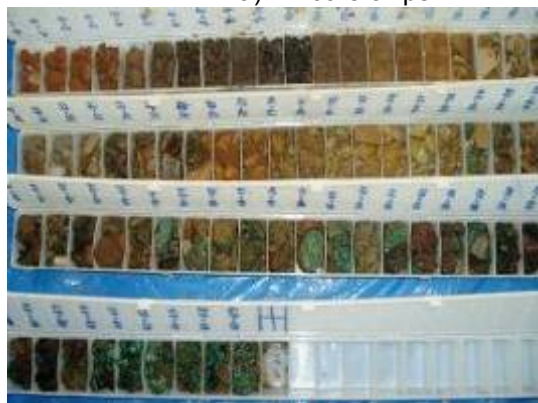
Oxide ore mineralogy at Tshifufia, Tshifufiamashi and Kinsevere Hill is composed predominantly of malachite ($\text{Cu}_2\text{CO}_3(\text{OH})_2$) and pseudomalachite ($\text{Cu}_5(\text{PO}_4)_2(\text{OH})_4$), with minor chrysocolla ($\text{Cu}_2\text{Al}_2\text{H}_2\text{Si}_2\text{O}_5(\text{OH})$) and rare intergrown heterogenite ($\text{CoO}(\text{OH})$). These occur as disseminations and/or in veins and veinlets, which sometimes coalesce into prominent “clots” (Figure 11.1 and Figure 11.2).

Figure 11.1 Brittle Vein Network of Supergene Cu Mineralisation in GRAT, Pit Wall



Figure 11.2 Mineralised Intercepts from Phase II Drilling Programme

a) Air core chips



(b) DDH core – oxide zone



(c) DDH core – sulphide zone



Malachite commonly occurs as coarse (0.2 to 0.7 mm) variably bladed crystals. A significant proportion is also observed within goethite, quartz and phyllosilicate gangue mineralogy. Noticeably finer-grained pseudomalachite is also identified within principal gangue phases together with heterogenite. Primary sulphide mineralisation is not observed at the surface. Minor copper sulphide (typically chalcocite) can occur within oxide ore zones, although it's often restricted to fresh

rock beneath, or within 10 m to 20 m thick transition zones, separating weathered and unweathered material.

Additional secondary Cu and Co phases (chalcocite, cuprite and inter-grown heterogenite) are preserved in weathered domains as either disseminated, fracture- or cleavage-controlled phases within favourable host assemblages.

The base of oxidation forms an irregular surface which varies with rock fabric, especially beneath deformation zones. Transitional horizons are not defined by appreciable increases in Cu mineralisation, native Cu and other Cu oxides. These tend to form erratic, laterally discontinuous layers, which transect bedding planes.

Stratiform, copper sulphide mineralisation, including chalcopyrite, chalcocite, and bornite typically occurs as finely disseminated, bedding parallel layers, stratiform veins or as replacement of pyrite nodules. Lesser cross cutting quartz-carbonate-sulphide veins are also observed. Sulphide remobilisation during faulting has also led to the development of variably mineralised breccias.

Several diamond drill holes at Tshifufia Central and Tshifufiamashi intersected primary sulphide mineralisation at depth. Local bedding and drill hole orientation mean all such holes will intersect this mineralisation in the CMN. Observations regarding sulphide mineralisation indicate:

- A more balanced volumetric distribution between disseminated stratiform mineralisation and fracture and breccia controlled mineralisation in the CMN.
- Breccia and vein controlled mineralisation tend to overlap spatially with stratiform mineralisation, whereas stratiform mineralisation may occur locally as the sole form of mineralisation.
- Two sub-types of stratiform mineralisation occur including:
 - o discrete stratiform veins filled predominantly with chalcopyrite and only minor quartz and calcite with very fine disseminated chalcopyrite in a black fine-grained calcareous carbonaceous siltstone; and
 - o broad intervals (>3 m) of light grey coloured coarse crystalline (probably recrystallised) silty dolomite with coarse grained disseminated chalcopyrite which coalesce locally to form discontinuous stratiform boudinaged veins.
- Mineralised veins in the first sub-type differ from those in the second sub-type in that they contain volumetrically more calcite and quartz as gangue minerals.

Although sulphide mineralisation is localised within laterally continuous stratigraphic intervals, supergene mineralisation observed in core and outcrop appears restricted to faults, and associated fracture / breccias zones.

Within the CMN, the TOB mineralisation is predominantly localised in the coarse-re-crystallised dolomitic layers which in the oxide zone occurs strongly friable partly unconsolidated dolo-psammite.

Further drill core and outcrop investigations suggest that three types of overprinting mineralisation can be identified at the Kinsevere Project:

- The earliest phase of mineralisation includes stratiform disseminated sulphide mineralisation, overprinted by deformation-controlled vein and fault-zone copper-quartz-carbonate mineralisation.

- This process is likely to have led to re-mobilisation and re-precipitation of early-stage stratiform mineralisation.
- Supergene processes, in deeply weathered terrains, have imposed a third phase of Cu remobilisation of Cu in the mineralised zone.

ITEM 12 Exploration

Information on the Company's exploration programmes is largely **unchanged since the March 2008 Technical Report** (Anvil, March 2008), **except for continuing work carried out in 2008.**

Following preliminary exploration and shallow mining conducted in the 1990's (by Gécamines and EXACO), Anvil has continued its exploration strategy at Kinsevere with the aim of defining and systematically upgrading the Company's Mineral Resource inventory. Anvil's exploration programmes have had three principal objectives:

1. to verify the results of previous explorers;
2. to improve the understanding of the oxide Mineral Resources for Anvil's Stage II Development Project (see Items 13 to 19); and
3. to identify other local Mineral Resource potential, including sulphide depth extensions at Kinsevere sufficient for the Project to extend to a Stage III development phase.

To date, the following has been achieved:

1. Trenching programme at Kinsevere deposits (November, 2004) - confirmed Gécamines surface results.
2. Kinsevere 2005 Drilling Programme (Phase I) resulting in a preliminary Mineral Resource estimate of 333,200 t copper metal defined at three deposits (Anvil, September 2005).
3. Feasibility Study Development Stage I (May, 2006) – confirmed viability of the HMS-EAF project designed to produce 23,000 tpa of black copper.
4. Kinsevere 2006 (Phase II) and 2007 (Phase III) Drilling Programmes (December 2006 and September 2007) resulting in an updated Mineral Resource at Tshifufia and Tshifufiamashi, the two largest of the three mineral deposits, and also at Kinsevere Hill.
5. Kinsevere 2008 Drilling Programme (Phase IV) resulting in a further updated oxide Mineral Resource for Tshifufia and Tshifufiamashi and a Mineral Resource for the Kinsevere Hill deposit merged with the Kinsevere Hill Extension (Kilongo). The Phase IV programme although not completed, enabled an updated sulphide Mineral Resource for Tshifufia and Tshifufiamashi.

RC infill and grade control drilling is planned for Tshifufia and Tshifufiamashi during 2010 and continuing during the Stage II operations phase. Once Stage II production is underway from 2011, it is expected that the Phase IV sulphide drilling programme will be completed and extended.

The Phase I to IV drilling investigations listed above were completed by various contractors and also by contract operators using Anvil drilling equipment. Details for each phase are listed in Item 13.

ITEM 13 Drilling

The following information provides an **update on the drilling work completed in 2008**, including details of collar and down-hole surveys. Information on sample length and mineralisation thickness is provided in Item 14.

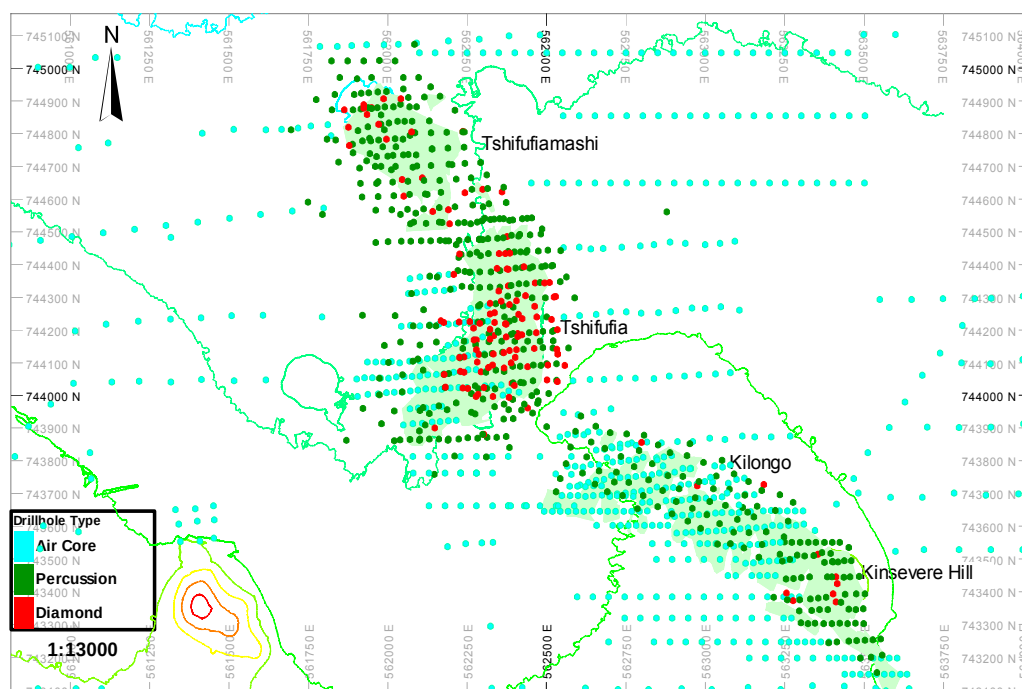
The Company's drilling programmes (Phases I to IV as described in Item 12) have concentrated on extensional and definition Mineral Resource drilling, with consideration for the collection of representative material for both metallurgical test work and for geotechnical studies, whilst ensuring that proposed infrastructure sites have been properly sterilised.

Table 13.1 summarises the number of holes and number of drilled metres by drill type and objective, including a cumulative total, project-to-date. Figure 13.1 shows the distribution of drillholes.

Table 13.1 Summary of Resource Drilling at Kinsevere

	2005 to 2007		2008		CUMULATIVE TOTAL	
	Metres	Number of Holes	Metres	Number of Holes	Metres	Number of Holes
Air-Core Drill	13,314	286	661	15	13,975	301
Diamond Drill	6,552	39	18,052	61	24,604	100
Reverse Circulation	26,540	318	22,414	177	48,955	495
Total	40,424	643	26,865	150	87,532	896

Figure 13.1 Plan Illustrating the Distribution of Drillholes Across Kinsevere



13.1 Drill Rigs and Operators

Drilling at Kinsevere has been completed by various contractors, but most recently by Ox and Capital Drilling using Anvil owned drill rigs, as listed in Table 13.2. Figure 13.2 shows RC drilling in the Tshifufia Pit during the Phase IV programme.

Table 13.2 Drill Rigs and Operators

Year	Drill Contractor	Type of Rig and Drilling
2005	Titan Drilling	Thor RC drill rig; 500 psi Compressor
2005	Resource Drilling International	Boart Longyear LF90 Diamond Drill rig
2006	Wallis Drilling (Anvil/Wallis)	Wallis Mantis 75 Air Core rig, with 150 PSI compressor
2006	Wallis Drilling (Anvil/Wallis)	Smith Capital Hotline 310 RC rig, with 350 PSI compressor
2006	Wallis Drilling (Anvil/Wallis)	Longyear LF90 Diamond rig x2
2006	Titan Drilling	Thor RC rig, with 360 PSI compressor
2006	Ox Drilling	Thor RC rig, with 360 PSI compressor
2007	Wallis Drilling (Anvil/Wallis)	Wallis Mantis 75 Air Core rig, with 150 PSI compressor
2007	Wallis Drilling (Anvil/Wallis)	Boart Longyear LF90 Diamond rig
2007	Titan Drilling	2 x Thor RC rig, with 435 PSI compressor
2008	Capital; Ox Drilling	4 x LF 90 DD rigs + 2 x Alton multi-purpose AC-RC-DD rig

Figure 13.2 In pit RC Drilling in the Tshifufia Pit in 2008



13.2 Collar Locations

All drill holes were initially located using a hand-held GPS or tape and compass from a surveyed grid, which though relatively accurate in the x and y planes (+/- 5 m) was not accurate in determining z co-ordinates. Following completion of each drilling programme, the collar locations were re-surveyed using a theodolite and the locations transferred into the Project database. Between the time of drilling and the time of surveying, some of the collar positions were disturbed and lost by mining and exploration activity, and so the only collar coordinates available were sourced from hand-held GPS estimates. Elevations of these holes were later defined more accurately using the results of a LIDAR survey (a detailed aerial survey). All survey data was included in the Project database.

13.3 Down- Hole Surveys

Downhole surveys (Table 13.3) were completed using an EZBQ multishot survey instrument and the diamond drillholes were surveyed as drilled. The RC holes were surveyed at the end of each year when a contractor was brought in to survey the holes.

Table 13.3 Down-hole Surveying and Operators

Year	Contractor	Survey type
2005	Richter Surveys	single shot down-hole surveying – 75% RC holes
2005	Anvil	Tropari survey instrument – Diamond drillholes
2006	Reflex	EZBQ Multishot survey tool (RC and Diamond)
2007	Reflex	EZBQ Multishot survey tool (RC)
2007	Wallis	EZBQ Multishot survey tool (Diamond)
2008	Capital	EZBQ Multishot survey tool (RC/Diamond)

ITEM 14 Sampling Method and Approach

Updated information is provided under this Item in respect of **sampling protocols, potential sampling bias and density measurement revisions.**

The drillhole locations where samples have been collected are illustrated in Figure 13.1. This figure is updated from a similar version provided in the March 2008 Technical Report (Anvil, March 2008).

14.1 Geological Logging

Chips from all RC and air core holes were logged at the drill site as drilling progressed. Diamond drill core was marked up at the drill site, but logged in the exploration camp on core racks. All logs were recorded on field sheets, which were subsequently captured into an MSExcel spreadsheet for uploading into Anvil's database.

All chip boxes and drill core were photographed prior to final storage/archiving.

14.2 Sampling Protocols

Sample intervals for diamond drill core were controlled by lithology and visible mineralisation (ie, samples were taken up to but not across lithological contacts and obvious high grade zones were sampled separately from lower grade intervals), to ensure that as much information as possible was collected on the controls of the mineralisation.

RC and air core chip samples were collected at 1 m intervals across visibly mineralised zones and at 4 m intervals across what appeared to be barren intercepts.

Drill chips (RC and air core) were split using a Jones riffle splitter to generate 2 kg samples, which were bagged, labelled and stored at the exploration camp, prior to dispatch to the laboratory.

Diamond core was split into halves and one half core was quartered. A quarter core was submitted to the laboratory for assay, a quarter was retained for SG measurements and possible metallurgical testwork, and the remaining half was retained in the core tray as a geological record.

Quality control and quality assurance measures adopted are described in Item 15 and 16 of this report.

14.3 Relationship Between Sample Length and Mineralised Thickness

There is no relationship between the sample length and the true thickness of the mineralisation. The mineralisation is significantly thicker than the average sample length.

14.4 Sample Representivity and Relevant Samples

It is Anvil's opinion that all samples are representative of their respective interval, and that no bias has been introduced by selective sampling. Further, it is Anvil's opinion that there are no other factors that are expected to result in any significant bias. Sample length was not based on rock type, widths of mineralised zones or any other geological controls, rather a standardised one metre sample length. Diamond core samples did, however, terminate at major lithological boundaries, but these small

samples do not have any relationship to the much greater thickness of the mineralised zones.

A list of best interval samples is presented in Table 14.1.

Table 14.1 Summary of Best Drill Intercepts (relevant samples)

Hole Number	From	To	Intercept (m)	% Cu	% Co	Orebody
TCAC001	22	63	31	3.54	0.84	TOB
TCAC002	6	69	63	7.83	0.58	TOB
TCAC003	59	69	10	2.51	0.11	TOB
TCAC005	32	68	36	6.92	0.37	TOB
TCAC006	16	69	54	6.67	0.39	TOB
TCAC007	32	69	37	4.92	0.28	TOB
TCAC008	39	67	28	5.59	0.23	TOB
TCAC013	19	69	50	6.22	0.84	TOB
TCAC014	47	69	22	2.83	0.06	TOB
TCAC016	38	67	29	4.92	0.03	TOB
TCAC017	32	55	23	3.45	0.06	TOB
TCAC018	10	48	38	4.84	0.13	TOB
TCAC019	48	69	19	4.24	0.02	TOB
TCAC020	28	61	33	4.10	0.08	TOB
TCAC021	23	67	44	9.24	0.36	TOB
TCAC023	28	69	41	6.28	0.14	TOB
TCAC024	12	43	31	5.20	0.09	TOB
TCAC025	13	63	50	7.42	0.12	TOB
TCAC027	44	70	26	12.80	0.11	TOB
TCAC030	24	35	11	5.79	0.02	TOB
TCAC031	31	75	44	6.27	0.25	TOB
TCAC032	27	66	39	6.10	0.16	TOB
TCAC033	26	32	6	11.97	0.01	TOB
TCAC034	18	27	9	5.52	0.12	TOB
TCAC037	24	50	26	6.41	0.17	TOB
TCRC040	24	63	39	2.84	0.08	LOB
	71	100	29	7.10	0.05	
TCRC041	53	91	38	7.98	0.04	TOB
TCRC042	21	46	25	5.15	0.16	TOB
	115	151	36	2.42	0.17	
TCRC043	29	57	28	2.39	0.08	TOB
	71	81	10	1.14	0.13	
TCRC044	44	59	14	2.77	0.09	TOB
	64	98	34	3.16	0.09	
TCRC045	18	24	6	4.91	0.04	TOB
	40	113	73	6.05	0.32	
TCRC046	17	40	23	7.84	0.48	TOB
	43	95	59	3.56	0.45	
TCRC047	36	71	35	3.32	0.21	TOB
TCRC048	55	90	35	6.94	0.08	UOB
TCRC049	25	73	48	1.90	0.54	TOB
TCRC050	41	89	48	1.82	0.41	TOB
TCRC051	24	77	53	2.26	0.44	TOB
TCRC052	38	90	52	5.50	0.22	TOB
TCRC056	51	90	39	4.73	0.34	TOB
TCAC047	25	33	8	1.68	0.81	TOB
	41	65	24	6.82	2.21	
TCAC053	25	45	20	9.11	0.32	TOB
TCRC057	54	72	18	10.15	0.09	TOB
TCRC059	16	64	48	1.29	0.47	TOB

Hole Number	From	To	Intercept (m)	% Cu	% Co	Orebody
TCRC060	20	44	24	0.97	0.83	TOB
	58	70	12	1.21	0.19	UOB
TCRC061	35	77	42	2.41	0.01	TOB, UOB
TCRC062	48	54	6	1.90	0.01	TOB, UOB
KPAC112	27	52	25	4.80	0.030	TOB
KPAC113	10	18	8	2.03	0.050	TOB
KPAC114	14	50	36	4.15	0.070	TOB
KPAC116	42	46	4	6.30	0.030	TOB
KPAC117	50	57	7	3.89	0.020	TOB
KPAC132	16	59	43	2.69	0.020	TOB
KPAC135	23	51	28	6.83	0.055	TOB
KPAC153	15	46	31	9.05	0.012	TOB
KPAC154	15	41	26	14.20	0.012	TOB
KPAC155	10	45	35	14.10	0.014	TOB
KPAC171	50	60	10	1.31	0.013	TOB
KPAC172	33	54	21	10.94	0.025	TOB
KPAC173	25	46	21	11.78	0.022	TOB
KPAC174	12	48	36	8.88	0.027	TOB
KPAC175	33	40	7	5.31	0.033	TOB
KPAC176	26	44	18	3.60	0.025	TOB
KPAC205	34	46	12	4.36	0.036	TOB
KPAC206	22	47	25	8.94	0.081	TOB
KPAC207	9	47	38	9.55	0.079	TOB
KPAC208	9	51	42	7.16	0.040	TOB
KPAC209	20	55	35	2.91	0.019	TOB
KPAC226	7	33	26	1.58	0.024	TOB
KPAC227	37	42	5	1.06	0.010	TOB
KPAC238	21	45	24	3.05	0.108	TOB
KPAC239	46	51	5	1.05	0.032	TOB
KPAC240	16	45	28	1.35	0.057	TOB
KPAC242	28	40	12	2.99	0.007	TOB
KPAC243	37	52	15	1.77	0.005	TOB
TFRC026	4	22	18	3.02	0.150	LOB
TFRC026	41	176	135	1.90	0.160	UOB
TFRC027	0	26	26	1.69	0.075	TOB
TFRC028	2	29	27	2.95	0.098	TOB
TFRC030	11	51	39	3.81	0.330	UOB/LOB
TFRC031	1	161	160	4.38	0.290	TOB/UOB/LOB
TFRC032	9	37	28	1.97	0.460	TOB
TFRC032A	3	44	41	2.28	0.290	TOB
TFRC032A	78	150	72	4.65	0.210	TOB/UOB
TFRC033	1	141	140	6.09	0.300	TOB/UOB/LOB
TFRC036	5	21	16	3.94	0.365	LOB
TFRC044	22	34	12	3.70	0.085	LOB
	87	131	44	1.91	0.063	UOB
TFRC045	52	59	7	3.77	0.297	TOB
	64	105	41	2.24	0.038	TOB
TFRC046	15	48	33	2.09	0.289	TOB
	53	82	29	4.81	0.726	TOB
TFRC047	9	20	10	1.44	0.018	TOB
TFRC051	101	120	19	2.14	0.210	TOB
TFRC052	24	61	37	2.11	0.590	TOB
TFRC054	36	56	20	1.45	1.350	TOB
TFRC054	86	107	21	1.92	0.190	UOB
TFRC056	4	34	30	5.18	0.420	LOB
TFRC057	0	72	72	4.18	0.068	UOB/LOB
TFRC057	115	146	31	2.50	0.260	TOB

Hole Number	From	To	Intercept (m)	% Cu	% Co	Orebody
TFRC058	19	38	19	4.30	0.076	TOB/UOB
TFRC058	41	90	49	4.64	0.300	LOB/TOB
TFRC059	68	84	16	1.37	0.081	TOB
TFRC060	17	73	54	3.71	0.230	TOB
TFRC061	44	61	17	4.02	0.017	UOB
TFRC061	115	132	17	3.53	0.036	UOB
TFRC062	30	89	59	1.81	0.390	TOB
TFRC063	68	122	53	0.78	0.100	TOB
TFRC066	71	78	7	3.83	0.005	UOB
TFRC067	79	89	10	1.70	0.120	TOB
TFRC072	54	73	19	0.86	0.049	TOB
TFRC073	105	124	19	2.43	0.019	TOB
TFRC075	20	66	46	1.95	0.810	TOB
TFRC082	61	150	86	4.28	0.110	TOB
TFRC084	27	74	46	1.73	0.052	TOB
TFRC085	33	141	104	2.17	0.055	TOB
TFRC088	84	97	13	1.42	0.019	TOB
TFRC089	95	105	10	3.47	0.008	TOB
TFRC095	62	150	86	2.40	0.250	UOB/TOB
TFRC096	46	150	104	1.69	0.061	UOB
TFRC097	56	150	94	2.27	0.150	UOB
TFRC098	31	52	21	2.11	0.012	LOB
TFRC101	44	100	56	2.45	0.059	TOB
TFRC103	4	106	102	3.17	0.240	TOB
TCRC143	50	101	51	6.92	0.076	UOB
TCRC144	89	99	10	6.21	0.396	TOB
TCRC146	67	79	12	3.51	0.103	UOB
TCRC147	107	140	33	1.52	0.258	TOB
KVRC027	50	62	12	3.88	0.003	TOB/UOB
KVRC028	54	128	74	2.34	0.029	TOB/UOB
KVRC029	34	56	22	1.47	0.110	TOB
KVRC034	18	39	21	1.81	0.260	TOB
KVRC035	5	30	25	6.47	0.062	TOB
KVRC036	1	13	11	3.28	0.014	TOB
KVRC038	4	16	12	1.45	0.015	TOB
KVRC039	24	46	22	1.41	0.023	TOB
KVRC039	60	111	51	2.18	0.024	TOB
KVRC040	23	63	35	2.48	0.018	TOB
KVRC044	7	41	34	0.91	0.012	TOB
KVRC045	41	118	67	5.80	0.036	TOB
KVRC046	55	106	45	3.70	0.099	TOB
KVRC047	65	72	7	1.92	0.024	TOB
KVRC047	91	104	13	4.44	0.020	TOB
KVRC049	52	64	12	2.09	0.004	TOB
KVRC054	24	49	25	3.75	0.016	TOB
KPRC017	60	99	37	1.45	0.007	TOB
KPRC018	18	59	39	4.23	0.037	TOB
KPRC019	13	36	23	1.60	0.010	TOB
KPRC022	58	66	8	1.53	0.001	UOB/TOB
KPRC022	79	87	8	1.64	0.003	TOB

14.5 Potential Sample Bias

Drilling and sampling methods were compared for potential bias across a similar Tshifufia volume using quantile-quantile (Q-Q) plots. As part of the updated Mineral Resource estimate described in Item 19, comparisons were restricted to 1 m sample lengths, to minimise the effects of sample size/mass. Diamond core has a smaller

proportion of the original sample volume in comparison to the reverse circulation and aircore samples, and has a lower risk of material loss and contamination, generally providing a more representative sample.

The comparison highlights the difference in Cu % values between diamond, reverse circulation (RC) and aircore drilling (AC) samples within the oxide domain (Figure 14.1, Figure 14.2 and Figure 14.3), with Cu values being 1.1 % higher at the 50th percentile for RC samples when compared to aircore drilled samples. Similarly, in comparing diamond core and RC samples, there is a positive bias towards the RC samples, with assays 1.2 % higher at the 50th percentile. Comparing the same diamond samples with aircore samples highlights comparable results (Figure 14.3). This suggests RC samples to be positively biased when compared to diamond and aircore samples.

As a result of the higher RC grades, current grade estimates may be slightly optimistic. This is more relevant to the oxidised material and to a lesser extent, the fresh sulphide mineralisation. Of the samples used during Mineral Resource estimation (Item 19), 57 % were from RC drilling, 31 % were diamond and 11 % aircore. It is expected that the quality of RC samples will deteriorate with depth, increasing risk of sample error to the fresh sulphide estimates. To a degree, this risk is offset by the increased number of better quality diamond core samples located within the sulphide domain.

This difference or potential bias will need to be investigated as a continuous improvement via a twinned drilling programme, heterogeneity testing and sampling observation for the various sources of sampling error. Investigation of the total sampling error will assist in defining the sampling variances for each of the methods employed, the true source of error or bias and enable an effective remedial action plan. This will support and or improve the current Mineral Resource classifications.

Figure 14.1 QQ Plot Comparing AC Cu Assay Results With RC Cu Results For Tshifufia Oxidised Material

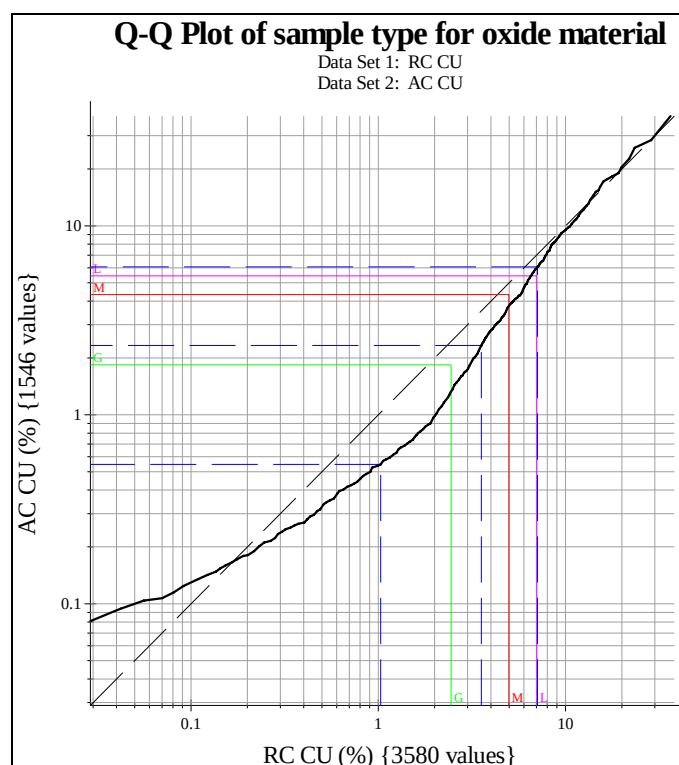


Figure 14.2 Q-Q Plot Comparing RC Cu Assay Results With Diamond Cu Results for Tshifufia Oxidised Material

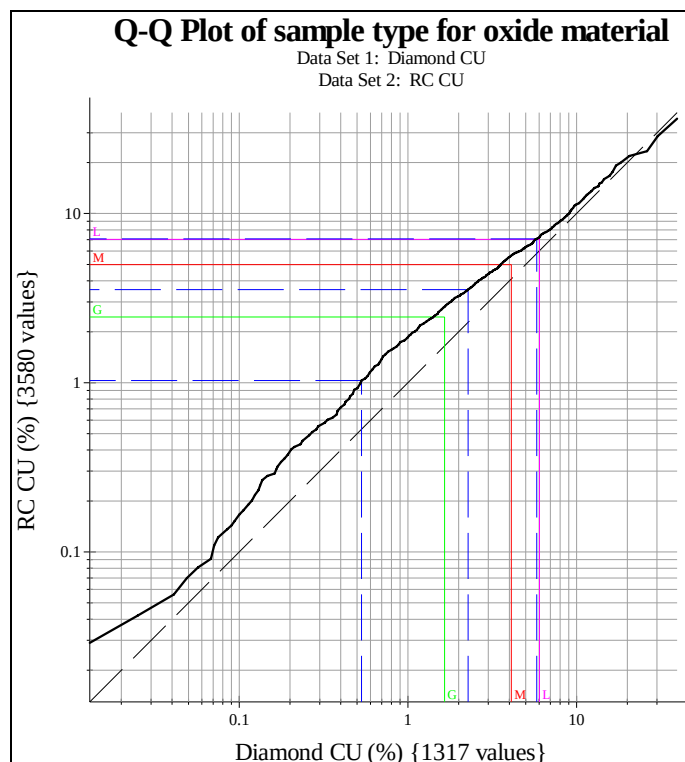
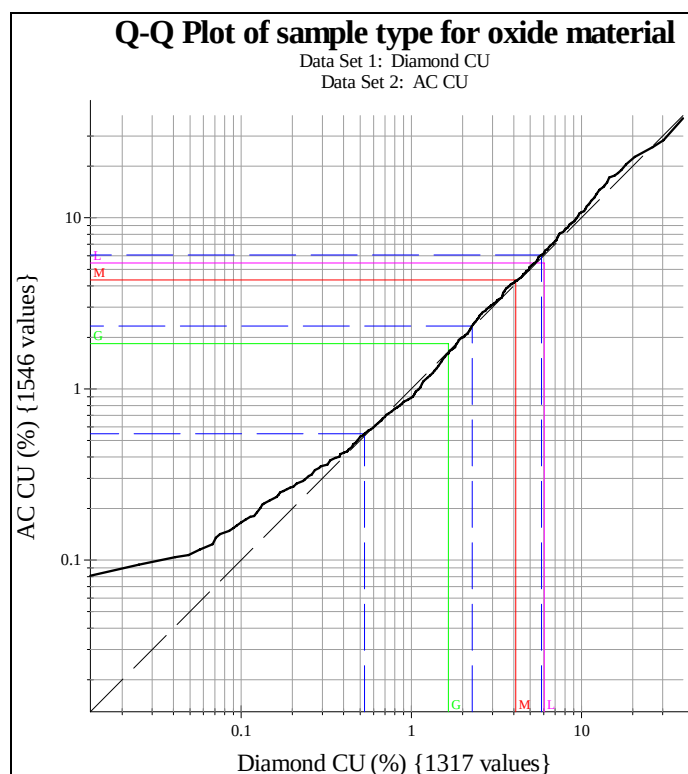


Figure 14.3 Q-Q Plot Comparing AC Cu Assay Results With Diamond Cu Results for Tshifufia Oxidised Material



14.6 Density Measurements

Historically, density was measured on site, at variable depth and across all lithologies from all cored drill samples. Intervals ranging from 10 to 20 cm in length were firstly weighed dry, and then again in water on a suspension cage, using an electronic scale. The density was then calculated using the following formula:

$$\text{Density} = \text{Dry Sample Weight} / (\text{Dry Sample Weight} - \text{Wet Sample Weight})$$

During 2008 it became apparent that the assigned densities (as determined from the density measurements and applied in the 2007 Mineral Resource models according to a linear regression relationship with copper grade) were at odds with those determined from mining and reconciliation studies.

The 2008 Technical Report subsequently commented (p267, Anvil, March 2008):

Following the identification of density measurements as a risk in the Kinsevere Project Feasibility Study in 2007, mining information has been reviewed which indicates there is a discrepancy between the actual bulk density and the measured bulk density. The information collected to date is not representative of the whole Mineral Resource. Therefore a programme is being instigated to collect and evaluate new and robust density information to be used in a third quarter 2008 update to the Mineral Resource estimate.

Investigations revealed that density applied in the Mineral Resource models up to 2007 had been based on density measurements made from core during the 2005/2006 drilling programmes. Unfortunately these density measurements were made on core that was selectively sampled (hard core as opposed to decomposed core), thereby introducing a bias in the density measurements.

Four in-pit bulk sample pits were subsequently excavated, surveyed and the excavated volume determined accurately. The excavated material was weighed on a weighbridge and samples from each pit were tested in the site laboratory to determine moisture content and dry bulk density. The results of the in-pit bulk sample tests are listed in Table 14.2.

Table 14.2 In-pit Bulk Density Sample Results

Location – depth below surface	Orebody	As-mined density (dry t/m ³)
Tshifufiamashi -25 m	TOB	1.65
Tshifufia – 25m	TOB	1.51
Tshifufia – 30m	TOB	1.51
Tshifufia – 30m	UOB	2.07

As a result of these investigations, the procedures for continuing density measurements with drill core were changed. For subsequent block modelling and estimation, newly measured density from 19 drillholes (about 1,700 samples) plus the four in-pit bulk sample estimates were used to assign density in the models (refer to Item 19.1.8).

14.7 Drill Data for Mineral Resource Modelling

Drill data was originally (ie, 2005 to 2006) stored using a DH logger database, managed by Anvil. This has seen progressive replacement by propriety ioGlobal and more recently “DataShed” systems, the latest of which is managed by Maxwell Geoservices Pty Ltd (Maxwell) in Perth Western Australia. To ensure appropriate

supervision and QAQC practices continue, Anvil has initiated a series of audits of this key corporate database.

A total of 630 drillholes were available for updated modelling as described in Item 19, comprising 174 (8,216 m) air core, 361 (35,991 m) reverse circulation, and 95 (23,925 m) diamond drilled holes, for a total metreage of 68,132 m for the Tshifufia and Tshifufiamashi deposits. This includes an increase of 32,294 m since the September 2007 Mineral Resource statement, and in total comprises 49,069 m of assays. For Kilongo and Kinsevere Hill there was a total of 159 air core, 144 reverse circulation and 9 diamond drilled holes for a total metreage of 21,291 m and 21,120 m sampled with assay results.

A further 8,991 m of drilling was being sampled and assayed at the time of Mineral Resource estimation and which should be available for another update during 2010.

Latest Phase IV drillholes were drilled along east west oriented lines spaced approximately 50 m apart with holes located every 25 m along each line. Drilling was angled at 50 to 60 degrees towards the east and west, minimising the risk of oblique ore body intersections and maximising true width intersections. Future infill (and proposed RC grade control) drilling between drill lines will assist to improve continuity of mineralised domains and copper mineralisation. The drilling coverage has resulted in the limits of the Kinsevere deposits being adequately closed to the north, south, east and west. Sulphide mineralisation is open at depth for both Tshifufia and Tshifufiamashi.

ITEM 15 Sample Preparation, Analyses and Security

The following information is essentially **unchanged from the March 2008 Technical Report** (Anvil, March 2008), **except for the reporting of certain QAQC issues** detected in mid 2009. Suspect sample results were not included in the Mineral Resource estimate update described in Item 19. QAQC monitoring data relating to Phase IV drilling is provided in Appendix A.

18.1 Sample Preparation

Samples from the 2008 drilling programme were prepared on site by Anvil's facility at Kinsevere. Sample materials submitted for sample preparation were crushed to minus 2 mm, split with a Jones riffle splitter to produce a 1 kg sample, and then pulverised to a 90% pass rate of 75 microns. The 2 mm sample residue was kept in storage at the laboratory as a reference sample. The 1 kg of pulp was reduced further, using a narrow aperture riffle splitter, to generate the analytical pulp.

These were subsequently dispatched to ALS Johannesburg or Actlab's Pacific laboratories for acid soluble copper (ASCu), total copper (TCu) and cobalt (Co) analysis. ALS Chemex is a leading supplier of analytical and assaying services to the mining industry worldwide, and the laboratory complies with the international standards ISO 9001:2000 and ISO 17205:2005. Actlabs Pacific is currently undergoing ISO17025 NATA certification. All sample preparation protocols, analytical methods used and security procedures adopted are considered to have been appropriate and suitable for the Kinsevere Project.

15.2 Analysis

All mineralised samples were submitted for total copper and acid-soluble copper. Total copper was determined by a four acid digest (HF-HNO₃-HClO₄ digestion, with HCl leach), followed by analysis by ICP-OES or AAS (detection range 0.01-40%). Acid soluble copper was determined by a sulphuric acid leach, followed by an AAS analysis. Results were reported via e-mail to the Data Manager in the Lubumbashi Exploration Office for update into Anvil's corporate (DataShed) database.

15.3 Quality Assurance and Control Measures

During 2008, three separate laboratories were used to analyse Kinsevere samples, including ALS Chemex (Johannesburg), Actlabs (Perth) and McPhar (Philippines).

Relevant QAQC protocols included the use of widespread sample and analytical duplicates, together with certified reference materials and in-house laboratory standards.

The 2008 QAQC data as used for the updated Mineral Resource estimate described in Item 19 were reviewed by Anvil and independently verified by Maxwell. Standards submitted indicate that sample assay results may be viewed as accurate. Duplicate analysis per pair and analytical methods suggest the sample assay results are precise. Blank standards reveal that contamination has been kept to a minimum across the various laboratories. Both blank and standard sample results have revealed some minor sample mislabelling.

Considering previous QAQC results and the number of samples available for estimation, Anvil believes that the accepted sample results will provide a representative estimate of Kinsevere's mineralisation.

All samples processed were analysed with certified reference materials (Table 15.1), sample and analytical duplicates, together with blanks as part of Anvil's quality control and quality assurance (QAQC) programme. Minimum standard and duplicate submission rates included 1 / 20 samples dispatched, with a blank every 50 samples. Periodic round robin testing was undertaken using Kinsevere's site based metallurgical laboratory, operated by SGS.

15.3.1 Standards

The standards or Certified Reference Materials (CRM's) used in the course of this QAQC program were purchased from Geostat Pty Ltd and African Mineral Standards. All CRM / standards were pre-packaged at the sample preparation facility into sample sachets, identical to normal pulverised samples.

Table 15.1 Mean Values (in ppm) of Certified Reference Materials

Standard	Copper	Lead	Zinc	Nickel	Arsenic	Cobalt	Silver
AMIS0038	20,730	-	-	-	-	44	-
AMIS0031	30,840	38	210	80	25	48	8.1
AMIS0032	57,500	-	276	116	30	53	13.9
AMIS0033	14,050	186	337	130	87	71	31.7
GBM301-7	5,542	30	150	29	17	14	1.8
GBM301-8	104,030	1,631	93	57	128	164	23.3
GBM304-16	22,721	728	718	16	-	-	-
GBM398-4	3,981	11714	5,117	4,071	12	1,974	48.7
GBM399-2	205	28	309	14	4	3	0.7
GBM903-13	28,990	21,531	9,373	24,387	-	-	-
GBM997-8	12,050	24	46	8	25		4.2

Standards used in the database generally fall within the analytical tolerance levels required for JORC level reporting for the Mineral Resource categories indicated within Item 19 of this report. As and when occasional analytical failures were observed, ($>\pm 2$ SD's outside of the accepted mean), individual batches were re-run to confirm analytical integrity. On several occasions, investigations revealed that isolated outliers were associated with incorrect sample labelling by field staff. As per previous drilling programmes, and except for the samples described in Item 15.4, no significant analytical issues were identified during this drilling phase and reported assays are considered to be both reliable and accurate.

15.3.2 Duplicates

Sample duplicates were inserted into each batch at irregular intervals, prior to dispatch from the Kinsevere exploration camp. Laboratories were instructed to create and insert laboratory duplicates every 20 samples. Mean average paired difference plots and associated scattergrams demonstrate high precision for lab, field and analytical duplicates, reflective of comparatively low cumulative preparation, sample and analytical errors.

15.4 Quality Control Issues

Assay results for 10,542 remaining Phase IV drilling samples were received from ActLabs (Perth) during April and May 2009. QAQC results from these sample batches were analysed for sub standard sampling and assaying practice. Of the 45 sample batches submitted, nine batches failed due to poor accuracy (standards) and

contamination (blanks) and twelve due to poor precision (duplicates). Analysis revealed that these QAQC issues were likely due to deficient Anvil preparation and dispatch procedures at the time of operations shutdown and staff retrenchments.

The **defective data is not included in the Mineral Resource estimate** compiled for this Technical Report and hence is not material to the Mineral Reserve inventory, production schedules and the cash flow modelling described in Item 25.

The 2009 and 2010 resampling programme will enable this data to be included in the final oxide Mineral Resource before commencement of Stage II production.

15.5 Security

A geologist or geo-technician was on site during the entire drilling programme, and the handling of all sample material (chips and core) was under the direct control of the project geologist. All samples were bagged and labelled on site.

Samples for sample preparation were transported from site to the Lubumbashi Exploration Office (or to Kitwe) by an Anvil vehicle, escorted by a geologist or geo-technician. They were dispatched to the ALS Chemex laboratory in Johannesburg or Actlabs in Australia by air freight (under normal airline security), after clearance through DRC customs facilitated by SDV-Agetraf (an agent well known to Anvil).

ITEM 16 Data Verification

The following information is **unchanged since the March 2008 Technical Report** (Anvil, March 2008).

The three main sample types collected at the Kinsevere Project are samples from diamond drill core (DD), from air core (AC) and from reverse circulation cuttings (RC). Diamond core is generally acknowledged as providing the best sample, although drilling practice and the treatment of the core can sometimes invalidate this generalisation. The assumption is therefore made that the diamond drill data has provided the best quality sample, and the RC data and AC data is evaluated against that data.

Anvil has attempted to compare DD data composites with other drill composites within 2.5 m of each other for evaluation. Unfortunately, insufficient pairs of composites have been available for this evaluation, so only RC samples could be compared with AC samples to check for sample robustness.

During 2006 Anvil completed a twin-hole programme where AC drillholes were twinned by RC holes. Prior to evaluation, the data was prepared by cutting all hole lengths to the shortest length of the twinned holes, removing duplicate samples, and removing intervals which had no sample in one of the twinned holes. The resultant data set comprised 624 m of RC sample data and 624 m of AC sample data. The length-weighted average grade for the complete dataset was 2.61% Cu for the air core data and 2.99% Cu for the RC data.

In order to understand the lower grade for the air core, the data was split into three horizons for further analysis. These were 0 m to 20 m depth, 20 m to 40 m depth and deeper than 40 m. The 0 m to 20 m data was characterised by a generally low grade and only 65 data points with varying sample intervals. Statistics other than a length-weighted average were therefore not considered appropriate. The RC data gave a slightly higher average grade than the AC data. The 20 m to 40 m data was quite consistent with all of the statistics being relatively similar (Table 16.1).

Table 16.1 Summary Statistics – 20 m to 40 m Depth

	Air Core	RC
Number	141	141
Mean	1.86	1.88
Median	0.80	0.88
75th Percentile	2.97	2.81
90th Percentile	5.52	6.22
95th Percentile	8.26	8.76

The data for the twinned holes at depths greater than 40 m (Table 16.2) showed the most variation between the two data sources. Notably, the median is similar in the RC data to the AC data, but the higher grades are not reproduced in the AC data.

It is evident that the air coring produces a sample that is biased low when compared to the RC data. This is obvious in both the AC-RC twin drill programme and the database review.

Review of the twinned hole data indicates that the AC works reasonably well up to about 40 m depth, but deeper than 40 m, there is a bias in the AC data with the grades being lower than the RC. It is unclear whether this is just an issue with high grade mineralisation not being lifted at depth from the AC holes, or that the only

significant high-grade mineralisation was tested at depth, or a combination of the two issues.

Table 16.2 Summary Statistics – Greater than 40 m Depth

	Air Core	RC
Number	223	223
Mean	5.42	6.70
Median	3.65	3.73
75th Percentile	7.94	8.59
90th Percentile	12.68	14.58
95th Percentile	17.97	19.20

The data described under this Item has been verified by Dr Geoffrey Booth acting as QP.

ITEM 17 Adjacent Properties

The nearby Nambulwa tenement PE 539 (Figure 9.2) has been part of the Kinsevere Project. An internal document on the 2007 Nambulwa exploration programme prepared by the exploration team of AMS (AMS, October 2007) was followed by a high-level technical analysis by Anvil in January 2008 (Anvil, January 2008). This analysis considered the Nambulwa copper mineralisation as incremental feed to the Stage II SX-EW processing facility. The analysis only considered potential plant feed from PE 539 and not from the surrounding PR 1884 and PR 5326 tenements (PR, Research or Exploration Permit) in which AMCK had a joint venture exploration interest with the Lufupa Company (Lufupa).

The defined Nambulwa grade/tonnage estimate, based on an indicative estimate of potential incremental feed to the Kinsevere Stage II SX-EW plant, was concluded to provide minimal contribution to the overall economics of the Base Case cash flow model presented in the Feasibility Study (AMCK, December 2007).

The estimated grade/tonnage only represented about six months of feed to the 60,000 tpa capacity plant. The average grade is less than that of the combined Kinsevere deposits. Hence, as incremental feed with an additional road haulage cost, the Nambulwa copper mineralisation was considered as unlikely to be mined and processed at Kinsevere within the next ten years.

Following this exploration and evaluation programme, **the Nambulwa tenement is in the process of being returned to Gécamines.**

ITEM 18 Mineral Processing and Metallurgical Testing

The following information is **unchanged since the March 2008 Technical Report** (Anvil, March 2008).

Flowsheet development for Stage II of the Kinsevere Project was based upon a comprehensive metallurgical testwork programme performed on drill core samples from the oxide zones of the Tshifufia, Tshifufiamashi and Kinsevere Hill deposits. Anvil collected the samples for this testwork in a manner that made the samples as representative as possible including average grade and rock type.

The majority of the testwork was performed by Independent Metallurgical Laboratories (IML) in Perth Western Australia, with some comminution testing at both Amdel in Adelaide (South Australia) and the JKTech facility in Brisbane Australia. The remaining testwork components were performed by other reputable laboratories and leading equipment and reagent manufacturers such as Outokumpu and Cognis.

Particular attention was paid to the results from the Tshifufia (Central and South) samples as this deposit was deemed to be potentially the largest and was expected to supply ore feed during the early Project years.

Orway Mineral Consultants, a subsidiary of Lycopodium Limited that specialises in comminution circuit design, were enlisted to interpret this area of the testwork and make recommendations on the crushing and grinding circuits.

Graeme Miller, of Miller Metallurgical Services, was retained by Anvil as a hydrometallurgical consultant to provide advice on the leaching, solvent extraction and electrowinning facets of the process flowsheet.

18.1 Metallurgical Samples

The testwork was performed on two sets of samples:

- Existing crushed material stored at Independent Metallurgical Laboratories. These samples had previously been used for metallurgical testwork relating to Kinsevere Stage I in early 2006.
- Fresh drill core obtained in mid-2006.

The reference numbers of the drill cores used for the metallurgical testwork are listed in Table 18.1.

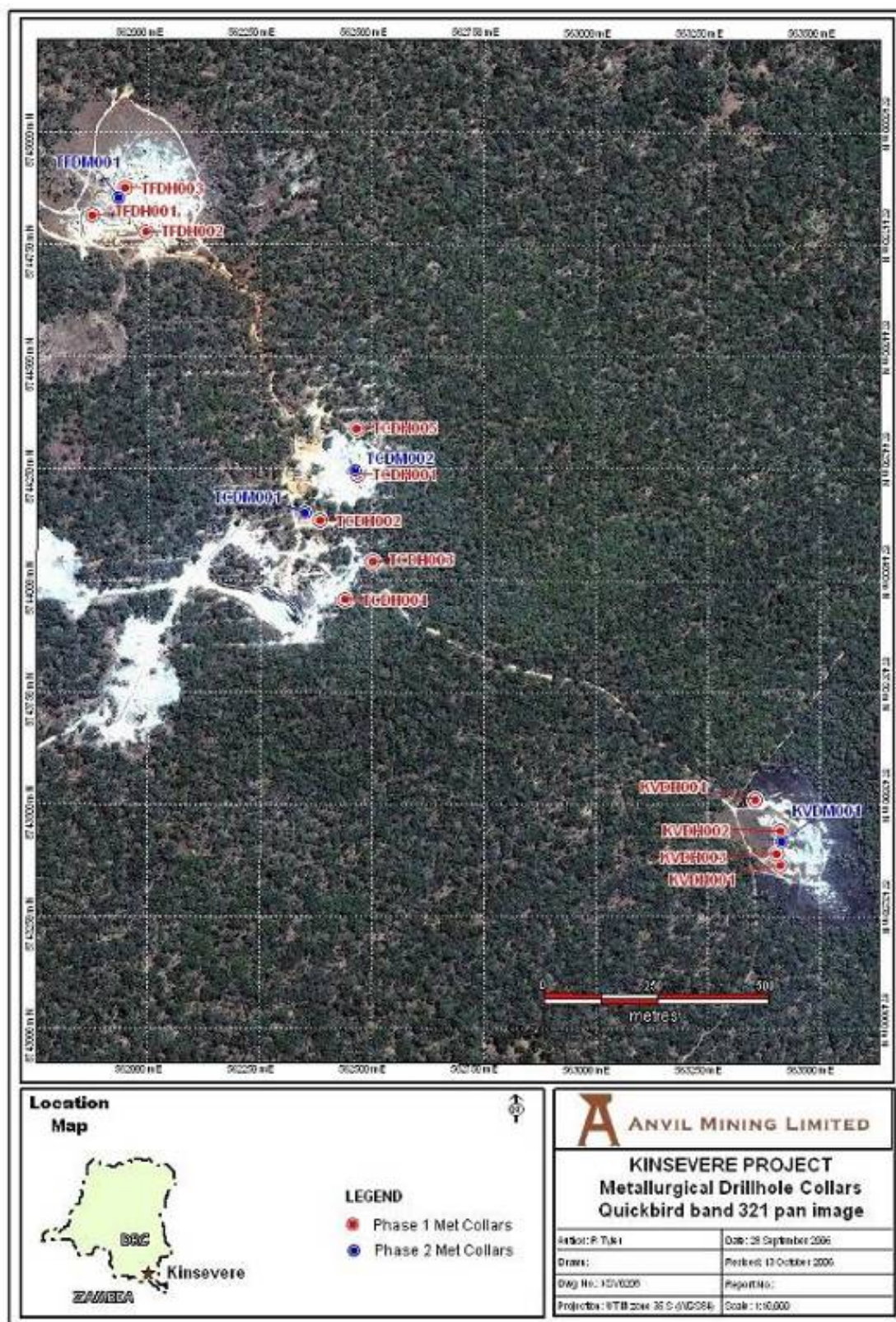
Table 18.1 Drill Core Samples Used for Metallurgical Testwork

	Tshifufia	Tshifufiamashi	Kinsevere Hill
Existing Core	TCDH001 TCDH002 TCDH003 TCDH004 TCDH005	TFDH001 TFDH002 TFDH003	KVDH001 KVDH002 KVDH003 KVDH004
Fresh Core	TCDM001 TCDM002	TFDM001	KVDM001

The locations of the drill hole collars are illustrated in Figure 18.1.

Limestone and burnt lime samples for neutralisation testwork were obtained from Gécamines' Likasi operation.

Figure 18.1 Metallurgical Drill Hole Collar Locations



The Feasibility Study report for Stage I of the Kinsevere Project (Anvil, March 2006) included a mining schedule for the Tshifufia, Tshifufia South, Tshifufiamashi and Kinsevere Hill deposits. The average grades for each deposit were calculated for the period 2008 to 2016 (amalgamating the data provided for Tshifufia and Tshifufia South).

The existing metallurgical samples from the Kinsevere Project consisted of individual samples representing short intervals (typically 1 to 3 m) each with a recorded analysis for Total Copper (TCu), Acid Soluble Copper (ASCu), Cobalt (Co) and Calcium Oxide (CaO).

Master Composites were prepared for each deposit by sequentially eliminating the individual samples with the highest copper grades until the average Total Copper grades for each deposit matched the average mining grades. No attention was paid to the other analyses relating to each sample.

Details of the Master Composite samples are shown in Table 18.2, including the comparison of the actual composite assays in relation to the preliminary mine plan.

Table 18.2 Master Composite Samples

Deposit	No. of Holes	Depth Range (m)	No. Samples	Cu %		Co %	
				Mine Plan	Actual	Mine Plan	Actual
Tshifufia	5	12 - 120	147	4.11	3.94	0.15	0.19
Tshifufiamashi	3	0 - 114	172	3.98	3.97	0.15	0.15
Kinsevere Hill	4	39 - 104	105	2.94	2.41	0.02	0.01

The fresh HQ full diamond core that was drilled specifically for the Project Study metallurgical testwork was inspected on arrival at IML. All of the core was extremely soft and had to be handled with care to prevent breakage for future comminution testwork.

Variability composite samples for the second stage of testwork were selected from the new core samples. The intervals included are detailed in Table 18.3.

Table 18.3 Variability Composite Samples

Deposit	Hole	Sample	From (m)	To (m)	Cu (%)	Co (%)
Tshifufia	1		34.9	52.5	10.78	0.29
	2	A	26	39.6	4.28	0.27
	2	B	59.8	73.8	7.09	0.35
Tshifufiamashi	1	A	6.9	23.8	3.47	0.29
	1	B	49	63	7.24	0.43
Kinsevere Hill	1	A	15.4	35.9	4.15	0.01
	1	B	78.5	91	1.98	0.01

More detailed information on metallurgical testwork is presented in Item 25.4.

18.2 Summary of Metallurgical Testwork Results

The major conclusions drawn from the testwork programme can be summarised as follows:

- Malachite $[\text{Cu}_2(\text{OH})_2\text{CO}_3]$ is the predominant copper mineral in all of the Kinsevere oxide deposits.
- The ore is highly weathered and SAG milling testwork characterised it as being “very soft”.
- The average Crushing Work Index and Unconfined Compressive Strength values determined for Tshifufia, Tshifufiamashi and Kinsevere Hill ores were between 1.4 to 3.5 kWh/t and 7.6 to 16.4 MPa, respectively.
- The Ball Mill Work Indices varied both between and within the deposits, but were generally moderate in the range 7 to 12 kWh/t.
- All deposits recorded very low Abrasion Indices (less than 0.03).
- Excellent Acid Soluble copper recoveries were attainable by leaching in sulphuric acid, with extraction percentages ranging from the mid to high 90's.
- Leach kinetics were rapid with only 4 to 8 hours required to effect near maximum acid soluble copper dissolution.
- Lowering the leach pH from 2.0 to 1.5 slightly improved acid soluble copper recovery.
- Tshifufiamashi samples responded marginally favourably to grinding to a P_{80} of 150 μm with copper recoveries slightly enhanced compared to a coarser 212 μm grind. The other deposits appeared less sensitive to the grind size.
- Gangue acid consumption levels were generally very low ranging from 3.1 to 17.7 kg/t.
- Analysis of leach solutions did not identify any elements deleterious to solvent extraction or electrowinning that were present in significant quantities. The filtered leach liquors were visually similar to pure blue copper sulphate solution.
- Both pre-leach and post-leach slurries responded well to commercial flocculants and high clarity overflows were obtained, particularly when highly diluted flocculant solution was used. Underflow pulp densities approaching 60%w/w were achieved with optimised test conditions.
- Solvent extraction testwork indicated that the Kinsevere leach solutions were amenable to processing using standard commercial copper extractants. Chemical transfer of iron to the electrolyte was expected to be minimal.
- Rheology testwork indicated that nearly all slurries exhibited shear thinning characteristics with viscosity declining as shear rate increased. The test slurries exhibited significantly increased viscosity levels at the higher pulp densities (55% to 65%w/w).

ITEM 19 Mineral Resource and Mineral Reserve Estimates

Detailed technical information provided under this Item relates specifically to the **Mineral Resource and Mineral Reserve estimate updates**.

Following publication of the Mineral Resource estimate for Tshifufia, Tshifufiamashi and Kinsevere Hill in September 2007, and reported in the Feasibility Study and 2008 Technical Report, approximately 3.4 million tonnes of ore have been mined from the Tshifufiamashi and Tshifufia Pits up to the suspension of mining in late 2008. In addition to mining depletion, the inclusion of assay data from extensional and delineation drilling completed since September 2007, and the revised density estimation, is deemed material to the current Kinsevere Mineral Resource status.

The Mineral Resource estimate for Tshifufia and Tshifufiamashi was updated by Anvil in February 2009 using the reliable assay data received up to December 2008 from the Phase IV exploration programme, and from 2008 and 2009 open pit mining information. Assay data which did not pass QAQC checks (as described in Item 15.4) was not included in the database for the updated estimate described herein. Samples will be reassayed during 2010. The estimate was updated again in January 2010 to account for mining deletion to the end of 2009.

It is noted that the update described in this Item 19 will benefit from the inclusion in 2010 of the outstanding assay information, together with further revisions to geological interpretations.

The Mineral Resource estimate for Kinsevere Hill and Kilongo was prepared by independent consultants CSA Global Pty Ltd (CSA, June 2008) using information collected by Anvil in the Phase III and Phase IV exploration programmes. There has been no mining depletion at Kinsevere Hill or Kilongo.

The principal Mineral Resource report authors are:

- Mr. Gerry Fahey – of CSA, who is responsible for the commentary on the content relating to Kinsevere Hill and Kilongo, and who is responsible as QP for the overall technical content of the Mineral Resource estimates. Mr Fahey has a BSc (Hons) in Geology from the National University of Ireland (1975) and is a Member of the Australasian Institute of Mining and Metallurgy and a Member of the Australian Institute of Geoscientists.
- Mr. David Gray – formerly of Anvil, and the author of the content relating to Tshifufia and Tshifufiamashi. Mr. Gray has a BSc (Hons) in Geology from Rhodes University in South Africa (1988) and is a registered Professional Natural Scientist (Geological Science) of the South African Council for Natural Scientific Professions, Registration No. 400018/04.

The primary sources of data for the updated Mineral Resource estimation are:

- Anvil exploration drillhole data collected across the Kinsevere Project area;
- Geological data collected by Anvil relevant to Kinsevere's geological continuity;
- LIDAR surface survey as at 13 April 2007 and the 31 December 2009 pit survey for the Kinsevere project area.

Based on the Mineral Resource models available at February 2009, open pit optimisations were then completed by DumpSolver Pty Ltd (DumpSolver), whilst mine design, production scheduling and Mineral Reserve estimation was completed by A & J Cameron & Associates Pty Ltd (AJC).

The principal Mineral Reserve report authors are:

- Mr. Anthony Cameron – of AJC, who completed the mine planning and Mineral Reserve estimation process and is responsible as QP for this component of the Technical Report. Mr Cameron has a BEng (Mining Engineering) degree from the University of Queensland (1988) and is a Fellow of the Australasian Institute of Mining and Metallurgy.
- Mr Michael Lawlor – of Anvil, who collated and reviewed the mining related commentary in Item 19. Mr Lawlor has a BEng (Honours) degree in Mining Engineering from the Western Australian School of Mines (1985), an MEngSc degree in Mining Geomechanics from the James Cook University of North Queensland (1993), and other post-graduate qualifications in mineral economics and project management. Mr Lawlor is a fellow of the Australasian Institute of Mining and Metallurgy and a Chartered Engineer of Engineers Australia.

19.1 Mineral Resource Estimate

19.1.1 Estimate Database

Since the September 2007 Mineral Resource estimate, an additional 32,294 m of infill and extensional drilling has been completed across the Tshifufia and Tshifufiamashi deposits, of which, 52% has been sampled and assayed. A total of 49,069 m of sampled drillhole data was available for Tshifufiamashi and Tshifufia (Figure 13.1) from 630 drillholes. Kilongo and Kinsevere Hill has a total of 17,605 m of samples from 266 drillholes (Figure 13.1). No additional drilling has been completed since December 2008.

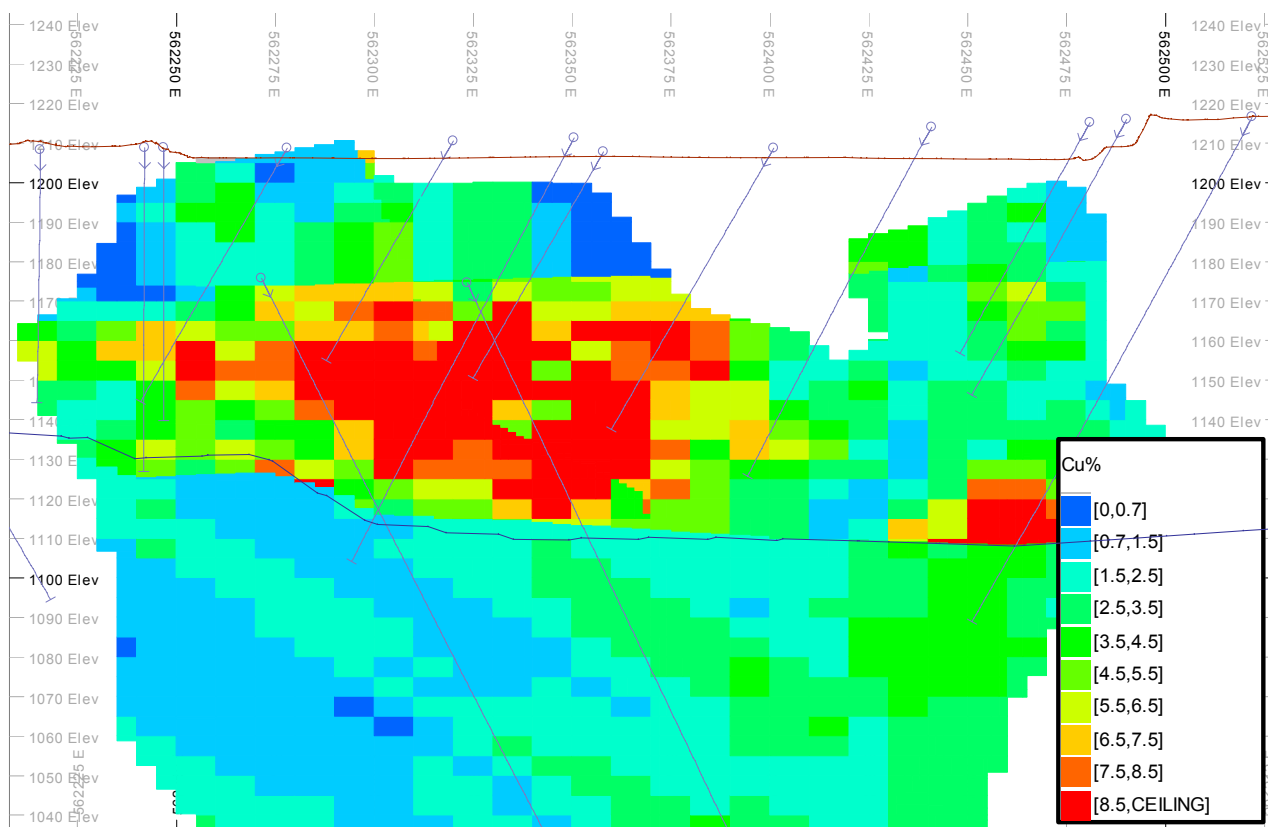
Tshifufia and Tshifufiamashi

The Anvil estimate for this Technical Report, updated in January 2010 in respect of Tshifufia and Tshifufiamashi, reflects all available and reliable assay data as at the end of December 2008, and mining depletion up to 31st December 2009. Upon receipt of all remaining assays from the resampling work described in Item 15.4, Anvil plans to publish a further update to these Mineral Resources during 2010.

The results of in-pit mapping, blast hole sample data and updated geological cross section interpretations were included in the revised Mineral Resource estimate for Tshifufia and Tshifufiamashi, together with 1,696 density determinations from Tshifufia and Tshifufiamashi. The density data was subdivided by domain and assigned to model volumes, to provide a more robust estimate of in situ Mineral Resource tonnes.

The local geology at Kinsevere is complex, and mineralisation is often affected by faulting, folding and varying degrees of weathering and alteration. The copper mineralisation between Tshifufiamashi and Tshifufia is displaced by an east west striking fault. High grade zones occur as discrete lodes and subcrop close to surface. Tshifufiamashi exhibits a regional north northwest trend and Tshifufia a north northeast trend. The Kinsevere Hill and Kilongo deposits have a more north westerly orientation. Typically, oxide mineralisation occurs within 110 m of surface and is characterised by supergene enrichment (Figure 19.1).

Figure 19.1 West to East Cross Section at Tshifufia, Typical of Kinsevere Mineralisation



Thusfar, economic mineralisation has been intersected at depths of up to 400 m at Tshifufia. In Figure 19.1, note the copper enriched upper oxidised zone relative to the underlying sulphide zone. Total copper (TCu) mineralisation is in percent units.

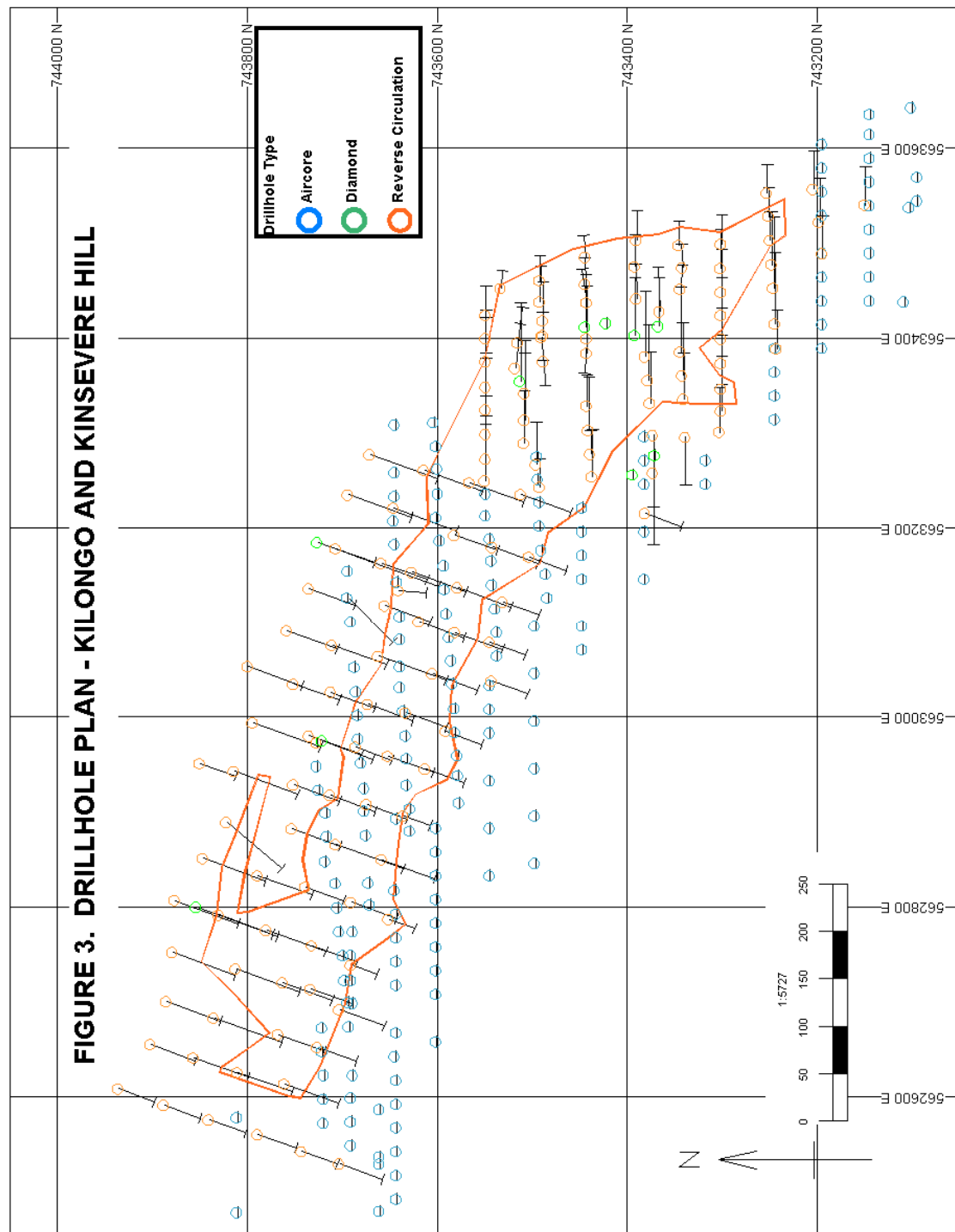
Kinsevere Hill and Kilongo

The CSA estimate for this Technical Report, completed in June 2008 in respect of Kinsevere Hill and Kilongo, reflects all available assay data up to 11th March 2008. There has been no mining depletion at either of these deposits.

Table 19.1 Boreholes and Sample Metres for Kinsevere Hill (KV) and Kilongo (KP)

Hole Type	Hole Prefix	Number of Holes	Total Metres Sample
Air-core	KPAC	159	6,870.0
Diamond	KPDH	3	563.6
	KVDH	6	506.0
RC	KPRC	77	7,934.0
	KVRC	67	5,445.0

Figure 19.2 Drillhole Plan of the Kinsevere Hill and Kilongo Project Areas



19.1.2 Geological and Mineralisation Models

Tshifufia and Tshifufiamashi

Stratigraphic and weathering profiles were developed using 50 m spaced cross-sections. Previous wireframe interpretations were updated according to the latest available information, with reference to Katangan system stratigraphy (Table 9.1). For modelling purposes, the stratigraphy (Table 9.1) was subdivided into the Lower Orebody (LOB), the Upper Orebody (UOB) and the Third Orebody (TOB). Across Kinsevere, the TOB is associated with the Kambove dolomite (previously CMN) lithologies, the UOB with the underlying dolomitic shales (SD) and the Kamoto dolomite (DSTR) lithologies. The LOB is associated with the lowermost grey green siltstones (GRAT) of the Kamoto dolomite lithologies (Table 9.1). The weathering domains comprise soil, oxidised material and fresh rock categories.

Strings/perimeters representing the boundaries of relevant stratigraphy and weathering profiles were digitised in three dimensions from two dimensional geological cross-sections and plans (Figure 9.3). These were linked to form wireframes (Figure 19.3), as a basis for solid volumes for the lower, upper and third ore bodies. Wireframe surfaces were also produced for the base of the soil cover, the 35 m depth below surface, the 70m depth below surface and the base of weathering (BOW). Delineation of the mineralised volume was guided by a 0.3% to 0.7% TCu threshold, based on log probability statistical analysis (Figure 19.4).

In addition to geological cross sections, plans and drillhole data, the mineralisation models were guided by an extensive set of blast hole sample data (Figure 19.5) from the Tshifufia and Tshifufiamashi Pits. This data has improved delineation of mineralised domains inclusive of internal waste volumes together with verification of variography orientations.

Figure 19.3 3-D View of Tshifufia and Tshifufiamashi Wireframed Ore Domains

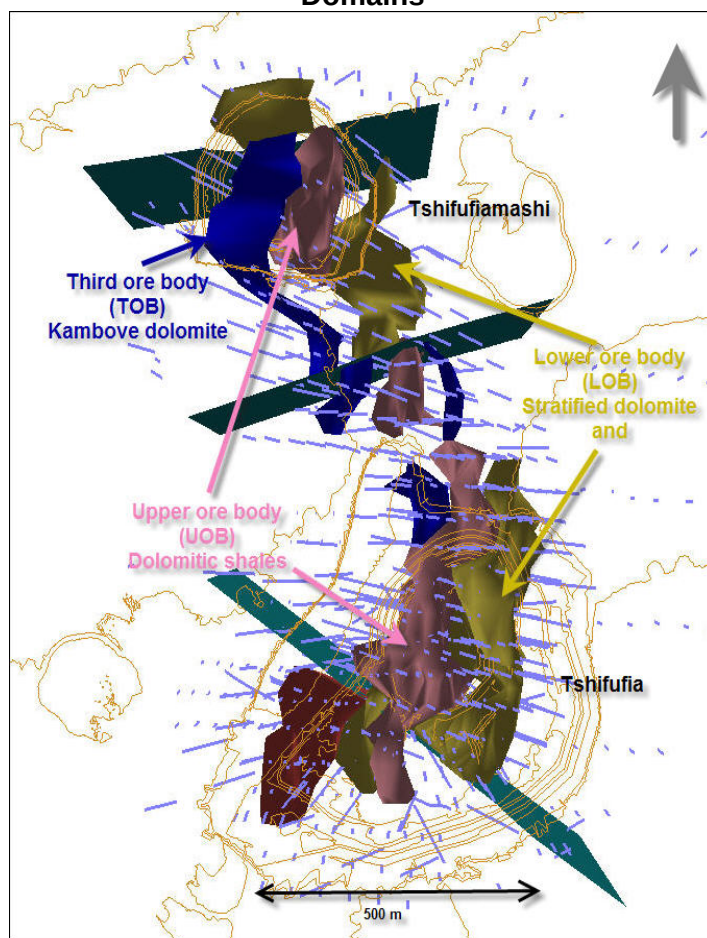


Figure 19.4 Log Probability Plot of All TCu Sample Values Highlighting the Inflection Points Between Waste and Mineralisation

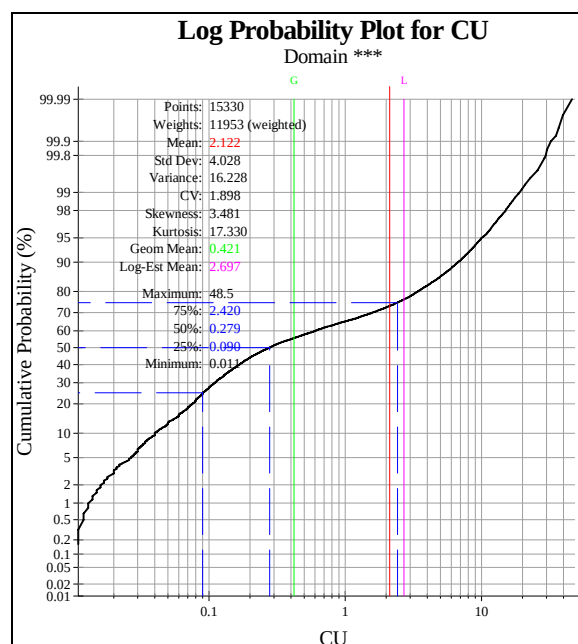
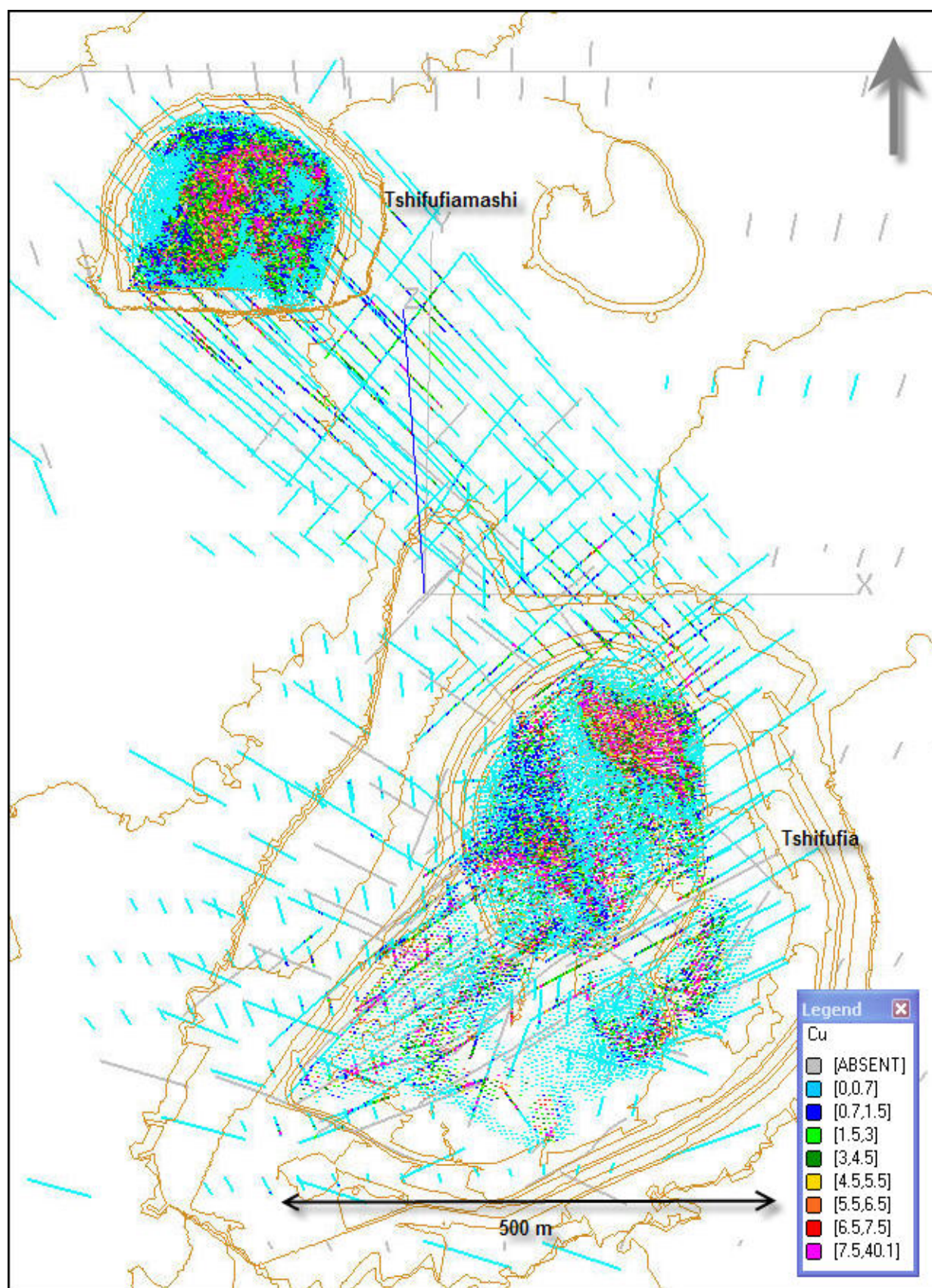


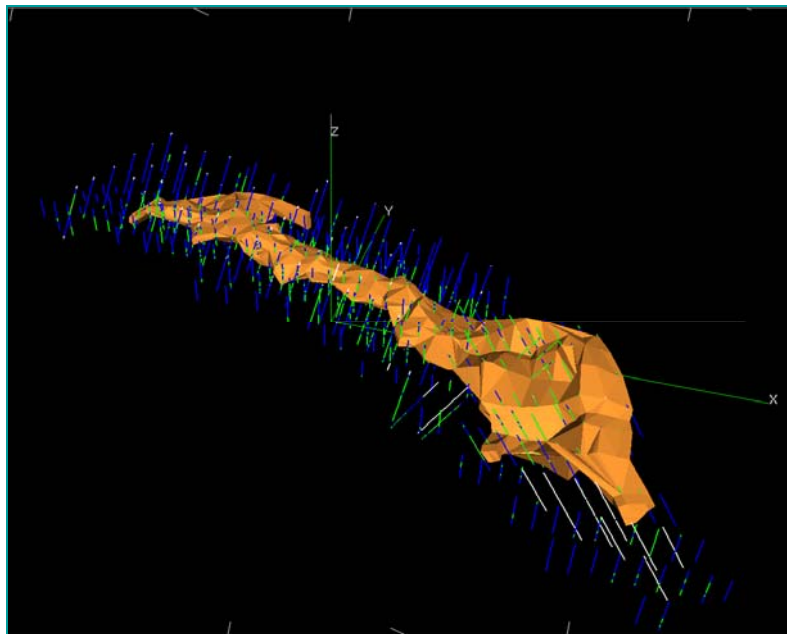
Figure 19.5 Oblique Northward 3-D Aerial View of Blast Hole Sample Data (closely spaced) and Exploration Drillhole Sample Data



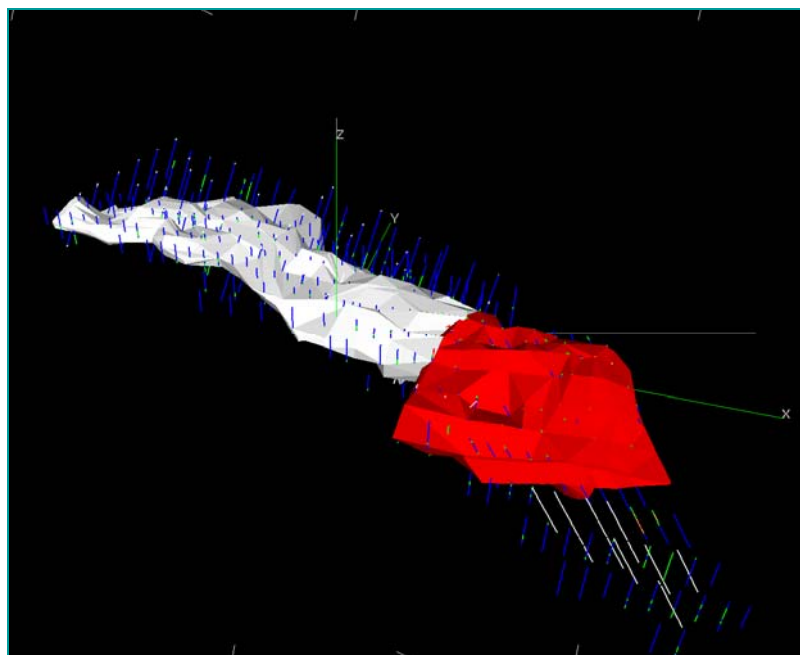
Kinsevere Hill and Kilongo

Mineralised outlines or envelopes were digitised on section and linked with wireframes. These volumes were based firstly upon a 0.1%Cu cutoff and then 0.5% Cu cutoff. The lower cutoff volumes included waste material which has resulted in a change to the mineralised boundaries. Interpolation parameters were selected to best represent low grade or waste volumes from the drillhole assays and waste included in the model.

Figure 19.6 Oblique View of Cu>0.5% Mineralised Envelope, Looking NW



**Figure 19.7 Oblique View of Cu>0.1% Mineralised Envelope, Looking NW
(Red is Kinsevere Hill, White is Kilongo)**



19.1.3 Data Validation

Collar Elevation Validation and Adjustment

In April 2007 Anvil acquired a detailed topographic survey over the Kinsevere Project area using LIDAR technology (Light Detection and Ranging) an airborne based surveying method, with an expected elevation accuracy of 15 cm. The close spacing of reflection points, commonly within 20 cm of each other in the horizontal plane, permits a high level of accuracy, especially with respect to elevation estimation.

Many of the drillhole collars were found to have elevations with differences in excess of 1 m from the LIDAR survey. These were attributed to the collar coordinates being located by hand-held GPS or where the collar elevations had been surveyed, the use of a different topographic datum.

All modelling for these latest estimates was completed using the Gecamines mine grid. Prior to modelling, all collar elevations of surveyed holes were checked against the elevation in the mine grid where the X and Y was the same as WGS84, but, Z was based on the local Gecamines mine datum. Accordingly, a constant elevation value of 22.3m was subtracted from the LIDAR surveys Z value.

All collar elevations which were determined using a hand-held GPS, were adjusted to the mine grid using the LIDAR survey elevation which had been adjusted to fit the Gecamines datum.

Other Data Validation

A series of data validations were completed by Anvil prior to de-surveying the drillhole data into a three dimensional format using the Datamine HOLES3D process. This included:

- Investigation and correction of all data entry overlaps or gaps for the downhole survey, sampling and geological logging data.
- Samples with acid soluble copper (ASCu) values but missing total copper had TCu set to the ASCu value.
- Where the ASCu value was higher than the total TCu value, the ASCu value was reset to the total Cu value (a result of minor precision issues).
- Negative assay values representing codes such as 'insufficient sample' were set to absent values.
- TCu values coded for 'below detection range' set to 0.01%.
- ASCu values coded for 'below detection range' set to 0.001%.
- Co values coded for 'below detection range' set to 0.01 ppm.
- Duplicate samples were located and corrected in the database.
- One interval of 44 m at 23.9 % Cu was found in diamond hole TCDH008 from 94 m to 138 m. On investigation it was found that the 44 m interval had only 1.05 m of core recovered. It was set to absent value (no sample) in the file.

The above issues were corrected and the database was re-extracted on the 21 January 2009 for Tshifufia and Tshifufiamashi. The Kilongo and Kinsevere Hill database was based on results as at 11th March 2008.

19.1.4 Data Preparation for Modelling

The de-surveyed 3-D assay drillhole file was coded and selected within the mineralisation and lithological 3-D wireframes. Each sample interval was coded with a fault block, lithological and weathering profile providing mineralised domain codes for estimation (Table 19.2 and Table 19.3).

Table 19.2 Domain Codes for Tshifufia and Tshifufiamashi

Field Name	Domain	Code
FBLOCK	Tshifufiamashi	2000
	Tshifufia	1000
OREZONE	LOB	0.4
	UOB	0.3
	TOB	0.1
	Fresh mineralisation	1.1
WEATH	Soil	5
	Waste	0
	0 m – 35 m	4
	35 m – 70 m	3
	70 m – Base of weathering	2
	Fresh	1

Table 19.3 Domain Codes for Kinsevere Hill and Kilongo

Field Name	Domain	Code
LITH	Kilongo TOB	2002
	Kilongo LOB-UOB	2003
	Soil	2009
	Hill HG RAT	2100
	Hill TOB	2102
	Hill LOB-UOB	2103
	Hill Southern Area	2105

Additional flagging codes and domains for the Kinsevere Hill and Kilongo model are listed in Tables 19.4 to 19.7.

Table 19.4 Stratigraphic Flagging Codes

Stratigraphic Unit	Flag field FSTRAT
Soil	9
SD	3
CMN	2
DSTR-RSF	1
RAT	0

Table 19.5 Weathering Flagging Codes

Weathering unit	Flag field WEATH
Oxide 0-35m	2
Oxide 35-100m	1
Fresh sulphide	0

Table 19.6 Area Flagging Codes

Area	Flag field KINHILL
Kinsevere Hill	100
Kilongo	0

Table 19.7 Mineralised Envelope Flagging Codes

Ore Flag	Flag field MIN
Waste - outside	0
Inside Cu>0.1% mineralised envelope	1000
Inside Cu>0.5% mineralised envelope	2000

The envelope defining the mineralisation for Tshifufia and Tshifufiamashi was previously interpolated (September 2007) using an indicator model which estimated the probability of any block exceeding a value of 0.5% TCu. The 2007 wireframes were used as a guideline for this phase of modelling. East-west vertical drillhole sections were interpreted with new mineralisation envelopes. The vertical section envelopes were linked to form three dimensional mineralised wireframes (Figure 19.3).

Samples that fell outside of the ore envelope were assigned a code value of zero and those within the ore wireframe were assigned values according to their lithological (OREZONE) domain. The coded drillhole file was then submitted for subsequent geostatistical analyses and grade interpolation.

19.1.5 Preparation of Composite data for Geostatistics and Estimation

Tshifufia and Tshifufiamashi

In order to determine the most common sample length for the Tshifufia and Tshifufiamashi data, the distribution of raw sample lengths was plotted for the samples. Approximately 60% of the data had a sample length of 1 m, 23% had a 4 m sample length and 10% had a 2 m sample length with only a few samples with lengths greater than 4 m (Figure 19.8). All data was composited to 1 m sample lengths, ensuring that intervals provided good resolution across domain boundaries.

Kinsevere Hill and Kilongo

The drillhole samples were of mixed length (Figure 19.9), as most below-grade samples (Cu<0.1%) were 4 m composites. The samples returning above a cut-off had been assayed mostly at 1 m intervals. However, a few 3 m composites (mostly at the ends of holes) and 2 m composites also occur in the sample database. The composite assay at which 1 m samples were submitted varied from around 0.1% Cu to 0.2% Cu.

For samples where Cu>0.5%, 83% were 1m length (Figure 19.10). All samples were composited to 1 m intervals for statistics and interpolation.

Figure 19.8 Sample Lengths from Raw Assay Data, Tshifufia and Tshifufiamashi

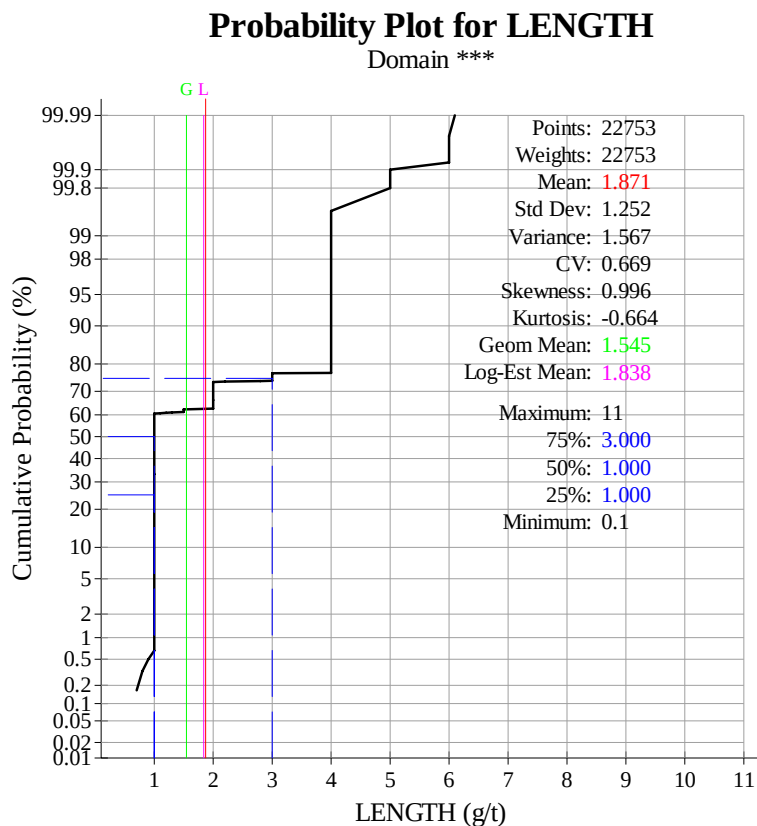


Figure 19.9 Histogram of Drillhole Assay Length – All Samples, Kinsevere Hill and Kilongo

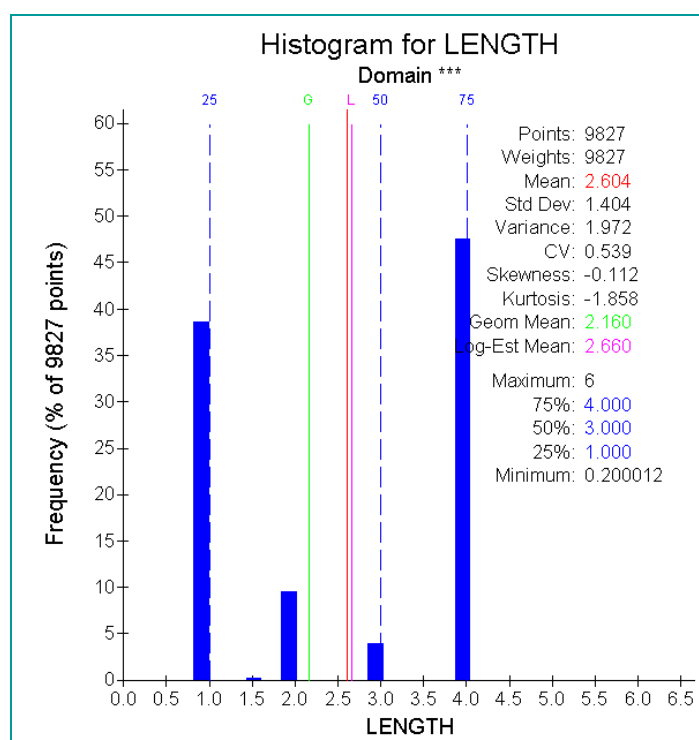
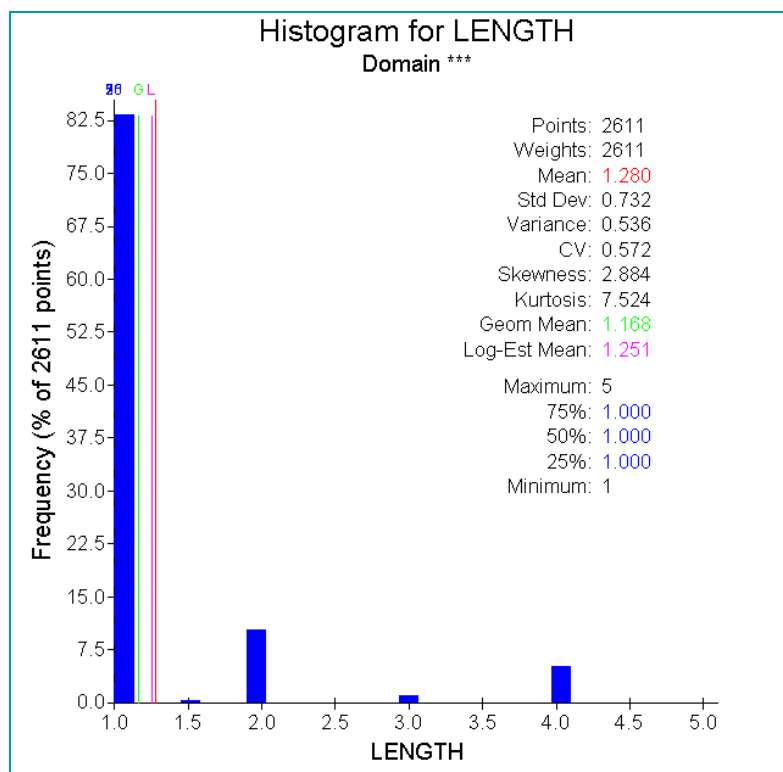


Figure 19.10 Histogram of Drillhole Assay Length – Only Samples Where Cu>0.5%, Kinsevere Hill and Kilongo



19.1.6 Statistics

Tshifufia and Tshifufiamashi

Statistical analyses of the data, including spatial statistics, were carried out using Snowden's Supervisor software. The statistical analysis of composite copper grades was completed for each of the final domains. Summary results are tabulated in Table 19.8.

Statistics for TCu, ASCu and Co were investigated by domain with histograms and probability plots. The objective of the domain selections was to reduce internal variability and domain mixing in order to assist with spatial analysis and provide a more robust estimate.

Several of the selected domains still have a degree of mixing as illustrated in Figure 19.11. This is due to difficulties in identifying the various domains, with their complex overlaps between geology, weathering and mineralisation. Future data capture and Mineral Resource estimates will benefit from separating these mixed domains. In contrast, the TOB domain tends to exhibit a single population (Figure 19.12).

Table 19.8 Univariate Statistics for TCu by Domain Code

Deposit	Domain code	Samples	Minimum	Maximum	Mean	Standard Deviation	CV	Variance
Tshifufiamashi	0	6,542	0.01	13.10	0.15	0.44	3.05	0.20
Tshifufiamashi	0.1	1,618	0.04	26.88	3.54	3.75	1.06	14.09
Tshifufiamashi	0.3	1,187	0.02	26.70	3.48	4.01	1.15	16.10
Tshifufiamashi	0.4	1,189	0.01	15.30	2.08	2.06	0.99	4.26
Tshifufiamashi	1	1,121	0.01	2.40	0.10	0.20	1.94	0.04
Tshifufiamashi	1.1	415	0.01	14.45	1.90	1.88	0.99	3.52
Tshifufia	0	18,754	0.01	25.40	0.21	0.85	3.95	0.72
Tshifufia	0.1	331	0.05	27.00	2.07	3.37	1.63	11.35
Tshifufia	0.3	5,172	0.01	47.68	4.35	5.17	1.19	26.74
Tshifufia	0.4	3,488	0.01	38.35	4.66	5.17	1.11	26.71
Tshifufia	1	3,346	0.01	9.56	0.24	0.62	2.55	0.38
Tshifufia	1.1	2,266	0.01	26.20	2.83	2.76	0.97	7.59
Kinsevere Hill	2002	1,707	0.04	45.84	3.72	5.18	1.39	26.83
Kinsevere Hill	2003	278	0.01	9.07	0.89	1.33	1.49	1.78
Kinsevere Hill	2009	6	0.61	2.39	1.50	0.98	0.65	0.95
Kinsevere Hill	2100	412	0.15	28.00	1.73	2.72	1.57	7.40
Kinsevere Hill	2102	694	0.02	27.30	2.32	3.07	1.32	9.40
Kinsevere Hill	2103	1,090	0.01	23.80	2.24	3.02	1.35	9.14
Kinsevere Hill	2105	11	0.64	2.05	1.13	0.51	0.45	0.26

Figure 19.11 Log Histogram of the total TCu % Values for the UOB Domain, Tshifufiamashi

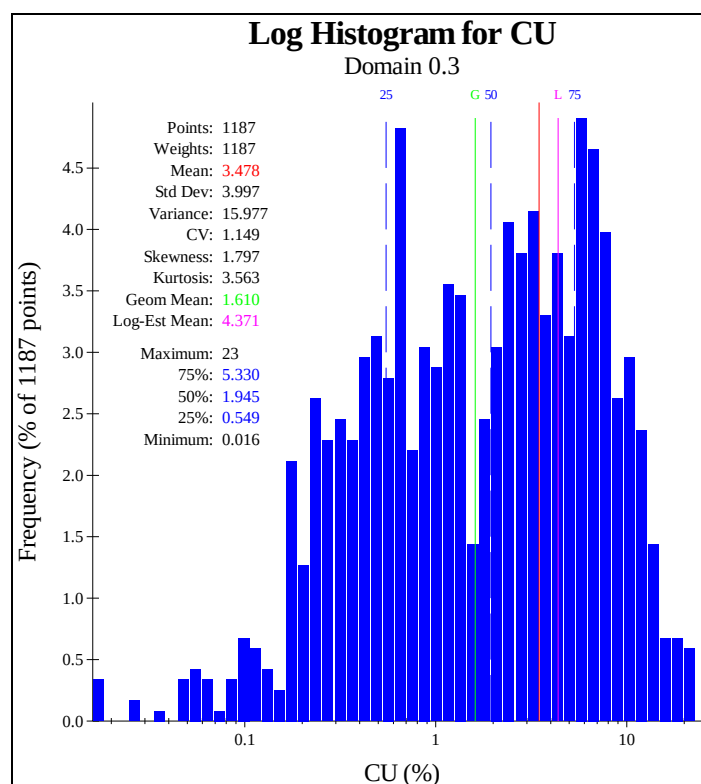
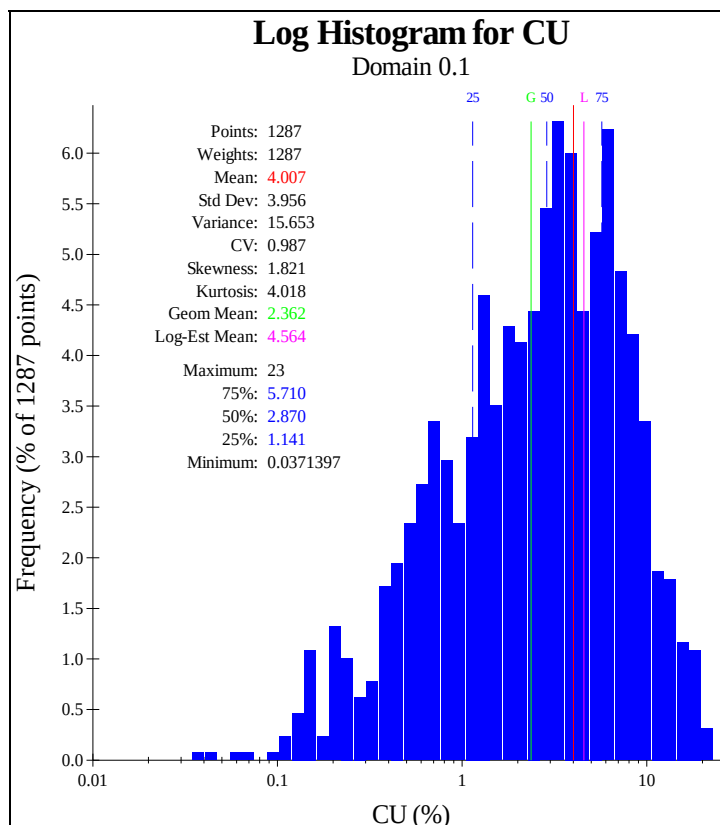


Figure 19.12 Log Histogram of the Total TCu % Values for the TOB Domain, Tshifufiamashi



Kinsevere Hill and Kilongo

The assays were coded for each combination of mineralisation envelope, stratigraphy and area in a field labelled WFSTRAT. Summary statistics, histograms, and cumulative frequency plots were reviewed. Table 19.9 summarises the coding and Table 19.10 to Table 19.13 present the summary statistics.

Histograms are shown in Figures 19.13 and 19.14.

Table 19.9 Orezone codes subdivide mineralised zone by stratigraphy, grade of mineralisation and area

WFSTRAT	Description	Site
<i>Unmineralised Cu<0.1</i>		
0	RAT	Kilongo (KP)
2	CMN	Kilongo
3	SD-RSF	Kilongo
9	Soil	Kilongo
100	RAT	Kinsevere Hill (KV)
102	CMN	Kinsevere Hill
103	SD-RSF	Kinsevere Hill
105	South of KV	Kinsevere Hill
109	Soil	Kinsevere Hill
<i>Weakly Mineralised Cu<0.5</i>		
1000	RAT	Kilongo
1002	TOB	Kilongo
1003	LOB-UOB	Kilongo
1009	Soil	Kilongo
1100	RAT	Kinsevere Hill
1102	TOB	Kinsevere Hill
1103	LOB-UOB	Kinsevere Hill
1109	Soil	Kinsevere Hill
<i>Mineralised Cu>=0.5%</i>		
2002	TOB	Kilongo
2003	LOB-UOB	Kilongo
2009	Soil	Kilongo
2100	RAT	Kinsevere Hill
2102	TOB	Kinsevere Hill
2103	LOB-UOB	Kinsevere Hill
2105	South of KV	Kinsevere Hill

Table 19.10 Summary of TCu Statistics, Mineralised Zone

TCu	2002 Kilongo TOB	2003 Kilongo LOB- UOB	2009 Soil	2100 Kinsv Hill HG RAT	2102 Kinsv Hill TOB	2103 Kinsv Hill LOB- UOB	2105 South of Kinsv Hill
Samples	1707	278	6	412	694	1090	11
Minimum	0.04	0.01	0.61	0.15	0.02	0.01	0.64
Maximum	45.84	9.07	2.39	28	27.3	23.8	2.05
Mean	3.72	0.89	1.50	1.73	2.32	2.24	1.13
Standard deviation	5.18	1.33	0.98	2.72	3.07	3.02	0.51
CV	1.39	1.50	0.65	1.57	1.32	1.352	0.46
Variance	26.83	1.78	0.95	7.40	9.40	9.14	0.26
Skewness	2.40	3.24	0.00	5.85	3.26	2.60	1.19
Geometric mean	1.49	0.41	1.21	1.10	1.19	0.88	1.04
10%	0.22	0.12	0.61	0.42	0.27	0.13	0.64
20%	0.36	0.15	0.61	0.57	0.51	0.28	0.85
30%	0.59	0.21	0.61	0.63	0.64	0.48	0.85
40%	0.92	0.28	0.61	0.76	0.83	0.72	0.85
50%	1.48	0.36	2.39	0.99	1.14	1.03	0.85
60%	2.27	0.48	2.39	1.28	1.65	1.52	1.06
70%	3.70	0.69	2.39	1.55	2.43	2.39	1.06
80%	6.33	1.35	2.39	2.18	3.71	3.72	1.49
90%	10.85	2.29	2.39	2.85	5.65	5.81	2.05
95%	15.00	3.42	2.39	5.55	8.38	8.62	2.05
97.5%	18.15	4.83	2.39	8.61	10.10	10.70	2.05
99%	22.30	7.28	2.39	14.50	15.25	14.85	2.05

Table 19.11 Summary of ASCu Statistics, Mineralised Zone

ASCu	2002 Kilongo TOB	2003 Kilongo LOB- UOB	2009 Soil	2100 Kinsv Hill HG RAT	2102 Kinsv Hill TOB	2103 Kinsv Hill LOB- UOB	2105 South of Kinsv Hill
Samples	1707	278	6	412	694	1090	11
Minimum	0.001	0.001	0.55	0.07	0.014	0.001	0.52
Maximum	43.70	8.87	2.11	26.70	23.21	21.66	1.91
Mean	3.17	0.72	1.33	1.49	2.04	2.03	1.00
Standard deviation	4.69	1.29	0.85	2.20	2.86	2.88	0.51
CV	1.48	1.80	0.64	1.48	1.40	1.42	0.51
Variance	22.03	1.67	0.73	4.86	8.19	8.31	0.26
Skewness	2.55	3.57	0.000	5.78	3.38	2.64	1.13
Geometric mean	1.04	0.23	1.08	0.90	0.97	0.61	0.90
10%	0.14	0.04	0.56	0.31	0.22	0.06	0.52
20%	0.22	0.06	0.56	0.41	0.36	0.16	0.71
30%	0.37	0.09	0.56	0.54	0.49	0.31	0.71
40%	0.58	0.15	0.56	0.63	0.66	0.55	0.71
50%	1.04	0.20	2.11	0.88	0.94	0.89	0.71
60%	1.75	0.31	2.11	1.12	1.41	1.34	0.96
70%	3.02	0.50	2.11	1.42	2.10	2.17	0.96
80%	5.61	1.00	2.11	2.05	3.27	3.54	1.39
90%	9.59	1.93	2.11	2.72	5.09	5.27	1.91
95%	12.92	3.07	2.11	4.93	7.56	8.02	1.91
97.5%	16.70	4.83	2.11	7.15	9.19	10.30	1.91
99%	20.20	7.28	2.11	10.30	14.60	13.55	1.91

Table 19.12 Summary of TCu Statistics, Unmineralised and Weakly Mineralised Zone

TCu	All un-mineralised	1000 Kilongo RAT LG	1002 Kilongo TOB LG	1003 Kilongo LOB-UOB LG	1009 LG Soil	1100 Kinsv Hill RAT LG	1102 Kinsv Hill TOB LG	1103 Kinsv Hill LOB-UOB LG	1109 Soil in LG
Samples	10805	172	2152	316	74	396	1111	615	11
Minimum	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.11
Maximum	2.75	0.76	1.77	0.94	0.76	0.98	2.65	13.5	2.65
Mean	0.06	0.17	0.16	0.10	0.23	0.19	0.21	0.20	0.39
Standard deviation	0.11	0.14	0.12	0.10	0.16	0.14	0.23	0.61	0.75
CV	1.95	0.82	0.75	1.01	0.69	0.76	1.06	3.12	1.93
Variance	0.01	0.02	0.02	0.01	0.03	0.02	0.05	0.37	0.56
Skewness	10.98	2.12	5.20	3.21	1.63	1.13	5.66	17.84	3.30
Geometric mean	0.03	0.12	0.13	0.06	0.19	0.12	0.16	0.12	0.20
10%	0.01	0.02	0.06	0.01	0.10	0.02	0.07	0.04	0.13
20%	0.01	0.05	0.09	0.01	0.12	0.05	0.10	0.07	0.13
30%	0.01	0.11	0.11	0.04	0.14	0.10	0.12	0.11	0.13
40%	0.02	0.12	0.12	0.06	0.15	0.14	0.14	0.13	0.13
50%	0.03	0.14	0.13	0.08	0.18	0.16	0.16	0.14	0.17
60%	0.05	0.15	0.15	0.09	0.20	0.19	0.18	0.15	0.17
70%	0.06	0.18	0.18	0.12	0.24	0.23	0.21	0.17	0.20
80%	0.08	0.23	0.22	0.16	0.36	0.31	0.27	0.19	0.20
90%	0.12	0.33	0.29	0.21	0.43	0.38	0.37	0.25	0.26
95%	0.17	0.41	0.37	0.25	0.47	0.44	0.49	0.32	2.65
97.5%	0.23	0.47	0.45	0.32	0.76	0.47	0.71	0.47	2.65
99%	0.42	0.76	0.48	0.47	0.76	0.48	1.12	1.75	2.65

Table 19.13 Summary of ASCu Statistics, Unmineralised and Weakly Mineralised Zone

ASCu	All un-mineralised	1000 Kilongo RAT LG	1002 Kilongo TOB LG	1003 Kilongo LOB-UOB LG	1009 LG Soil	1100 Kinsv Hill RAT LG	1102 Kinsv Hill TOB LG	1103 Kinsv Hill LOB-UOB LG	1109 Soil in LG
Samples	10805	172	2152	316	74	396	1111	615	11
Minimum	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.04
Maximum	1.97	0.32	1.50	0.85	0.69	0.69	2.25	13.50	2.41
Mean	0.04	0.11	0.10	0.04	0.17	0.12	0.15	0.13	0.33
Standard deviation	0.09	0.08	0.10	0.08	0.18	0.12	0.18	0.60	0.69
CV	2.39	0.72	0.94	1.86	1.08	0.94	1.26	4.57	2.08
Variance	0.01	0.01	0.01	0.01	0.03	0.01	0.03	0.36	0.48
Skewness	11.38	0.98	6.68	4.98	1.20	1.19	6.64	18.62	3.27
Geometric mean	0.01	0.08	0.07	0.02	0.07	0.06	0.10	0.06	0.15
10%	0.01	0.02	0.02	0.01	0.01	0.01	0.04	0.02	0.04
20%	0.01	0.04	0.04	0.01	0.01	0.02	0.06	0.02	0.10
30%	.01	0.06	0.06	0.01	0.02	0.04	0.07	0.03	0.12
40%	0.01	0.08	0.07	0.01	0.03	0.06	0.09	0.04	0.12
50%	0.02	0.11	0.09	0.01	0.11	0.09	0.11	0.05	0.12
60%	0.03	0.12	0.10	0.02	0.17	0.12	0.12	0.09	0.12
70%	0.04	0.13	0.12	0.04	0.22	0.15	0.15	0.12	0.18
80%	0.06	0.15	0.14	0.07	0.33	0.21	0.18	0.14	0.18
90%	0.09	0.21	0.19	0.10	0.39	0.31	0.28	0.18	0.24
95%	0.12	0.30	0.24	0.15	0.43	0.37	0.34	0.26	2.41
97.5%	0.18	0.30	0.30	0.29	0.69	0.40	0.51	0.43	2.41
99%	0.34	0.32	0.38	0.43	0.69	0.42	0.95	1.54	2.41

Figure 19.13 Cu Histograms for Zones Inside the 0.5%Cu Envelope

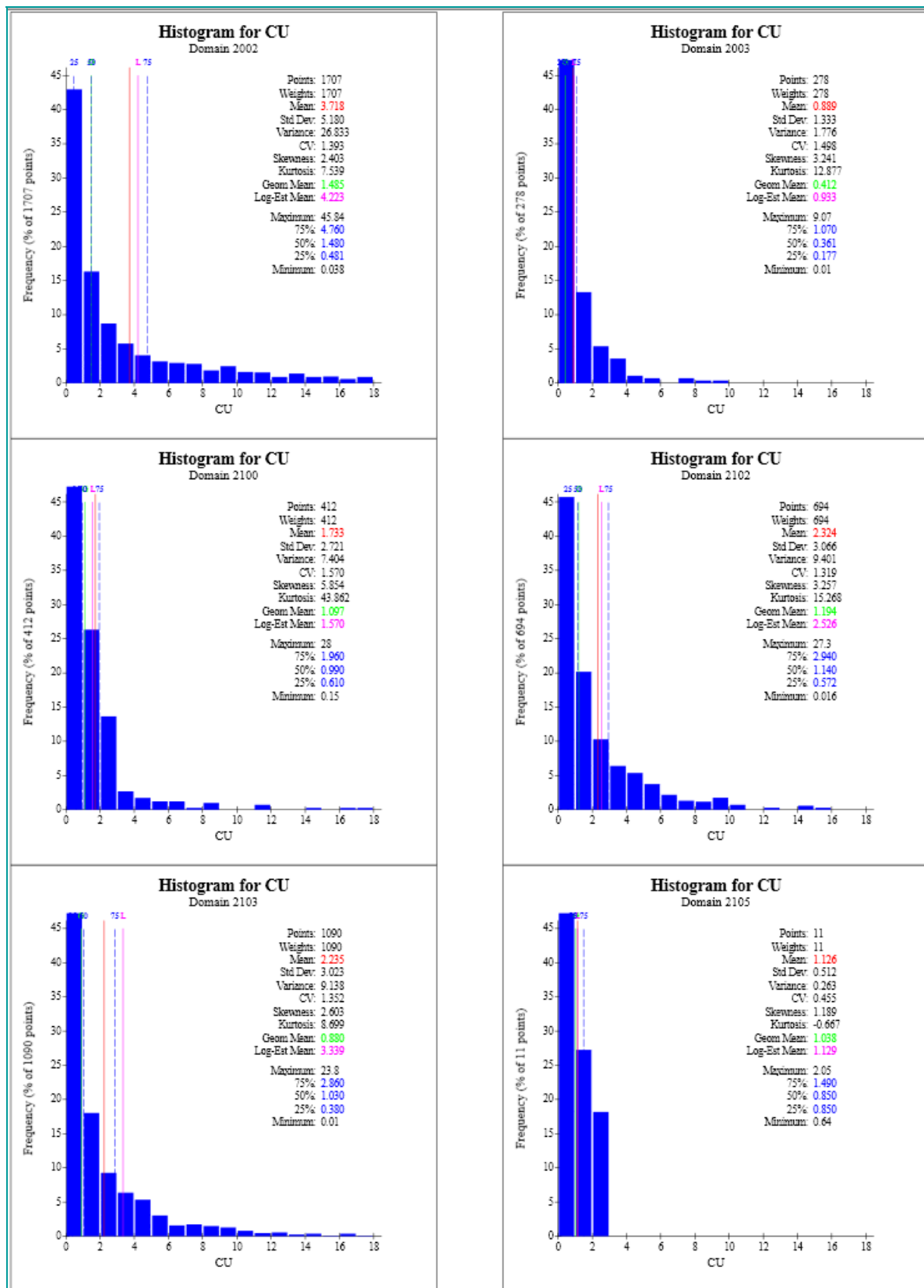
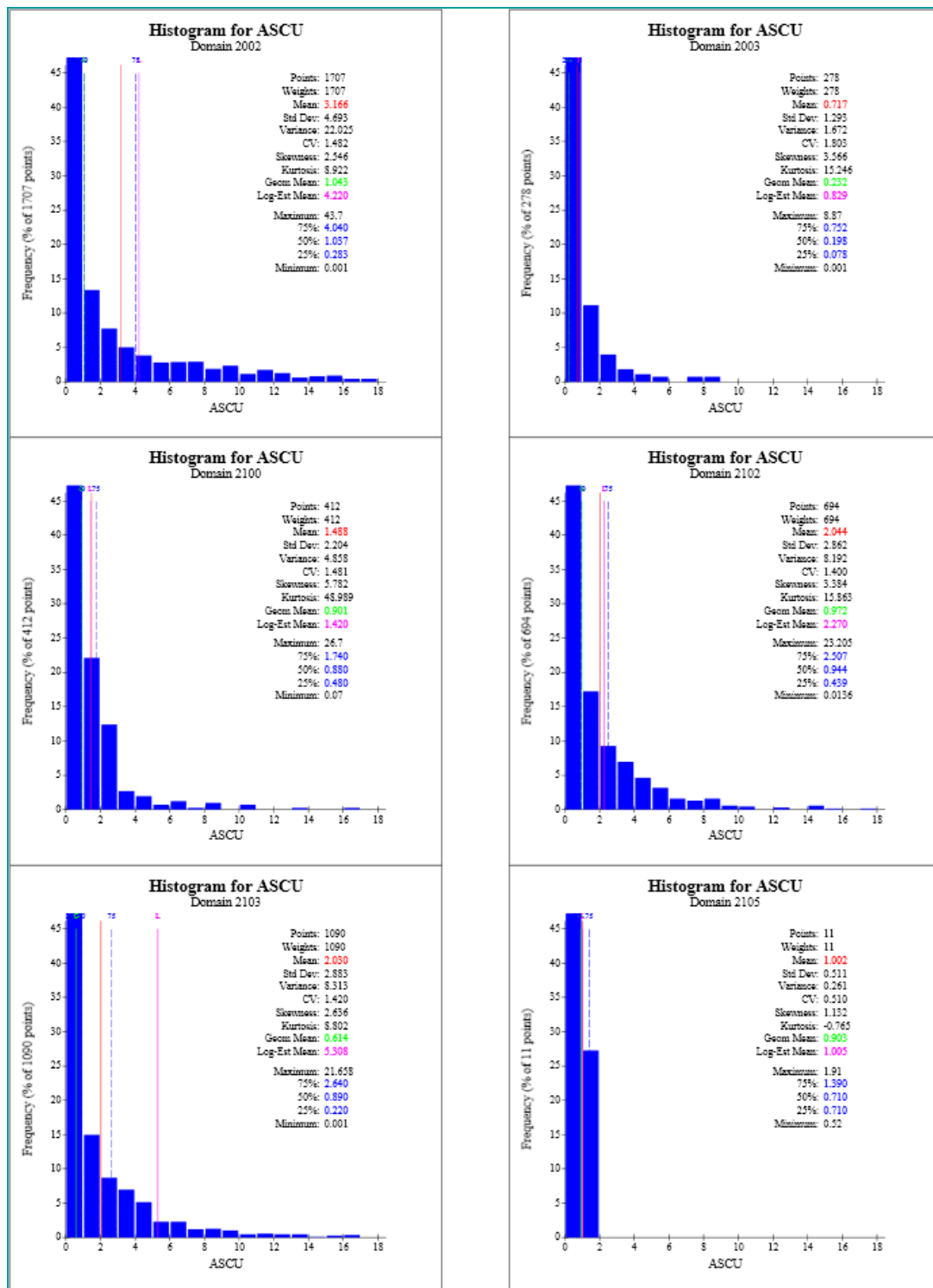


Figure 19.14 ASCu Histograms for Zones Inside the 0.5%Cu Envelope



19.1.7 Spatial Statistics

Variography was analysed using composited data located within the mineralised envelopes of each domain, based on the following methodology:

- the principal axes of anisotropy were determined using semi-variogram (variogram) fans based on normal scores variograms;

- normal scores variograms were calculated for each of the principal axes of anisotropy;
- down hole normal scores variograms were modelled for each domain and adjusted to determine the normal scores nugget effect;
- variogram models were then determined for each of the principal axes of anisotropy using the nugget effect from the downhole variogram; and
- the variogram models were then back-transformed to the original distribution and used to guide search parameters and complete ordinary kriging estimation.

Orientations were largely controlled by the strike of mineralisation and downhole variography. Variogram models for ASCu were linked to those of TCu where there was insufficient confidence in the variogram structure to interpret a suitable model. Variogram orientations were verified with those derived for the closely spaced blast hole data. Omni-directional variogram models were derived for the fresh sulphide mineralization and waste due to the inability of the irregularly spaced deeper intersections to provide clearly defined orientations of continuity.

Tshifufia and Tshifufiamashi

Tshifufia and Tshifufiamashi variography details are listed in Tables 19.14 to 19.17.

Table 19.14 Axes of Anisotropy for Tshifufia Variogram Models
(VAXIS=3 is Z and VAXIS=1 is X)

Domain	Grade variable	VANGLE1	VANGLE2	VANGLE3	VAXIS1	VAXIS2	VAXIS3
0	TCU	0	0	0	3	1	3
0	ASCU	0	0	0	3	1	3
0	CO	0	0	0	3	1	3
0.1	TCU	-70	30	0	3	1	3
0.1	ASCU	-70	30	0	3	1	3
0.1	CO	-70	30	0	3	1	3
0.3	TCU	140	30	180	3	1	3
0.3	ASCU	140	30	180	3	1	3
0.3	CO	140	30	180	3	1	3
0.4	TCU	0	0	0	3	1	3
0.4	ASCU	0	0	20	3	1	3
0.4	CO	0	20	50	3	1	3
1	TCU	0	0	0	3	1	3
1	ASCU	0	0	0	3	1	3
1	CO	0	0	0	3	1	3
1.1	TCU	-100	60	-70	3	1	3
1.1	ASCU	-100	60	-70	3	1	3
1.1	CO	105	60	-150	3	1	3

Table 19.15 Axes of Anisotropy for Tshifufiamashi Variogram Models
(VAXIS=3 is Z and VAXIS=1 is X)

Domain	Grade variable	VANGLE1	VANGLE2	VANGLE3	VAXIS1	VAXIS2	VAXIS3
0	TCU	0	0	30	3	1	3
0	ASCU	30	25	0	3	1	3
0	CO	30	25	10	3	1	3
0.1	TCU	-70	30	0	3	1	3
0.1	ASCU	-70	30	0	3	1	3
0.1	CO	-70	30	0	3	1	3
0.3	TCU	-80	20	10	3	1	3
0.3	ASCU	-80	20	10	3	1	3
0.3	CO	-80	20	10	3	1	3
0.4	TCU	170	50	180	3	1	3
0.4	ASCU	170	50	180	3	1	3
0.4	CO	170	50	180	3	1	3
1	TCU	90	90	90	3	1	3
1	ASCU	90	90	90	3	1	3
1	CO	90	90	90	3	1	3
1.1	TCU	90	90	90	3	1	3
1.1	ASCU	90	90	90	3	1	3
1.1	CO	90	90	90	3	1	3

Table 19.16 Variogram Parameters for Tshifufia using Spherical Models

Domain	Grade	Nugget	Structure 1				Structure 2			
			Sill1	Range1	Range2	Range3	Sill2	Range1	Range2	Range3
0	TCU	0.25	0.49	42	42	42	0.26	238	238	238
0	ASCU	0.37	0.53	23	23	23	0.10	136	136	136
0	CO	0.13	0.62	42	42	42	0.25	225	225	225
0.1	TCU	0.12	0.30	10	36	6	0.58	104	56	58
0.1	ASCU	0.12	0.27	11	55	5	0.61	106	55	53
0.1	CO	0.07	0.41	25	29	6	0.52	67	40	31
0.3	TCU	0.30	0.33	41	13	27	0.37	251	74	65
0.3	ASCU	0.30	0.34	46	17	20	0.36	251	74	59
0.3	CO	0.39	0.37	66	106	83	0.24	194	119	128
0.4	TCU	0.18	0.51	30	23	7	0.31	50	180	115
0.4	ASCU	0.19	0.51	35	34	11	0.30	76	121	58
0.4	CO	0.26	0.57	56	63	27	0.17	120	65	122
1	TCU	0.19	0.59	29	29	29	0.22	106	106	106
1	ASCU	0.61	0.29	29	29	29	0.10	232	232	232
1	CO	0.41	0.36	34	34	34	0.23	120	120	120
1.1	TCU	0.16	0.38	123	66	20	0.46	275	173	41
1.1	ASCU	0.21	0.46	83	66	28	0.33	301	173	78
1.1	CO	0.2	0.42	85	72	117	0.38	275	92	151

Table 19.17 Variogram Parameters for Tshifufiamashi using Spherical Models

Domain	Grade	Nugget	Structure 1				Structure 2			
			Sill1	Range1	Range2	Range3	Sill2	Range1	Range2	Range3
0	TCU	0.12	0.67	29	29	29	0.21	219	219	219
0	ASCU	0.14	0.8	25	24	24	0.07	145	143	143
0	CO	0.09	0.73	62	63	62	0.18	254	254	255
0.1	TCU	0.12	0.30	10	36	6	0.59	104	56	58
0.1	ASCU	0.12	0.27	11	55	5	0.61	106	55	53
0.1	CO	0.07	0.41	25	29	6	0.53	67	40	31
0.3	TCU	0.18	0.39	95	25	9	0.43	116	81	42
0.3	ASCU	0.18	0.40	95	25	9	0.42	116	81	42
0.3	CO	0.21	0.29	95	62	9	0.50	95	103	70
0.4	TCU	0.18	0.30	12	84	12	0.52	112	154	26
0.4	ASCU	0.20	0.31	12	84	12	0.49	115	134	26
0.4	CO	0.23	0.34	12	84	88	0.44	132	154	90
1	TCU	0.33	0.38	23	23	23	0.28	144	144	144
1	ASCU	0.36	0.38	13	13	13	0.27	144	144	144
1	CO	0.33	0.40	23	23	23	0.27	117	117	117
1.1	TCU	0.24	0.41	5	5	5	0.35	147	147	147
1.1	ASCU	0.10	0.57	43	43	43	0.33	126	126	126
1.1	CO	0.19	0.38	6	6	6	0.43	98	98	98

Kinsevere Hill and Kilongo

Kinsevere Hill and Kilongo variography details are listed in Tables 19.18 and 19.19, and shown in Figures 19.15 to 19.19.

Table 19.18 Axes of Anisotropy for Kinsevere Hill Variogram Models
(VAXIS=3 is Z and VAXIS=1 is X)

Domain	Grade variable	VANGLE1	VANGLE2	VANGLE3	VAXIS1	VAXIS2	VAXIS3
2002	TCU	-160	90	-90	3	1	3
2003	TCU	-160	90	-90	3	1	3
2100	TCU	-120	90	-90	3	1	3
2102	TCU	-120	90	-90	3	1	3
2103	TCU	60	90	0	3	1	3

Table 19.19 Variogram Parameters for Kinsevere Hill using Spherical Models

Domain	Grade	Nugget	Structure 1				Structure 2			
			Sill1	Range1	Range2	Range3	Sill2	Range1	Range2	Range3
2002	TCU	0.2	0.40	27	35	15	0.40	27	35	15
2003	TCU	0.4	0.40	6	60	40	0.20	30	60	40
2100	TCU	0.3	0.44	3	25	25	0.26	25	35	25
2102	TCU	0.2	0.20	5	5	5	0.60	25	30	30
2103	TCU	0.2	0.20	85	4	50	0.60	85	8	50

Figure 19.15 Variogram Models for Kilongo TOB within the 0.5% Cu Envelope

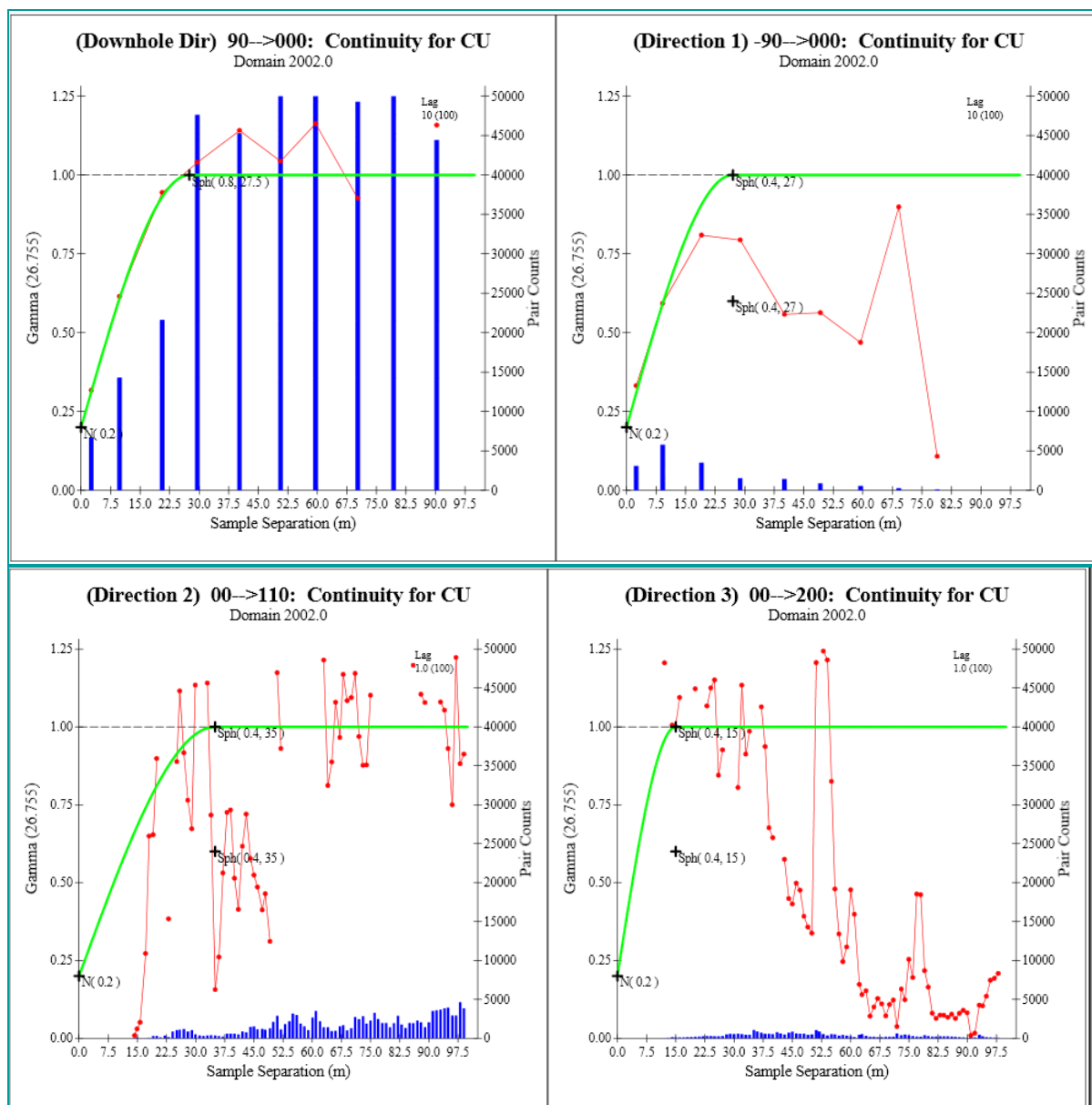


Figure 19.16 Variogram Models for Kilongo UPB/LOB within the 0.5% Cu Envelope

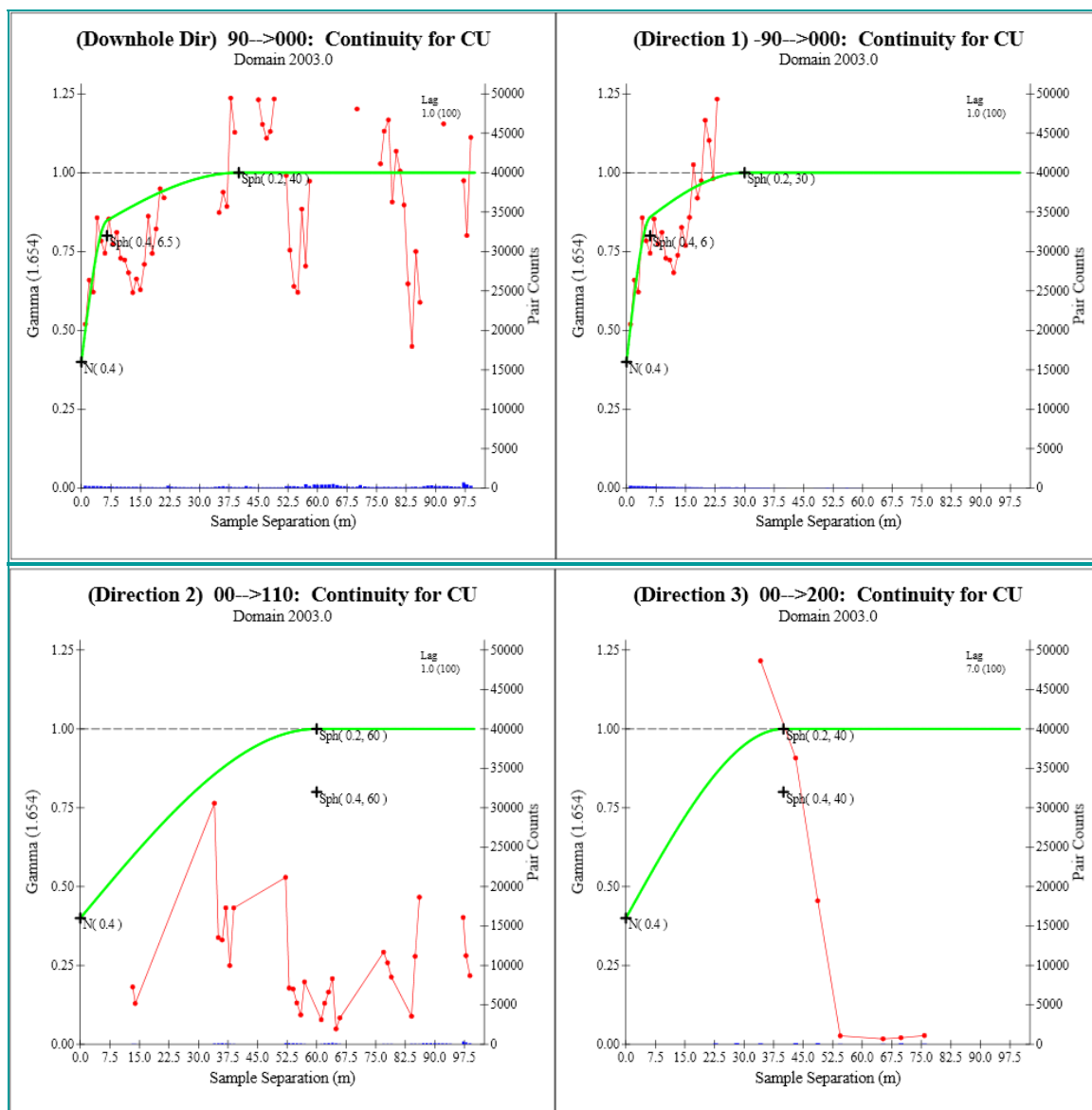


Figure 19.17 Variogram Models for Kilongo RAT within the 0.5% Cu Envelope

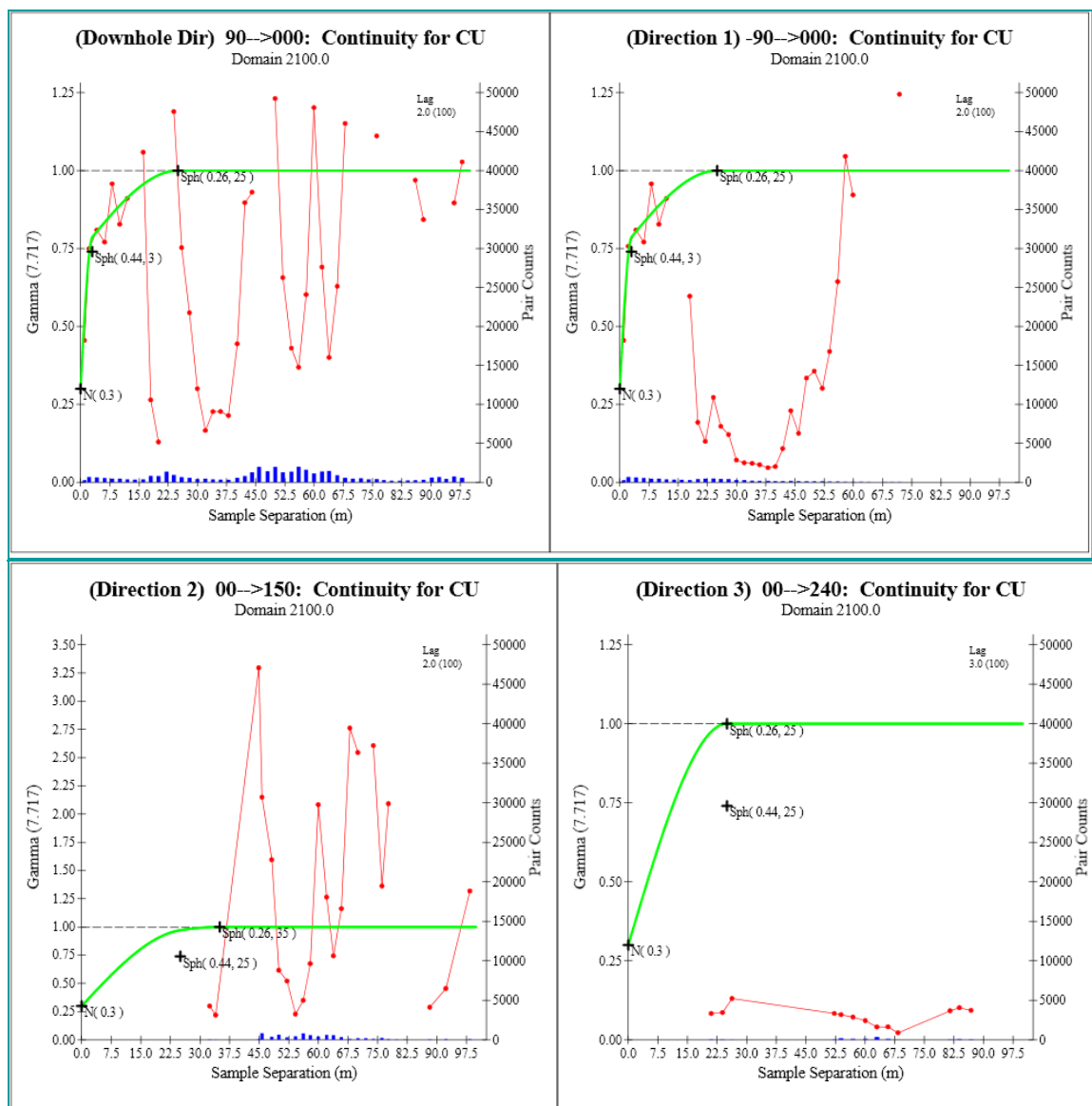


Figure 19.18 Variogram Models for Kinsevere Hill TOB within the 0.5% Cu Envelope

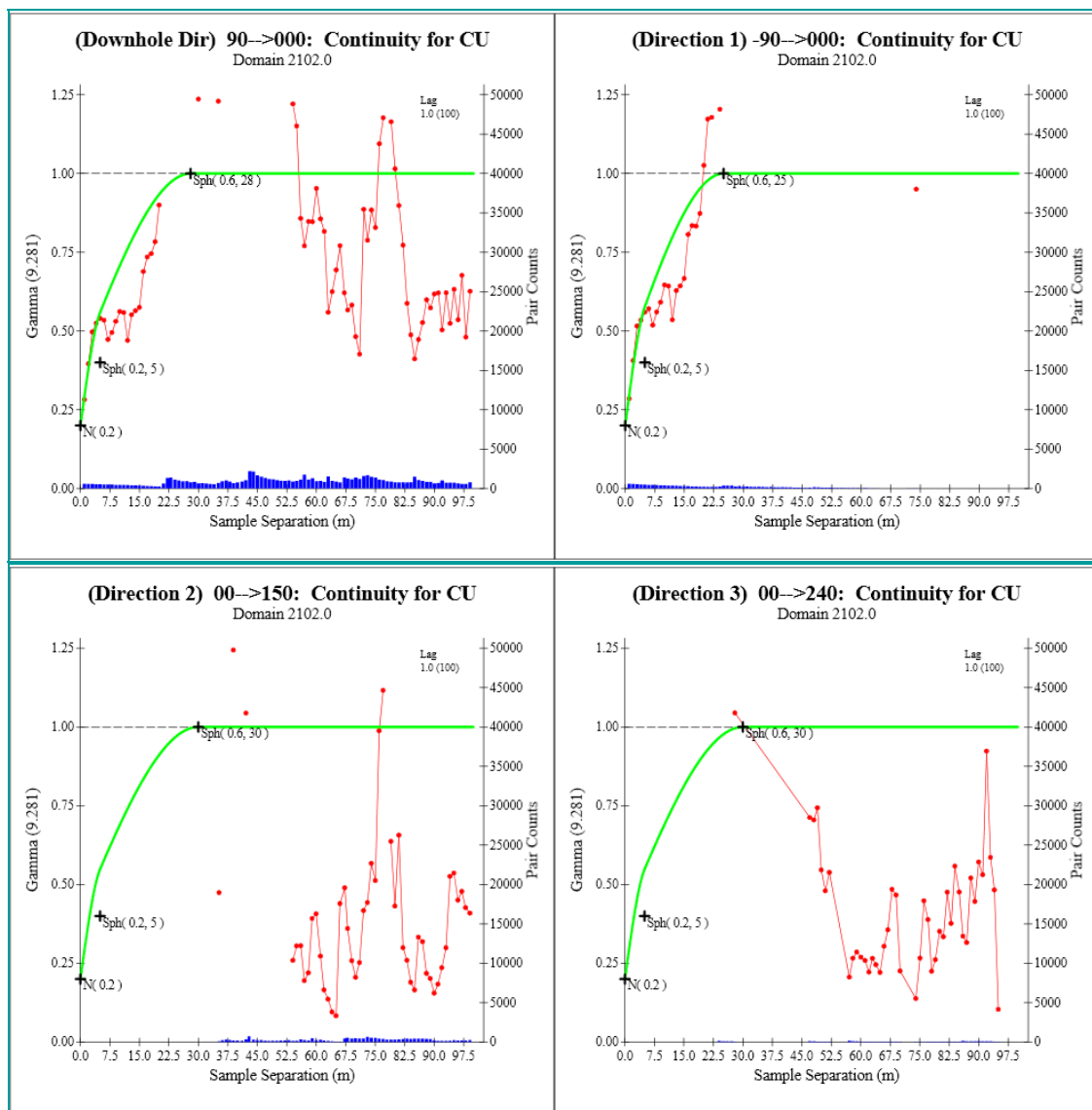
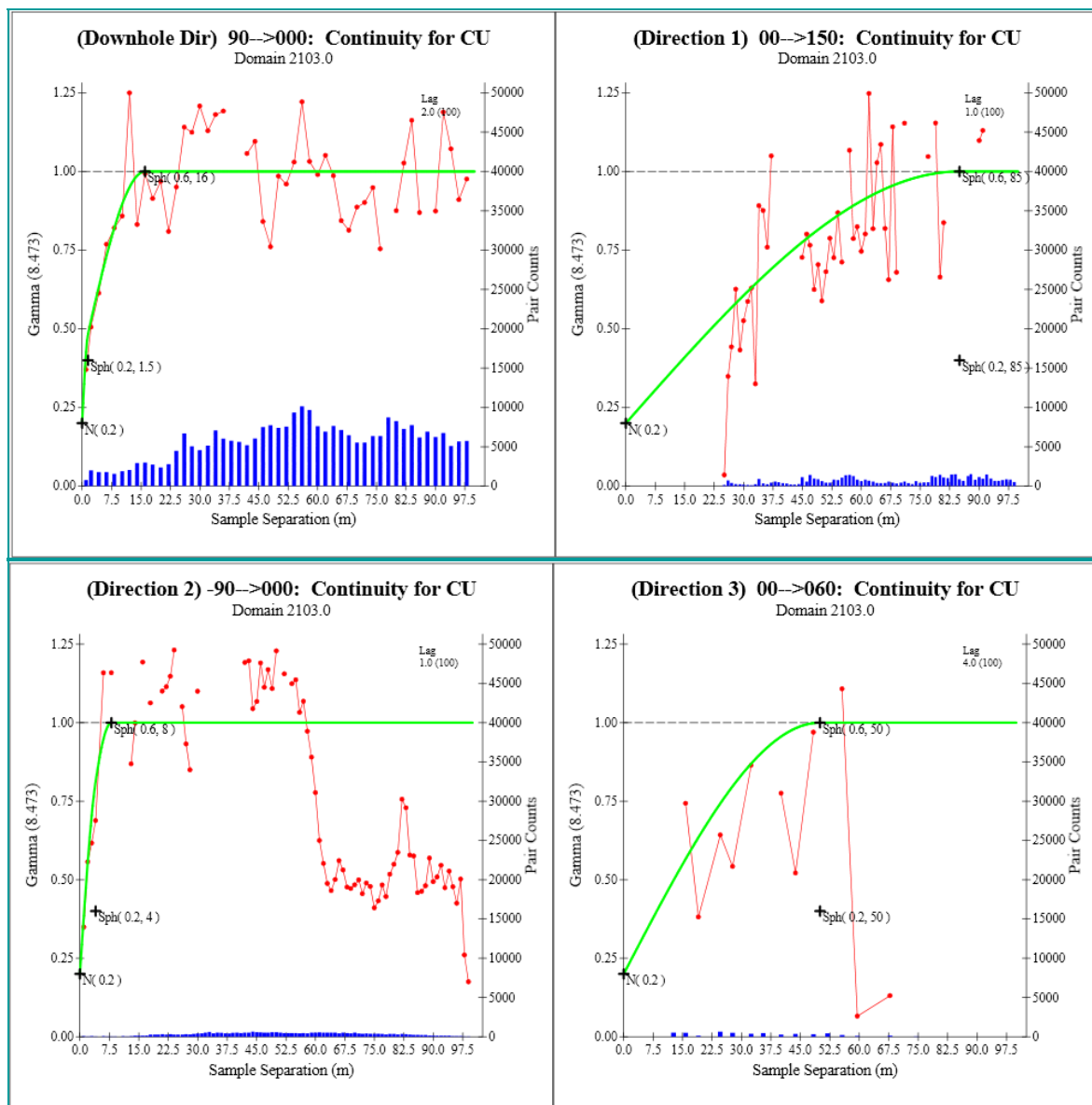


Figure 19.19 Variogram Models for Kinsevere Hill UPB/LOB within the 0.5% Cu Envelope



19.1.8 Block Model

The block model dimensions and parameters were based on the defined geographic boundaries and the average drill grid spacing. Sub-blocks were used to ensure that the block model honoured the domain geometries. Block estimates were controlled by the original parent block dimension.

Tshifufia and Tshifufiamashi

Tshifufia and Tshifufiamashi's individual parent block dimensions were 10 mE by 25 mN by 5 mRL with sub-blocks allowed. Kilongo and Kinsevere Hill's individual block dimensions were 25 mE by 25 mN by 5 mRL with sub-blocks allowed. The block dimension was supported by a kriging efficiency study (2007) which demonstrated little change in the kriging efficiency or slope of regression (a measure of bias) from this block size to larger block sizes.

Kinsevere Hill and Kilongo

Kinsevere Hill and Kilongo block model details are listed in Table 19.20.

Table 19.20 Block Model Dimensions and Cell Sizes

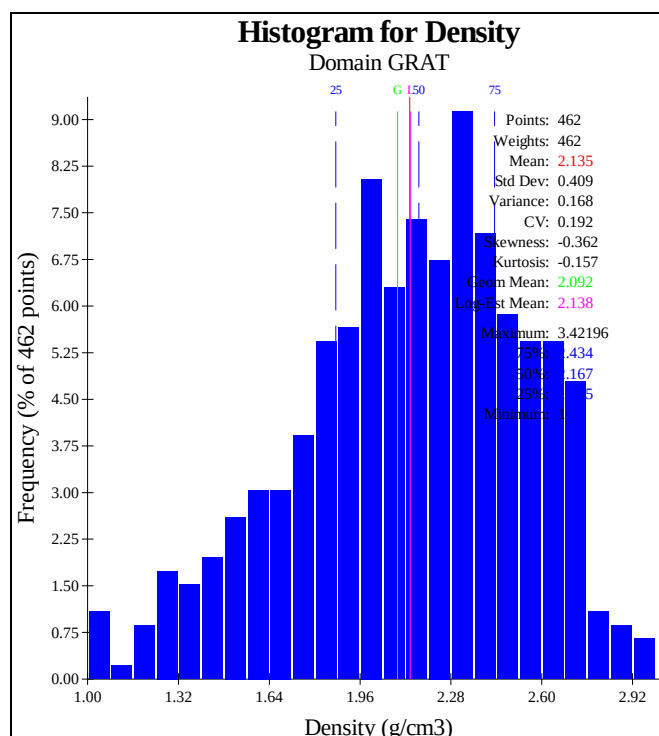
	X	Y	Z
Origin	562,300	743,000	700
Maximum value	563,800	744,100	1300
Extent	1,500	1,100	600
Block Size (Sub-block)	25m (3.125)	25m (3.125m)	5m (0.625m)

19.1.9 Density Estimates in the Models

Tshifufia and Tshifufiamashi

Factors influencing density include lithology, weathering and oxidation, together with Cu grade and sulphide mineral concentration. While density is expected to increase with grade, no specific correlation was identified at Kinsevere. Density data was restricted to 19 drillholes or a total of 1,696 samples and four in-pit bulk density samples. The drillhole data was analysed for statistical robustness and was determined to exhibit minimal domain mixing with normal distributions (Figure 19.20).

Figure 19.20 Histogram of Diamond Core Density Values for the GRAT (or LOB)



Diamond core samples, used for density measurements, become less representative with increasing degrees of oxidation, rock porosity and fracturing. This is reflected in the higher than expected average values of the more oxidised material (Table 19.21). As a result, only diamond sample average densities for the fresh to partially weathered material were used. In addition, only the GRAT (LOB) and CMN (TOB) lithologies had sufficient diamond sample data to provide reasonable estimates for the fresh material (Table 19.21). At Tshifufia and Tshifufiamashi, the mean density

value for fresh mineralisation was determined to be 2.32 t/m^3 . Regrettably, insufficient, representative drillhole data existed to model the SD (UOB) and the shallower highly oxidised domains.

Table 19.21 Average Density Values for Diamond Core Samples per Domain and Zone of Weathering

Depth	CMN (TOB)		SD(UOB)		GRAT(LOB)		Average density (g/cm ³)
	Samples	Density(g/cm ³)	Samples	Density(g/cm ³)	Samples	Density (g/cm ³)	
0 to 35 m	28	1.91	24	1.94	69	1.77	1.87
35 to 70 m	54	1.97			79	1.98	1.98
70 to 110 m	58	2.14			87	1.95	2.05
Fresh	994	2.58	76	1.96	227	2.37	2.30

In addition to the diamond core sample values, four in-pit bulk samples were taken for Kinsevere at approximately 25 to 30m below surface during 2008. The results are presented in Table 19.22.

Table 19.22 In-pit Bulk Density Sample Results

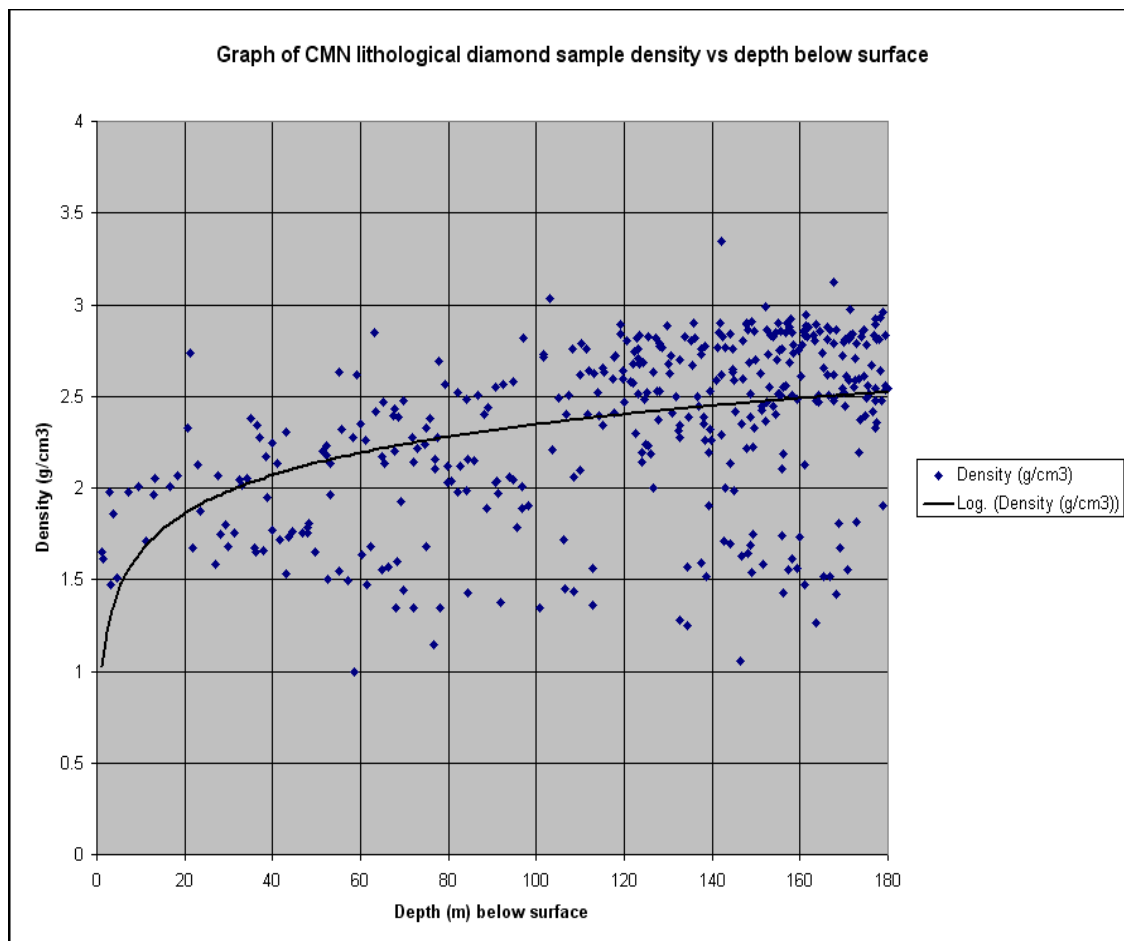
Location – depth below surface	Orebody	As-mined density (dry t/m ³)
Tshifufiamashi -25 m	TOB	1.65
Tshifufia – 25m	TOB	1.51
Tshifufia – 30m	TOB	1.51
Tshifufia – 30m	UOB	2.07

As a result, an estimate for the oxidised material to a depth of 35 m below surface was obtained from the large bulk sample values together with cumulative reconciliation data. Accordingly, the highly weathered material within 35 m of surface was assigned a density value of 1.55 t/m^3 .

Density values for the remaining material between 35 m and the base of weathering may be expected to be between 1.55 t/m^3 to 2.32 t/m^3 . An X-Y scatter plot of diamond core density values and depth below surface highlights an exponential decrease in density with decreasing depth below surface. Figure 19.21 illustrates this relationship for the CMN (TOB). The GRAT exhibits a similar density relationship with depth. Assigning density values for these remaining domains has attempted to honour this relationship, in addition to the fact that weathered diamond core sample averages are optimistic.

For the 35 m to 70 m zone, a density value of 1.75 t/m^3 was selected by reducing the average diamond core value by 0.23 t/m^3 . The 70 m to 110 m transition zone was assigned a value of 1.9 t/m^3 by reducing the average diamond core value by 0.15 t/m^3 . This approach is believed to be slightly conservative, but does consider rock type, the inability of diamond core samples to capture representative oxidised density values, the variable nature of oxidised material within weathered zones, and the rapid increase in density values with depth below surface.

Figure 19.21 X-Y Scatter Plot of Density and Depth Below Surface



Kinsevere Hill and Kilongo

Anvil supplied CSA with conservative fixed density values for use in the Kinsevere Hill and Kilongo modelling exercise (Table 19.23) based on bulk density values as per measurements for Tshifufia and Tshifufiamashi. Density values are fixed, not calculated from regressions of copper grades, and whilst are likely to be on the conservative side, are considered more realistic than the earlier density measurements from the core due to the potential selective sampling of more solid pieces of core.

Table 19.23 Density Values Used in Kinsevere Hill/Kilongo Model

Domains	Zone code FSTRAT	Density (tonnes/BCM)
CMN	2	1.55
SD	3	2.05
DSTR / RSF	4	2.05
RAT	0	1.55

19.1.10 Determination of Top Cuts

Tshifufia and Tshifufiamashi

Top cuts were employed to describe the maximum reasonable metal grade for a composite sample value within a given domain. If the grade of a sample exceeded this value, the grade was reset to the top cut value. The objective of applying top cuts is to minimise the risk of uniquely high metal concentrations from biasing individual block estimates, especially those located within areas of low sample support.

Top cuts for Tshifufia and Tshifufiamashi were established by investigating univariate statistics and histograms of sample values by domain. To assist with top cut selection, a disintegration analysis of the TCu, ASCU and Co values per domain was undertaken. This process determined the point or value where the difference between consecutive sample values of a sorted data set began to increase dramatically when compared to the preceding data differences. For Kilongo and Kinsevere Hill, top cut selection was guided by the value at the 99.5 percentile. A top cut was selected if it reduced the sample variance and did not materially change the mean value. The following top cuts were applied to the data for Mineral Resource estimation (Table 19.24).

Kinsevere Hill and Kilongo

The top cuts for interpolation were assigned by inspection of the grade histograms for a suitable value near the top of the distribution, generally above the 99.5 percentile (Table 19.25).

Table 19.24 Table of Top Cuts per Domain, Tshifufia and Tshifufiamashi

	Domain	TCu (%)	ASCu (%)	Co (ppm)
Tshifufiamashi	0	5.2	5.2	15,000
Tshifufiamashi	1	1.2	-	13,000
Tshifufiamashi	0.1	23	23	23,500
Tshifufiamashi	0.3	23	22.5	12,500
Tshifufiamashi	0.4	13.5	13.5	12,500
Tshifufiamashi	1.1	9	7.5	25,000
Tshifufia	0	8.8	9	21,000
Tshifufia	1	5.9	4.5	10,200
Tshifufia	0.1	25	25	18,000
Tshifufia	0.3	32	32	-
Tshifufia	0.4	-	32	-
Tshifufia	1.1	-	18	-
Kinsevere Hill	2002	25	25	-
Kinsevere Hill	2003	5	5	-
Kinsevere Hill	2100	9	9	-
Kinsevere Hill	2102	15	15	-
Kinsevere Hill	2103	15	15	-
Kinsevere Hill	Waste	0.5	0.45	-

Table 19.25 Top Cuts Used in Modelling, Kinsevere Hill and Kilongo

Ore Zone (WFSTRAT)	Zone Description	Cu CUT	ASCu CUT	Co CUT
2002	Kilongo (KP) TOB High-grade	25	25	-
2003	Kilongo (KP) LOB-UOB High-grade	5	5	-
2100	Kinsevere Hill (KV) RAT High-grade	9	9	-
2102	KV TOB High-grade	15	15	-
2103	KV LOB-UOB High-grade	15	15	-
All <109; 1009, 1109, 2009, 2109	All waste outside Cu>0.1 envelope; all soil.	0.5	0.45	-

19.1.11 Grade Estimation

Grades for TCu, ASCu were estimated into parent blocks of an empty domain coded block model using ordinary kriging (OK). In addition, and for completeness, Co was estimated for the Tshifufia and Tshifufiamashi block model. OK was deemed an appropriate interpolation technique owing to near normal data distributions and differentiable grade ranges. Kriging with negative weights was allowed as a result of the large number of samples available for a domain's block estimate. Estimation into parent blocks uses a discretisation of two (X points) by five (Y points) by two (Z points) in order to better represent to block volumes estimated.

Estimation parameters for kriging were based on variography, geological continuity and the average spatial distribution of data. The first pass search radius was set within half to two thirds of the variogram range to improve the quality of the local

block grade estimate for areas of close spaced drilling and to ensure that grade was not smeared laterally. Most blocks were estimated within the first search radius. Subsequent search radii were set to ensure that remaining blocks within the mineralised domain were interpolated with a Cu grade.

Soft boundaries were created between the oxidised and fresh weathering domains, in order to represent the variable nature of this boundary and the transition in values between the supergene enriched mineralisation and the underlying relatively depleted sulphide domain typical to Tshifufia and Tshifufiamashi. All other domain boundaries were hard and data between domains was not included for estimation.

Tshifufia and Tshifufiamashi

For the Tshifufia and Tshifufiamashi ore domains, a minimum of eight samples were required for a single block estimate and maximum of 30 samples in order to limit grade smoothing. The surrounding waste material used a minimum of 20 samples and a maximum of 40. Estimates were limited to a maximum of ten samples per drillhole.

Kilongo and Kinsevere Hill

Kilongo and Kinsevere Hill were limited to a maximum of 64 samples. Due to the large number of downhole samples available, estimates were also constrained to a maximum of ten samples per drillhole. Table 19.26 lists the grade estimation search parameters. Table 19.27 lists the variogram model details.

Table 19.26 Grade Estimation Search Parameters – Cu and ASCu

Domain	Description	Strike Radius	Cross Strike Radius	Perp. to Plane Radius	Rotation 1 around Z	Rotation 2 around X	Rotation 3 around Z	Factor for Pass 2	Factor for Pass 3	Min Samp	Max Samp	Max per Drill hole	Min Samp Pass 2	Max Samp Pass 2	Min Samp Pass 3	Max Samp Pass 3
2002	KP TOB HG	27	35	15	-160	90	-90	2	10	8	20	8	8	32	8	64
2003	KP UOB / LOB HG	30	60	40	-160	90	-90	2	10	8	20	8	8	32	8	64
2100	Kinsevere Hill RAT search	25	35	25	-120	90	-90	2	10	8	20	8	8	32	8	64
2102	KP TOB HG	25	30	30	-120	90	-90	2	10	8	20	8	8	32	8	64
2103	KP UOB / LOB HG	85	8	50	60	90	0	2	10	8	20	8	8	32	8	64

Table 19.27 Variogram Models for Grade Estimation – Cu and ASCu

Domain	Description	Major	Semi-Major	Minor	Nugget	Sill1	Range1	Range2	Range3	Sill2	Range1	Range2	Range3
						First Structure				Second structure			
2002	KP TOB HG	-160	90	-90	0.2	0.4	27	35	15	0.4	27	35	15
2003	KP UOB / LOB HG	-160	90	-90	0.4	0.4	6	60	40	0.2	30	60	40
2100	Kinsevere Hill RAT	-120	90	-90	0.3	0.44	3	25	25	0.26	25	35	25
2102	KP TOB HG	-120	90	-90	0.2	0.2	5	5	5	0.6	25	30	30
2103	KP UOB / LOB HG	60	90	0	0.2	0.2	85	4	50	0.6	85	8	50

19.1.12 Model Validation

Tshifufia and Tshifufiamashi

The first pass of model validation included:

- Checks for ASCU grade estimate greater than or equal to the TCu grade estimate in the same cell. Where this was encountered, the ASCU values were reset to the TCu estimate value.
- Checks for negative estimates. If there were any, they were reset to a minimum 0.01 grade.
- Checks to ensure that only blocks distant from the drillholes remained without grade estimates.

An example of a visual spatial comparison (Figure 19.22) of block estimates to drillhole data sample values, demonstrates a good representation of model grades to those of the drilling data.

The model was further validated by statistical comparison of mean composite grades and model grades, in addition to visual comparisons with drillholes (Figure 19.33). A table comparing the mean values for the estimate with those of the data (Table 19.28) illustrates acceptable correlation.

Figure 19.22 West to East Section (Line 23 - Tshifufia) Showing TCu Block Estimates Compared With the Drillhole Data

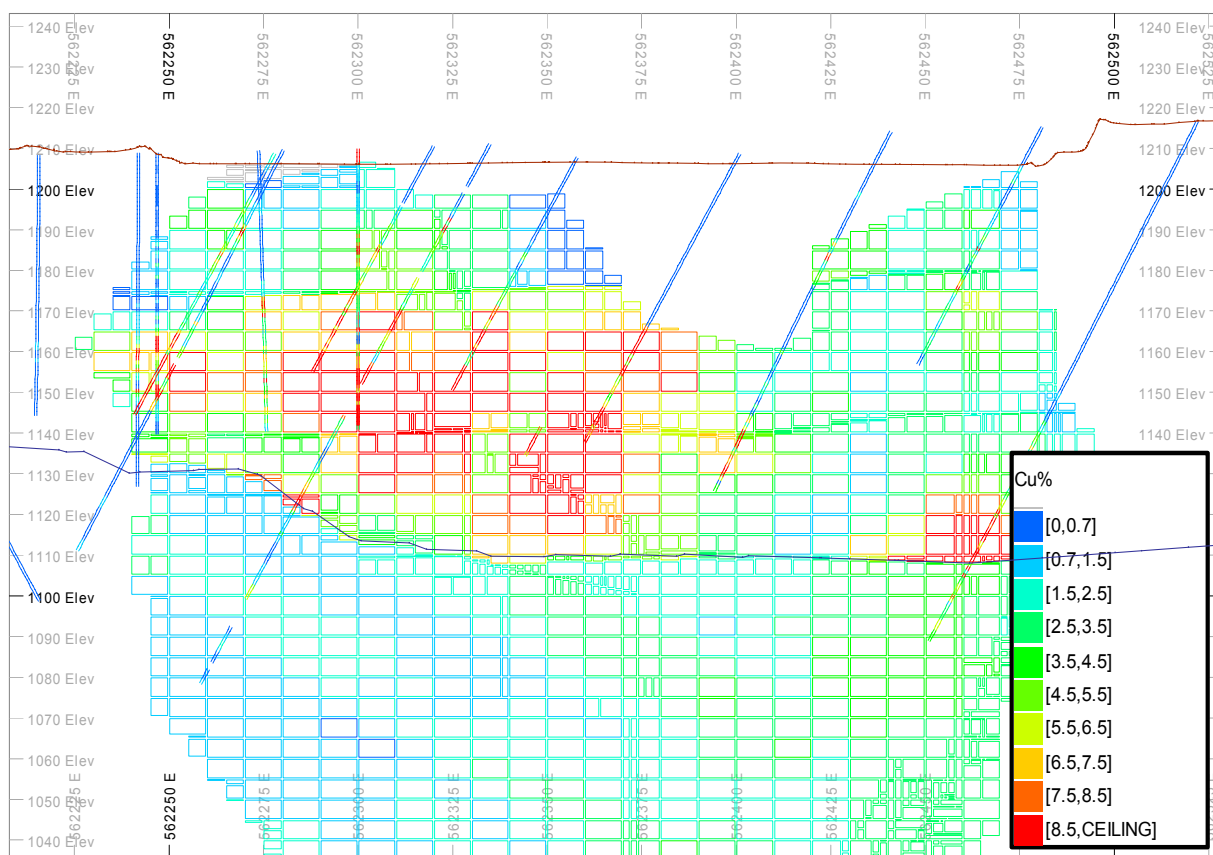


Table 19.28 Table of Mean Statistics per Domain Comparing Model Estimates with Data Values

Deposit	Domain	Field	Data	Model	Difference
Tshifufia	CMN (0.1)	TCU (%)	2.23	1.96	-0.27
Tshifufia	GRAT (0.4)	TCU (%)	4.55	4.88	0.33
Tshifufia	SD (0.3)	TCU (%)	4.31	4.47	0.16
Tshifufia	SULPHIDE (1.1)	TCU (%)	3.75	3.67	-0.08
Tshifufiamashi	CMN (0.1)	TCU (%)	3.51	3.42	-0.09
Tshifufiamashi	GRAT (0.4)	TCU (%)	2.04	1.88	-0.16
Tshifufiamashi	SD (0.3)	TCU (%)	3.79	4.14	0.35
Tshifufiamashi	SULPHIDE (1.1)	TCU (%)	2.42	2.35	-0.07
Tshifufia	CMN (0.1)	ASCU (%)	2.03	1.77	-0.26
Tshifufia	GRAT (0.4)	ASCU (%)	3.91	3.64	-0.27
Tshifufia	SD (0.3)	ASCU (%)	3.81	3.92	0.11
Tshifufia	SULPHIDE (1.1)	ASCU (%)	2.44	1.95	-0.49
Tshifufiamashi	CMN (0.1)	ASCU (%)	3.17	3.09	-0.08
Tshifufiamashi	GRAT (0.4)	ASCU (%)	1.24	1.21	-0.03
Tshifufiamashi	SD (0.3)	ASCU (%)	3.48	3.77	0.29
Tshifufiamashi	SULPHIDE (1.1)	ASCU (%)	1.54	1.5	-0.04

Statistical plots by domain were used to compare the mean model and drill grades data by northing (Figure 19.23 and Figure 19.24). These illustrate that model estimates respond well to changes in the composite grade data.

Figure 19.23 Northing Slices of the Tshifufiamashi Domain, SD (UOB/ 0.3), for TCu and ASCu, Comparing Data and Model Means

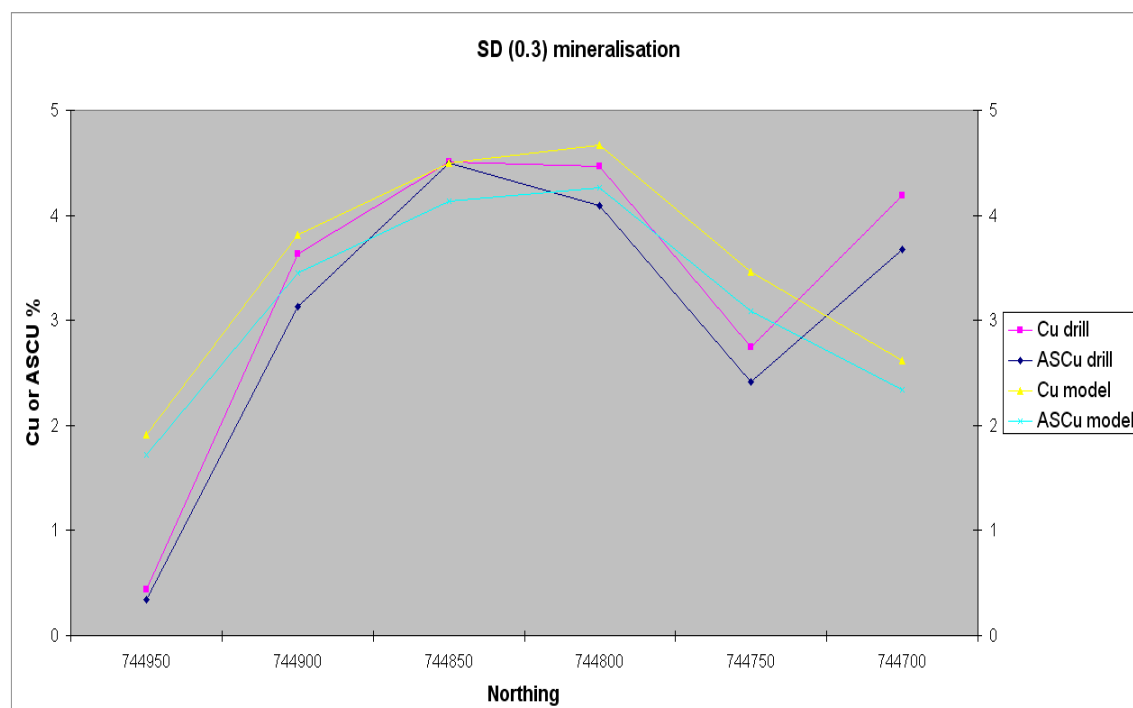
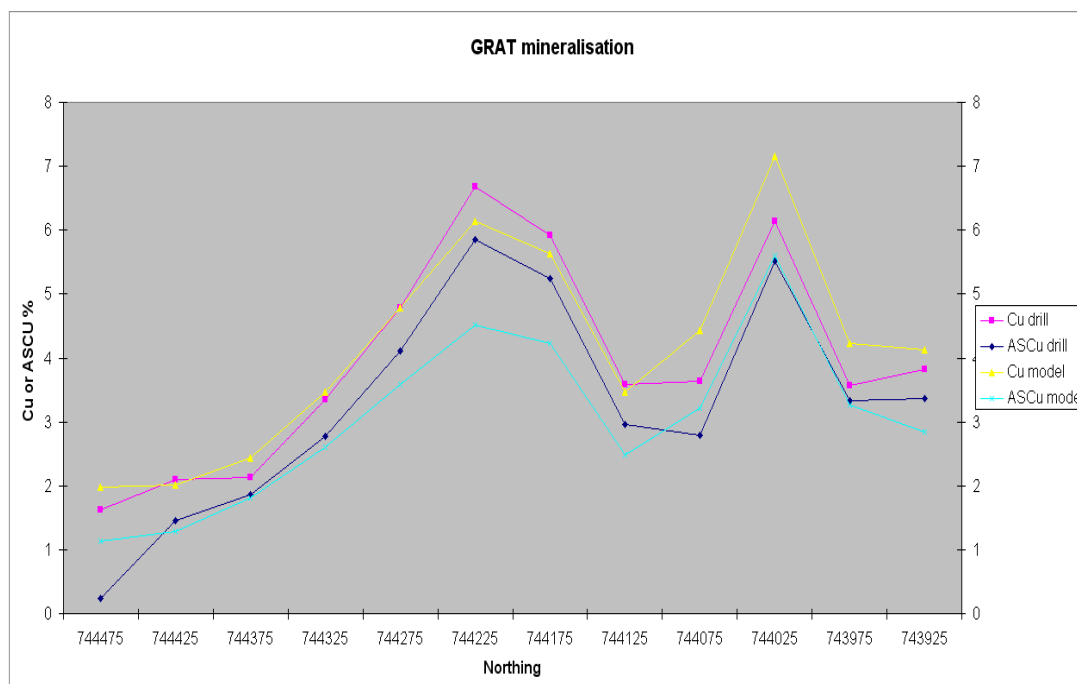


Figure 19.24 Northing Slices of the Tshifufia Domain, GRAT (LOB/ 0.4), for TCu and ASCu, Comparing Data and Model Means



From the summary statistics, visual validations and graphical plots, the OK estimates are consistent with the drillhole composites and are believed to be a fair representation of the mineralisation across the deposits.

Kinsevere Hill and Kilongo

The model was validated by statistical comparison of composite grades and model grades, and visual comparison with drillholes stepping through sections for each WFSTRAT individually (Figures 19.25 to 19.28)

Figure 19.25 Grade Comparison by Easting for Cu

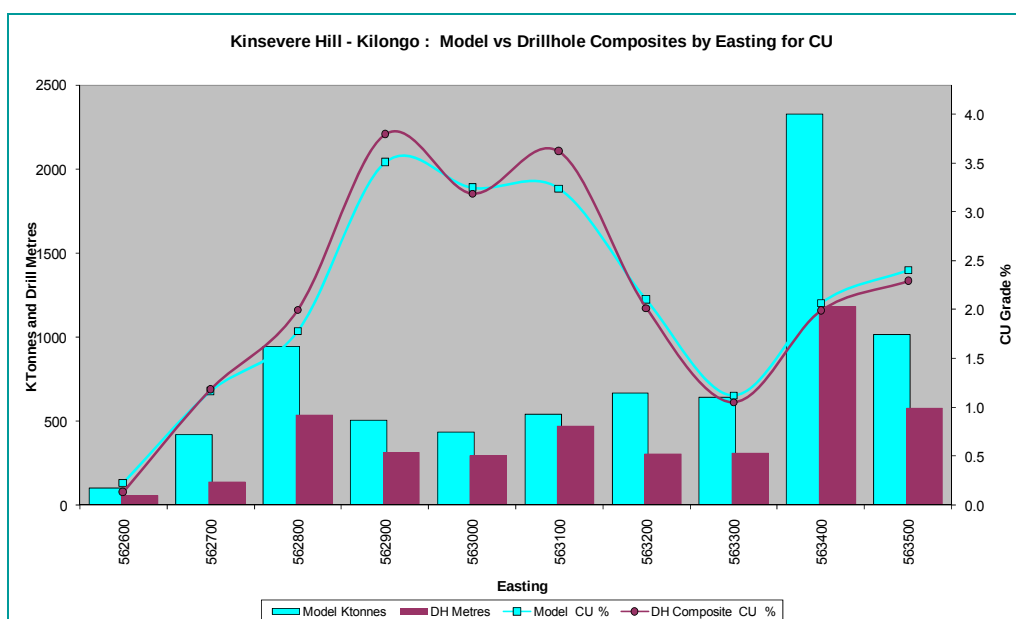


Figure 19.26 Grade Comparison by Northing for Cu

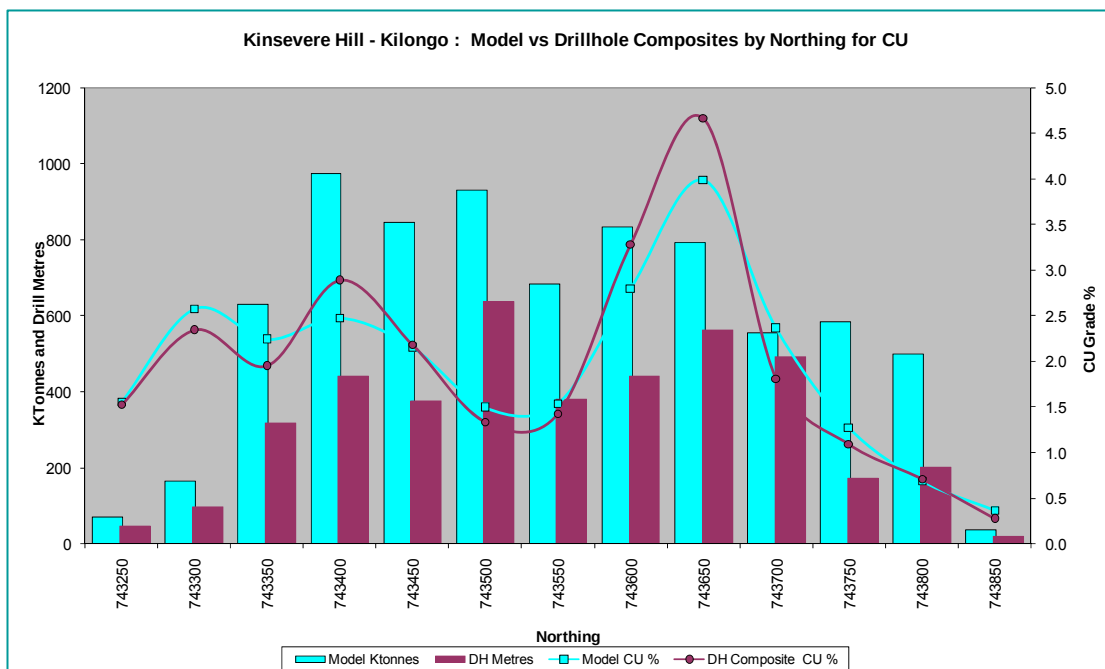


Figure 19.27 Grade Comparison by 10 m Bench for Cu

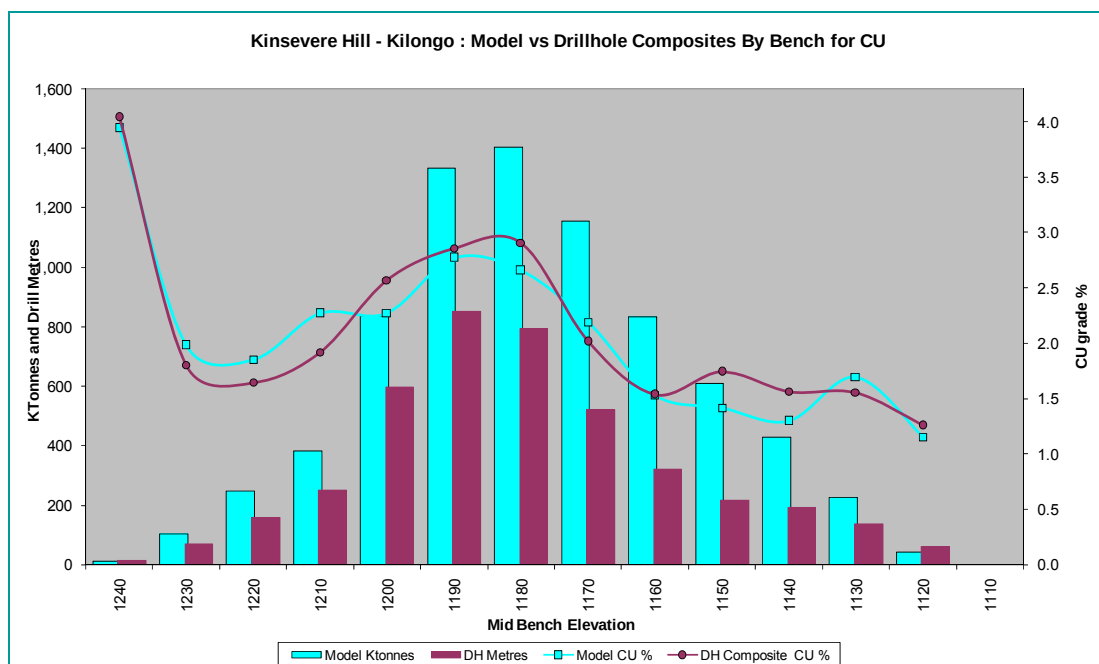
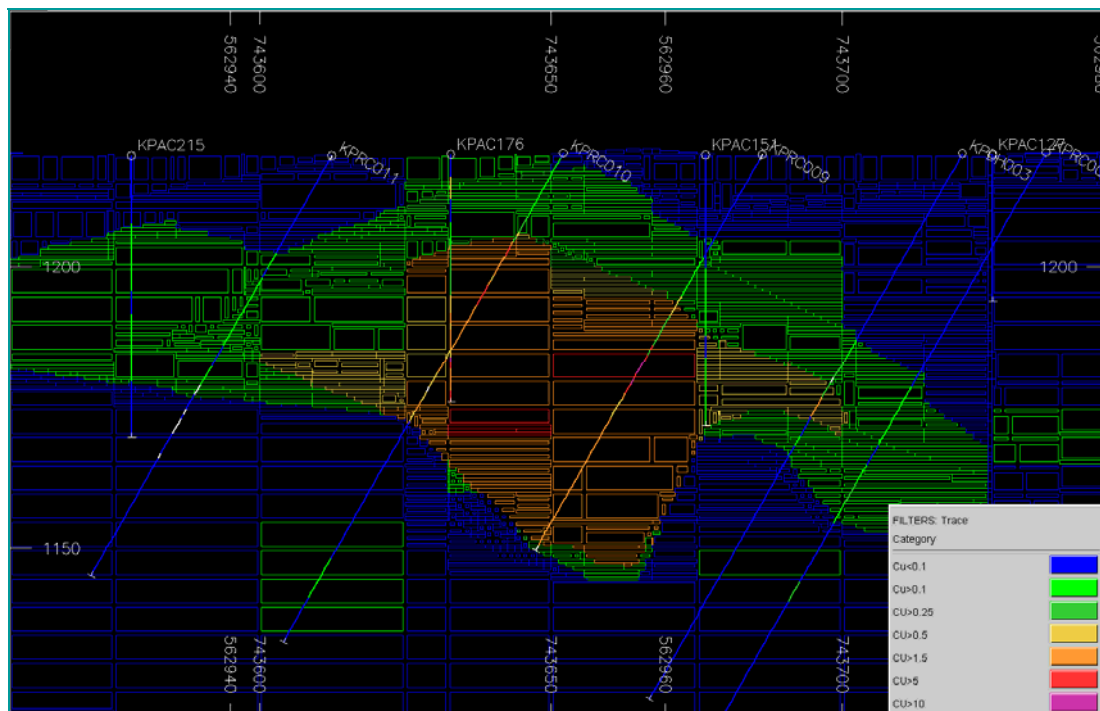


Figure 19.28 Section Showing the Close Correlation Between High Grade Composites and High Grade Zones in the Model



19.1.13 Mineral Resource Classification

The Mineral Resource has been classified and reported using the guidelines of the JORC Code (JORC, 2004), which comply with the CIM Standards on Mineral Resources and Reserves (CIM, 2000) of the Canadian Institute of Mining, Metallurgy and Petroleum (CIM).

Tshifufia and Tshifufiamashi

Classification of the Mineral Resource was primarily based on confidence in assayed grade, geological continuity and the resulting kriged estimates and efficiencies.

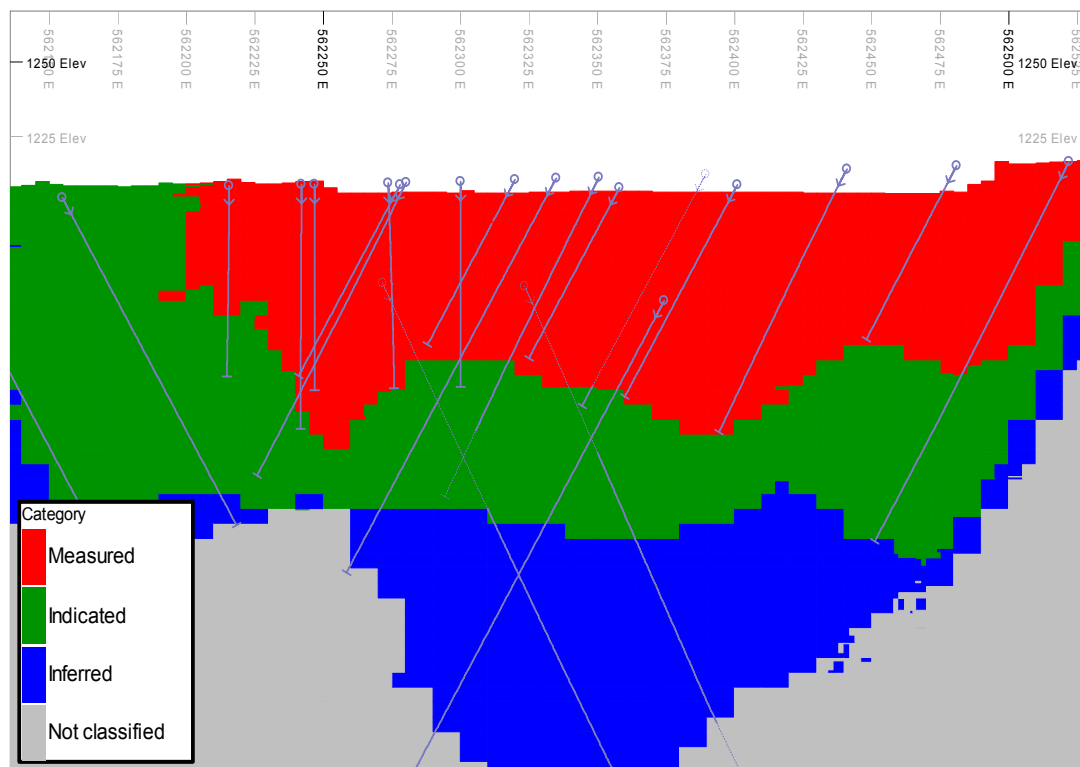
Geological confidence is supported by extensive open pit exposures, geological mapping and blast hole data, which in turn support drillhole logging and domain volumes.

Confidence in the kriged estimate is associated with drillhole coverage, analytical data integrity, kriging variance and efficiency and regression slope values. Specifically, kriging variances below 0.2, kriging efficiencies above 80% and regression slope values above 0.8 were considered appropriate for a Measured Mineral Resource category of classification. Whereas the use of mean domain density values is appropriate, subsequent models should make use of increased density data for more robust estimates.

Regarding drillhole spacing, a Measured Mineral Resource category was considered appropriate where there was a hole spacing of about 25 m along the drill lines which were between 25 m to 50 m apart. An Indicated Mineral Resource category was considered appropriate where there was a drill spacing of about 50 m to 75 m along drill lines and a line spacing of approximately 50 m. An Inferred Mineral Resource category was considered where there was a drill spacing of about 75 m to 100 m along drill lines and where the line spacing was around 100 m.

Most of the Measured and Indicated Mineral Resources are located within the oxidised mineralisation, with some Indicated Resource extending into fresh, sulphide domains.

Figure 19.29 West - East Section at Tshifufia Illustrating the Resource Classification in Relation to Drill Grid Spacing



Kinsevere Hill and Kilongo

The Mineral Resource at Kilongo and at Kinsevere Hill was classified based on:

- A copper cutoff of 0.7% TCu.
- Drillhole spacing, section spacing and sample intervals which define a coherent mineralised envelope, and grade continuity as indicated by variogram ranges of the order of the section and drillhole spacing.
- Base of drilling is a limit to the envelope – the few deeper intersections do not have mineralisation attributed.
- 1120 mRL base of model.
- Variogram ranges average around 30 m by 30 m by 15 m, and about 2/3 of cells interpolate with samples of their own WFSTRAT closer than these ranges when used as search radii.
- QAQC indicates a few issues exist, but assays show consistent overall results.
- Density estimates are consistent with Tshifufia and Tshifufiamashi modelling work, and considered conservative.

Because the drillhole data density is relatively consistent across the deposits, and areas of lower information density generally quite small and on the margins, the entire modelled lens was assigned one classification.

Although additional mineralisation exists below the bottom level of this model, it has not been adequately delineated to estimate a Mineral Resource.

19.1.14 Mineral Resource Tabulation

The oxide Mineral Resource at Tshifufia and Tshifufiamashi is derived from that portion of the block model which occurs above the BOW. Similarly, the Mineral Resource for the sulphide zone is derived from the block model volumes located below the BOW.

The oxide Measured and Indicated Mineral Resources for Tshifufia, Tshifufiamashi and Kinsevere Hill (including Kilongo) total 25.7 million tonnes at 3.78 % TCu, and were determined according to an economic cutoff grade of 0.7% TCu (Table 19.29). This is composed of:

- 9.3 million tonnes at 4.12 % TCu in the Measured Mineral Resource category; and
- 16.4 million tonnes at 3.59 % TCu in the Indicated Mineral Resource category.

The estimate accounts for mining depletion at Tshifufia up to the end of December 2009.

The sulphide Measured and Indicated Mineral Resources for Tshifufia and Tshifufiamashi total 3.76 million tonnes at 3.7% total TCu and were similarly determined according to a 0.7% total TCu cutoff grade. This is composed of:

- 0.06 million tonnes at 2.16 % TCu in the Measured Resource category; and
- 3.7 million tonnes at 3.73 % TCu in the Indicated Resource category.

Table 19.29 Kinsevere Project Measured and Indicated Mineral Resource Statement, at 31st December 2009

Kinsevere oxide Mineral Resource statement using a 0.7% TCu cutoff grade as at 31 st December 2009						
Area	Category	Volume (M m ³)	Density (t/m ³)	Tonnes (M t)	TCu (%)	ASCu (%)
Tshifufia	Measured	4.19	1.76	7.37	4.30	3.64
	Indicated	3.10	1.88	5.81	5.00	3.95
Tshifufiamashi	Measured	1.15	1.71	1.97	3.45	3.06
	Indicated	2.28	1.84	4.18	2.70	2.08
Kinsevere Hill	Measured	-	-	-	-	-
	Indicated	3.79	1.69	6.41	2.88	2.53
Total	Measured & Indicated	14.51	1.77	25.74	3.78	3.14
Kinsevere sulphide Mineral Resource statement using a 0.7% TCu cutoff grade as at 31 st December 2009						
Area	Category	Volume (M m ³)	Density (t/m ³)	Tonnes (M t)	TCu %	ASCu %
Tshifufia	Measured	0.03	2.32	0.06	2.16	0.97
	Indicated	1.33	2.32	3.09	3.99	2.60
Tshifufiamashi	Measured	-	-	-	-	-
	Indicated	0.26	2.32	0.61	2.40	1.59
Total	Measured & Indicated	1.62	2.32	3.76	3.70	2.41

The oxide and sulphide Inferred Mineral Resources are listed in Table 19.30.

Table 19.30 Kinsevere Project Inferred Mineral Resource Statement, at 31st December 2009

Kinsevere oxide Mineral Resource statement using a 0.7% TCu cutoff grade as at 31 st December 2009						
Area	Category	Volume (M m ³)	Density (t/m ³)	Tonnes (M t)	TCu%	ASCu %
Tshifufiamashi	Inferred	0.33	1.86	0.62	2.55	1.91
Tshifufia	Inferred	0.49	1.90	0.92	4.73	3.42
Kinsevere Hill	Inferred	-	-	-	-	-
Total	Inferred	0.82	1.88	1.54	3.85	2.81
Kinsevere sulphide Mineral Resource statement using a 0.7% TCu cutoff grade as at 31 st December 2009						
Area	Category	Volume (M m ³)	Density (t/m ³)	Tonnes (M t)	TCu%	ASCu %
Tshifufiamashi	Inferred	0.45	2.32	1.05	2.37	1.49
Tshifufia	Inferred	4.99	2.32	11.59	3.65	1.85
Total	Inferred	5.44	2.32	12.64	3.54	1.82

19.2 Mineral Reserve Estimate

As part of the updated Mineral Reserve estimation process, pit optimisation aspects were completed by DumpSolver (March, 2009), whilst pit designs and Mineral Reserve estimation were completed by AJC. The overall work programme was supervised by Anvil, who also supplied cost, recovery and revenue information for the optimisations, in addition to operational requirements for the pit designs.

19.2.1 Pit Optimisation Inputs

Whittle Four-X software was used to determine optimal pit shells for the Kinsevere deposits defined by the separate Mineral Resource models for Tshifufia/Tshifufiamashi (Anvil) and Kinsevere Hill/Kilongo (CSA). Analysis was undertaken for Measured and Indicated Resource categories and oxide mineralisation only. Optimisations were run with a fixed annual, long-term copper metal price value. Operating cost, revenue and pit slope design parameters were input to define conceptual staged and ultimate pit geometries suitable for detailed mine design.

For pit optimisation and subsequent analysis, operating costs were divided into fixed and variable components for each activity centre, including mining, processing and metal production. Variable costs were allocated according to their respective activity centres.

Metal Prices

The base copper price used for pit optimisation and subsequent analyses was \$1.43/lb (or \$3152.61/tonne). This was the same price as adopted in the Feasibility Study (AMCK, December 2007). The run-of-mine cobalt grade was tracked but recovery and hence revenue, was ignored.

Fixed Costs

Fixed costs are interpreted to be those which cease if and when the operation itself stops. For the purposes of the optimisation, they comprise mining contractor over heads, site mining staff, dewatering, light vehicles, laboratory, administration as well as fly-in-fly-out expenses. The manner in which they are handled depends on the Project production limit, which for Stage II is defined by the tank-house capacity, or 60,000 tonnes of LME A-grade cathode copper.

The projected fixed costs for Stage II were based on estimates made in the Feasibility Study (AMCK, December 2007), adjusted from actual operating experience during Stage I and again with a subsequent review in January 2009. The Stage II costs relating to 1.6 Mtpa process throughput for production of 60,000 tpa of copper cathode are listed in Table 19.31.

Table 19.31 Fixed Costs – Stage II

	\$/year	\$/tonne Cu
Administration	14,193,000	236.55
Technical Services	3,472,000	57.85
Product Marketing	183,000	3.05
Plant		
Salaries/Labour	2,854,000	47.56
Sundries	1,488,000	24.81
Infrastructure Maintenance		
Salaries/Labour	4,317,000	71.94
Sundries	875,000	14.59
Mining Contractor Overheads	1,361,000	22.69
Mobile Plant Maintenance ¹	-	-
Mining Overheads	4,157,000	69.28
Perimeter Dewatering Bores	449,000	7.48
In-Pit Dewatering	180,000	3.00
Total Fixed Costs	33,528,000	558.81

1. Stage II Mobile Plant Maintenance included in Infrastructure Maintenance

The total fixed costs used for optimisation are equivalent to \$520.17/tonne or \$0.24/lb of copper.

Variable Mining Costs

The variable mining cost inputs were based on a reconciliation of actual charges invoiced against the mining contract, in addition to actual AMCK operating cost figures gathered to date. The load and haul contract provided for a uniform rate of \$5.00/BCM for ore and waste rock. Based on AMCK reconciliations of work completed in 2008, this rate was escalated by 3.2% to provide an allowance for day works. Costs were assigned to each block in the model and divided by the specific block density to convert to \$/tonne mined.

A (hypothetical) owner mining study was completed by Anvil in 2008 (Anvil, February 2009), inclusive of unit cost estimates built-up from first principles by an independent mining cost estimation consultant. Assuming the same fleet as contracted by MCK, the same scheduled ore and waste hauls, and after allowing for a contractor profit margin, this study showed that there was little basis for adopting depth-incremented haulage costs.

Drill and blast costs were determined with reference to 2008 actual cost and performance data. Although previously included in the optimisation inputs, a rehabilitation allowance was not included in this instance due to the removal of these non-cash amounts from the Project cash flow model.

A revised grade control cost of \$0.12/tonne was added to the variable processing cost as an ore cost. These costs are summarised in Table 19.32.

Table 19.32 Variable Mining Costs – Stage II

Rock Mining Costs	Units	Cost
Load and Haul	\$/BCM	5.00
Day works (3.2% Load & Haul)	\$/BCM	0.16
Drill and Blast	\$/BCM	0.18
Rehabilitation	\$/BCM	0.00
Default Waste Mining Cost	\$/BCM	5.34
Differential Ore Mining Costs	Units	Cost
Grade Control	\$/tonne ore	0.12

The unit costs were converted to \$/tonne on a block-by-block basis in the SURPAC block model using the formula:

$$\text{Mining Cost (\$/tonne)} = \text{IF}(\text{density} > 0, (\text{LH} + \text{DW} + \text{DB} + \text{RB}) / \text{density}, 0)$$

The \$/tonne mining cost was then carried across to the Whittle model as the mining cost adjustment factor (MCAF) value and the reference mining cost set to 1.00.

Variable Processing Costs

Variable processing cost estimates were collated by Anvil for the Stage II plant (Table 19.33). The Stage II process costs comprise crushing, grinding and acid leach but not SX-EW costs which are calculated as a metal cost. Process consumables usage considers an average estimate for gangue sulphuric acid consumption (GAC). Testwork for the Feasibility Study confirmed that GAC is expected to be relatively low due to the absence of carbonaceous gangue minerals in the host rock. Since the Feasibility Study, the variable plant operating costs have been revised downwards from \$12.86/tonne to \$10.69/tonne (ie, excluding the differential ore mining cost). The main difference arises from a reduction in plant consumables and spares costs which were \$7.69/tonne and \$2.31/tonne, respectively.

Metal Costs

Metal costs for Stage II were determined by Anvil and are summarised in Table 19.34. Metal costs were defined in three sub-categories; SX-EW, cathode marketing and royalties. Since the operation is limited by SX-EW capacity, fixed costs were expressed in terms of \$/tonne of cathode and added to the SX-EW sub-category.

Export duties and royalties were treated as price-dependent. Export duties and the DRC Government royalty were calculated as 1% and 2.0% of copper price less cathode transport and assay costs, respectively. The Gécamines royalty used to be calculated on a sliding scale with specified maxima and minima. As an outcome of the Mining Agreements renegotiation process (refer to Item 20.2), a revised fixed rate of 2.5% of the copper price was adopted. Earlier studies used the sliding royalty which was capped at \$70/tonne. Therefore the revised approach represents a modest increase at the base case metal price of \$1.43/lb.

Table 19.33 Variable Processing Costs – Stage II

	Cost (\$/tonne ore)
Plant ROM Feed	0.81
Long term Stockpile Reclaim ¹	0.07
Plant Power	1.40
Plant TSF Operation	0.66
Plant Laboratory	0.08
Plant Process Water	0.01
Plant Consumables	6.16
Plant Maintenance Spares	1.39
Maintenance TSF	0.05
Maintenance Consumables	0.03
Differential Ore Mining	0.12
Total Processing Cost	10.78

Table 19.34 Metal Costs at \$1.43/lb – Stage II

	Cost (\$/tonne Cu)	Cost (c/lb Cu)
SX-EW		
Time Costs (As Fixed)	558.81	25.3
Power	114.64	5.2
Consumables	170.35	7.7
Sub-total	843.79	38.3
Marketing/Transport/Export		
Cathode Copper Cartage	216.00	9.8
Sea-freight Participation	94.20	4.3
Infrastructure Levy	30.00	1.4
Assay Costs	0.00	0.0
Export Duties	28.42	1.3
Export Documentation	96.88	4.4
Sub-total	465.50	21.1
Royalties		
Gécamines Royalty	78.82	3.6
DRC Government Royalty	56.85	2.6
Sub-total	135.66	6.2
Total Metal Cost	1,444.96	65.5

Net Copper Price

The net copper price is the price less metal costs. In Whittle jargon the metal costs are referred to as Selling Costs. Since the limiting production factor (or target) at Kinsevere is metal production, time costs are also included as a selling cost. At Kinsevere, the formula is thus:

$$\text{NETPRICE} = \text{PRICE} - \text{SELLCOST}$$

At \$1.43/lb, the net price used for cut-off grade calculation becomes:

$$\text{NETPRICE} = \$3,152.61 - \$1,444.96 = \$1,707.65/\text{tonne}$$

To ensure the parametric pit shells generated by Whittle are correct using the range of revenue factors, price dependent metal costs must be applied during pit optimisation as expressions.

Processing Recovery

Based on metallurgical process modelling completed for the Feasibility Study, a fixed recovery figure of 92.14% was adopted.

Mining Dilution and Recovery

The pit optimisation and analysis adopted a mining recovery factor of 1.00 and dilution factor of 1.05.

Slope Parameters

Sensitivity analyses completed for the Feasibility Study indicated that the final pit geometry of optimal pit outlines is relatively insensitive to recommended inter-berm slope angles. Earlier studies had tried to finesse slope parameters by including final ramp layouts within the inter-berm slope angles. The sensitivity analyses also indicated that each deposit is relatively insensitive to mining costs and therefore for simplicity and for the purposes of optimisation, the inter-berm angles specified in the Feasibility Study were adopted for the overall slope specification, as listed in Table 19.35.

Table 19.35 Pit Optimisation Slope Angles

Pit Wall Location	Surface to 40m	Below 40m
Tshifufiamashi Pit west wall	40°	32°
All other pit walls	40°	40°

Marginal Cut-Off Grade

Whittle uses the following simplified formula to calculate the marginal cut-off grade as listed in Table 19.36:

$$\text{Marginal COG} = \text{PROCOST} \times \text{MINDIL} / ((\text{PRICE} - \text{SELLCOST}) \times \text{RECOV})$$

To avoid confusion, the NETPRICE (i.e. PRICE – SELLCOST) must be expressed in units of metal grade. Since the grade is % Cu (or % ASCu), this is \$/10kg. In other words, the \$/tonne must be divided by 100. The PROCOST is the sum of the processing cost plus the ore mining cost differential.

Table 19.36 Marginal Cut Off Grades for Stage II

	Units	at \$1.43/lb
PROCOST	\$/t proc	10.78
RECOV	% ASCu	92.14%
PRICE	\$/lb	1.43
PRICE	\$/10kg	31.53
SELLCOST	\$/10kg	14.45
NETPRICE	\$/10kg	17.08
MINDIL		1.05
COG	% ASCu	0.72%

19.2.2 Pit Optimisation Results

The optimisation methodology was to run each of the reblocked Mineral Resource models (ie, the combined Tshifufia and Tshifufiamashi model, and the separate Kinsevere Hill and Kilongo model) through the Four-X software using the long-term fixed price of \$1.43/lb and with only oxide Measured and Indicated Mineral Resources directed to processing. Oxide Inferred Mineral Resources and all sulphide mineralisation were treated as waste. The staged and ultimate pit shells were selected on the basis of:

- Real discount rate of 10%
- Production limit of 60,000 tpa cathode
- Up to three push-backs

In each metal price scenario, the ultimate pits were selected on the forward side of the cash flow curve in order to provide flexibility in the event of increased operating costs. This approach is particularly relevant to the Kinsevere Hill model, due to the lower average grade of the deposit (ie, relative to Tshifufia and Tshifufiamashi) and its inability to meet the required head grade to sustain 60,000 tpa of copper output. This selection process differed from that adopted in the Feasibility Study (AMCK, December 2007), where the ultimate pits were selected further out on the cashflow curve, providing a relatively longer life operation.

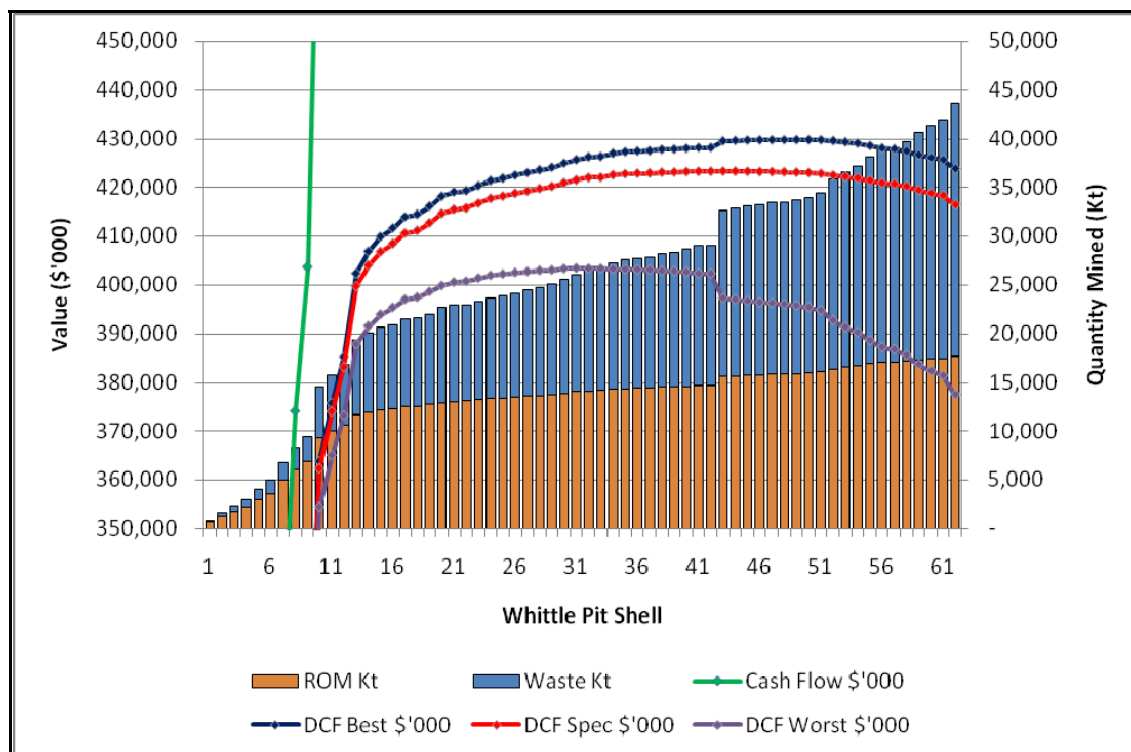
Tshifufia and Tshifufiamashi

The optimisation results for Tshifufia and Tshifufiamashi are summarised in Table 19.37, whilst the corresponding graphical results are shown in Figure 19.30. The ultimate pit selected for \$1.43/lb scenario was Pit 32.

Table 19.37 Pit Optimisation Results Tshifufia & Tshifufiamashi

Copper Price	\$1.43/lb
Ultimate Pit	32
Push-backs	6,10
Cash Flow (\$'000)	632,545
DCF Specified (\$'000)	421,892
Mined (Kt)	26,516
Waste (Kt)	12,348
Strip Ratio	0.9
Mill Feed (Kt)	14,167
Avg. Grade ASCu %	3.86
Avg. Grade Cu %	4.57

Figure 19.30 Tshifufia & Tshifufiamashi Optimisation at \$1.43/lb



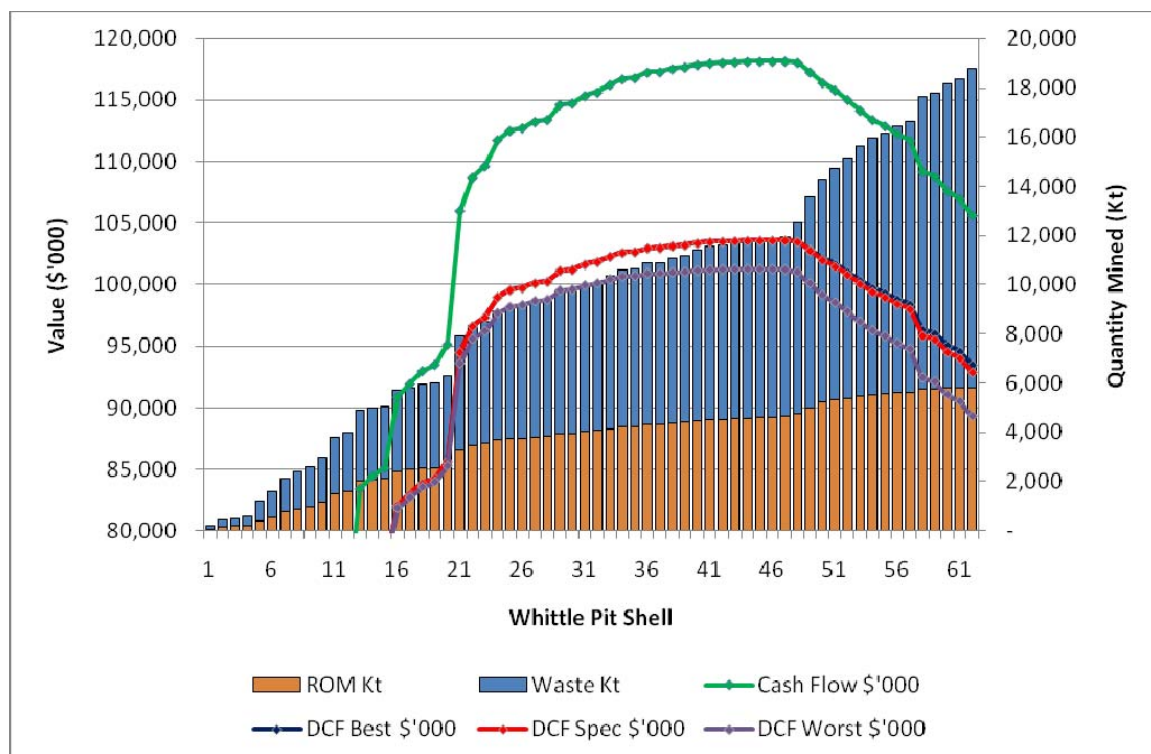
Kinsevere Hill and Kilongo

The optimisation results for Kinsevere Hill and Kilongo are summarised in Table 19.38, whilst the corresponding graphical results are shown in Figure 19.31. The Ultimate pit selected for \$1.43/lb scenario was Pit 40.

Table 19.38 Pit Optimisation Results Kinsevere Hill & Kilongo

Copper Price	\$1.43/lb
Ultimate Pit	40
Push-backs	12
Cash Flow (\$'000)	117,897
DCF Specified (\$'000)	103,377
Mined (Kt)	11,414
Waste (Kt)	6,955
Strip Ratio	1.6
Mill Feed (Kt)	4,459
Avg. Grade ASCu %	2.94
Avg. Grade Cu %	3.29

Figure 19.31 Kinsevere Hill - Kilongo Optimisation at \$1.43/lb



The Kinsevere Hill and Kilongo deposits are lower grade than Tshifufia and Tshifufiamashi. If the plant throughput is limited to 1.6 Mtpa, then the combined deposits could not sustain cathode production at 60,000 tpa. The deposits are economic but assuming that Tshifufia and Tshifufiamashi are mined first, then subsequent mining and processing of Kinsevere Hill ore would need to be supplemented with feed from other sources (eg, low grade oxide ore stockpiles, HMS plant floats/effluent, and/or Stage III sulphide ore) in order to maintain the 60,000 tpa output.

19.2.3 Pit Optimisation Sensitivity Analyses

No further work has been carried out on sensitivity analyses since the Feasibility Study. The Project is relatively insensitive to changes in mining costs, processing costs and overall pit slope angles. Due to the relatively high average grade of the deposits, the most sensitive variable is copper metal price.

19.2.4 Mine Design

A series of pit designs were developed using the selected staged and ultimate pit shells and according to the design parameters listed below.

In terms of haul ramp exits, the following information was considered:

- waste dumps are already located on the eastern side of the pits;
- the TSF (for which waste will be used over the life of mine for the construction of downstream raised embankments) is to be located to the south; and
- the processing plants are located to the west and south west of the pits.

Design and Planning Parameters

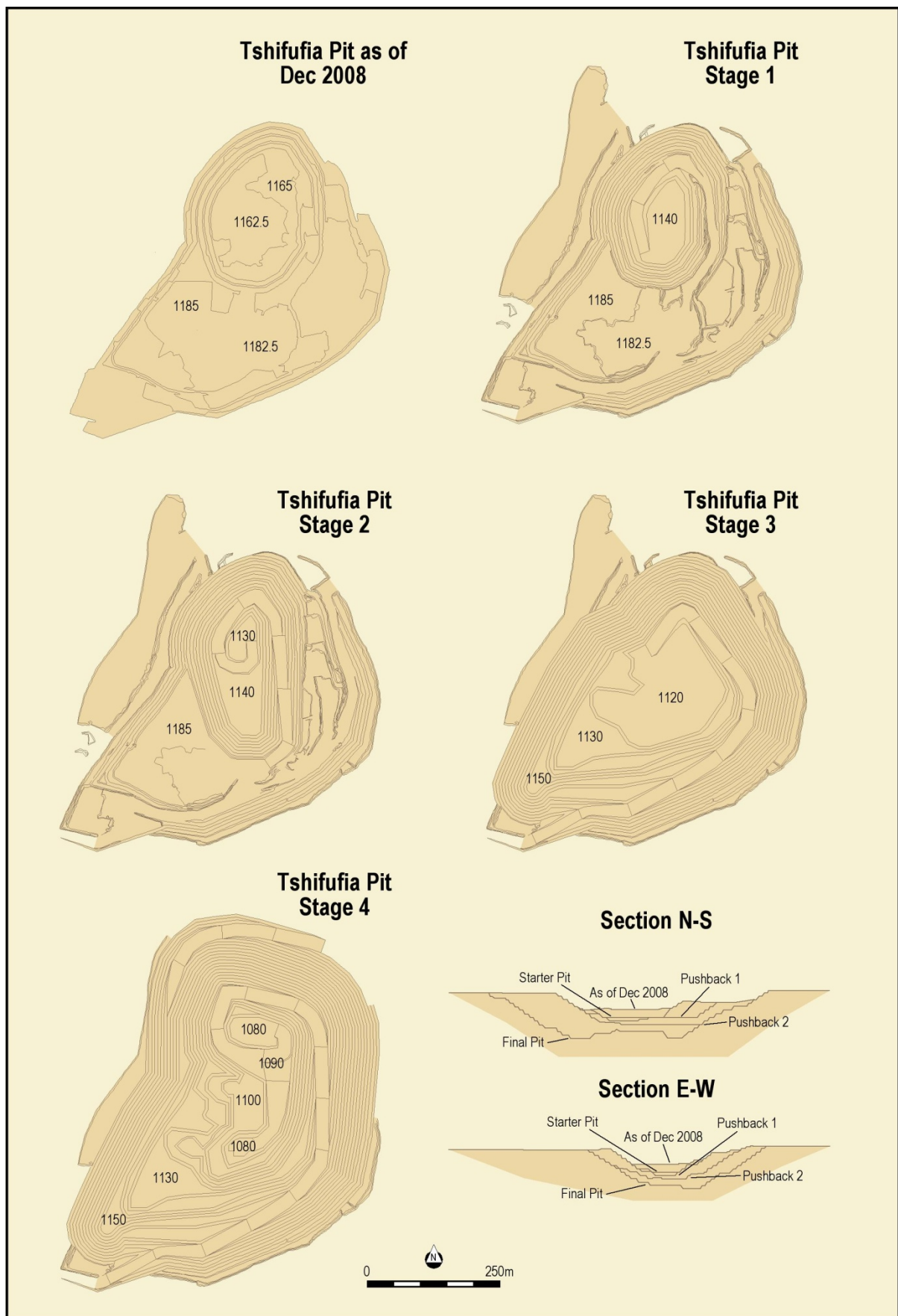
The smallest selective mining unit adopted in the pit design process was 25 m x 25 m. This is considered to be the minimum dimensions required to enable truck turning and positioning. An excavator can dig below this level, however, and top-load into the trucks. Design bench/fitch heights, ramp widths and gradients were as follows:

- Access Ramps
 - o Gradient = 1 in 9
 - o Width = 20 m
- Pit Walls
 - o Inter-ramp slope = 40° (32° for Tshifufiamashi Pit west wall)
 - o Batter Height = 10 m (5 m for Tshifufiamashi Pit west wall)
 - o Batter Angle = 60°
 - o Berm Width = 6.2 m (5 m for Tshifufiamashi Pit west wall)

Tshifufia Pit Designs

The four stages of Tshifufia pit designs (oxide pits) are shown in Figure 19.32. The ultimate oxide pit extends to a depth of 130 m below surface (1080 mRL) at the northern end and 80 m below surface (1130 mRL) at the southern end.

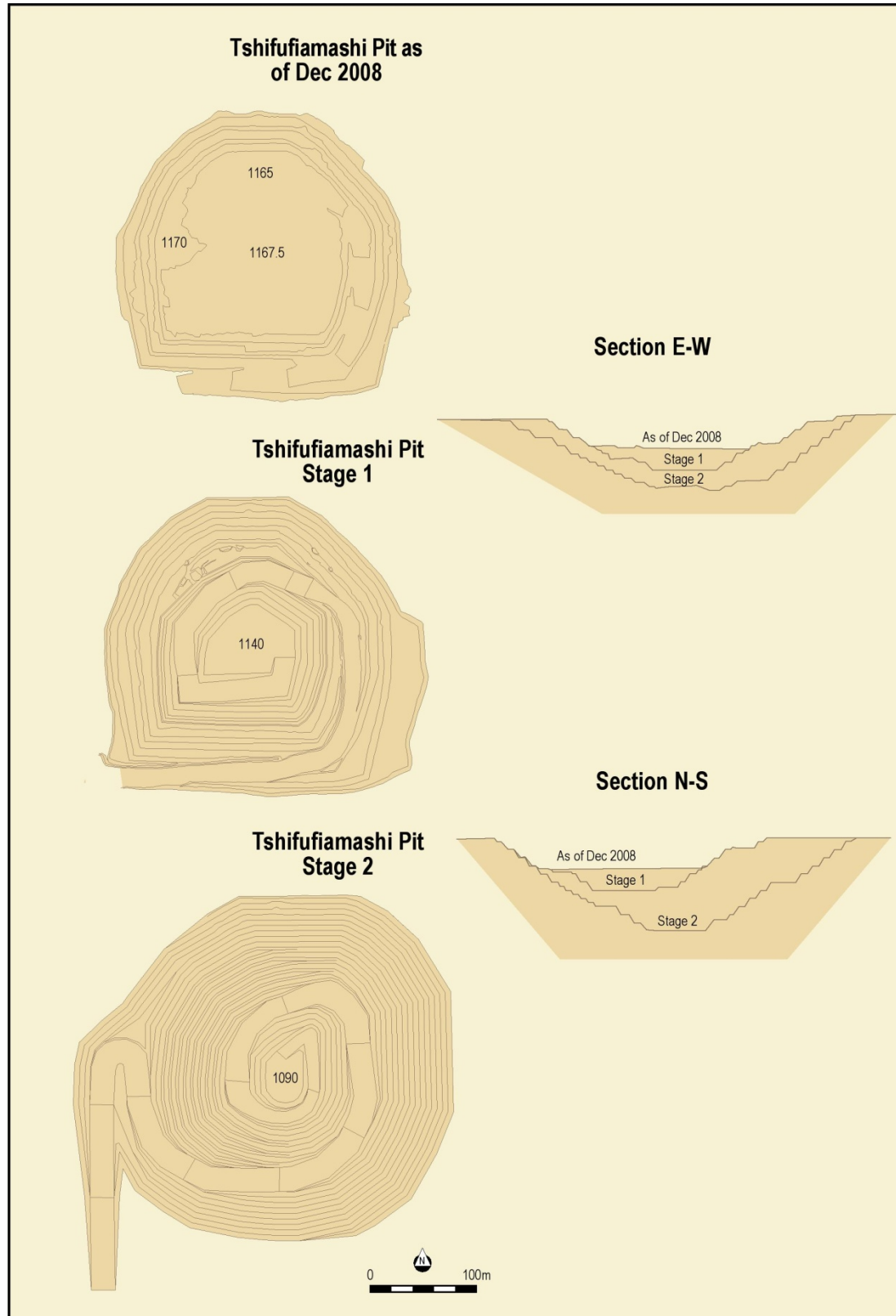
Figure 19.32 Tshifufia Design Pit, Showing Intermediate Stages



Tshifufiamashi Pit Designs

The Tshifufiamashi pit designs (oxide pits) are shown in Figure 19.33.

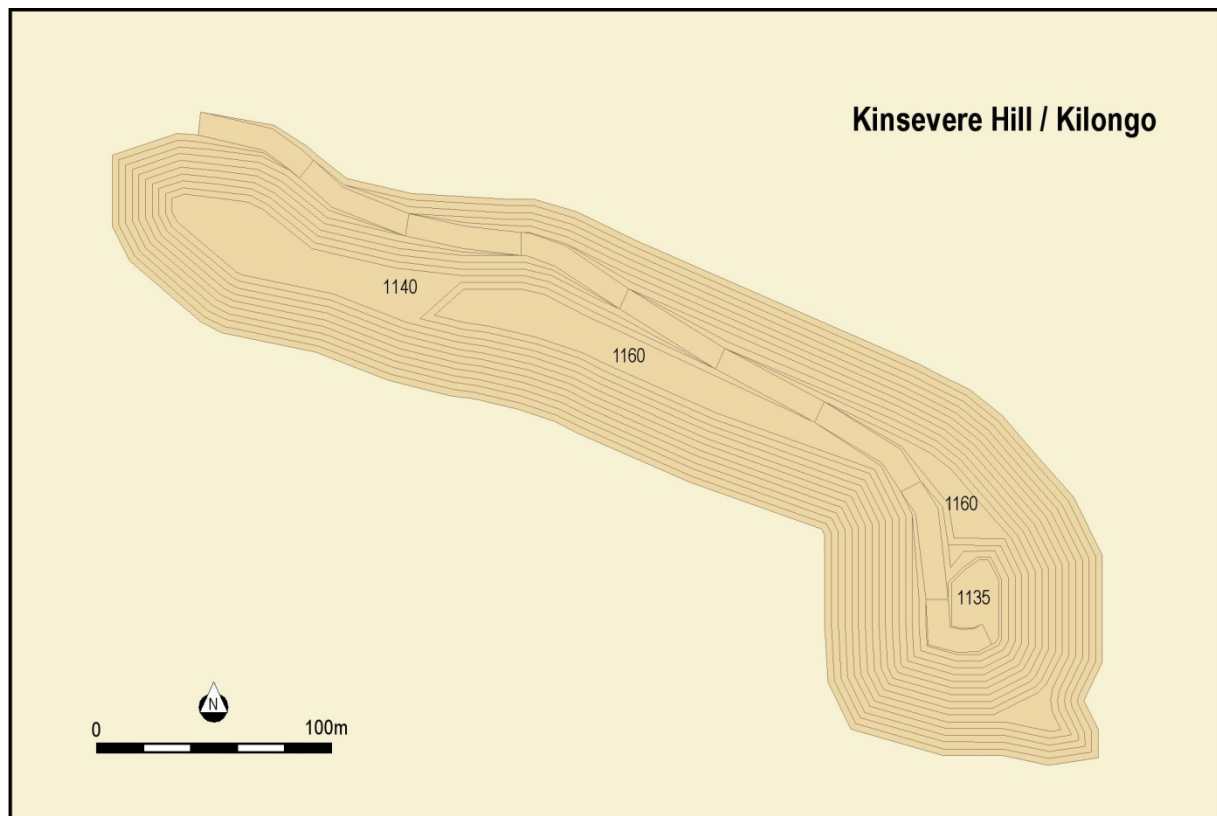
Figure 19.33 Tshifufiamashi Design Pit, Showing Intermediate Stages



Kinsevere Hill – Kilongo Pit Designs

The single stage Kinsevere Hill/Kilongo Pit (oxide) is shown in Figure 19.34. The Kinsevere Hill deposit is at the south eastern end, and the Kilongo deposit extends to the north west. The proposed sequence of mining is from Kilongo towards Kinsevere Hill.

Figure 19.34 Kinsevere Hill/Kilongo Design Pit



19.2.5 Mineral Reserve Tabulation

The total Mineral Reserve is estimated as 18.4 million tonnes at 3.53% ASCu. The estimate is entirely within the oxide Measured and Indicated Mineral Resource estimate reported in Table 19.29. A breakdown by pit and classification is provided in Table 19.39. Table 19.40 lists the additional Mineral Reserve contained in ROM pad stockpiles, as at the end of December 2009.

The reported Mineral Reserve is based on an economic cut-off grade which accounts for a longer-term copper metal price projection of \$1.43/lb (\$3,153/t).

Table 19.39 Kinsevere Project Mineral Reserve Statement, at 31st December 2009

Kinsevere oxide Mineral Reserve statement as at 31 st December 2009 (based on \$1.43/lb Cu)					
Deposit	Classification	Tonnes (Mt)	TCu (%)	ASCu (%)	Contained ASCu (k t)
Tshifufia	Proven	7.14	4.12	3.51	250.5
	Probable	4.40	5.09	4.05	178.3
Tshifufiamashi	Proven	1.65	3.56	3.21	52.8
	Probable	1.20	4.02	3.68	44.3
Kinsevere Hill	Proven	-	-	-	-
	Probable	4.02	3.42	3.07	123.5
Total	Proven & Probable	18.41	4.14	3.53	649.5

Table 19.40 Kinsevere Project Stockpile Reserve Statement, at 31st December 2009

Kinsevere oxide Stockpiles as at 31 st December 2009 (based on \$1.43/lb Cu)					
Stockpile	Classification	Tonnes (Mt)	TCu (%)	ASCu (%)	Contained ASCu (k t)
High Grade (+4% ASCu)	Proven	-	-	-	-
	Probable	0.03	8.56	7.18	2.1
Medium Grade (2% to 4% ASCu)	Proven	-	-	-	-
	Probable	1.11	2.90	2.43	27.0
Low Grade (<2% ASCu)	Proven	-	-	-	-
	Probable	1.26	1.04	0.87	11.0
Total	Proven & Probable	2.40	1.99	1.67	40.1

The Mineral Reserve listed in Table 19.39 represents mining recovery of 72% and 81% of the Measured plus Indicated Mineral Resource tonnage and metal estimate, respectively. In total, 42% of the combined Mineral Reserve tonnage (ie, including stockpiles) is classified as Proven with the remainder classified as Probable.

The Proven and Probable Mineral Reserve estimate has been reported using the guidelines of the JORC Code (JORC, 2004) and is consistent with reserves reported using the CIM Standards on Mineral Resources and Reserves (CIM, 2000) of the Canadian Institute of Mining, Metallurgy and Petroleum (CIM). Table 19.41 sets-out the JORC Code checklist responses for estimation and reporting of an Ore Reserve, which is equivalent to a Mineral Reserve under the CIM standards.

Table 19.41 Kinsevere Project Mineral Reserve Summary, Checklist of Assessment and Reporting Criteria

Criteria	Explanation
Mineral Resource estimate for conversion to Mineral Reserves	The Mineral Resource estimates for Tshifufia and Tshifufiamashi were prepared initially in February 2009 and updated for mining depletion to the end of December 2009. The estimates and the associated Mineral Resource models have been the basis for mine planning, production scheduling and Mineral Reserve estimation for these deposits, allowing for mining depletion as at 31 December 2009. The Mineral Resource estimate for Kinsevere Hill was prepared in June 2008. This estimate and the associated resource model have been used for mine planning, production scheduling and Mineral Reserve estimation for the deposit. The oxide Measured and Indicated Mineral Resource estimates are reported inclusive of the Mineral Reserves.
Study status	The level of mine planning completed for this Study is commensurate with the requirements for a Feasibility Study to bankable standard. Mine planning and design has included conventional Whittle Four-X optimisation of the Mineral Resources, detailed staged and ultimate pit designs inclusive of geotechnical design recommendations, and comprehensive life of mine production schedules (on a monthly and annual basis). The mine plan is considered to be technically achievable and economically viable, based on the DCF analyses that have been completed subsequent to the Feasibility Study. In the conversion of Mineral Resources to Mineral Reserves, modifying factors such as mining/metallurgical parameters, economic, product marketing, legal, environmental (in respect of waste dump design and site rehabilitation allowances), social and government considerations have been taken into account.
Cut-off parameters	Marginal cut-off grades have been considered during the Whittle Four-X optimisation process.
Mining factors or assumptions	A conventional approach has been adopted in the use of optimisation followed by detailed mine design and production scheduling. The selected excavator/truck mining method is considered to be appropriate to the style of mineralisation and to the operating conditions at the DRC site. The method is proven at the site for production to date. Topsoil has already been pre-stripped and stockpiled for future rehabilitation works. Access into the design pits has taken account of the ROM pad location, optimised waste dump locations and the location of the TSF (which will be built partly from inert mine waste). Geotechnical investigations and analyses for pit slope stability have been carried out, and these have included drilling/logging investigations, mapping and inspections by a geotechnical consultant, laboratory testing and comprehensive kinematic analyses. The work has been supported by equally comprehensive groundwater investigations, all of which have led to the recommendation and adoption of relatively flat, drained overall pit slope angles. Grade control practices are being developed for the newly commenced Project. In time, the information from campaigned RC grade control drilling and assaying will be used to refine the Mineral Resource models. The Mineral Resource models used for optimisation and design are current as at 31 December 2009. Adopted mining dilution and recovery factors are considered to be appropriate for this level of study, based on the size of the mining equipment, the mining flitch height and the geometry of the ore/waste contacts. This also applies to the minimum mining dimensions of 25 m x 25 m. All infrastructure to support the contracted excavator/truck mining method is in place.
Metallurgical factors or assumptions	Comprehensive metallurgical testwork based on representatively selected samples has been carried out to support SX-EW processing.

Criteria	Explanation
	The choice of processing method has been endorsed by independent metallurgical advice, and a number of positive technical/processing studies were completed leading up to the Feasibility Study. SX-EW processing is considered to be well-tested and proven world wide, and has also been successfully adopted for similar projects in the DRC and nearby Zambia. The metallurgical recovery factor has been based on representative oxide and mixed ore samples and testwork from each of the three deposits, supported by metallurgical simulation. There are no seriously deleterious elements, and the available water supply for processing is suitable without treatment. No pilot scale testwork has been carried out. However the behaviour of the ore types as presented to a crushing and milling circuit is well understood from HMS operations to date.
Cost and revenue factors	Mining operating costs are based partly on a combination of contractor fixed and variable costs, which are already known. Process operating costs have been built up from first principles and take account of detailed estimates of manning numbers, consumables usage and cost. Mining capital costs are minimal. Lump sum processing capital costs have been estimated in detail by the Company's SXEW construction contractor, Ausenco. Optimisations have been completed on a conservative long-term copper price projection. No revenue is attributable to cobalt mineralisation. Product transport and marketing charges have been considered in the optimisations and cashflow models. These are conservative relative to indications during negotiations with future customers. Royalties payable to Gécamines and the DRC government have been considered in the optimisations and cashflow models.
Market assessment	The Company's research indicates that medium to longer term demand exceeds supply. Competition from other suppliers in the DRC is a threat, but it is expected that there will be a ready market for the sale of Stage II product. The Company has an established network of customers and sales negotiations are already in progress.
Other	As part of the Feasibility Study, a thorough risk analysis was completed on all technical and non-technical risks which could have an impact on the success of the Project. Risk mitigation strategies have been identified which will continue into the Project implementation and operations phase. There are potential government, social and security related risks for a company operating in the DRC. The Company considers that, by virtue of its significant in-country experience to date, the social and security risks are manageable.
Classification	The Mineral Reserves have been classified as Proven or Probable, according to the corresponding Measured and Indicated Mineral Resources.
Audits or reviews	The mine planning leading up to the Mineral Reserve estimates was completed by the Company's mining consultants. Audits and reviews of the technical input to and progress of this work were completed by the Company's Manager, Group Technical Services. An independent review was also carried out, as part of the Project funding investigations in 2009, by consultants SRK Consulting (UK) Ltd.
Discussion of relative accuracy/confidence	The average grades that are evident in the Mineral Resource models adopted for mine planning and Mineral Reserve estimation are comfortably higher than the marginal cut-off grades apparent in the optimisation process. Optimisation sensitivity analyses have shown that the Project DCF is relatively insensitive to +/- 20% changes in mining or process operating costs. Whilst metal pricing will be the more sensitive variable, it is noted that conservative price projections have been adopted in the Mineral Reserve process.

ITEM 20 Other Relevant Data and Information

The following information summarises events which have **impacted on the Project development plan and Cash Flow Model as envisaged in the Feasibility Study and 2008 Technical Report**. This information is in addition to the impacts arising due to the Mineral Resource and Mineral Reserve updates as reported in Item 19.

Also reported under this Item are the findings of several **technical studies impacting on the progression of the Stage II Project to a proposed Stage III operation**. This information is presented in the context of the reported oxide and sulphide Mineral Resources in Item 19.1.

Preliminary evaluation of cobalt recoverability supports the discontinuation of reporting cobalt grades in the Mineral Resource statements. In the economic analysis described in Item 25, no value is attributed to cobalt that passes through the Stage II circuit.

Several potential Project enhancements are also described under this Item.

Since the cessation of initial Stage II construction activities in late 2008, Ausenco has now been recontracted on a Lump Sum Turn Key (LSTK) basis. Capital cost increases arising from the time lag and renegotiated contract are described in Item 25.11.

20.1 Stage I Mining and Processing

In November 2008, the Company placed the fabrication and construction works associated with the Stage II Project on hold, pending improvements in the global financial and commodity markets, and also until Project funding was in place. At this time, and in an effort to conserve cashflows, the MCK mining contract was terminated.

The Stage I HMS processing activities were also halted in November 2008 and the plant was placed under care and maintenance. Meanwhile, the first of the two proposed furnaces at the Stage I EAF facility had been commissioned in August 2008 and it became the sole source of copper production for the remainder of 2008. This first furnace operated at below its design capacity of 12,000 tonnes of 'black copper' per year due to a combination of design, logistical and technical issues. Commissioning of the second furnace was consequently postponed until these difficulties could be overcome.

By the end of 2008, the Company had identified a number of EAF process flow issues associated with copper metal handling and furnace feed limitations. A number of materials handling improvements were put in place in an attempt to increase production output and a re-line of the furnace was undertaken in February and March 2009. A technical review also commenced in early 2009 with the aim of improving the feed blend and better managing the operating temperature of the furnace.

An opportunity to sell HMS concentrate presented itself in March 2009, the economic return from which appeared to be more favourable than persisting with the EAF and producing black copper under the prevailing technical circumstances. Under this opportunity, the HMS plant was subsequently recommissioned to process ROM stockpiled ore that would otherwise have been reclaimed as feed to the Stage II SX-EW plant. The EAF was shut down and the HMS concentrate sold at the mine gate, rather than smelted.

A new mining contract was put in place with MCK in July 2009 and mining was resumed on a reduced scale to enable continued HMS concentrate production from August 2009 to December 2010.

The forecast production plan for 2010 involves:

- The mining of approximately 610,000 tonnes of ore from the Tshifufia Central_2 stage pit, at an average grade of 3.6% TCu.
- The coincident mining of approximately 300,000 tonnes of waste, which will be delivered to the Stage II TSF embankments.
- An average ore and waste mining rate of about 2,150 bcm/day, mined on a five day per week, single shift basis.
- Mining focussed on the Central_2 cutback benches between 1185 mRL and 1175 m RL levels, in dry ground above the water table.
- Mining of these higher level benches yields lower grade ore than would otherwise be available from mining of the existing faces at lower elevations in the Central_1 Pit.
- Drilling and equipping of new groundwater bores to lower the water level below the Central_1 Pit floor before larger scale mining (for Stage II production) proceeds at these elevations from 2011.
- Feed to the crusher at an average head grade of 3.8% TCu, and at a rate up to the maximum crushing rate of 2,000 tonnes per day.
- An overall HMS recovery of 60%, yielding a total concentrate tonnage (combined HMS sinks and spirals product) of about 60,000 tonnes at an average grade of 25%, for approximately 15,000 tonnes of copper metal recovered.

This production plan provides just sufficient cashflow to sustain the Stage I operations and corporate overheads throughout the 2010 Stage II construction programme.

20.2 DRC Government Review

The mineral rights to Kinsevere are held by Gécamines. AMCK has a lease agreement with Gécamines to mine and process ore from Kinsevere until 2024, followed by an automatic 15 year extension.

In February 2008, the Company was advised by the Minister of Mines that, as a consequence of a Government commission, the terms of the Kinsevere *Contrat d'Amodiation* (Lease Agreement) were under review. The Company completed its negotiations with Gécamines and the Government in January 2009 and the key details of the amended agreement are set out in Table 20.1.

20.3 Potential Pit Expansions for Stage III

During 2008, preliminary mine planning was carried out to indicate the potential for open pit mining depth extensions into sulphide ore (ie, the conceptual Stage III Kinsevere Project). The objective of this exercise was to assess the possible western pit limits and proximity to the Stage II plant infrastructure, in addition to assessing the potential scale of Stage III processing.

The Stage III development concept has to date considered a parallel comminution circuit, high pressure acid leach plant to treat sulphide ores, followed by a 30,000 tpa expansion to the SX-EW cathode capacity. The geometry of the sulphide deposit appears to favour a sequential treatment of oxide and then sulphide inventories. However, a preliminary assessment of the extent of currently defined sulphide

material indicates that the expansion of cathode capacity to 90,000 tpa would not be justifiable at this time. Hence, the Stage III optimisation considered the building of a flotation plant and high pressure leach circuit to treat sulphide ore to maintain cathode output at 60,000 tpa. Fixed costs associated with Stage III were therefore assumed to be the same as Stage II.

Table 20.1 Key Amendments to the Kinsevere Lease Agreement

Term	Original Agreement	Amended Agreement
Rent payments	Rent paid to Gécamines on a sliding scale, from a floor price of \$35 per tonne of copper at a LME copper price of \$2,200 per tonne (or \$1.00/lb copper) to a ceiling price of \$70 per tonne of copper at an LME copper price of \$4,000 per tonne (or \$1.80/lb copper) on each tonne of commercially viable copper metal extracted at Kinsevere.	Rent paid to Gécamines based on 2.5% of gross turnover.
Cash payments	A <i>pas de porte</i> (entry premium) payment of \$5 million of which \$1 million was paid in December 2005 and \$4 million was paid in January 2007.	An additional “ <i>pas de porte</i> ” (entry premium) payment of \$15 million is to be made, of which \$10 million is to be paid within six months of the amended agreement coming into effect and the balance within 12 months. ¹
Advisory committee	Nil.	An Advisory Committee, comprised of three members from Gécamines and three from AMCK, will be responsible for following the progress of the Kinsevere mine.
AMCK management	Nil.	Gécamines to have one member in the management of AMCK to fill such position as may be determined by AMCK.
Social development	Nil.	AMCK will continue the implementation of economic and social development programs for the benefit of the communities surrounding the Kinsevere mine.
Employment	Nil	AMCK has agreed to provide former Gécamines employees priority opportunities for employment provided they possess equivalent qualifications and capabilities.
Provision of services and materials	Nil.	AMCK has agreed to provide Gécamines with a preferential opportunity to supply AMCK with services and materials on commercially competitive terms.

1. In the event that Anvil arranges financing of \$125 million or more, the timing of payments will be revised such that \$10 million shall be paid within 14 days of the receipt of funds of such financing and \$5 million paid within six months of the date of payment of the \$10 million.

Pit optimisation included Inferred Mineral Resources as potential mill feed, hence conceptual pit outlines cannot be used to report Mineral Reserves. Only preliminary

testwork has been carried out to determine recovery of copper from sulphide mineralisation via the conceptual Stage III plant. Based on this information, an overall recovery of 85% of total copper was assumed.

20.3.1 Pit Optimisation Inputs

Pit optimisation inputs were based on Stage II parameters for oxide Mineral Resources and estimated Stage III parameters for sulphide Mineral Resources, at copper prices of \$1.43/lb, \$1.60/lb and \$1.95/lb.

The Stage III process for treatment of sulphide feed was assumed to comprise crushing/grinding, flotation, high pressure leach (HPL) then SX-EW. Compared with Stage II, crushing costs were increased slightly to take account of the higher work index likely to be encountered for sulphide feed. The plant consumables estimate for Stage III no longer needs to account for gangue acid consumption. The flotation and high pressure leach costs were estimated by difference, using published data on mixed float ores treated at First Quantum Minerals' Kansanshi Mine in Zambia. The Kansanshi costs for a similar crush/grind/flotation/HPL/SX-EW circuit are reported to be \$0.55/lb copper. At Kinsevere, crush/grind costs are expected to be \$7.54/tonne (or \$0.09/lb assuming throughput of 1.6 Mtpa producing 60,000 tpa cathode). Also, SX-EW costs at Kinsevere are expected to be around \$0.35/lb. By difference, the flotation/HPL component would have a projected cost of \$0.10/lb or \$7.89/tonne.

The marginal cut-off grades applicable in the Whittle Four-X optimisation process, and based on the estimated and assumed input parameters described above, are listed in Table 20.2.

Table 20.2 Marginal Cut-Off Grades at \$1.43/lb

	Units	Stage II	Stage III
PROCOST	\$/t proc	14.05	15.89
RECOV	% Cu	92.14% ASCu	85% Cu
PRICE	\$/lb	1.43	1.43
PRICE	\$/10kg	31.53	31.53
SELLCOST	\$/10kg	13.62	13.62
NETPRICE	\$/10kg	17.91	17.91
MINDIL		1.05	1.05
COG	%	0.89% ASCu ¹	1.10% Cu

1. The COG for Stage II was revised to 0.72% ASCu in subsequent optimisations

20.3.2 Pit Optimisation Results

Pit optimisation results are presented in Table 20.3 for a scenario inclusive of Measured, Indicated and Inferred Mineral Resources at Tshifufia Central and Tshifufiamashi. Mineral Resources identified as oxide were directed to the conventional Stage II comminution/acid leach/SX-EW circuit, whilst the sulphide material would be crushed and milled and then treated via a float/high pressure acid leach/SX-EW plant to recover cathode copper.

A Stage III optimisation of Kinsevere Hill/Kilongo was carried out for completeness. Currently no sulphide mineralisation has been modelled beneath the Kinsevere Hill oxide Mineral Resources. However for consistency, the optimisation was re-run with

Inferred Mineral Resources added as potential mill feed. Results for Kinsevere Hill/Kilongo are summarised in Table 20.4.

Figure 20.1 shows the conceptual ultimate pit shell for Stage III, based on the 2008 defined inventory for all Mineral Resource categories. Under these circumstances, the western pit crest of Tshifufia does not encroach on the SX-EW plant.

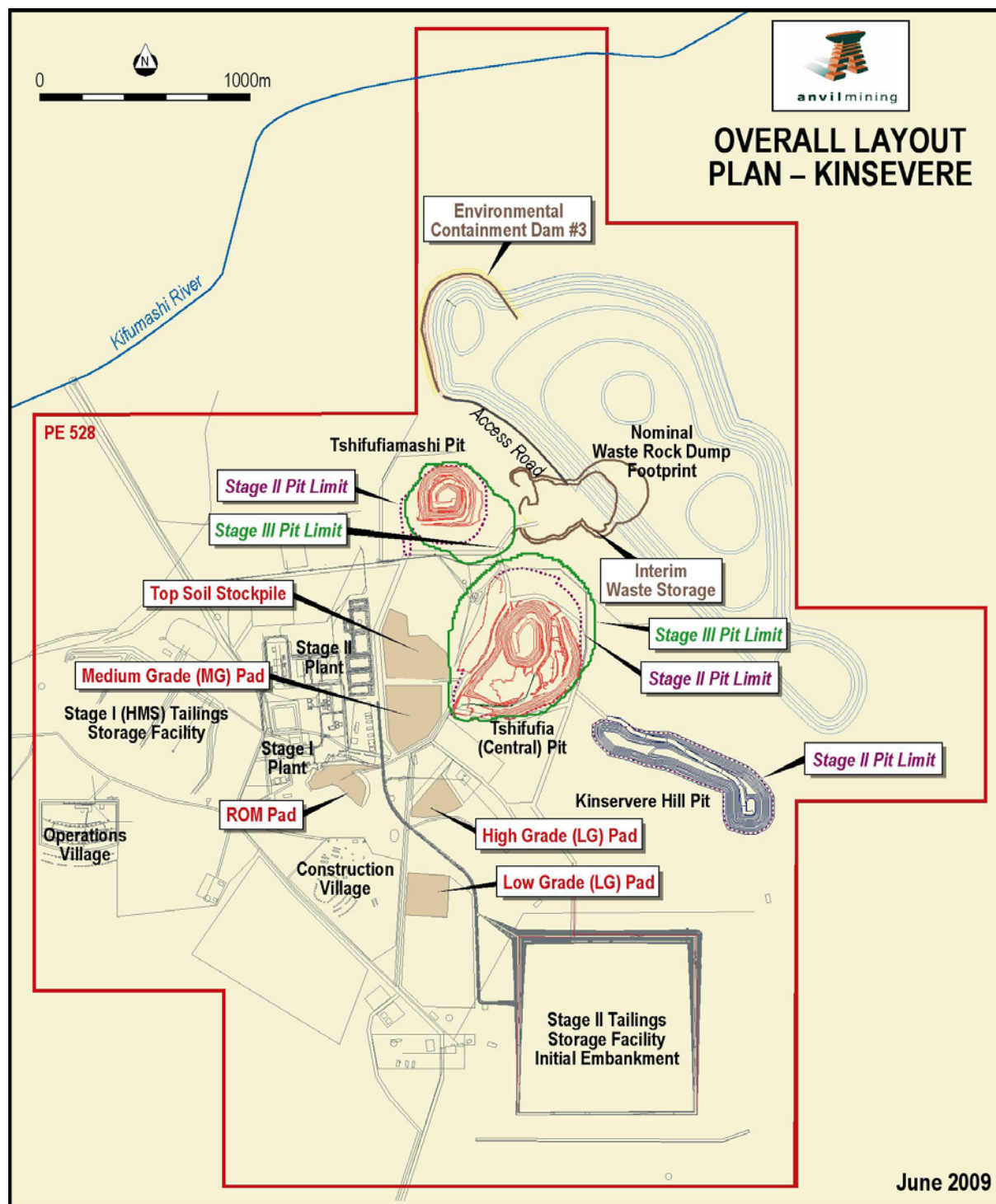
Table 20.3 Stage III Pit Optimisation Results - Tshifufia Central & Tshifufiamashi

		\$1.43/lb
Mineral Resource Categories		Meas+Ind+Inf
Life	Years	9.6
Mined	Mt	47.74
Waste	Mt	25.46
Oxide		
Mill Feed	Mt	15.39
Cu Head Grade	% Cu	4.10
ASCu Head Grade	% ASCu	3.62
Sulphide		
Mill Feed	Mt	6.89
Cu Head Grade	% Cu	2.60
ASCu Head Grade	% ASCu	0.99
Cathode	kt	665.4
Total Op Cost	\$M	1422.87
Cash Flow	\$M	674.94
NPV @ 10%	\$M	429.77
Cash Cost	\$/lb	0.97

Table 20.4 Stage III Pit Optimisation Results – Kinsevere Hill/Kilongo

		\$1.43/lb
Mineral Resource Categories		Meas+Ind+Inf
Life	Years	2.00
Mined	Mt	11.61
Waste	Mt	7.39
Oxide		
Mill Feed	Mt	4.22
Cu Head Grade	% Cu	3.44
ASCu Head Grade	% ASCu	3.08
Cathode	kt	119.8
Total Op Cost	\$M	265.40
Cash Flow	\$M	112.43
NPV @ 10%	\$M	98.83
Cash Cost	\$/lb	1.00

Figure 20.1 Conceptual Stage III Pit Limits (Green Perimeter)



20.3.3 Assessment

The optimisation was carried out using marginal cut-off grades appropriate for each of the oxide and sulphide processing routes. Following the optimisations, preliminary schedules showed that even with staged pits, only a small proportion of sulphide material would be mined before all economic oxide mineralisation is exhausted.

This indicated that with the then defined Mineral Resource, there would be little benefit in running parallel oxide and sulphide circuits. Sulphide material could be treated once all oxide material has been mined and processed. The analysis also showed that the sulphide head grade is significantly lower than for the oxide material. Therefore, to maintain cathode copper output at 60,000 tpa, an expansion of the crushing and milling circuit (and probably filtration capacity) would be required to boost plant throughput.

On the basis of the adopted plant recovery and average head grades, the throughput would need to be boosted to between 2.8 and 3.0 Mtpa, depending on the selected long-term copper price. An alternative approach would be to raise the minimum cut-off grades, but this would decrease the quantity of run-of-mine ore and likely increase low grade stockpiles.

If the average head grade of sulphide material is 2.6% Cu (ie, refer Table 20.3) and assuming an 85% recovery, with throughput boosted to 2.8 Mtpa, then the sulphide inventory as then currently defined, would only extend the Project by another 2.5 years.

Cathode output from treating 6.9 Mt of sulphide material amounts to about 152,000 tonnes. Assuming cash costs are \$1.00/lb, the potential cash flow from cathode sourced from the sulphide plant would be about \$140 to \$150 million. At this stage, the Mineral Resource inventory is insufficient to justify expenditure to expand comminution from 1.6 Mtpa to 2.8 Mtpa and construct a high pressure leach circuit.

20.4 Metallurgical Testwork for Kinsevere Stage III

During 2008 and as part of the conceptual planning study described in Item 20.3, preliminary metallurgical testwork for the Stage III Project was performed, predominantly at the Perth laboratory of Amdel Limited, and with specialised testwork and analyses organised at other facilities in Australia.

The testwork scope was primarily planned to provide an indication of the mineralogical characteristics and flotation response of sulphide ore. However the sample was also intended to investigate other key metallurgical aspects of the sulphide ore, including comminution.

20.4.1 Testwork Samples

In mid-2008 an intensive drilling programme was being conducted on the Kinsevere sulphide deposit to determine the extent and grade of the orebody (ie, the Phase IV programme).

Two drill holes were selected as being likely to intersect the sulphide mineralised zones. The core samples consisted of quarter core, which were shrink-wrapped in plastic film to minimise oxidation in transit and packed in individual bags representing one metre intervals.

Details of the samples selected are shown in Table 20.5. Only samples located within the sulphide zone were sent for testing.

Table 20.5 Details of Drill Core Intervals Selected for Metallurgical Testwork

Drill Hole	Depth from (m)	Depth to (m)	Mineralisation
TCDH 032	106	240	Chalcopyrite
TCDH 035	118	450.5	Pyrite

The geological log of the core identified the predominant mineralisation in these zones as chalcopyrite and pyrite for drill holes TCDH 032 and TCDH 035, respectively.

It was decided to focus the initial testwork on samples down to 200 metres in depth, particularly as copper mineralisation had not been identified in TCDH035, according to the core log, down to 450 metres. The testwork commenced in September 2008 based on a programme developed between Amdel and Anvil, and with input from GRD Minproc Johannesburg.

The two drill cores (TCDH032 & TCDH035) were divided into five metre intervals and suitable samples were taken for SMC comminution testing to determine ore characteristics relevant to SAG milling. The remaining core was then crushed and assayed. The average copper grades for each drill core were:

- TCDH 032 - 1.53% T Cu
- TCDH 035 - 0.43% T Cu

As it was important to produce a composite sample with a reasonable copper grade for the flotation testwork, only drill core TCDH 032 was used to generate the Master Composite. All core sections with a copper grade below 1.3% TCu were excluded. This provided a 50 kg sample assaying 2.38% TCu.

The work was based on only two drill holes and therefore may not be representative. Considerably more metallurgical samples from this mineralisation will need to be tested before reliable conclusions can be drawn.

Due to financial constraints, the testwork programme was halted at the end of October 2008 and the samples placed in cold storage until further notice.

20.4.2 Mineralogy

Mineralogical investigations including both X-ray diffraction and optical examination indicated that chalcopyrite was the major copper mineral present, with lesser amounts of pyrite.

20.4.3 Master Composite Chemical Analysis

A comprehensive ICP analysis was performed on the master composite. The key constituents are summarised in Table 20.6.

Table 20.6 Summary of Master Composite Analysis (TCDH 032)

Species	Units	Assay
Cu	%	2.38
Co	ppm	2040
Fe	%	2.61
S Total	%	2.34
S2-	%	2.25
Ca	%	0.49
Mg	%	17.4
Si	%	14.6
Mn	ppm	434
Al	%	2.72
As	ppm	53
Hg	ppm	0.06
Bi	ppm	3.0
Sb	ppm	0.6
Pb	ppm	10
Zn	ppm	15
U	ppm	3.2

20.4.4 Comminution Testwork

The SMC comminution testwork results are summarised in Table 20.7.

Table 20.7 Summary of SMC Test Results

Sample	Interval (m)	SG	Dwi kWh/m ³	Mia kWh/t	A	B	A*b	t ₁₀ @1kWh/t
TCDH 032	106 - 155	2.81	3.4	10.9	73.8	1.12	82.7	49.7
TCDH 032	160 - 200	2.71	5.2	15.8	68.3	0.82	52.6	36.7
TCDH 035	118 - 155	2.78	5.1	15.3	66.2	0.77	54.3	37.0
TCDH 035	160 - 200	2.70	5.3	16.1	62.0	0.82	50.8	34.7

The DWi values range from 3.5 to 5.3 which position them in the 20 to 44 percentile range in the JK Tech database (ie, the percentage of all ores tested that were softer than the current test samples).

The Mia parameter relates to the comminution energy required to grind coarse particles above 750 microns in size. This parameter can be used, in conjunction with corresponding Mib values that relate to fine grinding, to estimate conventional comminution circuit energy requirements.

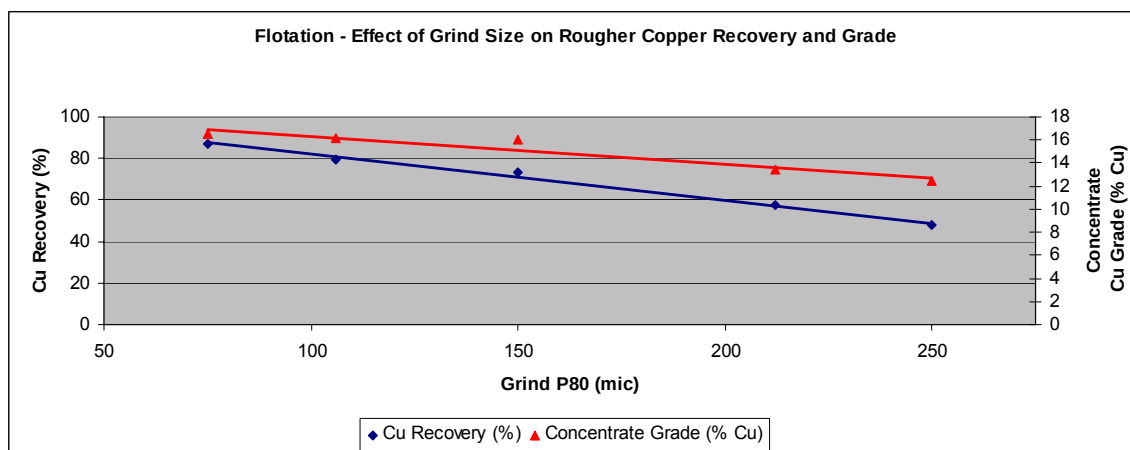
The lower the value of A*b, the harder is the ore and the more resistant it is to impact breakage. The lowest A*b value recorded, 50.8, is rated in the medium range. Overall the SMC test results indicate that the core samples can be rated, with respect to hardness, as soft to medium.

Lower t₁₀ values indicate that less material passes one-tenth of the original particle size and that there is a higher resistance to abrasion breakage. The results were assessed by JK Tech as soft to medium in relation to their database of previous testwork.

20.4.5 Flotation Testwork

A suite of sighter flotation tests were performed to investigate the effect of grind size on recovery. The collector used was Sodium Ethyl Xanthate (150 g/t) with MIBC frother. A graph of the results is shown in Figure 20.2.

Figure 20.2 Flotation Testwork Results – Kinsevere Sulphide Samples



Both the concentrate recovery and grade increased with fineness of grind. The copper concentrate generated at a P_{80} of 75 microns assayed 16.5% copper representing 87.2% recovery.

Frother addition was excessive at 19 to 29 drops and the resulting froth was brittle.

20.4.6 Conclusions

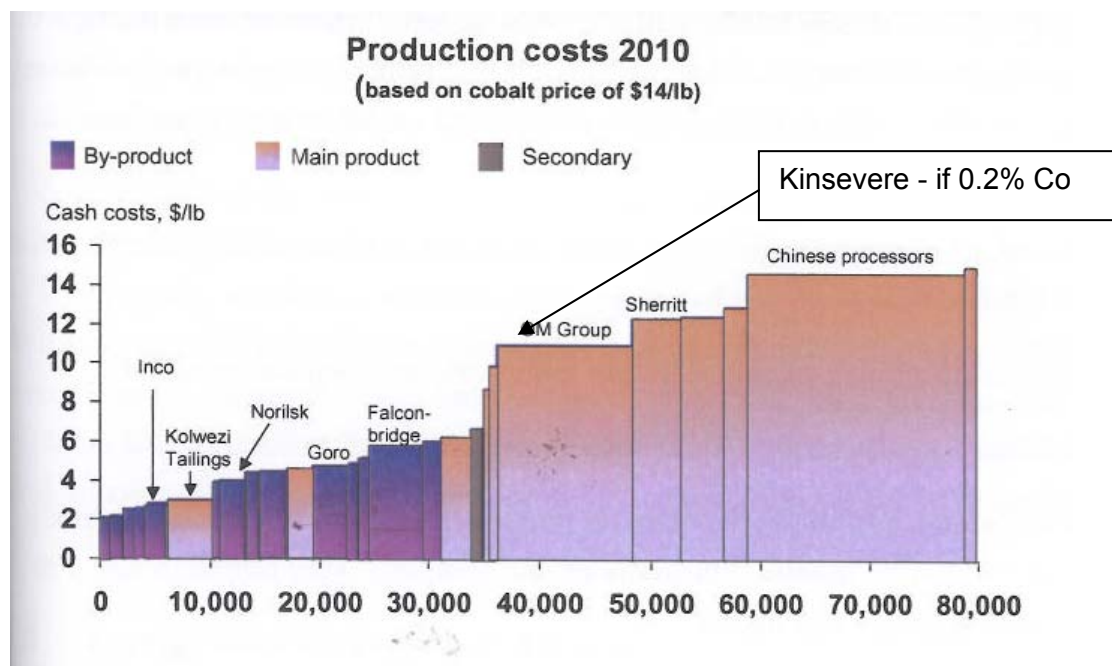
The preliminary testwork for Stage III sulphide processing indicates generally favourable comminution characteristics and supports the conceptual 85% recovery adopted in the planning studies described in Item 20.3.

20.5 Viability of Cobalt Recovery

When the Stage II operation is in progress it will have approximately 3,500 tonnes of contained cobalt passing through the plant directly to tailings each year. Approximately 50% of this cobalt is acid soluble and will already be in solution in the leach discharge.

A high-level internal assessment was completed in July 2008 (Anvil, July 2008) to assess the potential for cobalt recovery and concluded that at currently defined Co mineralisation grades, recovery of cobalt at Kinsevere in the longer term is likely to be at the high end of the production cost profile, as shown in Figure 25.15.

Figure 20.3 Estimated Production Cost for Cobalt Production at Kinsevere



20.6 Project Enhancements

20.6.1 Mining

Mining of the three open pits will continue to be by conventional excavator and truck mining methods, mostly within 'free-digging' oxide and mixed/transition horizons.

Previous internal studies have shown that there is the potential to reduce the mining costs through the adoption of fewer and larger trucks. More investigation on the trafficability of larger rigid body trucks needs to be carried out before going out to tender for the Stage II mining contract.

During 2010, work will be carried out to optimise the layout of the ROM pad, ready to receive ore for feed to the Stage II sizer. This will likely involve a redesign of the ROM stockpiles based on a finger configuration.

20.6.2 Processing

The inclusion of both 'milling in raffinate' and 'direct tailings disposal' in the Stage II flowsheet will provide considerable potential benefits to the Project. As previously noted in the 2008 Technical Report, these are expected to include:

- a reduction in operating costs for moderate additional capital outlay;
- higher copper recoveries, hence improved utilisation of the Mineral Resource;
- reduced acid, lime and limestone requirements (ie, reduced from around 225 to 80 t per day);
- social benefits, due to significant reductions in heavy vehicle movements on the surrounding roads;
- environmental benefits associated with reduced carbon emissions; and
- reduced Project risk due to minimisation of reagent imports.

With discharge of non-neutralised tailings, the savings in operating costs associated with eliminating lime and limestone addition to tailings is significant. Equally important will be the easing of supply problems, and associated Project risk, due to the large reductions in overall reagent tonnages required for the operation.

During 2010, work will proceed on a scoping study of options to heap-leach low grade material (eg, between 0.3%ASCu and 0.72%ASCu) as a supplement to the SX-EW plant feed. Supplementary plant feed from sources external to the Kinsevere tenement will also be considered.

ITEM 21 Interpretation and Conclusions

The following information summarises the interpretations and conclusions of the QPs in respect of updates to the Mineral Resources and Mineral Reserves.

Further conclusions are also drawn in respect to **uncertainty and Project risks impacting on the commencement of Stage II mining and processing in January 2011.**

21.1 Mineral Resource Modelling and Estimation

The Kinsevere Project may be interpreted as an example of a structurally modified stratiform copper deposit.

The local geology is complex, with stratigraphy often affected by faulting, folding and varying degrees of weathering and alteration. Good drillhole coverage and pit exposures have enabled continued improvements to the geological understanding.

Copper mineralisation is clearly visible within both the drill samples and the pit excavations. Sulphide mineralisation generally starts at average depths of 110 m below surface, but is dependent on the variable degrees of weathering and alteration. Kinsevere's oxide zone of mineralisation is characterised by supergene enrichment of copper metal, particularly for the TOB.

The results of the exploration drilling have contributed to the geological model, and the extent of mineralisation. The current geological model is considered robust as a basis for Mineral Resource estimation, and is supported by sample grid spacings which range from 25 m to approximately 100 m. The additional drillhole, sampling, density, geological and in-pit blast hole data available has enabled 332,560 Cu tonnes of previous Indicated and Inferred Mineral Resources to be upgraded to Measured and Indicated categories.

21.1.1 Mineral Resource Estimate Comparison

The January 2010 Mineral Resource estimates were compared to those of September 2007. Table 21.1 tabulates the oxide and sulphide Mineral Resources for September 2007.

The latest Mineral Resource estimates have been guided by additional drillhole, sampling, density, geological and in-pit blast hole sample data available as at end of December 2008. Additional drilling and blast hole sample data has enabled 332,560 Cu tonnes of previous Indicated and Inferred Mineral Resources to be upgraded to Measured and Indicated categories. A portion of oxide Mineral Resources from the southern areas of Tshifufiamashi were assigned to the northern portions of Tshifufia due to re-locating the east west striking fault separating these two ore bodies.

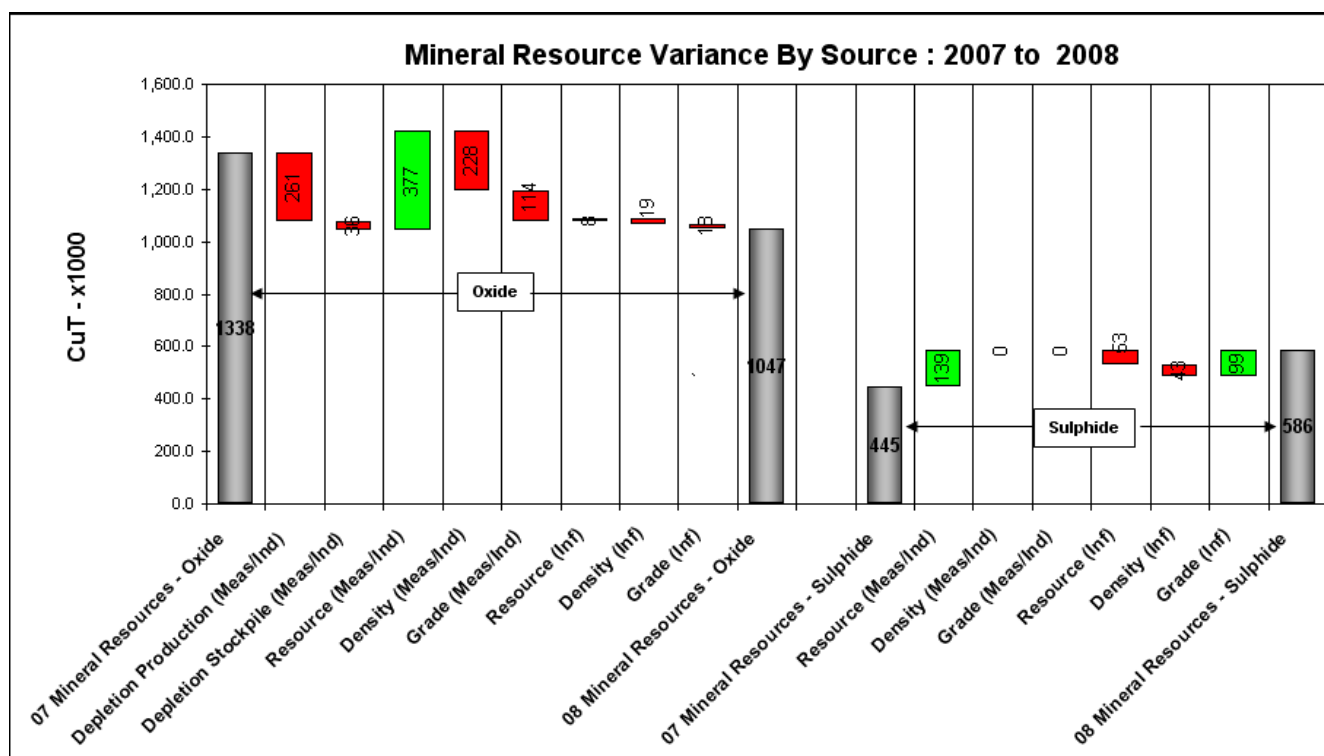
Figure 21.1 illustrates the relative and cumulative change in Cu tonnes from the 2007 to 2008 estimates for the oxide and sulphide Mineral Resources, respectively.

Since the 2007 estimate, the latest Mineral Resource estimate has been reduced by 9%, a total of 150,000 tonnes of Cu. Reductions are associated with mining depletion, density and to a lesser extent grade. The reductions are off-set by increases in the Indicated and Inferred Mineral Resources for the sulphide mineralisation.

Table 21.1 The 2007 Oxide and Sulphide Mineral Resource Estimates

Oxide Mineral Resource - 30 September 2007 at 0.7% TCu cutoff					
Deposit	Classification	Volume (M m ³)	Density (t/m ³)	Tonnes (M t)	Cu (%)
Tshifufiamashi	Measured	6.35	2.32	14.72	4.05
	Indicated	2.74	2.32	6.35	4.65
Tshifufia	Measured	1.65	2.32	3.82	3.61
	Indicated	3.31	2.32	7.69	2.46
Kinsevere Hill	Measured	-	-	-	-
	Indicated	0.38	2.32	0.88	2.60
Sub Total	Measured & Indicated	14.42	2.32	33.46	3.71
Tshifufiamashi	Inferred	0.64	2.32	1.49	3.70
Tshifufia	Inferred	0.33	2.32	0.77	2.50
Kinsevere Hill	Inferred	0.15	2.32	0.35	3.10
Sub Total	Inferred	1.12	2.32	2.61	3.27
Sulphide Mineral Resource - 30 September 2007 at 0.7% TCu cutoff					
Deposit	Classification	Volume (M m ³)	Density (t/m ³)	Tonnes (M t)	Cu (%)
Tshifufia and Tshifufiamashi	Inferred	6.09	2.52	15.34	2.90

Figure 21.1 Waterfall Chart of Mineral Resource Estimate Changes Between 2007 and 2008 (before allowing for 2009 mining depletion)



The current geological models are considered to be robust enough to form the basis of the Mineral Resource estimate described in this Technical Report, and the sampling density is sufficient to delineate the Mineral Resource, as classified. Continuing structural mapping is required to improve domain definition in the block models.

Additional data generated from planned grade control drilling, together with the results of other planned geological research (eg, SiroVision structural analysis), will lead to a future refinement or modification to the geological and block models.

21.2 Mineral Reserve Estimation

The updated Mineral Reserve estimate is based on the February 2009 Mineral Resource model, and as subsequently adjusted for mining depletion to the end of December 2009. The Mineral Reserve estimate relates specifically to the Measured and Indicated Mineral Resource categories and oxide mineralisation only. The oxide Measured and Indicated Mineral Resource estimate (Table 19.29) is inclusive of the Mineral Reserves listed in Table 19.39.

The Mineral Reserve estimate was derived from conventional mine planning processes including Whittle optimisation, detailed pit designs according to geotechnical design parameters and production scheduling. Hence, in the conversion of Mineral Resources to Mineral Reserves, modifying factors such as mining/metallurgical parameters, economic, product marketing, legal, environmental (in respect of waste dump design and site rehabilitation allowances), social and government considerations have been taken into account. In this context, the Mineral Reserve estimate is based on the following:

- a flat copper price projection of \$1.43/lb (ie, the same price adopted in the Feasibility Study);
- an overall Stage II processing recovery of 92.14% Cu;
- a mining recovery factor of 1.00 and mining dilution factor of 1.05;
- overall slope angles of 40° for all pits, except for the Tshifufiamashi West Wall, where the overall angle is flattened to 32° below 40 m from surface;
- a calculated marginal cut-off grade for Stage II of 0.72% ASCu;
- a smallest selective mining unit of 25 m x 25 m x 2.5 m (vertical); and
- allowance for access ramps at 1:9 gradient and 20 m width (to suit potential conversion to a rigid body truck fleet).

In addition to the Mineral Reserve contained within designed open pits, there is previously mined ore located on various stockpiles, available for HMS or SX-EW processing.

21.2.1 Mineral Reserve Estimate Comparisons

The Mineral Reserve estimate reported in Item 19.2.5 differs significantly from that previously reported in the Feasibility Study (AMCK, December 2007) and the 2008 Technical Report (Anvil, March 2008). A comparison between the two estimates is provided in Table 21.2.

Table 21.2 Mineral Reserve and Associated Mine Waste Comparisons

Estimates	Ore Tonnes (Mt)	TCu (%)	Contained TCu (kt)	Waste Tonnes (Mt)
Feasibility Study (2007)	25.75	4.03	1,037.5	43.68
Mineral Reserve Update (2009)	18.41	4.14	762.2	22.68
Difference	-7.34 -29%	3.75	-275.3 -27%	-21.00 -48%

The reasons for the difference can be listed as follows:

- changes to the Mineral Resource model as described in Item 21.1; and

- a different approach to the selection of the ultimate pit from the suite of optimal pit shells.

In relation to the second point listed above, it is worth noting that several years of mining have been subtracted from the Stage II mine life considered in the Feasibility Study. The ore mined in these years in fact contributed little extra value to the Project, considering the extra waste that needed to be mined (as listed in Table 21.2).

21.3 Uncertainty and Risk

21.3.1 Stage II Implementation

The Company has prepared a detailed Stage II Implementation Plan (Anvil, February 2010), the content of which is described in Item 25.6.

The LSTK contract with Ausenco was signed on 22 January 2010, although an Early Works Programme had already commenced in the fourth quarter of 2009.

Risks to the Project Implementation Plan include not achieving the LSTK schedule and poor performance by sub-contractors or suppliers. A Company "Client Representative" was appointed in February 2010, who will monitor Project budget and timeframes through strategising, planning and executing suitable responses as issues arise.

In their Independent Review, SRK (2009) were concerned about the delay in finalising the LSTK agreement and hence they considered that completion of construction and commissioning could extend for a further three months beyond the planned milestone date of 1 January 2011. In this respect, they opined that the remaining capital estimate could be understated by some 10% to 15% (refer also to Item 8.3.6).

This potential outcome is addressed in the economic sensitivity analyses described in Item 25.13.

21.3.2 Mining and Processing

Stability of the designed overall pit slopes will be reliant on the maintenance of drained slope conditions and a significant volume of water will need to be abstracted from in-pit and perimeter depressurisation bores over the life of the mine. The volume of water will be in excess of that required in the SX-EW plant and will need to be discharged into the Kifumashi River.

Success of the 2009 to 2010 (in progress) groundwater bore drilling and equipping programme is critical to enabling the Stage II mining programme to proceed below the current groundwater levels. Furthermore, efforts in 2010 need to be focussed on improving the availability of existing dewatering pumps and re-engineering the groundwater disposal and discharge system.

Testwork completed on drill core samples from each of the three Project deposits demonstrated that the mineralisation has favourable characteristics for SX-EW processing. Specifically, the ore exhibits excellent acid soluble recoveries and relatively rapid leach kinetics. Gangue acid consumption is also very low. The site water is of good quality and is free of dissolved elements or anions that could be deleterious to processing.

The Feasibility Study (AMCK, December 2007) noted that metallurgical testwork showed a general absence of any deleterious elements or ions. Notwithstanding this, additional elements including arsenic and manganese will be assayed as part of the proposed RC grade control drilling programme. This information will be used in short to medium term mine planning to delineate zones where any elevated and potential contaminant minerals will be encountered.

In their Independent Review, SRK (2009) considered that there remained a technical risk for 'milling in raffinate' and 'direct tailings disposal' due to the limited industry-wide application. Accordingly, they recommended that should a retro-fit be required, the associated capital costs would be of the order of \$20 million and would require some four to six months to install and commission (refer also to Item 8.3.6).

This is considered to be a "worst case" scenario.

21.3.3 Environmental Compliance

DRC Government approval of the EIA for Stage I and Stage II was issued in writing by CAMI on October 15 2007. Approval of a variance to the design and operation of the Stage II TSF was issued by the Directorate for the Protection of the Mining Environment (DPEM) on 28 October 2008.

An additional Environmental and Social Impact Assessment (ESIA) was completed by Knight Piésold in October 2009 (Knight Piésold, October 2009). The ESIA provides a more comprehensive review and management plan for the undertaking of Stage II of the Project. It is a document required by Project financiers to ascertain the Company's compliance with the requirements of the Equator Principles and the Performance Standards of the International Finance Corporation (IFC).

A number of deficiencies were identified in the Independent Engineer review of the Company's ESIA document (SRK, 2009). Further work is required for full Equator Principles (EP) and IFC Performance Standards (PS) compliance, as set out in Item 25.10. It is noted that if Project refinancing is required, and depending on the selected financier's particular requirements, full compliance or evidence of progress towards full EP and PS compliance may be required.

Areas of environmental uncertainty that require management plans are as follows:

- A flora survey within the Project site identified the presence of various plants of conservation importance, including the particularly rare *Gladiolus robiliartianus*. To mitigate the Project impact, AMCK has established a floral reserve and is cultivating specimens for replanting. Furthermore, biodiversity studies planned for 2010 will assess whether these rare plants are located elsewhere, or indeed only within the Project tenement.
- Geochemical testwork on oxide waste rock has shown that it has low acid-generating potential; however SRK (2009) considers that there is insufficient evidence to support this conclusion and therefore more sampling is required. It is possible that the proposed oxide waste dumps and embankments may pose an acid-rock drainage (ARD) risk to the surrounding environment. Further testwork is to be undertaken during 2010, in addition to the redesign, if required, of the exposed waste slopes.
- Depending on the hydrogeological modelling to be carried out in 2010 (Item 25.1.7), it may be necessary to prepare an addendum to the EIA to address the discharge of increased volumes of abstracted groundwater.

21.3.4 Social and Political Risk Management

In the context of the DRC, social and political risk management is a critical component of the business. These are also the areas in which the greatest latent threat to the Company's reputation lies. The Company's 'social licence to operate', which is essentially underpinned by community support and constructive government relations, is a critical component to mitigating this potential risk.

Broadly speaking the risk factors can be divided into the following categories:

- Social
- Political
- Religious
- Economic

Economic issues such as poor living standards, lack of foodstuffs and an inability for local people to generate at least a subsistence wage, are catalysts for creation of instability within the other areas. Economic dynamics within the region and surrounding communities can significantly influence the volatility of the other factors, both positively and negatively.

Economic improvement within a given area can significantly mitigate these factors and clearly the greater the level of vested interest the government and local communities have, the better the position of the Company. Generating 'political capital' of this kind is invaluable though often difficult to quantify. This has the capacity to detract from a collective understanding of the importance of political and social capital within such an environment; not only as a risk management tool but also in terms of creating a strategic competitive advantage for the Company and providing an authentic commitment to uplifting host communities.

The commencement of Kinsevere Stage II has resulted in the Company embarking on a further public consultation project, the bulk of which will run over the months of April and May 2010. The objective of this is to improve communication with the surrounding communities as well as to manage their understanding and expectations surrounding the Stage II project.

Anvil has followed a path of comprehensive and committed corporate social investment and will continue to do so. The strategy of this will be reviewed and improved over the course of 2010. To date investment initiatives have included health care projects, agricultural assistance, the construction and staffing of schools, providing sources of drinking water to local communities, and other significant infrastructure developments.

21.3.5 Economic Sensitivity Analysis

The economic analysis for the Project is described in detail in Item 25.12. However, in the context of economic uncertainty and risk, a number of sensitivity analysis results are reported under Item 25.13, to confirm the relative robustness of the Project.

These analyses indicate that the impact of 10% changes in operating and capital costs is far less significant than 10% fluctuations in copper price. Due to the relatively high average copper grade of the Kinsevere Mineral Resource, a 10% change in price represents an approximate change of \$120 million in NPV.

As mentioned above, SRK (2009) noted a number of aspects which could impact on the reported Project value, and these can be summarised in Table 21.3.

Table 21.3 SRK Cash Flow Model Sensitivity Scenarios

Test Scenario	Description
1	Base Case at current LME Forward Copper Price, averaging \$2.44/lb (real)
2	Reflecting Base Case with the following adjustments: • Allowance for work-in-progress (allowance for 10 days in pipeline) Inclusion of environmental liability provisioning as ongoing contributions to operating expenditures rather than a capital expenditure item on closure • Inclusion of terminal benefits liabilities on closure
3	Reflects Scenario 2 with: • Construction completion and commissioning delay of 3 months later than 1 January 2011 • Increased capital expenditure of +15% • Increased dewatering expenditure from \$50k/month to \$100k/month • Increased capital expenditure for additional bores of \$3.0 million in 2010
4	Reflects Scenario 3 with: • Downward adjustment of metal content (grade) by -10%
5	Reflects Scenario 3 with: • Incorporation of additional other corporate costs within the Kinsevere Project
6	Reflects Scenario 5 with: • Retrofitting of the SX-EW plant to replace MIR/DTD flowsheet, thus incurring additional capital expenditure of \$20 million and a long term increase to SX-EW operating expenditure of \$9.00/t

In relation to the Stage II Project, and including sunk costs prior to January 2010, the sensitivity analysis results showed that:

- including adjustments as per Scenario 3 (reflecting a three month commissioning delay and increased capital expenditure) reduces the Base Case NPV by 13%; and
- as a worst case, including additional corporate costs, further adjustments for plant retro-fitting and increased SX-EW operating costs reduces the Base Case NPV by 49%.

ITEM 22 Recommendations

The following work programmes are recommended for the Kinsevere Project during 2010, in respect of aspects relating to the Mineral Resource and Mineral Reserve. Other work items are identified and a breakdown of budgeted costs is provided.

22.1 Oxide Resampling and Assaying

In Item 15.4 it was noted that QAQC analysis of assay sample batches received during April and May 2009 revealed an issue with deficient sample preparation and dispatch procedures. This suspect data **was excluded** from the February 2009 Mineral Resource estimate for Tshifufia and Tshifufiamashi, and the December 2009 update reported in Item 19.1. The impact is immaterial to the Mineral Reserve estimate and the cash flow model outcomes described in Item 25.

Considering the QAQC results, sampling practices and associated risks, the following actions were internally recommended:

1. Select an appropriate portion of the failed batch samples for re-assay, including appropriate submission of blind standards, blanks and repeats.
2. Analyse the re-assayed results for acceptable QAQC. If there are no QAQC risks then:
 - a) accept the new results/old results, otherwise
 - b) identify the risk and implement a more comprehensive re-sampling programme, and
 - c) improve sampling practice

10% of the failed batch samples (ie, 289 samples including duplicates) were dispatched from site in December 2009 for the initial re-assay, and the results were pending at the time of preparing this Technical Report. Receipt of this information will enable the Mineral Resource estimate to be further updated in 2010.

The follow-up re-assaying (the more comprehensive programme, if required) would involve a further 12,361 samples, including duplicates. An allowance of \$105,800 is provided in the current cash flow model as a Stage I (2010) operating cost, for the initial programme only.

22.2 Sulphide Assaying

Following the cessation of operations in late 2008, a quantity (ie, 22 batches of 203 samples each = 4,536 samples) of Phase IV core samples had been pulverised and packaged ready for shipment to the ActLab facility in Perth.

Based on the QAQC status of oxide samples prepared at the same time, it was internally recommended that 10% of the samples (using pulverised remaining core) be reassayed as an initial check. In addition to this an additional 2% of the samples should be inserted as duplicates.

Depending on the results of this initial check, the remaining 90% of the sulphide core samples may need to be reassayed (plus allowances for duplicates).

The initial re-sample batch was dispatched from site in December 2009, and the results were pending at the time of preparing this Technical Report. Receipt of this information will enable the Mineral Resource estimate to be further updated in 2010.

An allowance of \$106,000 is provided in the current cash flow model as a Stage I (2010) operating cost, for the initial programme only.

22.3 Density Measurements

In order to supplement the bulk density measurements produced in 2008 (Item 14.6), it is recommended that additional bulk density pit measurements should be taken. The location of these test pits should be according to the key geological units identified, whilst the frequency should be dependant on change in weathering.

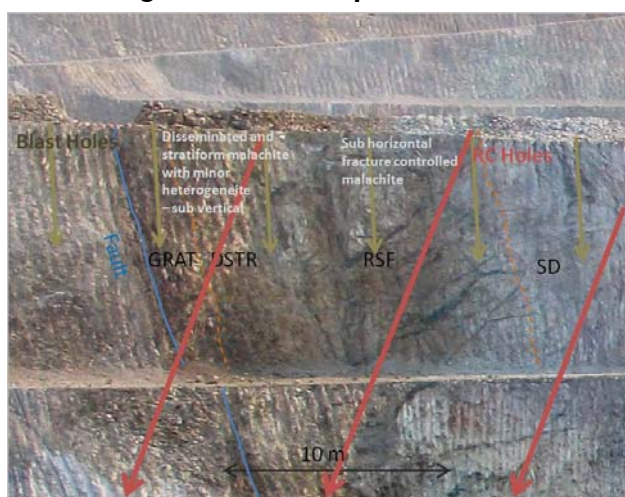
No specific budget allowance is made for this work; the earthmoving, survey and measurement costs would form part of the usual mine operating costs.

22.4 Grade Control Drilling

During 2010 it is recommended that the grade control methodology be changed-over from blasthole sampling to RC drilling (over multiple benches) and sampling.

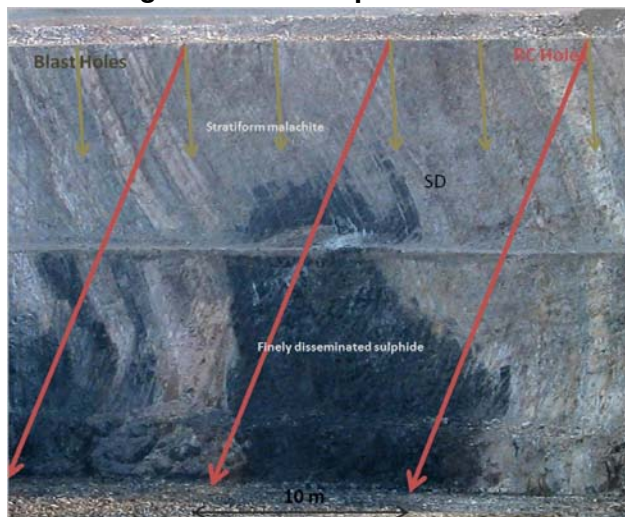
The recommended grade control RC drilling programme should be focused on delineating the three main zones of mineralisation as well as the three stages of mineralisation associated with each zone (see Figures 22.1 and 22.2).

Figure 22.1 Proposed Grade Control Drilling in Lower Orebody



Lower Ore Body (GRAT, DSTR and RSF) mineralisation as influenced by fault zone fracturing and jointing. RC drilling at 60 degrees to the east is best suited for this style of mineralisation. Blast hole drilling (5 m by 5 m grid) will provide additional local detail.

Figure 22.2 Proposed Grade Control Drilling in Upper Orebody



The Upper Ore Body (SD), with stratiform and disseminated supergene enriched (black) mineralisation within shales of the SD unit. RC grade control drilling (10 m by 10 m grids) at 60 degrees towards the east provides best intersections. Blast hole drilling (5 m by 5 m grid) will provide additional local detail.

The Third Ore Body, associated with the CMN dolomites, is characterised by a combination of stratiform malachite, irregular veinlets and nodules of malachite in addition to black heterogenite (“Noir”) associated with the malachite. Here, RC drillhole orientation will focus on attaining highest intersection angles with the adjacent SD layering or the Upper Ore Body. Planned guidelines for RC drilling include:

- high angle of intersection with dominant strike and dip of lode margins/contacts taking into consideration local scale of mineralisation. Hole angles are planned at 60 degrees east or west according to layer strike and dip
- optimal model block dimensions to adequately define contacts and transition zones with adjacent/surrounding waste. This requires grid spacing of 10 m by 10m with 20 – 30 m depths
- optimal sample recovery for different lithologies - sample composites every 2 m
- QAQC: rate of insert is one CRM, blank and duplicate for every 20 to 30 samples. An umpire laboratory will be used for every 50th sample
- using existing holes to avoid sample overlap.

The current Central Pit floor will be drilled in several stages. Stage A will focus on the shallow southern area first, followed by Stage C located in the deepest pit area. Stage B, located along the eastern most benches will be drilled in 2011, whilst later drilling at Tshifufiamashi is scheduled as Stage D.

Table 22.1 Proposed Number of Drillholes and Total Drilling Metres for each Stage of the 2010 Grade Control Programme

	# of holes	metres
Stage A	430	15,480
Stage B	-	-
Stage C	206	7,488
Stage D	-	-
TOTAL	636	22,968

A Services Agreement was executed with Titan Drilling Congo s.p.r.l. in February 2010, to conduct grade control RC drill work and to also supply a cyclone, riffle splitter and sampling crew. A provision of \$640,000 was made in the the current cash flow model as a Stage I (2010) corporate cost. The value of the Services Agreement is now approximately \$785,500 (capitalised, and based on a dry rate/metre of RC drilling). Additional costs apply for fuel consumption and assaying.

Further information on the drilling programme is provided in Item 25.1.

22.5 Data Management

Drilling data, including assay results, are currently recorded in an MSExcel spreadsheet format and then verified at site. The following data is captured for each drill hole:

- geology for each metre logged
- two downhole surveys for each drillhole
- DGPS collar survey before and after drilling
- sampling in 2 m composites

- water strikes
- assays for acid soluble and total copper, cobalt and manganese
- Niton analyses for multiple elements.

Once verified, the spreadsheets are sent to Perth to be loaded into the corporate database system DataShed.

From April 2010, it is recommended that the Centric mine management system (Item 22.8) supersede MExcel documents, and that data should be recorded in web-based forms which will also be verified at site and loaded into DataShed.

No specific budget allowance is made for routine data management services.

22.6 Grade Control Modelling

Internal reconciliation reviews have indicated the importance of representing the inherent variability in mineralised domains, grades and local geology during grade control. Hence, locally accurate grade control block model estimates are recommended, based around the development of a tailored grade control modelling system within Surpac (for compatibility with site-based mine planning software usage).

It is recommended that the Company improves, standardises and automates as much of the grade control process as possible, with procedures including:

- RC drilling (Item 22.4) – hole design, sample sheet generation, bag preparation, drillhole layout, drilling, sample pick-up and hole surveying;
- data management (Item 22.5) – chip logging, sample and assay tracking, loading of drilled data, validation, QAQC and sign-off;
- data interpretation – logging, pit floor mapping (including Sirovision data), 3D modelling and updating of geology and mineralised domains, grade interpolation, block design and bench mark-up plans;
- ore management – daily dig plan, ore model map for drilling and blasting, ore block mark-up, ore spotting, reconciliation input and daily block status reporting (Item 22.8); and
- mine planning – scheduling grade control drilling and short term mine planning.

Coding of a macro-driven Surpac grade control modelling system actually commenced in February 2010, with a budget of \$50,000.

22.7 Photogrammetry

In order to improve the definition of structural boundaries within the Company's grade control and Mineral Resource estimation models, it is recommended that a terrestrial photogrammetry programme be carried out within the two existing Kinsevere Pits. The deliverables should be DTM files of the pits suitable for structural interpretation by the geology team and consultant geotechnical engineer.

This work package actually commenced in March 2010 with a budget of \$40,000.

22.8 Production Tracking and Reconciliation

It is recommended that a web-based mine production tracking system be developed for managing the mine to crusher metal balance and to enable the implementation of routine reconciliation measures.

Following a site scoping study in December 2009, the first stage of implementation will commence in March 2010. This first stage will focus on grade control data management and interface with DataShed, to be followed by a second stage implementation of systems to manage excavator and truck material movements.

The value of this implementation for 2010 is approximately \$170,000 (ie, capitalised). Further expenditure is likely required in following years to enhance the system. Further information on the proposed tracking system is provided in Item 25.1.

22.9 Updated Mineral Resource Modelling

It is recommended that further work be carried out in 2010 to update the Mineral Resource model and estimate, involving the incorporation of outstanding oxide and sulphide assay data from the 2008 Phase IV drilling programme, in addition to the incorporation of grade control data gathered since the resumption of mining in August 2009. Completion of the updated Mineral Resource model and estimate is contingent on completion of these tasks.

The budgeted provision in the current cash flow model (ie, a Stage I operating cost) for this update was \$35,000. A further \$55,000 will likely be required to cover ongoing consulting fees.

22.10 Geotechnical Engineering

In respect of mine geotechnical engineering activities, it is recommended that work be carried out in 2010 to update the drilling database with additional geotechnical information from the mapping of pit exposures. Where possible, geotechnical domains should be identified around the various design pit perimeters and the existing slope design parameters refined accordingly.

This work should be linked to the photogrammetry exercise described in Item 22.7. The budgeted allowance for this update is \$20,000.

22.11 Updated LOM Plan

It is recommended that following the completion of the Mineral Resource update, a new Life of Mine (LOM) Plan and Production Schedule should be developed. This will be the last opportunity to update the plan before commencing Stage II operations, and will be contingent on completion of the updated Mineral Resource model and estimate, and preceeding tasks.

The recommended work programme should follow the usual sequence of mine planning steps involving pit optimisation, detailed stage pit designs and production scheduling (mining and processing, integrated with a stockpile building and reclaim strategy).

It is possible that potential mining and processing enhancements involving the use of larger trucks and the adoption of low grade heap leaching (Item 20.6), respectively, can be considered at this time.

The budgeted allowance for this update is \$75,000.

22.12 Short-Medium Term Planning

Short-medium term mine planning should follow after the LOM Plan update and involve:

- a monthly mining, stockpiling and ROM feed schedule for 2011;

- an associated waste stripping schedule, identifying waste destined for the TSF embankments or the waste dump;
- a stockpile reclaim plan; and
- revisions to the site layout plan, showing revised infrastructure.

Completion of the short-medium term planning will be contingent on the Stage C grade control model,

An initial budget provision of \$25,000 was made for the current cash flow model, but this will need to be increased by a further \$44,000 to cater for previously unforeseen site layout revisions.

22.13 Groundwater Bore Drilling, Aquifer Testing and Equipping

The drilling and equipping of additional groundwater bores is considered to be a critical and essential technical item to be addressed in 2010, enabling sufficient groundwater drawdown to be achieved ahead of Stage II mining in 2011. Specific recommendations on the programme of work (which has already commenced in November 2009) are provided in Item 25.1.7.

The budgeted allowance for this work in the current cash flow model is \$720,900 (ie, a Stage I corporate cost), including drilling supervision by a Knight Piésold hydrogeologist. Due to drilling delays and the requirement for extended site supervision, plus revised equipment quotes, a further \$560,000 will be required. In the original estimate, the requirement for in-pit and plant site monitoring bores was unforeseen.

The adequacy of this budget provision will need to be reviewed again in 2010, once the drilling programme and subsequent modelling work is complete and the size of the individual bore pumps and associated pipework and electricals can be quantified.

22.14 Groundwater Discharge

Item 25.1.7 further describes the recommended groundwater modelling programme, whilst Item 25.1.8 describes the recommended work programme to address the potential discharge to the Kifumashi River of up to 600 L/sec of abstracted groundwater.

The engineering and procurement of the enhanced surface discharge facilities was not budgeted in the initial cash flow model for 2010, nor was the allowance for an addendum to the EIA to address the impact to the Kifumashi River.

The adequacy of a new budget provision of about \$320,000 will need to be reviewed in 2010, once the drilling programme and subsequent modelling work (Item 22.13) is complete.

22.15 Groundwater Drawdown and Quality Monitoring

In addition to the routine groundwater drawdown monitoring currently being carried out by Knight Piésold, it is recommended that a similar level of external scrutiny be carried out in respect of mine discharge and community water quality.

This was previously an unbudgeted expense. Following receipt of a proposal from Knight Piésold, it is recommended that a new operating cost provision of \$132,260 be made for 2010.

22.16 Geotechnical Investigations for the TSF

The Independent Engineer review recommended that a number of cored geotechnical holes be drilled inside the footprint of the Stage II TSF, in order to collect samples for mechanical testing and confirm the embankment design parameters.

An allowance of \$100,000 has been made for this work.

22.17 Surface Water Management Planning

A number of environmental control dams have been designed to manage surface run-off across the site. The recommended location and construction requirements for these are described in Item 25.1.8.

The scope and costs for the civil earthworks will need to be included in the new mining contract to be tendered in late 2010 for Stage II mining (Item 22.19).

This planning will be carried out as part of the short-medium term mine planning scope. No further budget provision is required.

22.18 ARD Testwork and Waste Dump Planning

A requirement for additional acid rock drainage (ARD) testwork was recommended by the Independent Engineer (SRK, 2009) and a budget allowance of \$60,000 has been made for testwork on oxide and sulphide samples.

The current Life of Mine Plan (Item 25.1.9) shows that all waste available to be mined from the Kinsevere Pits is likely to be required for TSF embankment construction. It is recommended that further mining engineering work be carried out to assess the suitability of the mined waste according to the zoned construction requirements and specifications defined for these embankments.

The availability of oxide waste may become an operations issue for future Stage III sulphide mining and processing. Hence, engineering studies are recommended for 2010 to optimise the supply of oxide waste for the TSF and to build retaining walls on the waste dump site as structures to contain potentially acid forming rock during future mining.

A budget allowance for \$25,000 has been allowed for TSF/waste dump engineering and optimisation.

22.19 Stage II Mining Contract

The current mining contract with MCK Mining s.p.r.l. extends only to the reduced production plan for the Stage I operations. It is recommended that a new scope of work be prepared for the Stage II mining programme based on the updated Life of Mine Plan and Production Schedule.

Item 25.1.1 describes a 2008 internal evaluation on the use of larger mining equipment for Stage II. Against this background, mining tender schedules should be prepared, featuring Stage II ore and waste volumes by pit, stage and bench, in addition to haulage distance information. It is recommended that tenders conforming to the supply of rigid body trucks and suitably matched excavators should then be invited from contractors. As was the case for the Stage I resumed mining contract, an independent tender estimate should be completed for comparison purposes.

An allowance of \$13,600 has been made for the independent tender estimate.

22.20 Environmental Compliance

It is recommended that the Independent Engineer review recommendations be included in the Company's existing environmental action plan (the ESAP) (Anvil, November 2009), with specific work required on aspects such as:

- additional baseline studies for the assessment of biodiversity; and
- a retrospective impact assessment for the access road/powerline corridor.

A programme of work was actually commenced in January 2010 to gather wet season information on the above aspects. Further work will be carried out in the 2010 dry season.

A 2010 budget of approximately \$102,000 has been allowed for this further environmental work by Knight Piésold.

22.21 Social Development

It is recommended that the Independent Engineer review recommendations be included in the Company's existing environmental action plan (the ESAP) (Anvil, November 2009), with specific work required on the development of community management, social management plans and stakeholder engagement plans.

A budget allowance of \$18,000 has been allowed for assistance from Knight Piésold.

22.22 Closure Plan

A conceptual closure plan and rehabilitation cost estimate was prepared for the 2008 Feasibility Study. The amount of the provision in the current cash flow model is \$23.4 million. This plan requires updating and alignment with the commitments contained in the new ESIA (KP, 2009), and an allowance of \$16,000 has been made to re-assess the adequacy of the closure provisions.

22.23 Project Enhancement Studies

Item 20.6 noted a number of Project enhancements which are recommended to be carried out in 2010.

Subject to a recommended detailed review, it is expected that Stage II mining will feature the use of fewer and larger trucks than currently in use for Stage I. Completion of this work will enable the scope and schedules to be prepared for a new Stage II mining contract (Item 22.19). A budget allowance of \$13,600 has been allowed for this work.

A further capital allowance of \$250,000 will be required for ROM pad earthworks to prepare for Stage II feed to the mineral sizer. In addition, another capital allowance of \$1.1 million will be required for heap leaching studies including testwork and field trials.

22.24 Summary of Cost Estimates for Recommended Work Programme

Table 22.2 summarises the cost estimates for the recommended work programme. Note that the production tracking and reconciliation provision was previously budgeted as an operating expense. It has since been re-classified as a capital expense.

Table 22.2 Summary of Cost Estimates for Recommended Work Programme

Item	2010 Budget Allowance		Required Additional		Total Provision	
	Operating	Corporate	Operating	Capital	Operating	Corp+Cap
Oxide Resampling and Assaying	\$105,800				\$105,800	
Sulphide Assaying	\$106,000				\$106,000	
Density Measurements						
Grade Control Drilling		\$640,000		\$437,950		\$1,077,950
Data Management						
Grade Control Modelling	\$50,000				\$50,000	
Photogrammetry	\$40,000				\$40,000	
Production Tracking and Reconciliation	\$170,000					\$170,000
Updated Mineral Resource Modelling	\$35,000		\$55,000		\$90,000	
Mine Geotechnical Engineering	\$20,000				\$20,000	
Updated LOM Plan	\$75,000				\$75,000	
Short-Medium Term Planning	\$25,000		\$44,000		\$69,000	
Groundwater Bore Drilling, Aquifer Testing and Equipping		\$591,646		\$558,956		\$1,150,602
Groundwater Drilling Supervision and Modelling		\$129,254				\$129,254
Groundwater Discharge Engineering and EIA				\$319,456		\$319,456
Groundwater Drawdown Monitoring	\$90,000				\$90,000	
Groundwater Quality Monitoring			\$132,260		\$132,260	
Geotechnical Investigations for the TSF				\$100,000		\$100,000
Surface Water Management Planning						
ARD Testwork	\$60,000				\$60,000	
Waste Dump Planning and Optimisation	\$25,000				\$25,000	
Stage II Mining Contract	\$13,600				\$13,600	
Environmental Compliance	\$102,000		\$10,000		\$112,000	
Social Development	\$18,000				\$18,000	
Closure Plan	\$16,000				\$16,000	
Project Enhancement Studies						
Equipment Review	\$13,600				\$13,600	
ROM pad earthworks				\$250,000		\$250,000
Leaching Studies				\$1,100,000		\$1,100,000
	\$965,000	\$1,360,900	\$241,260	\$2,766,362	\$1,036,260	\$4,297,262

ITEM 23 References

- African Mining Consultants, 2007. Kinsevere Copper – Cobalt Project, Environmental Impact Assessment. Report to Anvil Mining Ltd, July.
- AMCK Mining s.p.r.l., 2007. Kinsevere Copper Project Stage II – Feasibility Study Preliminary Report. Internal Report, April.
- AMCK Mining s.p.r.l., 2007. Kinsevere Copper Project Stage II Feasibility Study, Internal Report, December.
- Anvil, 2005. Kinsevere Evaluation Study. Internal Report, September.
- Anvil, 2006. Kinsevere Project, Lubumbashi Region, Democratic Republic of Congo. Options Study. Internal Report, March.
- Anvil, 2006. Kinsevere Project, Lubumbashi Region, Democratic Republic of Congo. Feasibility Study, Stage 1: HMS-EAF Plant. Internal Report, May.
- Anvil, 2008. Nambulwa Project Scoping Study. Internal Report, January.
- Anvil, 2008. NI 43-101 Technical Report, Kinsevere Copper Deposit. March.
- Anvil, 2008. Proposed Design for Kinsevere Stage 2 Tailings Facility. Letter to DPEM, July.
- Anvil, 2008. Kinsevere Stage III Cobalt Processing. Internal Memo, July.
- Anvil, 2008. Kinsevere Stage II Tailings Management Proposal. Letter to DPEM, September.
- Anvil, 2009. Anvil Announces the Following Updates...News Release, February.
- Anvil, 2009. Owner Mining Study. Internal Memo, February.
- Anvil, 2009. Optimal Block Size Selection. An Interpretation of a Suitable Selective Mining Unit. Internal Memo. May.
- Anvil, 2009. Kinsevere Copper Project Stage II, Environmental and Social Action Plan (ESAP). Internal Report, November.
- Anvil, 2010. Anvil Reports Updated Economic Analysis of Kinsevere Stage II. News Release, January.
- Anvil, 2010. Kinsevere Stage II Project, Implementation Plan. Internal Report, February.
- Anvil Mining Services, 2007. Nambulwa Project Scope Drilling Programme, June – September 2007. Internal Report, October.
- Ausenco, 2008. Kinsevere Copper Project, Engineering Closeout Report. Report to Anvil Mining Ltd, December.
- CIM, 2000. CIM Standards on Mineral Resources and Reserves. Prepared by The CIM Standing Committee On Reserve Definitions, August 2000.

CSA Global Pty Ltd, 2008. Resource Modelling Report, Anvil Mining Ltd. Kinsevere Hill and Kilongo. Report to Anvil Mining Ltd, June.

Direction de Protection de l'Environnement Minier (DPEM), 2008. Agreement in Principle on the Proposal for the Management of the Kinsevere Stage 2 Tailings. Letter to AMCK Mining sprl, August.

Direction de Protection de l'Environnement Minier (DPEM), 2008. Proposed Management of Kinsevere Stage 2 Tailings. Letter to AMCK Mining s.p.r.l., October.

DumpSolver Pty Ltd, 2009. Kinsevere Copper Project, Addendum to February 2009 Open Pit Optimisation Technical Report. Report to Anvil Mining Ltd, March.

JORC, 2004. The Australasian Code for Reporting of Exploration Results, Mineral Resources and Ore Reserves. Prepared by the Joint Ore Reserves Committee of the Australasian Institute of Mining and Metallurgy, the Australian Institute of Geoscientists and the Minerals Council of Australia, December.

Knight Piésold Consulting, 2007a. Kinsevere Copper Mine Draft Hydrogeological Feasibility Report. Report No. 5148/20/01 to Anvil Mining, October.

Knight Piésold Consulting, 2007b. Kinsevere Copper Mine Draft Hydrogeological Modelling Report. Report No. 5148/20/01 to Anvil Mining, October.

Knight Piésold Consulting, 2007c. Kinsevere Copper Project Stage 2, Site Report 16th to 21st September, 2007. Letter Report to Anvil Mining, October.

Knight Piésold Consulting, 2008. Kinsevere Copper Cobalt Project, Streamflow Monitoring Report. No. 5337/10 to Anvil Mining, July.

Knight Piésold Consulting, 2008. Kinsevere Copper Cobalt Project, Streamflow Monitoring Report to Anvil Mining, August.

Knight Piésold Consulting, 2008. Tailings Storage Facility Basin Permeability Testing and Seepage Analysis. Report to Anvil Mining, September.

Knight Piésold Consulting, 2008. Kinsevere Copper Mine Draft Hydrogeological Feasibility Report. Report No. 5337/02 to Anvil Mining, December.

Knight Piésold Consulting, 2009. TSF Construction Schedule. Memo Report to AMCK Mining, June.

Knight Piésold Consulting, 2009. Final Environmental and Social Impact Assessment Report. Report No. 301 00135 to AMCK Mining s.p.r.l., October.

SRK Consulting (UK) Ltd, 2009. An Independent Engineers' Report on the Kinsevere Copper Project, Democratic Republic of Congo. Report to Anvil Mining Ltd, September.

Turner Mining and Geotechnical Pty Ltd, 2008. Kinsevere Site Visit. Letter Report to Anvil Mining, August.

ITEM 24 Date and Signature Page

This report is dated 31st March 2010

(Signed) "*Geoffrey Booth*"

Geoffrey Booth
Qualified Person

(Signed) "*Anthony Cameron*"

Anthony Cameron
Qualified Person

(Signed) "*Gerry Fahey*"

Gerry Fahey
Qualified Person

(Signed) "*Michael Lawlor*"

Michael Lawlor
Qualified Person

*Geoffrey Booth
43 Whitecliffe Square
Iluka, Western Australia, 6028
Tel +61 8 9305 3072
geoff.booth@citadelrg.com.au*

I, Geoffrey Booth, do hereby certify that:

1. I was previously the Group Feasibility Study Manager for Anvil Mining Limited.
2. This certificate applies to the technical report entitled "Anvil Mining Limited NI 43-101 Technical Report Kinsevere Copper Project Katanga Province, Democratic Republic of Congo" dated 31st March 2010 (the "Technical Report").
3. I am a professional geologist having graduated with a Bachelor of Science degree with Honours (1979) and a Master of Science degree (1983) in Geology from the University of Toronto. I was subsequently awarded a PhD in Applied Geology and Geochemistry from the University of New South Wales (1988).
4. I am a Chartered Professional and Fellow of the Australasian Institute of Mining and Metallurgy.
5. I have worked as a geologist for a total of thirty one years since my graduation from university.
6. I have read the definition of "qualified person" as set out in National Instrument 43-101 – Standards of Disclosure for Mineral Projects ("NI 43-101") and certify that, by reason of my education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, I am a "qualified person" for the purposes of NI 43-101.
7. I most recently personally inspected the Kinsevere Copper Project property described in the Technical Report between 7th and 15th October 2008.
8. I am responsible for the preparation of those portions of the Technical Report relating to Exploration, Drilling, Sampling Method and Approach, Sample Preparation, Analyses and Security, and Data Verification, namely Items 12, 13, 14, 15 and 16.
9. I am independent (as defined by Section 1.4 of NI 43-101) of Anvil Mining Limited.
10. I have had prior involvement with the property that is the subject of the Technical Report. The nature of my prior involvement has been in the preparation and validation of data supporting the Mineral Resource estimates, commencing in 2008.
11. I have read NI 43-101 and Form 43-101F1 and the Technical Report has been prepared in compliance with that instrument and form.
12. As of the date of this certificate, to the best of my knowledge, information and belief, the Technical Report contains all scientific and technical information that is required for be disclosed to make the Technical Report not misleading.

Signed and dated this 31st day of March, 2010 at Iluka, Western Australia, Australia.

(Signed) "Geoffrey Booth"

Geoffrey Booth

*Anthony Cameron
A & J Cameron & Associates Pty Ltd
PO Box 8046
South Perth, Western Australia, 6151
Tel +61 8 9474 1313
anthony.r.cameron@bigpond.com*

I, Anthony Cameron, do hereby certify that:

1. I am the proprietor and principal mining consultant of A & J Cameron & Associates Pty Ltd, mining consultants to Anvil Mining Ltd.
2. This certificate applies to the technical report entitled "Anvil Mining Limited NI 43-101 Technical Report Kinsevere Copper Project Katanga Province, Democratic Republic of Congo" dated 31st March 2010 (the "Technical Report").
3. I am a professional mining engineer having graduated with a Bachelor's Degree in Mining Engineering from the University of Queensland in 1988. In addition I was awarded a Graduate Diploma in Business by Curtin University in Western Australia in 2000 and a Masters in Commercial Law by the University of Melbourne in 2005.
4. I am a Fellow of the Australasian Institute of Mining and Metallurgy (membership number 108264) and have been a member since 1994.
5. I have worked continuously as a mining engineer for a period of 22 years since my graduation from university, and have held positions in open pit mining production and mine planning. I have been a consultant engineer for a period of 9 years.
6. I have read the definition of "qualified person" set out in National Instrument 43-101 – Standards of Disclosure for Mineral Projects ("NI 43-101") and certify that, by reason of my education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, I am a "qualified person" for the purposes of NI 43-101.
7. I most recently personally inspected the Kinsevere Copper Project property described in the Technical Report between 6th and 7th March 2008.
8. I am responsible for the preparation of those portions of the Technical Report relating to Mineral Reserve estimation and Mining/Production Schedules, namely Items 19.2 and 25.1.9.
9. I am independent (as defined by Section 1.4 of NI 43-101) of Anvil Mining Limited.
10. I have had prior involvement with the property that is subject of the Technical Report. The nature of my prior involvement has been in mine planning and the preparation of Mineral Reserve estimates, commencing in 2006.
11. I have read NI 43-101 and Form 43-101F1 and the Technical Report has been prepared in compliance with that instrument and form.
12. As of the date of this certificate, to the best of my knowledge, information and belief, the Technical Report contains all scientific and technical information that is required for be disclosed to make the Technical Report not misleading.

Signed and dated this 31st day of March, 2010 at Perth, Western Australia, Australia.

(Signed) "Anthony Cameron"

Anthony Cameron

*Gerry Fahey
CSA Global Pty Ltd
Level 1, 47 Burswood Road
Burswood, Western Australia, 6100
Tel +61 8 9355 1677
gerry.fahey@csaaus.com.au*

I, Gerry Fahey, do hereby certify that:

1. I am a Director of CSA Global Pty Ltd.
2. This certificate applies to the technical report entitled "Anvil Mining Limited NI 43-101 Technical Report Kinsevere Copper Project Katanga Province, Democratic Republic of Congo" dated 31st March 2010 (the "Technical Report").
3. I am a professional geologist having graduated with a Bachelor of Science degree with Honours in Geology from the National University of Ireland in 1975.
4. I am a Member of the Australasian Institute of Mining and Metallurgy, a Member of the Australian Institute of Geoscientists and a Member of the Joint Ore Reserve Committee (JORC).
5. I have worked as a geologist for a total of thirty three years since my graduation from university.
6. I have read the definition of "qualified person" as set out in National Instrument 43-101 – Standards of Disclosure for Mineral Projects ("NI 43-101") and certify that, by reason of my education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, I am a "qualified person" for the purposes of NI 43-101.
7. I most recently personally inspected the Kinsevere Copper Project property described in the Technical Report in January 2008.
8. I am responsible for the preparation of those portions of the Technical Report relating to Mineral Resource estimation, namely Item 19.1.
9. I am independent (as defined by Section 1.4 of NI 43-101) of Anvil Mining Limited.
10. I have had prior involvement with the property that is the subject of the Technical Report. The nature of my prior involvement has been in the preparation of Mineral Resource estimates previous to that reported in this Technical Report.
11. I have read NI 43-101 and Form 43-101F1 and the Technical Report has been prepared in compliance with that instrument and form.
12. As of the date of this certificate, to the best of my knowledge, information and belief, the Technical Report contains all scientific and technical information that is required for be disclosed to make the Technical Report not misleading.

Signed and dated this 31st day of March, 2010 at Perth, Western Australia, Australia.

(Signed) "Gerry Fahey"

Gerry Fahey

*Michael Lawlor
Anvil Mining Limited
Level 1, 76 Hasler Road
Osborne Park, Western Australia 6017
Tel +61 8 6465 2200
michaella@anvilmining.com*

I, Michael Lawlor, do hereby certify that:

1. I am the Manager Group Technical Services, of Anvil Mining Limited.
2. This certificate applies to the technical report entitled "Anvil Mining Limited NI 43-101 Technical Report Kinsevere Copper Project Katanga Province, Democratic Republic of Congo" dated 31st March 2010 (the "Technical Report").
3. I am a professional mining engineer having graduated with an undergraduate degree of Bachelor of Engineering (Honours) from the Western Australian School of Mines in 1986. In addition, I have obtained a Master of Engineering Science degree from the James Cook University of North Queensland (1993), and subsequent Graduate Certificates in Mineral Economics and Project Management from Curtin University (Western Australia).
4. I am a Fellow of the Australasian Institute of Mining and Metallurgy, and a Member and Chartered Engineer of the Institution of Engineers, Australia.
5. I have worked as mining and geotechnical engineer for a period in excess of twenty years since my graduation from university.
6. I have read the definition of "qualified person" as set out in National Instrument 43-101 – Standards of Disclosure for Mineral Projects ("NI 43-101") and certify that, by reason of my education, affiliation with a professional association (as defined in NI 43-101) and past relevant work experience, I am a "qualified person" for the purposes of NI 43-101.
7. I most recently personally inspected the Kinsevere Copper Project property described in the Technical Report between 7th and 18th December 2009.
8. I am responsible for the preparation of those portions of the Technical Report not covered specifically by the other qualified persons.
9. I am not independent (as defined by Section 1.4 of NI 43-101) of Anvil Mining Limited.
10. I have had prior involvement with the property that is the subject of the Technical Report. The nature of my prior involvement has been in mine planning and the preparation of Mineral Reserve estimates, commencing in 2007. I also collated the Project Feasibility Study in December 2007.
11. I have read NI 43-101 and Form 43-101F1 and the Technical Report has been prepared in compliance with that instrument and form.
12. As of the date of this certificate, to the best of my knowledge, information and belief, the Technical Report contains all scientific and technical information that is required for be disclosed to make the Technical Report not misleading.

Signed and dated this 31st day of March, 2010 at Perth, Western Australia, Australia.

(Signed) "*Michael Lawlor*"

Michael Lawlor

ITEM 25 Additional Requirements for Technical Reports on Development Properties and Production Properties

The following additional information is provided to provide the **technical background on aspects covered in Items 19 to 22**.

25.1 Mining Operations

25.1.1 Proposed Mining Method

The proposed mining method is unchanged from the description in the March 2008 Technical Report. Conventional open pit mining methods will be used, based on articulated dump trucks and 85 t class hydraulic excavators.

During 2010, a study will be completed to assess the feasibility of changing the contracted mining fleet for Stage II to fewer and larger trucks (ie, rigid body trucks in place of articulated dump trucks) and larger hydraulic excavators. An (hypothetical) owner mining study was completed by Anvil in 2008 (Anvil, February 2009), in which potential cost savings were apparent.

An internal SMU (selective mining unit) investigation was also completed in May 2009 (Anvil, May 2009). This investigation indicated that a suitable SMU size is 5 m x 10 m x 10 m (east - west, north - south, RL), which may be increased in the north - south direction up to 25 m and reduced by 5 m in the RL direction. These block sizes appear well suited to optimum extraction and suggest that a larger excavator size is not a risk to mining dilution or loss of ore material.

The optimal block sizes for the Kinsevere deposits can be summarised as follows:

- current Mineral Resource model: 10 m (east) x 25 m (north) x 5 m (RL)
- grade control model (recommended): 10 m x 10 m x 5 m
- suitable SMU: 5 m x 10 m x 5 m (or 10 m)
- maximum SMU: 5 m x 25 m x 5 m (or 10 m)

25.1.2 Mining Dilution and Recovery

The pit optimisations completed for the Mineral Reserve estimate (Item 19) assumed a mining dilution factor of 105% (at nil diluent grade), and a mining recovery factor of 100% (cf, 95% in the Feasibility Study). These factors were carried through into the subsequent production scheduling. These assumptions reflect the size of the mining excavator buckets, the 2.5 m flitch height, the dip of the ore horizons and the good visual control between ore and waste rock types. In practice, there will be a dilution grade when mining to grade range boundaries marked on the pit floor.

At this stage of the Project, there is insufficient historical information to reconcile mining dilution and recovery.

25.1.3 Production Reconciliations

A metal mass balance was undertaken for Kinsevere's nineteen month mining history to December 2008, as part of an internal mining reconciliation investigation. Calculated metal movements for this period include:

<u>Mined:</u>	Total mined ex pit Resource Model)	145,419 Cu tonne (then current Mineral
<u>Stocks:</u>	Non Current Ore	55,219
	ROM Stockpiles	17,850
	Plant Stockpiles	<u>15,506</u>
	Total	88,575 Cu tonne
<u>Tails:</u>	TSF1 Dam	10,201 Cu tonne
<u>Production:</u>	Ex-Gate	37,014 Cu tonne

Difference = Stocks + Tailings + Production – Mining
(ie, Mining Loss)
 $(88,575 + 10,201 + 37,014) - 145,419 = < 9,629 > \text{Cu tonne}$

This physical material movement is illustrated in Figure 25.1.

Claim vs actual reconciled metal tonnage corresponds well with a loss of 9.6 kt over 145 kt Cu, determined against Kinsevere's then current Mineral Resource model (February 2009). This represents a reconciliation loss of about 6.6%, which is considered to lie within the upper quartile of best practise standards.

Notwithstanding this outcome, it remains difficult to reconcile mining production and stockpile inventories, and it is possible that the metal loss figure noted above is the median of larger variances.

25.1.4 Proposed Production Tracking System

In order to address the production reconciliation issues, a web-based production tracking system will be developed in 2010 for managing the mine to crusher metal balance chain, with the ability to:

- record and report production and productivity;
- monitor and manage stockpiles; and
- measure Mineral Resource/Reserve depletion.

A schematic of the proposed tracking process is shown in Figure 25.2.

The Stage II processing plant will have a reconciliation or Metal Accounting system that will be based around either an MS Excel or MS Access database. This package will draw data from a Laboratory Information System (LIMS) for the external contractor who will be responsible for the laboratory analysis.

Scoping work for implementation of the tracking system commenced on site in December 2009, with the system expected to be operational during Q2 2010.

25.1.5 Grade Control Method

Following the resumption of mining in August 2009, there was no change to the grade control method based on blast hole sampling as carried out in previous mining and as described in the March 2008 Technical Report. However modifications to future grade control programmes are planned (refer to Item 22.4), including campaigned RC drilling over multiple benches.

Figure 25.1 Kinsevere Mine-Mill Metal Flow

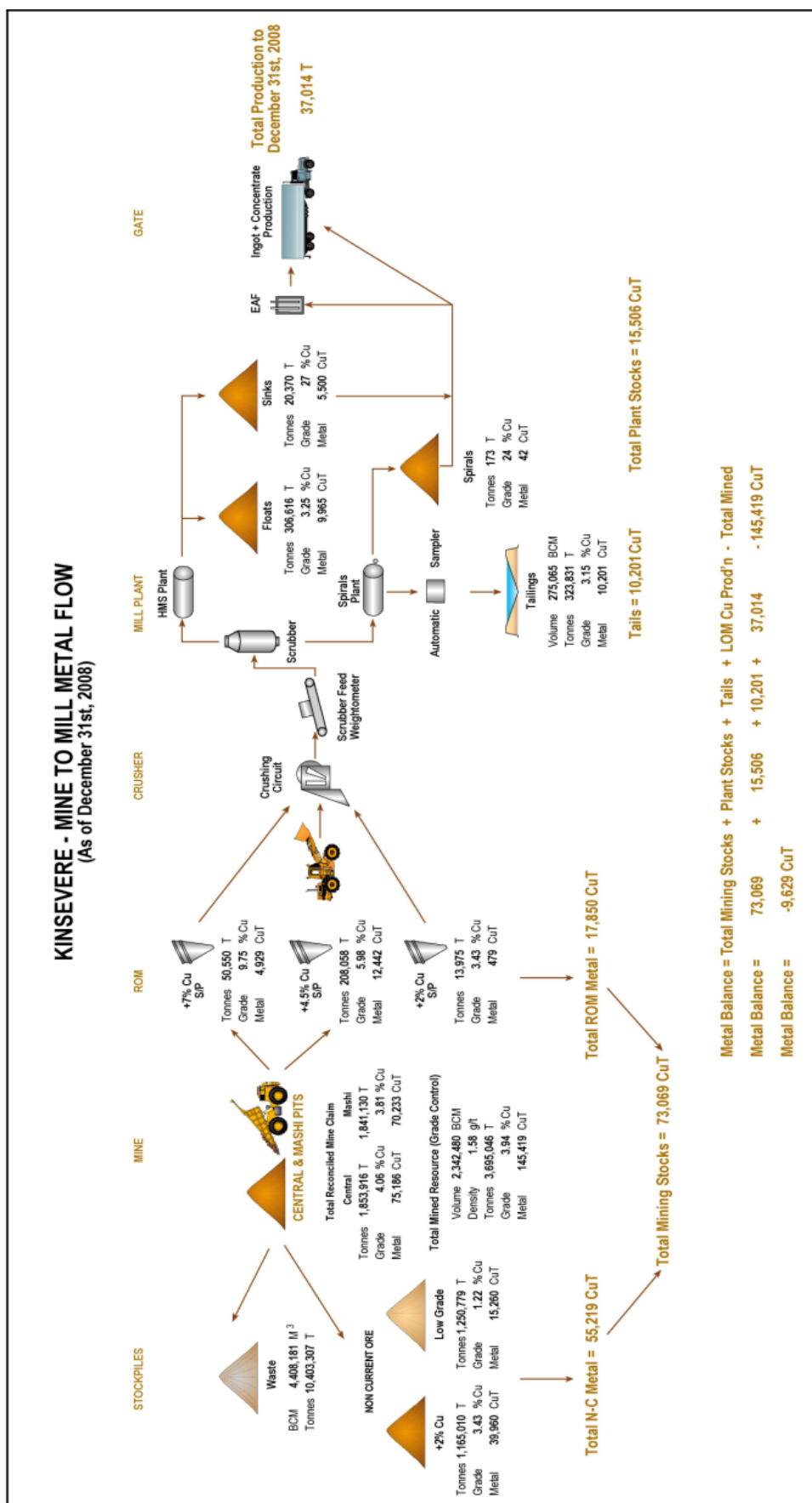
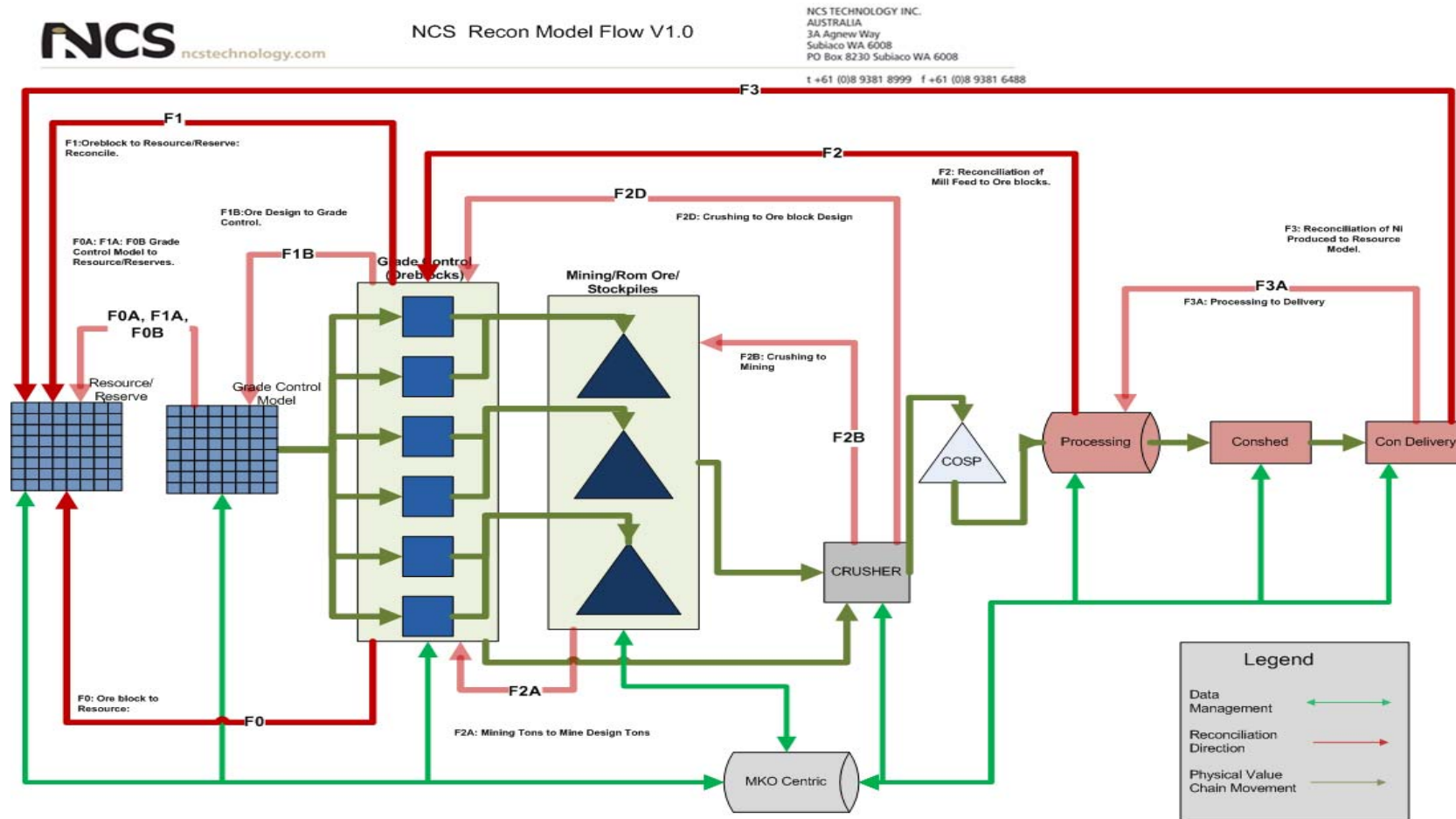


Figure 25.2 Proposed Production Tracking System

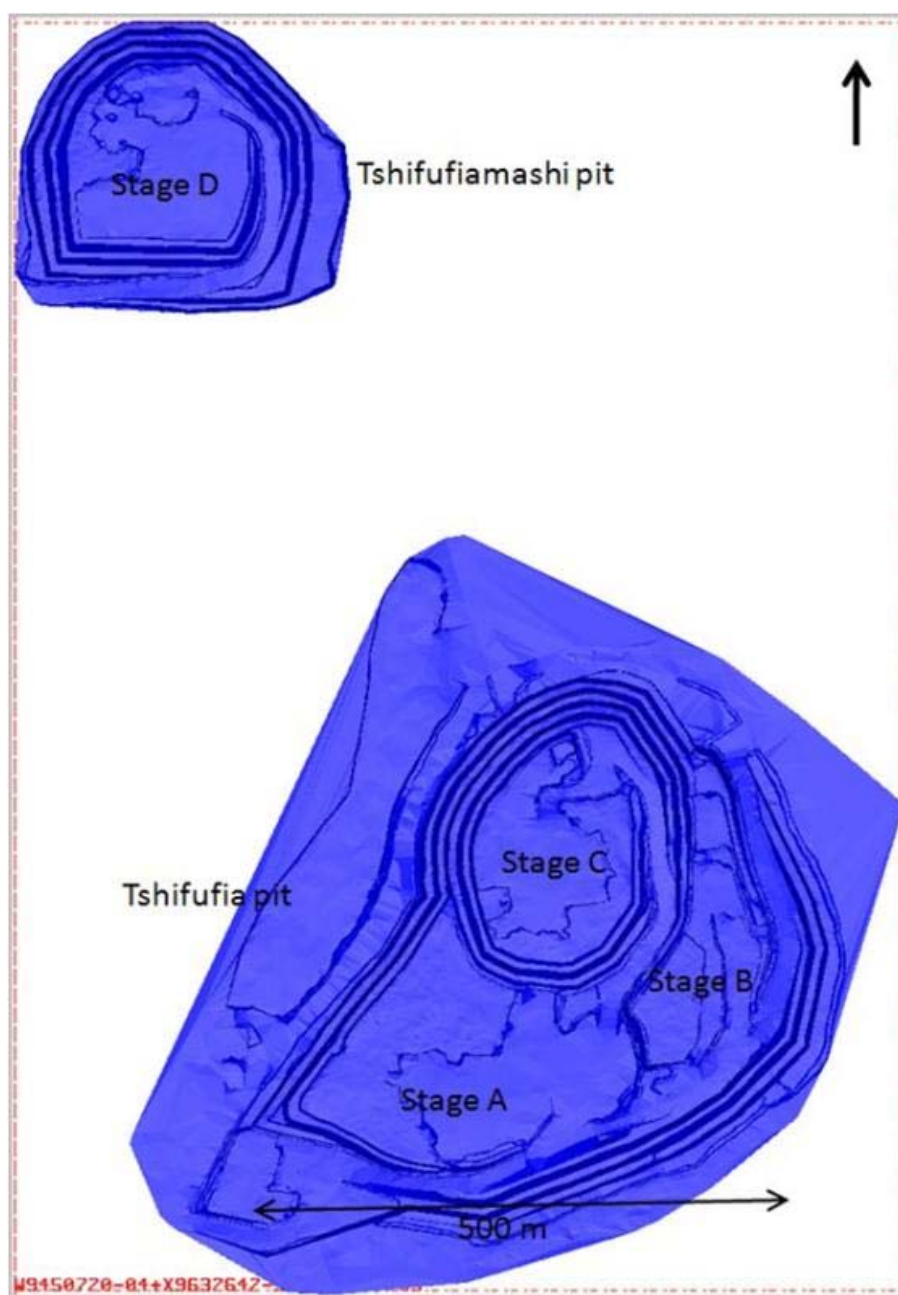


In February 2010 a contract was let to Titan Drilling Congo s.p.r.l. (Titan) for the drilling of 23,000 metres of RC grade control holes. The initial contract is for two drilling campaigns, shown as Stages A and C in Figure 25.3. The drilling of Stages B and D will be carried out in 2011.

The drilling programme involves the drilling of inclined RC holes over a 36 m length, and on a spacing of 10 m x 10 m. A trailer mounted cyclone and sample splitting device will be used to produce 2 m sample composites.

Stage A is expected to be completed by the end of March 2010, based on current progress of 180 m/day (ie, five holes). Assay samples are being processed on site and the data will ultimately be dispatched to Perth for DataShed database storage and QAQC review.

Figure 25.3 Proposed Grade Control Drilling Campaigns



25.1.6 Geotechnical Engineering

An update report by Turner Mining and Geotechnical Pty Ltd (TMG) detailing the findings of a site visit (TMG, August 2008) was produced in August 2008. Previous pit slope design recommendations were endorsed, as reiterated in Table 25.1, and were used as input for both the pit optimisation and pit design processes (Item 19.2).

Table 25.1 Revised Pit Slope Design Parameters, September 2007

Domain	Inter-ramp Slope Angle	Batter Angle	Batter Height	Berm Width
All Slopes Except Tshifufiamashi West Wall				
Soil (to 5 mbs)	Remove soil and weak alluvium			
East and West	40°	80°	5 m	5 m
5 m to 100 m	40°	55°	10 m	5 m
	40°	60°	10 m	6 m
	40°	64°	10 m	7 m
Tshifufiamashi West Wall				
Soil (to 1 mbs)	Remove soil and weak alluvium			
1 m to 40 m	40°	60°	10 m	6 m
Below 40 m	32°	60°	5 m	5 m

The slope design parameters in Table 25.1 could be adopted for Kinsevere Hill. Whilst the rocktypes and ground conditions are likely to be similar, geotechnical conditions in actual exposures within the northern pits could be used to update the Kinsevere Hill design parameters.

Further geotechnical work is required to confirm the slope design parameters, and in particular, ongoing logging of drill core and mapping of exposures is required to confirm the rock mass structural characteristics. Further slope stability analysis is also required. TMG was on site in December 2009 to commence this work programme.

The Independent Engineer Report by SRK Consulting (SRK, 2009) notes that, in their opinion, the overall slope angles as proposed remain conservative provided that dry slope conditions are maintained.

25.1.7 Hydrogeology

Groundwater Level Monitoring

Groundwater levels are now being monitored in 35 boreholes and monitoring wells across the site (refer to Figure 25.4) on a weekly and some times daily basis. Abstraction rate is recorded at two local supply wells and at all dewatering and supply boreholes at the mine site. Continuous pumping had been maintained whilst the operation was on care and maintenance, in order to avoid pit slope degradation and enable ready re-access when mining resumes. Flow rates at February 2009 were in the order of 265 L/s.

Due to a number of ongoing maintenance and pump power supply issues, the dewatering rate had reduced to less than 200 L/s in the second half of the year, as shown in Figure 25.5. Review of monitoring graphs indicates that there is no

significant drawdown trend at the current abstraction rates. Some bores exhibit only a localised drawdown effect. These graphs, however, also indicate a quick recovery after pump outages.

There are currently five water quality monitoring sites (MON07 to MON11) at the Kifumashi River which have been monitored on a monthly basis since October 2007 for physio-chemical parameters, some major ions and a limited suite of trace metals. The same suite of determinants has also been monitored at eight locations within the mine site, including sediment and water storage ponds, since August 2008, and at eight potable water outlet points since October 2008.

Figure 25.4 Location of Existing Groundwater Abstraction and Monitoring Bores

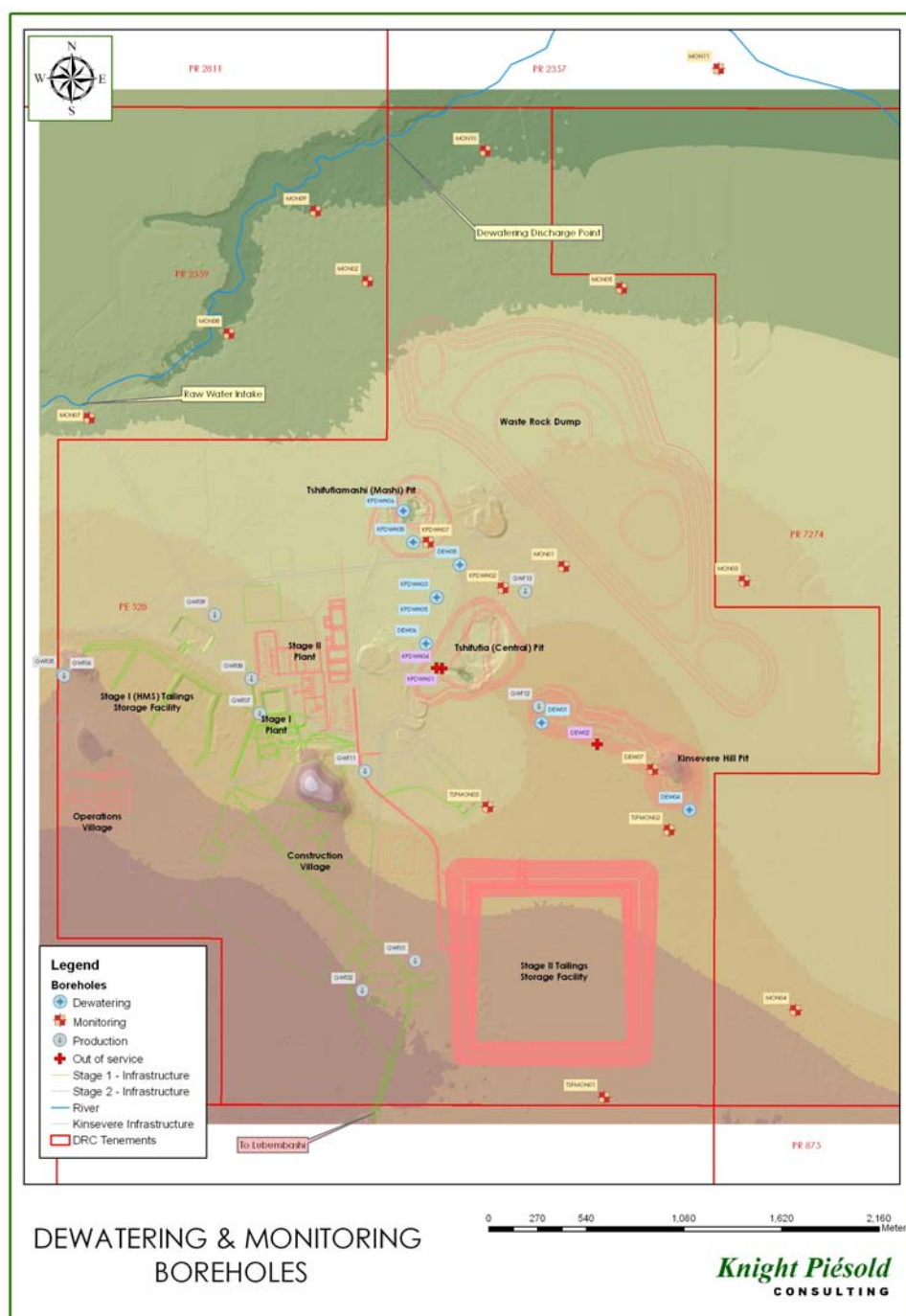
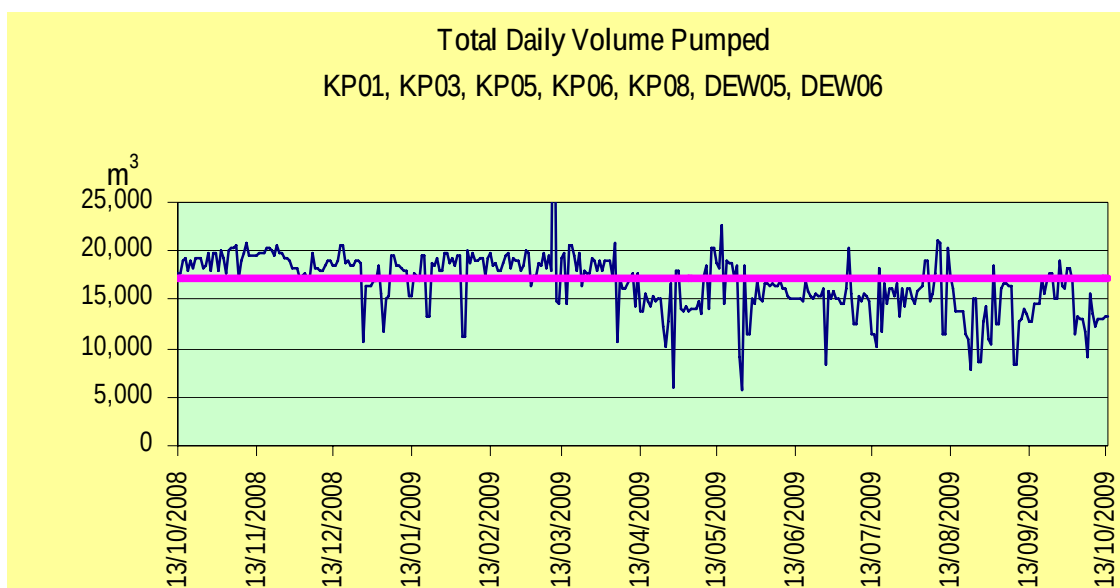


Figure 25.5 Groundwater Monitoring Trends



During 2010 an independent consultant will be appointed to routinely review the quality of water discharged to the Kifumashi River and supplied to “potable” water bores on the Project site and in the local community.

Updated Dewatering Requirements

In October 2007, Knight Piésold (KP) completed a detailed hydrogeological assessment for the Project area, with the objective of defining the interaction between mining activities and regional groundwater flows and levels. Further hydrogeological drilling, investigation and analysis was completed in 2008 resulting in the compilation of a Feasibility Report by KP (December 2008).

In the October 2007 KP report it was advised that nine boreholes (ie, four then existing, five to be drilled and equipped) would be required for ongoing pumping over the life of the mine, with a combined abstraction rate of 220 L/s.

However, with the inclusion of new geological information in 2008, the hydrogeological concept of aquifer boundaries has changed. In particular, the discovery of RAT breccia is now expected to influence the site hydrogeology due to its higher transmissivity. Groundwater inflow to the open pits is expected to increase which will require an increase in the dewatering rate in order to keep the pits dry. KP (December 2008) notes the risk of a hydraulic connection between the open pits and the Kifumashi River.

A series of analytical dewatering calculations were carried out by KP including a worst case scenario where the drawdown intercepts RAT breccia and the river. Under these circumstances, the pumping rate would need to be increased to between 550 L/s and 600 L/s, requiring the drilling/equipping of additional boreholes.

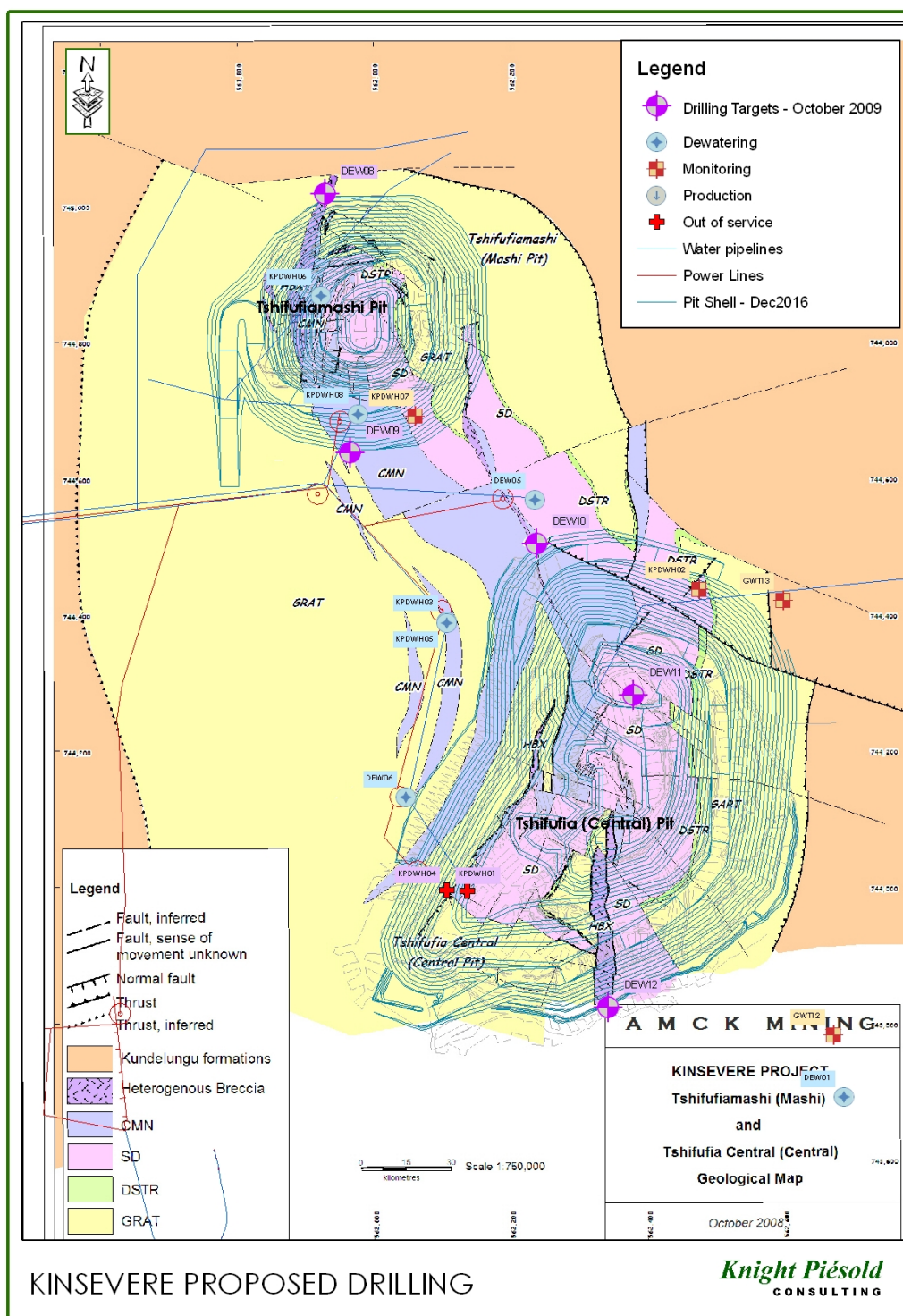
The proposed bore drilling programme for Q4 2009 to Q1 2010 is as follows:

- 60 m deep, 6.5” diameter probe (pilot hole) drilling to assess suitable locations;
- 5 x 16” diameter, cased and slotted boreholes drilled by rotary percussion, to depths as follows:

- o DEW08 at Tshifufiamashi, to 140 m;
- o DEW09 at Tshifufiamashi, to 145 m;
- o DEW10 at Central, to 150 m;
- o DEW11 at Central, to 135 m; and
- o DEW12 at Central, to 155 m

The proposed locations are shown in Figure 25.6.

Figure 25.6 Location Plan of Proposed New Boreholes



Eastern Drilling was contracted in November 2009 to complete the borehole drilling and casing programme. Each of the five newly completed bores is expected to be equipped and commissioned by mid 2010.

Groundwater Modelling

The numerical model developed by KP for the 2007 study (KP, October 2007) has not been updated. The Independent Engineer Report by SRK Consulting (SRK, September 2009) recommended that an updated model is required, in addition to further investigations on the hydrogeological performance of the RAT breccia. This model is required to assess the long term impact of pit dewatering and the groundwater connection between the river, the mine and the TSF.

Following completion of the drilling and aquifer testing programme as described above, KP will complete a programme of groundwater modelling to confirm the likely overall abstraction rate.

25.1.8 Hydrology

Kifumashi River

During 2008 KP completed a stream flow monitoring programme in order to quantify the hydrological conditions and the baseline of flows in the Kifumashi River. The KP site visit report (KP, July 2008) noted that the dewatering discharge from the mine to the river would be in the order of 145 L/s, potentially increasing to 250 L/s to 300 L/s. This projection was updated in December 2008 to potentially 550 L/s to 600 L/s.

Further engineering design work will be required in 2010 for the pipeline reticulation network to the river. Considerably larger pipes or a duplication of the network may be required to accommodate the increased discharge rate. The area downstream of the discharge point is a flat marsh/wetland environment which may have the ability to store and attenuate peak flows. KP recommends the installation of flow measurement weirs for monitoring baseline conditions.

In addition, the KP Feasibility Study (KP, December 2008) considers the potential impacts from pit dewatering activities on flows in the river. The groundwater returned to the river is expected to be of equal or better quality than the river water itself. Although there will be return discharge to the river, the pit dewatering programme over the longer term could result in a significant lowering of the river level.

Mine and Plant Site

KP has developed a Surface Water Management Plan for the operations site in order to manage and contain surface water runoff and to maintain sediment loadings (from disturbed catchments) at acceptable levels. KP produced a report (KP, August 2008) detailing the location and construction requirements for a number of strategically placed environmental control dams (ECDs) and diversion channels.

Two types of surface water run-off can be expected, ie:

- dirty water (sediment and contaminated) runoff from disturbed areas; and
- clean water from undisturbed catchment areas.

KP's design concept was to keep these water types separate from each other. Dirty water runoff would report to the ECDs via a series of diversion channels. Clean water runoff from most of the undisturbed catchments would be directed to natural watercourses. The adopted design storm water event criteria are as listed in Table 25.2.

Table 25.2 Design Storm Criteria

Parameter	Design Criteria
Site Diversion Structures - Flow Capacity	1 in 100 Year ARI / 24 hour duration
Culvert Structures – Flow Capacity	1 in 100 Year ARI / 24 hour duration
Environmental Control Dams (ECDs)	
Basin Storage	1 in 20 Year ARI / 24 hour duration
Emergency Spillway	1 in 100 Year ARI / 24 hour duration
Maximum suspended solids discharged from ECD)	50 mg/L
Design Storm Event	Storm Event Magnitude (mm)
1 in 2 year ARI, 24 hour storm event	86
1 in 5 year ARI, 24 hour storm event	104
1 in 10 year ARI, 24 hour storm event	114
1 in 20 year ARI, 24 hour storm event	124
1 in 100 year ARI, 24 hour storm event	145

ECD design parameters are summarised below:

- Upstream Embankment Slope = 2.0H:1V.
- Downstream Embankment Slope = 2.5H:1V.
- Crest Width = 4 m.
- Zone F1 chimney drain within the ECD embankment.
- Floating skimmer type abstraction system.
- PSWP consists of fill only (i.e. no local excavation).
- Situated within the proposed waste dump footprint to reduce site disturbance.

The proposed ECD locations are shown on Figure 25.7, along with the proposed alignments of the diversion channels. The ECDs are required downstream from any catchments where construction or mining activities are likely to lead to elevated levels of sediment in runoff reporting to local waterways. A typical ECD comprises a small homogeneous earthfill embankment structure (between 2 to 4 m high). It is required to impound enough water for a sufficient period of time to allow settlement of suspended solids.

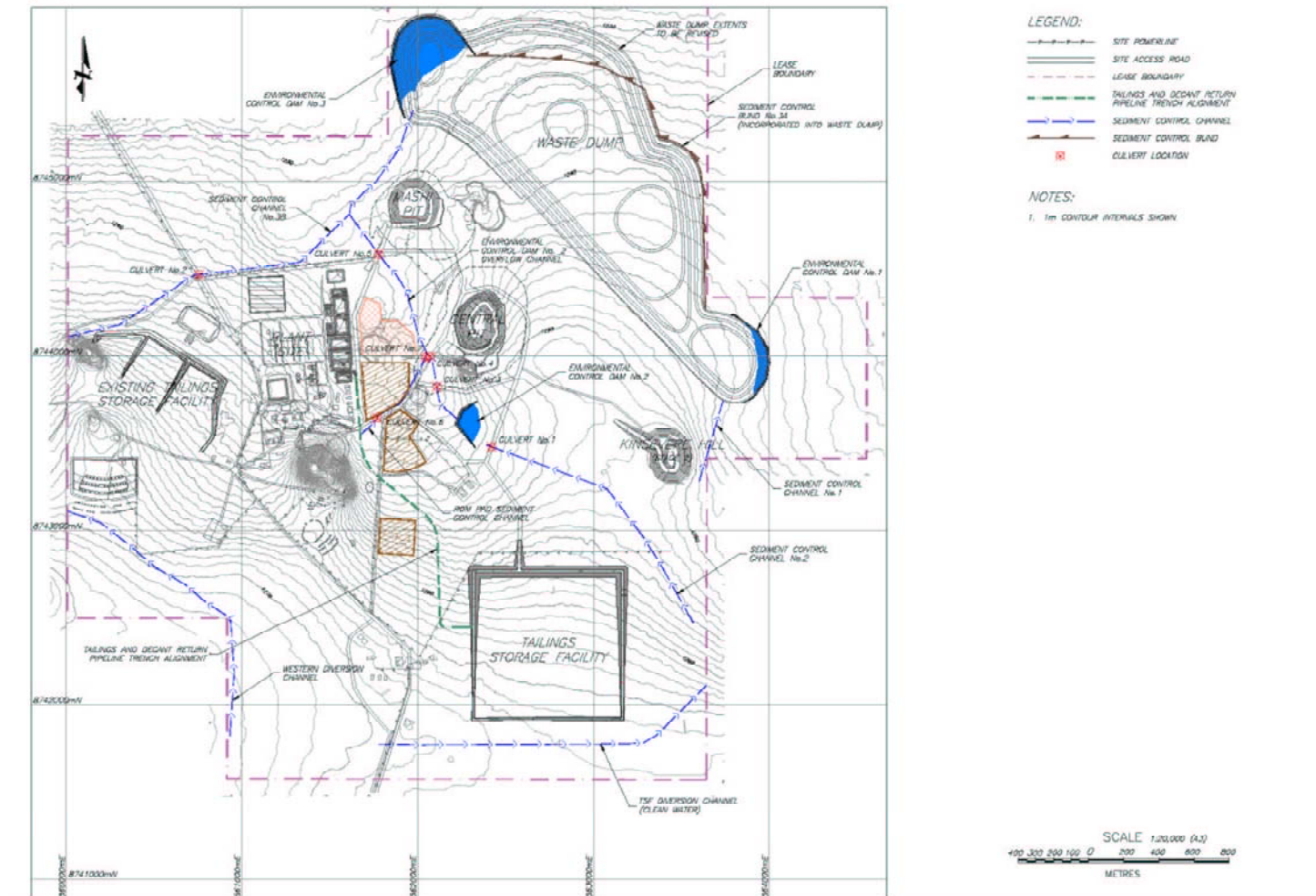
The ECD design is driven by the dual considerations of storage capacity (for runoff containment) and minimum surface area (for settling of particles in suspension). ECD design requirements and dimensions are summarised in Table 25.3.

Table 25.3 ECD Design Summary

Structure	Crest Elevation (m RL)	Maximum Height (m)	Storage Capacity (m ³)	Maximum Inundation Area (ha)
ECD No. 1	1243.3	1.4	7,300	2.0
ECD No. 2	1244.8	3.0	18,900	1.8
ECD No. 3	1231.3	4.2	189,000	11.6

Each ECD would be designed in detail to be equipped with two discharge structures, a decant tower with an adjustable inlet RL for routing flows under normal conditions and an overflow spillway for flows in excess of the containment capacity. The decant tower would comprise a vertical ring decant structure placed within a perforated concrete tower which, in turn, would be supported by coarse rockfill. The decant tower would discharge into the downstream river channel via an HDPE conduit placed through the embankment.

Figure 25.7 Arrangement Concept, Site Water Management Plan



Diversion structure design parameters are summarised below:

- Unlined earth channels, to be vegetated during operation.
- Triangular v-ditch (able to be excavated by grader) where practicable.
- When trapezoidal, minimum base width of 1 m.
- Minimum channel depth = 0.50 m.
- Minimum channel freeboard (from peak flow to top of channel) = 0.10 m.
- Minimum channel vertical grade = 0.25%.
- Sediment Control Bund 3A construction incorporated into waste dump development.

Diversion channel design information is summarised in Table 25.4.

Table 25.4 Diversion Channel Design Summary

Chainage		Base Width (m)	Depth (m)	Length (m)	Side Slopes
From (m)	To (m)				
CLEANWATER DIVERSION CHANNELS					
TSF Diversion Channel ¹					
0 (Start)	1,995 (End)	-	1.00	1,995	5H:1V, 2H:1V
Western Diversion Channel					
0 (Start)	1,760 (End)	-	1.00	1,760	5H:1V, 2H:1V
SEDIMENT CONTROL CHANNELS					
Sediment Control Channel 1					
0 (Start)	480 (End)	-	0.50	480	5H:1V, 2H:1V
Sediment Control Channel 2					
0 (Start)	1,552	-	0.75	1,552	5H:1V, 2H:1V
1,552	1,642 (End)	1.0	1.50	90	2H:1V
Sediment Control Channel 3A ²					
Sediment Control Channel 3B					
0 (Start)	781	-	0.50	781	5H:1V, 2H:1V
781	1,655	-	1.00	874	5H:1V, 2H:1V
1,655	2,527 (End)	1.0	1.50	872	2H:1V
ECD 2 Overflow Channel					
0 (Start)	179	1.0	0.75	179	2H:1V
179	1,367 (End)	1.0	1.50	1,188	2H:1V
ROM Pad Sediment Control Channel					
0 (Start)	25 (End)	1.0	1.50	625	2H:1V

¹ as shown on Figure 25.7.

² incorporated into waste dump development

Culvert design parameters are summarised below:

- Concrete pipe culverts.
- Culvert locations based on proposed site access road layout.
- Culvert trenches will require excavation into existing access roads where applicable.
- Concrete headwall structures at both inlet and outlet.
- Maximum headwater elevation above culverts = 0.6 m.

Culvert design information is summarised in Table 25.5.

Table 25.5 Culvert Design Summary

Channel ID	No. of Culverts	Culvert Diameter (mm)
Sediment Control Channel 2		
CLV 1	2	900
Sediment Control Channel 3B		
CLV 2	1	600
ECD 2 Overflow Channel		
CLV 3	4	900
CLV 4	6	900
CLV 5	6	900
ROM Pad Sediment Control Channel		
CLV 6	3	900
CLV 7	3	900

25.1.9 Mining and Processing Schedules

A Life of Mine production schedule was prepared using the SURPAC MineSched software package, and from the detailed pit designs and the inventories listed in Item 19. Various schedule scenarios were actually produced to test Project sensitivity and these are listed in Table 25.6.

Only Scenarios E and H are reported herein. All of the other scenarios yielded a lesser NPV. Scenario E (highlighted in Table 25.6) is the Life of Mine production schedule as reviewed by the Independent Engineer (SRK, 2009) in 2009. Scenario H (also highlighted in Table 25.6), however, is the production schedule that now applies to the updated Mineral Reserve estimate as reported in Item 19.2. It assumes that:

- in accordance with the Company's 2009-2010 Budget, mining in the base of the Central_1 Pit is postponed to allow more effective groundwater drawdown to take affect from a reinstated bore drilling and equipping programme;
- mining for HMS plant feed to the end of December 2010 therefore proceeds at higher elevations and as a cutback to the south of the Central_1 Pit (ie, the Central_2 cutback is brought forward);
- this results in the mining and HMS processing of lower grade ores than that projected in the Scenario E schedule, over the same period; and
- the revenue arising from this 2009-2010 Budget, however, is just sufficient to sustain the Stage I site operating costs, with a modest cash buffer.

Figure 25.8 shows a graph of the Scenario E production profile as determined in the 2009 Life of Mine Plan. Figure 25.9 shows a graph of the production profile for Scenario H reflecting the LOM Plan updated for the 2009-2010 Budget and depleted for production to the end of 2009. Points to note for Scenario H are the significant reduction in Stage I process feed grade, offset by slightly higher average stockpile reclaim grades from 2019.

Table 25.6 Life of Mine Plan Production Schedule Scenarios

LOM Plan Scenario	Scenario Description
A	All stockpiled ore as at December 2008 available for SII processing, ie no resumed Stage I processing. Mining and Stage II processing commences in January 2011. All stockpile grades factored by 85%.
B	Stockpiled (ROM) High Grade ore processed by Stage I plant, commencing April 2009 and completed in August 2009. Mining and Stage II processing commences in January 2011. Medium and Low Grade stockpiles available for Stage II. All stockpile grades factored by 85%.
C	All stockpiled ore as at December 2008 available for SII processing, ie no resumed Stage I processing. Mining and Stage II processing commences in January 2011. All stockpile grades are unfactored.
D	Stockpiled (ROM) High Grade ore processed by Stage I plant, commencing April 2009 and completed in August 2009. Mining and Stage II processing commences in January 2011. Medium and Low Grade stockpiles available for Stage II. All stockpile grades are unfactored.
E	Stockpiled (ROM) High Grade ore processed by Stage I plant, commencing April 2009 and completed in August 2009. Mining recommences in July 2009 providing continuing feed to the Stage I plant until December 2010. Mining (at an increased scale) and Stage II processing commences in January 2011. Medium and Low Grade stockpiles available for Stage II. All stockpile grades factored by 85%.
F	Stockpiled (ROM) High Grade ore processed by Stage I plant, commencing April 2009 and completed in August 2009. Mining recommences in July 2009 providing continuing feed to the Stage I plant until December 2010. Mining (at an increased scale) and Stage II processing commences in January 2011. Medium and Low Grade stockpiles available for Stage II. All stockpile grades are unfactored.
H	As for Scenario E. Mining commences in August 2009 according to the 2009-2010 budget projection. Mining (at an increased scale) and Stage II processing commences in January 2011. Medium and Low Grade stockpiles available for Stage II. All stockpile grades factored by 85%.

Figure 25.8 Scenario E Production Profile (Superseded)

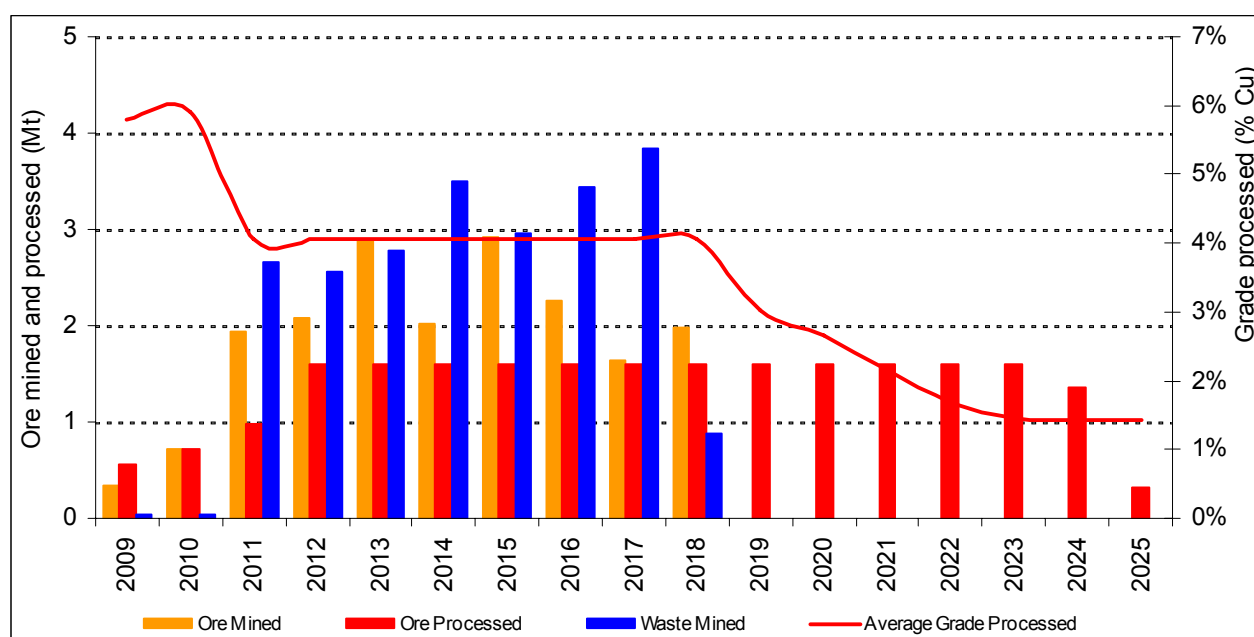
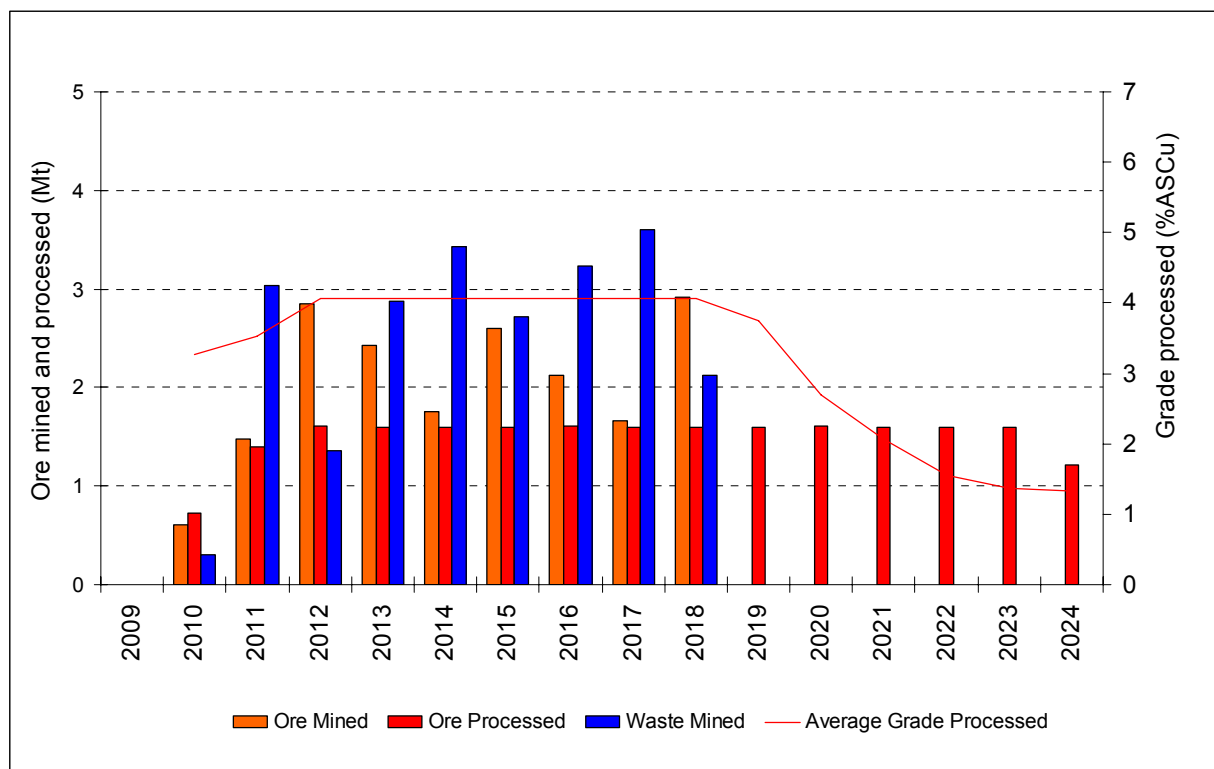


Figure 25.9 Scenario H Production Profile (Current)



The schedules were prepared using several grade range stockpiles in order to optimise the feed into either of the HMS or SX-EW plants, whilst maintaining a balanced ore and waste mining rate. These stockpile grade ranges and nomenclature are listed in Table 25.7.

Table 25.7 Scheduled Stockpile Grade Ranges

Stockpile	Grade Range ASCu (%)
High Grade (HGO)	+4
Medium Grade (MGO)	2 to 4
Low Grade (LGO)	0.72 to 2

The mining sequence was forced to target the highest grade ore whilst honouring the staged pit development, and limiting the vertical advance rate to no more than 25 m per year. The average total material movement rate was restricted to not more than 3.12 million bcm per year (equivalent to 260,000 bcm per month).

For the 'base case' schedule (Scenario H), an output production constraint of 60,000 t of copper metal was imposed. The design plant throughput of 1.6 million tonnes per annum was allowed a feed rate tolerance of +/- 5%. The production ramp-up limits imposed on SX-EW processing were as listed in Table 25.8.

Table 25.8 Production Ramp-Up Limits

Ramp-up Period	Recovery	Capacity
Month 1	74.6%	30%
Month 2	78.1%	60%
Month 3	80.9%	75%
Month 4	85.4%	85%
Month 5	89.4%	95%
Month 6	91.4%	99%
Month 7	92.14%	100%

HMS floats and effluent tonnes and grade were calculated in an MSExcel spreadsheet, at the completion of scheduling, based on an algorithm related to plant feed tonnes, grade and recovery. The material was assumed to report to a combined stockpile inventory which was then reviewed during the financial modelling and reclaimed into the final years of SX-EW processing, as appropriate.

Life of Mine Schedule

Table 25.9 presents an annual summary of the Life of Mine schedule, with SX-EW processing from January 2011 through to 2024. The processing inventory includes 1.7 Mt HMS floats and effluent stockpiled up to the end of Stage I processing in 2010, at an average grade of 2.4% ASCu. Table 25.10 shows the annual stockpile balances.

These two tables show that mining is completed in 2018, whilst SX-EW processing continues until 2023, drawing down on medium grade (MGO) and low grade (LGO) stockpiles. It is also possible within this period to draw down on the stockpiled HMS floats and effluent and improve on the grade going into the plant. This additional inventory is included as “opportunistic feed” in the processing schedule listed in Table 25.9.

The mining sequence for each of the three staged pits is shown in Figure 25.10, indicating:

- Tshifufia Stage 1 Pit is mined in 2010;
- Tshifufia Stage 2 Pit is mined between 2010 and 2013;
- Tshifufia Stage 3 Pit is mined between 2011 and 2013;
- Tshifufia Stage 4 Pit is mined between 2012 and 2016;
- Tshifufiamashi Stage 1 Pit is mined in 2012;
- Tshifufiamashi Stage 2 Pit is mined between 2013 and 2017; and
- Kinsevere Hill Pit is mined between 2014 and 2018.

Table 25.9 Life of Mine Mining and Processing Summary

Year	Ore Mined (Tonnes)	Waste Mined (Tonnes)	Total Mined (Tonnes)	Strip Ratio (w:o Tonnes)	Average Mined (BCM/month)	Ore Processed (Tonnes)	Ore Processed TCu (%)	Recovered Copper (Tonnes)
2010*	609,661	301,083	910,744	0.49	46,499	730,000	3.84	16,799
Year	Ore Mined (Tonnes)	Waste Mined (Tonnes)	Total Mined (Tonnes)	Strip Ratio (w:o Tonnes)	Average Mined (BCM/month)	Ore Processed (Tonnes)	Ore Processed ASCu (%)	Recovered Copper (Tonnes)
2011	1,483,027	3,035,823	4,518,850	2.05	228,374	1,392,099	3.53	43,834
2012	2,847,363	1,358,325	4,205,689	0.48	201,380	1,604,544	4.07	60,146
2013	2,429,063	2,881,082	5,310,145	1.19	258,863	1,600,160	4.07	59,975
2014	1,750,274	3,424,097	5,174,371	1.96	258,863	1,600,160	4.06	59,787
2015	2,593,959	2,713,971	5,307,931	1.05	258,863	1,600,160	4.06	59,898
2016	2,121,841	3,237,186	5,359,027	1.53	259,613	1,604,544	4.06	60,045
2017	1,664,351	3,602,486	5,266,837	2.16	258,863	1,600,160	4.06	59,872
2018	2,912,979	2,130,484	5,043,462	0.73	248,255	1,600,160	4.06	59,866
2019						1,600,160	3.74	55,191
2020						1,604,544	2.70	39,974
2021						1,600,160	2.08	30,625
2022						1,600,160	1.56	22,932
2023**						1,600,160	1.36	20,038
2024**						1,212,574	1.33	14,892
Subtotal	17,802,858	22,383,453	40,186,311	1.26	246,634	21,819,745	3.23	647,075
TOTAL	18,412,519	22,684,536	41,097,055	1.23	224,397	22,549,745		663,874

* Stage I HMS Processing

** Stage II processing extended by blending stockpile reclaim and Stage I floats and effluent between 2021 and 2024

Table 25.10 Life of Mine Stockpile Balances

Year	HGO Ore Tonnes Stockpile Balance	HGO (%ASCu)	MGO Ore Tonnes Stockpile Balance	MGO (%ASCu)	LGO Ore Tonnes Stockpile Balance	LGO (%ASCu)
2009	28,742	7.18	1,109,961	2.43	1,260,718	0.87
2010	27,439	7.18	822,973	2.45	1,428,670	0.95
2011	51,120	5.19	486,288	2.48	1,832,602	0.99
2012	662,762	7.17	863,709	2.61	2,086,359	1.04
2013	957,900	6.75	1,229,118	2.67	2,254,714	1.08
2014	602,062	6.68	1,353,229	2.65	2,636,555	1.11
2015	791,337	6.17	1,733,915	2.67	3,060,394	1.14
2016	804,103	6.08	1,966,652	2.67	3,332,188	1.16
2017	592,404	6.04	2,049,651	2.69	3,525,079	1.17
2018			2,511,782	2.70	4,317,513	1.21
2019			1,839,798	2.70	4,039,995	1.21
2020			235,254	2.70	4,039,995	1.21
2021					2,675,089	1.21
2022					1,074,929	1.21
2023						
2024						

Figure 25.10 Life of Mine Mining Sequence (Total Ore + Waste)

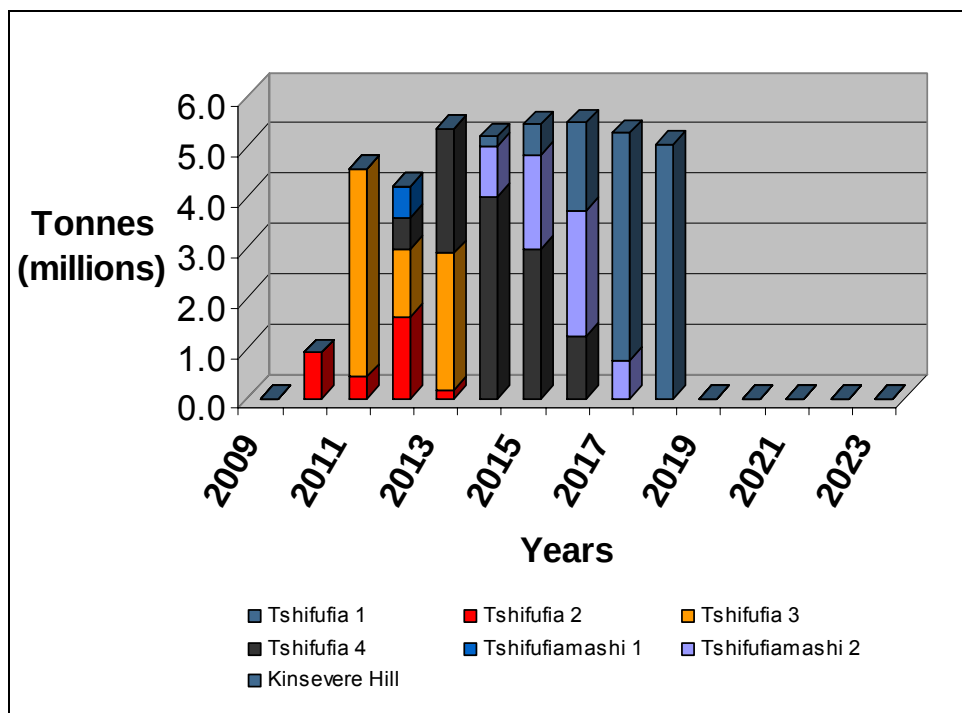


Figure 25.11 shows the ore movement profile for the Life of Mine schedule, with ore direct feed and stockpile reclaim making up the 1.6 mtpa of process plant feed. This figure excludes the Stage I floats and effluent reclaim.

Figure 25.11 Base Case Material Movement Profile

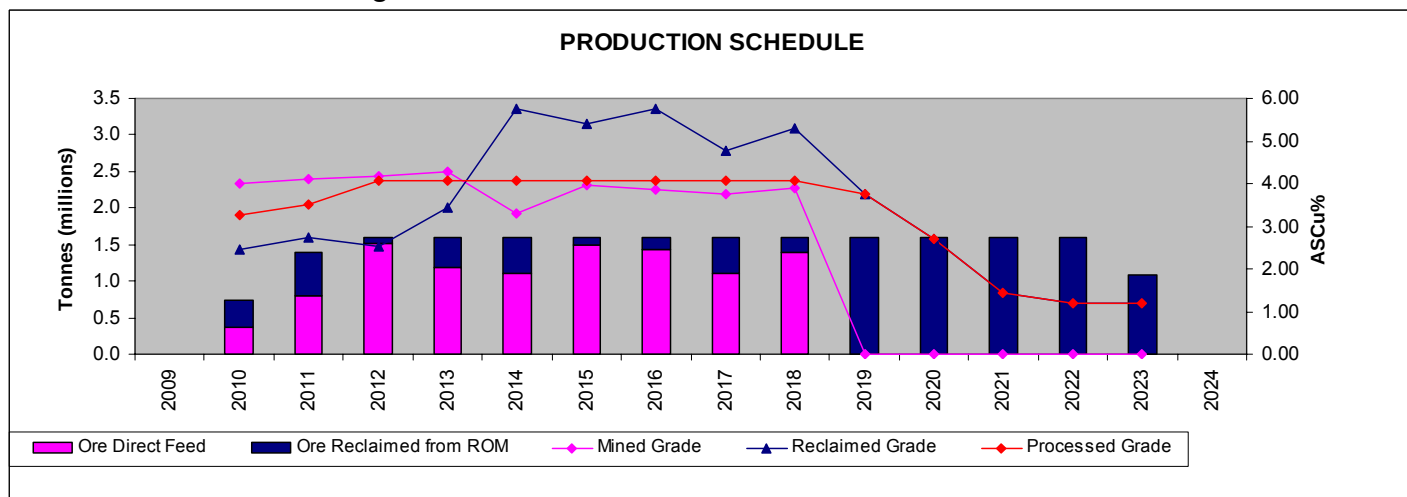
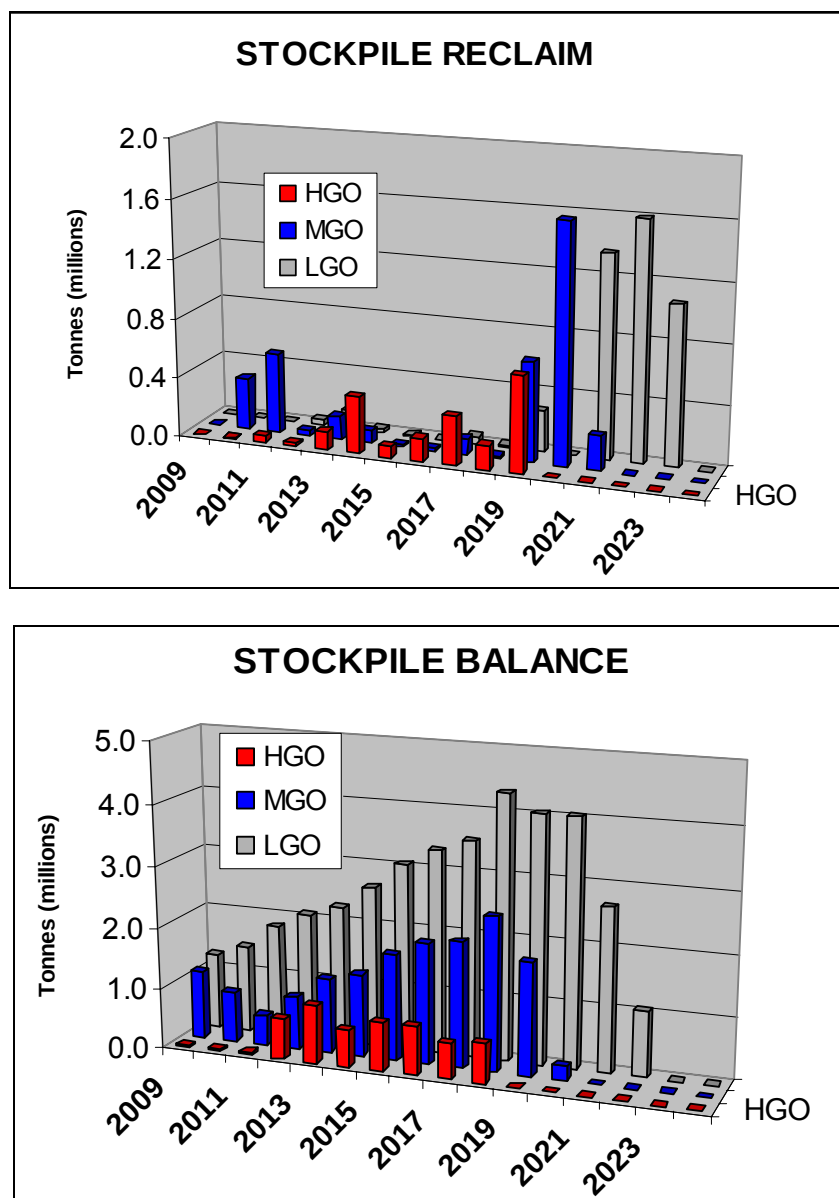


Figure 25.12 shows the annual stockpile balance over the life of the operation, and highlights the size of the medium grade and low grade stockpiles, which reach a maximum size of 2.5 million tonnes and 4.3 million tonnes, respectively, in 2018 (ie, the last year of mining). The high grade stockpile is mostly reclaimed before the end of mining.

Figure 25.12 Base Case Stockpile Reclaim and Balance Profile



Schedule Changes Since January 2010 News Release

Between the January 2010 News Release (Anvil, 2010) and this Technical Report, changes were made to the Stage I and Stage II production schedules resulting in a minor reduction to the total copper metal recovered in Stage II. Tables 25.11 and 25.12 show the metal balance differences.

There is no change to the Mineral Reserve reported in Item 19.2, only to the processing inventories for each stage. The F+E referred to in these tables means reprocessed Stage I floats and effluent (estimated/projected inventories). This additional inventory should be considered as “opportunistic” plant feed.

Table 25.11 Metal Balance, 27th January 2010 News Release

Mining		tonnes	TCu%	ASCu%	total metal	AS metal
Stage I mine		463,248	4.63	3.94	21,466	18,260
Stage I s/p		268,155	2.11	1.77	5,668	4,757
subtotal		731,403	3.71	3.14	27,134	23,017
Stage II mine		17,949,271	4.13	3.52	740,864	631,525
Stage II s/p		2,131,266	1.97	1.66	42,080	35,314
subtotal		20,080,537	3.90	3.32	782,944	666,839
Total mine		18,412,519	4.14	3.53	762,330	649,785
Total s/p		2,399,421	1.99	1.67	47,748	40,070
TOTAL		20,811,940	3.89	3.31	810,079	689,856
Processing	Recovery	Tonnes	TCu%	ASCu%	total metal	AS metal
Stage I mine	60.0%	463,248	4.63	3.94	12,880	10,956
Stage I s/p	60.0%	268,155	2.11	1.77	3,401	2,854
subtotal		731,403	3.71	3.14	16,280	13,810
Stage II mine	91.9%	17,949,271	4.13	3.52	700,505	580,308
Stage II s/p	91.9%	2,131,266	1.97	1.66	43,876	32,450
subtotal		20,080,537	3.90	3.32	744,380	612,758
Stage II (plus F+E)	92.1%	1,777,795		2.17		35,767
TOTAL Stage II	91.9%	21,858,332		3.23		648,525

Table 25.12 Metal Balance, This Technical Report

Mining		Tonnes	TCu%	ASCu%	total metal	AS metal
Stage I mine		609,661	3.63	3.19	22,161	19,471
Stage I s/p		120,339	4.85	3.99	5,837	4,801
subtotal		730,000	3.84	3.32	27,998	24,271
Stage II mine		17,802,858	4.14	3.53	737,623	627,796
Stage II s/p		2,279,082	1.84	1.54	41,877	35,206
subtotal		20,081,940	3.88	3.30	779,500	663,003
Total mine		18,412,519	4.14	3.53	762,330	649,785
Total s/p		2,399,421	1.99	1.67	47,748	40,070
TOTAL		20,811,940	3.89	3.31	810,079	689,856
Processing	Recovery	Tonnes	TCu%	ASCu%	total metal	AS metal
Stage I mine	60.0%	609,661	3.63	3.19	13,296	11,682
Stage I s/p	60.0%	120,339	4.85	3.99	3,502	2,880
subtotal		730,000	3.84	3.32	16,799	14,563
Stage II mine	91.9%	17,802,858	4.14	3.53	678,106	577,140
Stage II s/p	91.9%	2,279,082	1.84	1.54	38,498	32,366
subtotal		20,081,940	3.88	3.30	716,604	609,506
Stage II (plus F+E)	92.1%	1,737,805		2.35		37,569
TOTAL Stage II	91.9%	21,819,745		3.23		647,075

25.1.10 Waste Dump Optimisation and Design

No further work on the optimisation and design of the waste dump has been completed. Hence, there is no change to the design commentary as presented in the March 2008 Technical Report.

The current mine schedule indicates a total oxide waste movement of approximately 13.8 million bcm (insitu volume), mined between 2010 and 2018. Due to the change in the optimal shell selection philosophy, this is a significant reduction in waste mining since that reported in the March 2008 Technical Report.

The design waste dump capacity (the proposed dump outline is shown in Figure 25.7) is considerably in excess of this volume. The dump was designed before the TSF layout was updated (Item 25.5). In view of the fact that the TSF embankments will now be downstream raised, rather than upstream raised, most of the mined oxide waste will be required for the TSF, as suggested by the total embankment construction quantities listed in Table 25.13 (KP, June 2009).

Based on the results of preliminary mine planning for Stage III, it is likely that the Stage III operation could commence before Stage II mining and processing is complete. This being the case, the existing waste dump design needs to be reoptimised and designed to cater for the encapsulation of potentially acid forming waste rock, whilst maintaining a supply of waste material for TSF embankment construction.

Table 25.13 Revised TSF Construction Quantities, June 2009

	Top RL (m)	Storage (Mt)	Zone A (m ³)	Zone C (m ³)	Zone D (m ³)	Wear Course (m ³)	Total (m ³)
Stage 1	1,267.8	1.6	137,000	538,000	41,000	3,000	719,000
Stage 2	1,269.7	3.2	39,000	266,000	4,000	3,000	312,000
Stage 3	1,271.4	4.8	36,000	298,000	4,000	3,000	341,000
Stage 4	1,273.1	6.4	36,000	356,000	4,000	3,000	399,000
Stage 5	1,274.7	8.0	34,000	390,000	4,000	3,000	431,000
Stage 6	1,276.3	9.6	35,000	683,000	4,000	3,000	725,000
Stage 7	1,277.9	11.2	35,000	555,000	4,000	3,000	597,000
Stage 8	1,279.4	12.8	33,000	579,000	4,000	3,000	619,000
Stage 9	1,280.9	14.4	34,000	636,000	4,000	3,000	677,000
Stage 10	1,282.3	16.0	32,000	647,000	4,000	3,000	686,000
Stage 11	1,283.8	17.6	34,000	751,000	4,000	3,000	792,000
Stage 12	1,285.2	19.2	32,000	757,000	4,000	3,000	796,000
Stage 13	1,286.5	20.8	30,000	752,000	4,000	3,000	789,000
Stage 14	1,287.9	22.4	33,000	863,000	4,000	3,000	903,000
Stage 15	1,289.3	24.0	33,000	920,000	4,000	3,000	960,000
Stage 16	1,290.7	26.2	33,000	2,756,000	4,000	3,000	2,796,000
			646,000	11,747,000	101,000	48,000	12,542,000

25.2 Infrastructure

The Stage II Project will benefit considerably from the infrastructure already established to support the mining, construction, processing operations and other activities for the Stage I Project.

25.2.1 Electrical Power Supply

The DRC is well serviced by major grids operated and maintained by *Société Nationale d'Electricité* (SNEL) which link the major hydro-generating capacity of the

DRC to its regional centres and neighbouring countries. AMCK has entered into a binding agreement with SNEL to connect Kinsevere to the 220 kV grid which is used to export power to Zambia and Zimbabwe. This grid is the most reliable electrical distribution network operated by SNEL and has sufficient excess capacity to allow for the development of the 60,000 tpa SX-EW plant.

During Stage I a 120 kV powerline was constructed from the Lubumbashi airport to a 120-33 kV switchyard adjacent to the EAF site, over a distance of approximately 26 km. The line size and conductors used in the construction of this line were sized for a 60 MW load which is capable of providing power to both the 60,000 tpa SX-EW facility (40 MW) and the 25,000 tpa EAF (15 MW). This line was connected to the existing 120 kV grid via a temporary link.

As part of the Stage II development, the temporary link to the 120 kV grid has now been changed to a permanent link to the 220 kV grid which is much more reliable. This was achieved through the construction of a new 220 kV line from the Karavia sub-station in Lubumbashi adjacent to the Gécamines smelter which feeds a new substation constructed adjacent to the Lubumbashi airport.

Power would be stepped down to 120 kV before being fed to the Kinsevere 120 kV grid. The construction of the new 220 kV link and substation has been carried out through a Joint Venture between AMCK and Ruashi Mining owned by Metorex. The Ruashi Mine is also fed from the new substation. Though the Joint Venture has funded the construction of the link, it will recoup the cost of the 220 kV link through a tariff rebate.

25.2.2 Electrical Distribution

The 120 kV incoming line will be stepped down to 33 kV through 3 x 30 MVA 120 / 33 kV transformers at the mine site substation, where it will be reticulated to the various Motor Control Centres (MCC's) around the site by overhead and underground lines. Stage II will require the construction of a second 33 kV grid approximately 3.6 km in length at the mine site to service the SX-EW plant. This 33 kV grid will link up with the existing grid to form a ring feed.

All MCC buildings will be prefabricated modular style sealed from dust with insulation and airconditioning.

Step-down power transformers (33 kV to 525 V) will be located in a fenced enclosure adjacent to the associated MCC building and will feed the MCC incoming circuit breakers via short cables. The MCC's will be separated from the transformer compounds by concrete walls to provide a two hour fire rating. Smoke and/or thermal detectors will be installed in each substation, and connected to the central plant alarm siren.

25.2.3 Power Demand

The Stage II maximum demand for the 60,000 tpa SX-EW Plant is around 40 MW. The electrowinning cells contribute 72% of this demand while the 525 V MCC's load contributes 19% of the demand, and the SAG mill the remaining 9%.

Allowances have been made in the design and costing of the electrical distribution network for the installation of power factor correction which is designed to improve the power factor to 95%.

25.2.4 Plant Electrics and Instrumentation

The 33 kV supply from the 33 kV switchboard will be stepped down to 525 V for reticulation to the various MCC's around the site by overhead line and above ground cable runs. Two standard 33 / 0.525 kV transformers are proposed to minimise spare inventories. The 33 kV switchboard will also provide power for the electrowinning rectifiers as well as the power factor correction unit. An 11 kV voltage supply has been selected for the SAG mill motor.

The Kinsevere Stage II plant site will consist of the following electrical rooms:

- 33 kV main switchroom
- Switchroom 1
- Switchroom 2
- Switchroom 3
- Rectifier room 1
- Rectifier room 2
- Plant control room

By selecting 33 kV as the plant distribution voltage, the power supplies to all MCCs are transformed directly from 33 kV to 525 V. This arrangement is similar to Kinsevere Stage I which decreases the overall design cost and reduces maintenance and equipment spares requirements. The bulk of the plant load will be consumed by two sets of electrowinning cell units. The DC supplies to the electrowinning cell units will be supplied from the rectifiers. Each rectifier comprises 33 kV - 385 V one star, one delta primary, and dual star secondary transformers interconnected by busbar to a 12 pulse rectifier.

The existing Stage I power generators will become the Stage II emergency power, although these will not be adequate to continue production. Drives for equipment such as the CCD thickener rakes and slurry tank agitators require emergency power while other non-essential drives are switched off during power outage to meet the emergency load demand.

There will be an HV switchboard for the 33 kV plant distribution to the 525 V MCCs via 33 / 0.525 kV transformers, and another for the 11 kV supply to the SAG mill motor. Two sets of rectifier units will be designed to provide the DC supply to two separate electrowinning cell modules. There will be a number of variable speed drive (VSD) units ranging from 2.2 kW to 350 kW. These VSDs will be floor or wall mounted, dependant upon size along an internal wall of the respective MCC building.

The process plant will be controlled by Programmable Logic Controllers (PLCs). A total of 8 PLCs will be provided in the plant: one in switchroom 3, four in switchroom 1, two in switchroom 2 and one in the 33 kV main switchroom. All PLCs will interface to the SCADA system. The SCADA system will provide plant operators with the means to monitor and control the plant. The operator station will be located in the plant control room.

The plant will be instrumented only to the level needed for successful operation and control. The instrumentation estimate was based on the instrument list developed from the process flow diagrams (PFDs) and the plant Control Philosophy.

25.2.5 Water Supply and Management

Introduction

The various plants facilities and associated tailings storage facility have been designed such that all process water will be recycled where possible. Safeguards such as overflow dams and neutralization ponds have been incorporated into the design to ensure that zero discharge of toxic or polluted water occurs from the Project area.

Raw Water

Hydrogeological drilling in areas adjacent to the open pits has identified that a significant aquifer is available within the mining lease and that all raw water could be sourced from a dewatering borefield. It is planned that a number of large diameter bores will be drilled within the mining lease to supply most of the raw water requirements of the mine, which can be backed-up with water sourced from the Kifumashi River via a pipeline 200 mm in diameter and 1.8 km in length.

Process Water

Process water will be stored in a process water pond situated adjacent to the processing plant which was constructed during the Stage I development. The pond has a capacity of 50,000 m³ which will provide a reservoir sufficient for at least 48 hours production. Additional water storage capacity has also been constructed as part of the Stage I development namely the raw water pond (17,000 m³) and the return water dam (100,000 m³).

The process water pond leaks and consequently a remediation cost for lining has been included in the Stage II capital costs.

Potable Water

It is estimated that the Project will require in the region of 500 to 1,000 m³ of potable water on a daily basis. This will be sourced from the operating bores within the Project area and has been completed as part of the Stage I development. Raw water sourced from these bores is presently pumped to a 6,000 litre potable water tank where it is distributed to the camp, office block and process plant. As part of Stage II, a second potable water tank will be constructed to supply water to the main camp.

25.2.6 Communications

The satellite line service established for Stage I will be extended for Stage II. There is a local service provider in the area providing good cell phone communication systems using the tower established for Stage I. Communications (voice, fax, data and email) to and from the Project site will be through a PABX system. Within the plant/mine areas, an internal short wave radio communication system will be used.

25.2.7 Site Access

During Stage I development, an existing 8 m wide road was rehabilitated to link the Project area to the Lubumbashi – Likasi road. As part of the Stage II development, the existing maintenance road adjacent to the 120 kV powerline has been upgraded to all-weather status from the mine to the Lubumbashi airport. The road is 10 m wide and constructed from laterite. At this stage, no allowances have been made to cover the cost of bituminising the road in the capital estimate for Stage II.

The new road crosses the Muomba River approximately 10 km to the south of the mine and a tributary of this river approximately 15 km to the south of the mine. Armco culverts have been installed at these crossing points.

Due consideration has been given to the movement of transport vehicles in the plant to minimise the volume of heavy traffic through the main operational areas. All roads will be 6 m wide with a 1 m shoulder on each side and will have a granular surface.

The closest airport is at Lubumbashi about 30 minutes drive from the site, which has multiple connections with other airports in the DRC, Kenya, Zimbabwe, Zambia and South Africa.

25.2.8 Infrastructure Buildings

The following existing Stage I buildings will serve the Stage II Project without requiring further work:

- Administration offices
- Technical Services and Training Building
- Site laboratory
- Mine store
- Government inspection building
- Process plant workshop
- Light Vehicle Workshop
- Dayworkers mess (HMS Plant)
- Generator Building
- Electric Arc Furnace (EAF) store
- EAF workshop
- EAF laboratory
- EAF plant office
- Dayworkers mess (EAF site)
- Mining contractors workshop, office and store

Stage II development will require these buildings to be supplemented by the following buildings:

- Fitting workshop (12 m x 28 m)
- Electrical and instrumentation workshop (12 m x 28 m)
- Heavy vehicle workshop (12 m x 28 m)
- SX-EW metallurgical laboratory
- SX-EW plant change room and ablution block
- Expansion of the dayworkers mess
- Reagent storage area
- Plant operations offices
- Expansion of mine stores

Offices and change houses are pre-fabricated modular style of polycore foam panel construction for speed and economy. The stores and workshop buildings will be fabricated with corrugated zincalume (spandek) walls and roofing on a steel frame, similar to the Stage I buildings. Floors will be broom finish monolithic concrete. All buildings will be built very early in the Project such that they are available as offices and stores for the construction phase.

A mine equipment maintenance workshop is provided for the mining contractor.

25.2.9 Accommodation Camps

There are two accommodation camps at Kinsevere, namely the contractors' camp and the AMCK operations camp.

The contractors' camp is situated adjacent to the plant site and can accommodate the estimated Stage II construction work force and other subcontractors. The new AMCK operations camp is situated approximately 1 km from the mine site within the mining lease and consists of 40 x prefabricated wooden cabins, a mess, recreation facility with attached bar and laundry facility.

25.2.10 Fuel Storage

Diesel fuel for the stand-by power generator and site vehicles is stored in two above-ground horizontal tanks coupled to a kerb-side metered dispensing bowser for the vehicle refuelling. Fuel is delivered in conventional road tankers. Diesel off-loading uses a dedicated pump located close to the storage tank.

25.3 Processing Operations

25.3.1 Stage I Process Flowsheet

Items 8.3 and 20.1 describe the removal of the EAF from the Stage I flowsheet and the impact on current operations.

25.3.2 Stage II Process Flowsheet

Introduction

The physical and chemical properties of the Kinsvere oxide ore favour the adoption of the conventional process route of acid leaching, solvent extraction and electrowinning as practised at many of the nearby mines on the Zambian copperbelt. A design production capacity of 60,000 tpa has been considered appropriate for the current Mineral Resource and Mineral Reserve inventories, though the plant layout allows for future expansion to 120,000 tpa.

Design Features

A significant decision taken during the plant design process was to reduce the capital cost by eliminating most standby pumps from the process flowsheet. This would inevitably adversely affect plant utilisation and efficiency to some extent. It has subsequently been decided to have standby slurry pumps for the cyclone feed and tails disposal duties. However no standby thickener underflow pumps have been specified, as individual thickeners can be temporarily bypassed. Neither will standby units be provided for most solution pump duties.

In a copper leach plant, the water balance for the processing system has a significant effect on plant operating costs. Fresh ore has to be milled prior to leaching, and in a conventional milling circuit the cast iron mill requires that fresh water is used during the grinding process to prevent corrosion. This represents a continuous significant volumetric input to the system. The fresh water in the ore slurry, after passing to the leaching circuit has subsequently to be acidified to decrease the pH down to 1.5 to dissolve the copper. After leaching, the copper solution is decanted and washed from the slurry, and the residual acid in the tailings requires to be neutralised prior to disposal into a tailings dam.

There are also other points where fresh water is added to the system, including feed moisture, flocculant addition and pump gland service water. This water also becomes

acidified and any solution accumulation in the process plant will require excess liquor to be bled from the system, via the neutralisation plant, to the tailings dam.

The net result is that considerable quantities of sulphuric acid, lime and limestone would be required to operate the leach plant

To overcome this problem, the Stage II design allows acidic raffinate from the solvent extraction plant to be used as liquor addition to milling (i.e., 'milling in raffinate' (MIR)) which enables the addition of fresh water to be minimised and allows more productive use to be made of the quantity that must be added. Any raw water make-up to the system enters, not at the mill, but via the CCD wash circuit which lowers both the copper and sulphuric acid concentration of the slurry passing to neutralisation and tails disposal. Consequently the usage of acid, lime and limestone is reduced and the recovery of copper is slightly increased.

A process model developed for the Feasibility Study Preliminary Report (AMCK, 2007) indicated that the benefits of incorporating a corrosion resistant mill into the flowsheet included:

- reduced acid consumption (9,925 tpa saving);
- decreased soluble copper losses (Additional 54.2 tpa); and
- lower fresh water addition (reduction of 437,000 m³/a).

The Stage II plant is designed to discharge non-neutralised tailings directly to the tails dam. This 'direct tailings disposal' (DTD) negates the need for addition of lime and limestone for neutralisation and allows excess acidic solution from the dam to be returned to the plant. A major benefit of this flowsheet is that additional copper dissolution occurs in the tails dam which reportedly increases overall plant recovery by 1 to 2%.

Crushing

Fresh core samples of the three orebodies indicated that the ore is extremely weathered over most of the oxidised zone, even at depths of 100 m. Conventional crushing testwork indicated very low values for both the ICWi and UCS.

The material is also very fine and can be expected to have high water contents in the wet season (>20% moisture). This type of material is not suitable for size reduction in a conventional jaw crusher. The ore will tend to stick to the jaws and the throughput rate will be low and irregular.

The most suitable crushing device for these conditions is a mineral sizer. This consists of two rotating shafts equipped with rows of teeth. Unlike a jaw crusher, the unit does not rely on gravity for material to pass down through the crushing chamber. Instead the sizer grabs the ore with the rows of teeth and forces it between the shafts, similar to a pumping action. Also as per pumping, the throughput capacity is proportional to the shaft speed and consequently the presence of wet sticky material does not hinder the sizer performance. The machines can also crush all but the hardest rocks, coping with UCS values of 140 to 250 MPa, dependent on specification.

Sizer selection and the installation design are important. The sizer must be capable of gripping the ore in its teeth. If the ROM feed is too large it will sit on top of the shafts and can cause high tooth wear. However the Abrasion Index of the ore is very low so wear rates on the sizer are expected to be moderate. A 625 series sizer with three teeth per row has been selected for Stage II, with shaft centres 625 mm apart.

This will allow ore lumps up to 750 mm in diameter to be gripped. In contrast, the smaller 500 mm sizer can only handle rocks around 500 mm in diameter.

The Stage II sizer will be fed by a variable speed apron feeder which will provide effective feed-rate control even when the material is wet.

Originally, the preliminary plant design had a large coarse ore stockpile located between the crushing and grinding sections to help maintain a steady feed to the milling and leaching sections. However the live capacity of the stockpile would have been considerably reduced when the ore was wet and sticky. This stockpile facility was removed as part of the cost cutting exercise with a direct mill feed option being chosen.

Milling

Tests performed to size a semi autogenous grinding (SAG) mill indicated that all three deposits were characterised by very soft ore. Rod and ball mill work indices were generally modest.

The Tshifufia orebody consists of soft laminated siltstone shale of low competency. It was considered a fairly uniform orebody with little sign of increase in hardness with depth (although Tshifufiamashi does exhibit this to an extent). The ore is soft, clayey and friable. It will not provide good grinding media and therefore will need a high operating ball charge in the mill. The Abrasion Index is very low, hence wear is expected to be minimal. Grinding power requirements will be moderate to achieve the coarse product size selected for downstream processing.

Initially the throughput rate was set at 217 tph and the grind size P_{80} – 150 microns (as there was some evidence that Tshifufiamashi ores benefited from a finer grind). Also as EW current density can be expected to progressively improve as operating staff better understand the relevant processing issues, it was anticipated that the mill would be required to progressively treat higher tonnages later in the Project. Allowing the grind to coarsen to P_{80} of 212 microns would have enabled this to occur without significant copper loss in leaching.

For this original scenario, a 5.79 mØ x 6.69 m effective grinding length (EGL) mill fitted with a 3.25 MW variable speed drive was selected. The mill motor was expected to be power limiting and therefore operation at higher ball charges would have required a speed reduction to ensure the design motor power draw was not exceeded. Consequently a 3.5 MW drive was considered a better selection for operation at full speed and high ball charges.

Following a review to reduce capital costs, the specified grind P_{80} was revised to 212 microns as the leach recovery was not expected to be seriously affected. As a result, the recommended mill size was reduced to 5.79 mØ x 5.79 m (EGL) operating with a 15% ball charge and equipped with a 2.55 MW motor.

The ultimately selected unit is a Primary Ball Mill rather than a conventional SAG mill. The ball size was selected as 65 mm, which allowed rubber liners to be specified which will not require a liner handling machine. Larger diameter balls (up to 90 mm) can be used if the mill feed becomes more competent. To allow for milling in raffinate a corrosion resistant mill is required.

Control of Water Balance

In a conventional leach circuit a pre-leach thickener is used to control the input of fresh water into the leach circuit and limits the need for a liquid bleed from the

system. With the adoption of the milling in raffinate concept the water balance control was significantly enhanced (compared to conventional leach circuits) with the opportunity to re-cycle a portion of the raffinate into the milling circuit. Consequently the pre-leach thickener was removed from the original flowsheet.

Leaching

Leach testwork indicated that the ASCu recovery was slightly increased (~2.4%) if leaching was extended from 4 to 8 hours. Consequently 8 hours was selected as the initial residence time but this was subsequently decreased to 6 hours to reduce capital expenditure.

A leach pH of 1.5 was selected for the design criteria. This can be optimised after commissioning as leach recoveries were only slightly decreased at pH 2.0. This would also decrease acid consumption marginally.

A design leach extraction for acid soluble copper of 93% was selected for process design.

Post Leach Solid/Liquid Separation

The Kinsevere Options Study (Anvil, 2006) considered the requirements for the Stage II SX-EW plant and included belt filtration as the predominant means of post leach solid/liquid separation. However, this concept is not in vogue in many of the African leach/SX-EW plants. The modern leach plants, such as Kansanshi, tend to rely on CCD thickeners as they are more reliable, easier and cheaper to operate. Consequently thickeners have now replaced belt filtration in the Stage II equipment list.

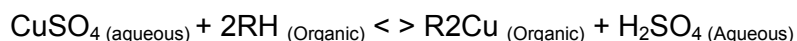
Clarification

Pinned bed clarifiers are used in most of the Zambian leach plants to minimise suspended solids in the PLS feed to the SX modules. For Kinsevere Stage II, one such unit will be installed ahead of both the high grade and low grade SX circuits to ensure that crud generation in the mixer-settlers is minimised.

Solvent Extraction

A recent process flowsheet development in solvent extraction is the 'split SX' concept designed to reduce acid consumption and increase copper recovery. The PLS is separated into high grade and low grade streams ahead of the SX plant which consists of two independent modules.

The high grade stream (containing 75% of the copper) has the copper extracted according to the following general SX reaction:



The reaction is reversible, showing *extraction* to the right and *stripping* to the left. As the copper is extracted from the aqueous PLS, sulphuric acid is formed. Consequently due to the elevated copper tenor in the high grade SX feed, after the copper is removed into the organic phase, the acid strength of the resulting raffinate is correspondingly high. This is therefore directed back to leaching.

The high copper feed grade also results in a relatively high raffinate copper tenor, but this is not important as the solution is recycled to leaching.

Conversely, the low grade PLS generates a raffinate with less acid and lower copper levels. This is better suited for use as wash solution in the CCD wash thickeners and

minimises the levels of acid and copper that are lost to neutralisation and tailings. In contrast, a standard SX plant treating a combined PLS would have higher acid and copper levels in the CCD wash liquor, directly leading to higher acid consumption and lower copper recovery.

The standard split circuit design uses two separate SX modules, which allows a significant additional benefit to the project layout and project risk profile to be realised. The modules can be located sufficiently far apart to provide assurance that if one module suffers a fire event, then the other unit will remain unaffected. The solution flow configuration of the remaining SX module can also be adapted to a 'Series/Parallel' arrangement (after the pipework has been modified), which will enable plant production to continue at around 75% of design capacity.

Crud Treatment

Until recently a plate and frame filter press has been standard equipment for both the recovery of organic from crud and the routine clay treatment of organic inventory to counter the effect of surface-active contaminants which interfere with the physical and chemical performance of solvent extraction. However the procedure is a rather inefficient batch process.

In Zambia there has been a move away from filter presses and Bwana Mkubwa, Kansanshi, Chingola and Mufulira have all installed tricanter centrifuges which efficiently separate the crud or organic/clay mix into solid, organic and water streams. The units are considered extremely reliable and can operate in a continuous mode.

Consequently a tricanter centrifuge has been included in the Stage II equipment list.

At Kansanshi an oleophilic rope mop skimmer is used to recover organic from raffinate. An absorbent polyethylene rope is attached to pulleys and stretched across the organic reclaim pond. It slowly and continually passes through a squeezing mechanism which removes the recovered organic which is then pumped to the centrifuge. The equipment was originally designed to recover oil floating on water. This will be a useful tool for the Stage II Project, as the extractant content of the organic phase is relatively high and maximising the organic recovery makes economic sense.

Electrowinning

Standard multi-media filters, incorporating anthracite and garnet, were included in the flowsheet to remove solids and residual organic from the strong electrolyte exiting the strip circuit. The design has included sufficient multi-media units to allow filtration operations to continue when one unit is performing a routine backwash cycle. The alternative use of flotation cells to recover organic have not proved popular on the Zambian plants that were visited by Anvil personnel.

The largest electrowinning cells recommended by Xstrata technology, with 69 cathodes, were used to minimise capital cost and reduce the length of the tankhouse building.

A conservative Current Density of 290 Amps per m² was selected to help ensure that high quality cathodes are produced at plant start-up. It is realised that as the operations crew gain a thorough understanding and control of the plant parameters which affect cathode quality, then higher current densities and correspondingly higher production rates will be feasible in later years (provided a sufficiently large rectifier is installed initially).

A manually operated crane and semi automatic stripping machine, complete with plate flexing device, have been selected to keep the technology and maintenance requirements relatively simple.

Neutralisation

With the adoption of direct disposal of non-neutralised tailings the tailings neutralisation circuit was removed from the original flowsheet.

Tailings Disposal

The initial starter embankment for the tailings storage facility (TSF) was completed in 2008.

The original design of the four-sided dam included initial starter wall construction, with subsequent 'downstream' wall raising on northern and eastern walls, and with the 'upstream' wall raising technique employed on the southern and western walls.

The design of the TSF has now changed to incorporate downstream raised embankments on all four sides. This is because, under the direct tailings disposal concept, a HDPE liner must be carried up the internal wall of the TSF (refer to Item 25.5.1).

Plant Layout

Consideration was given in the plant layout to minimise the problems associated with expanding the plant production capacity in future. Some major items, such as the mineral sizer, will have sufficient capacity to handle double the planned Stage II feedrate. Space has been left to retrofit a second grinding mill to operate in a SAG/Ball combination.

Similarly space has been left for installing additional thickeners, leach tanks and SX modules together with expanding the electrowinning tankhouse.

Figure 25.13 shows the latest site layout plan.

25.3.3 Process Design and Description

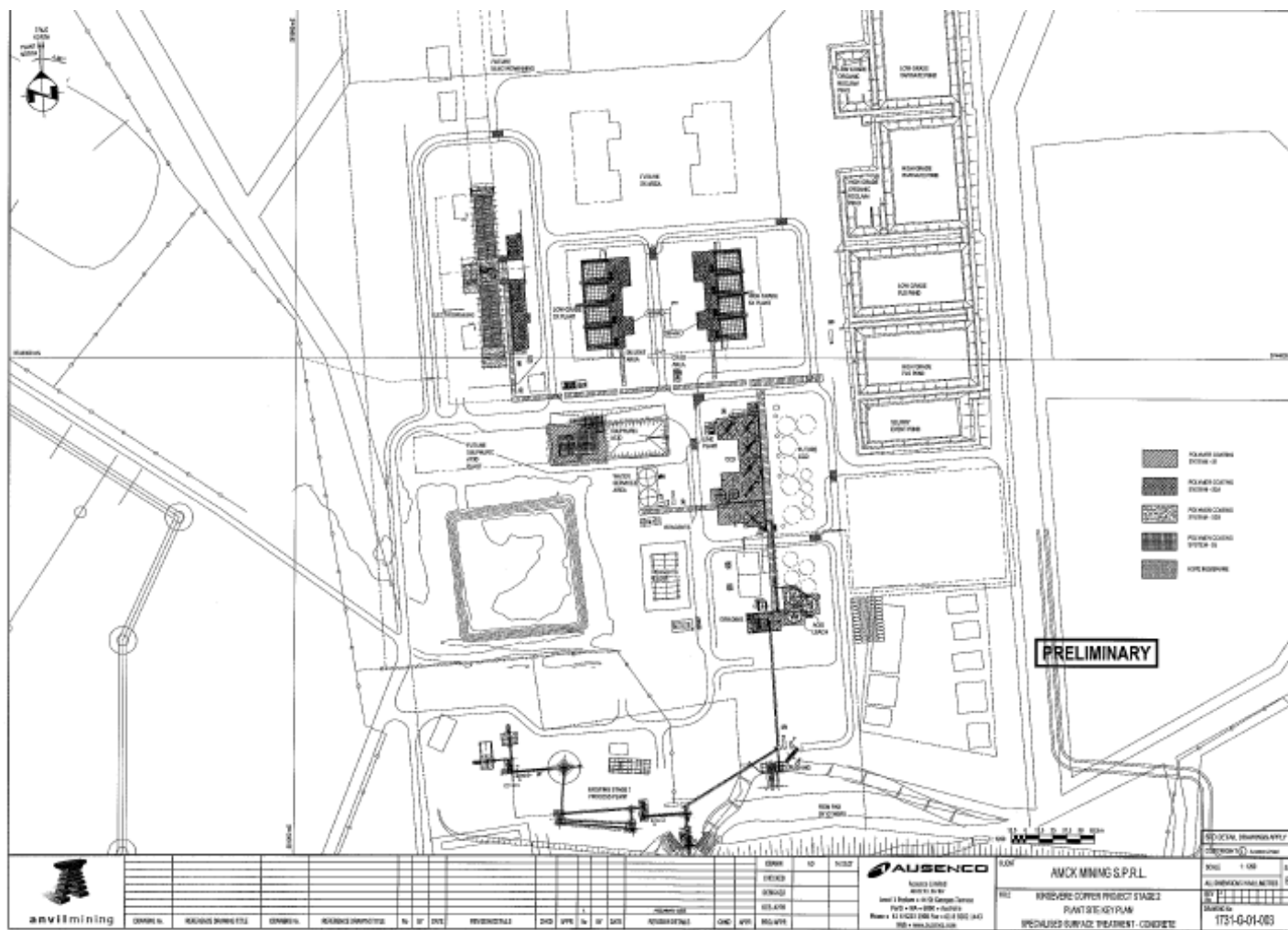
Process Design Criteria

The final key design parameters for Stage II include:

- | | |
|-------------------------|--|
| ▪ Ore Treatment Rate | 1,619,900 tpa |
| ▪ Ore Grade | 4.02% Acid Soluble Cu |
| ▪ Grind P80 | 212 microns |
| ▪ Plant Utilisations | 92% (Crushing)
92% (Milling & Leaching)
94% (Electrowinning) |
| ▪ Leach pulp density | 18.5% w/w |
| ▪ Leach Copper Recovery | 93% (Acid Soluble) |
| ▪ Leach Residence time | 6 hours |
| ▪ Leach pH | 1.5 |

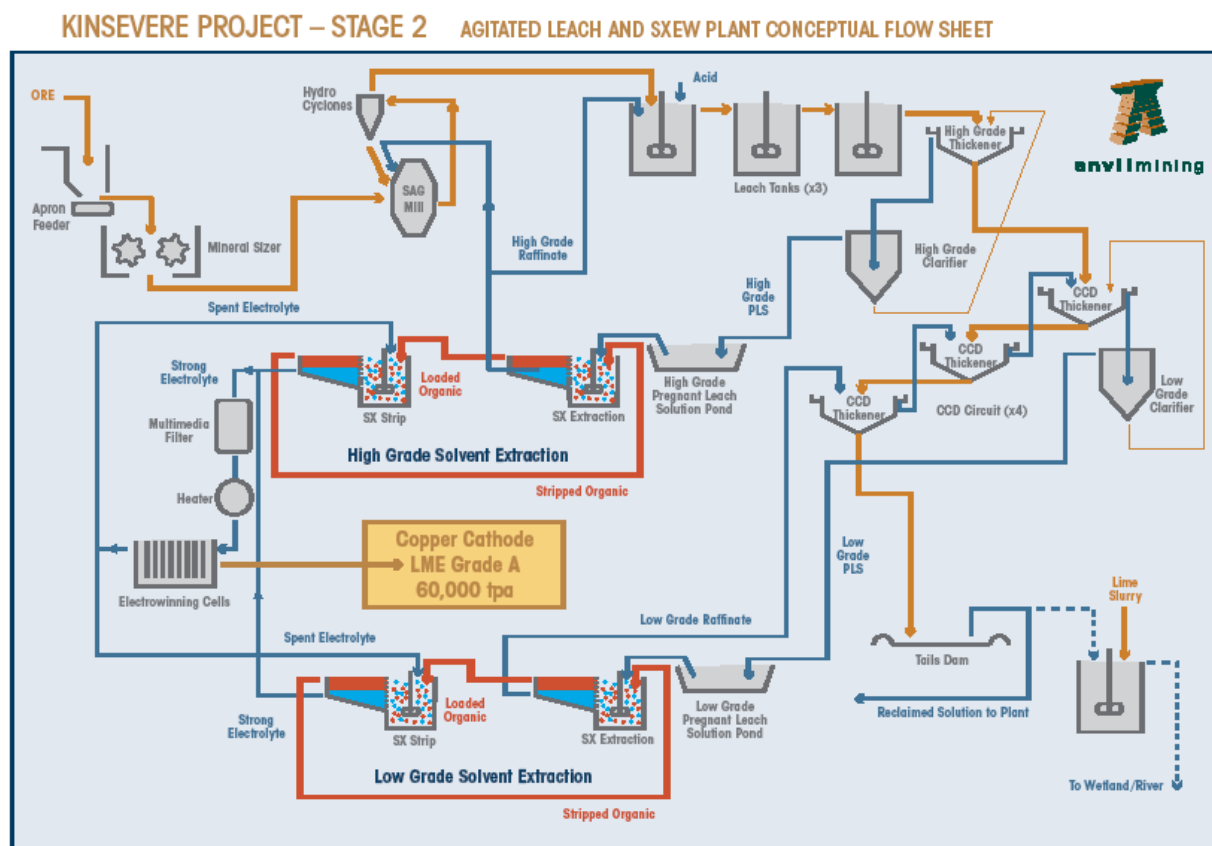
Based on metallurgical simulation results, the overall recovery as adopted for pit optimisation and economic analysis was 92.14%.

Figure 25.13 Stage II Plant Layout Plan



A schematic flowsheet of the Stage II plant is shown in Figure 25.14.

Figure 25.14 Stage II Schematic Flowsheet



The grinding mill will be a 2.55 MW variable speed Primary Ball Mill (5.79 mØ x 5.79 m EGL) containing 15%v/v ball load. The mill shell internals are coated with epoxy to provide a corrosion resistant lining enabling the ore to be slurried with high grade acidic raffinate. The mill will operate in closed circuit with hydrocyclones which will discharge cyclone overflow at 30% pulp density direct to the leach circuit.

Initially, the leach product will discharge to the High Grade Thickener which will generate high grade PLS as its overflow. This will be treated in a Pinned Bed Clarifier to minimise the levels of entrained solids reporting to the high grade SX plant.

High grade thickener underflow will then be pumped to a series of five CCD thickeners which will utilise low grade SX raffinate as the wash stream to maximise soluble copper recovery from the slurry. The overflow from CCD 1, termed low grade

PLS, will be clarified prior to treatment through the low grade SX plant. CCD 5 underflow will pass to the TSF. All five CCD thickeners will operate with underflows with pulp densities of approximately 60% solids which will also help maximise soluble copper recovery.

The underflow from CCD 5 will be diluted to 40 to 45% solids prior to discharge to the TSF using recycled solution from the tails dam. This is necessary to reduce the effect of rheology on the pumping system.

The solvent extraction circuit will consist of two separate modules operating in parallel. The high grade PLS (11 g/l copper) containing around 75% of dissolved copper will report to the high grade SX plant while the low grade PLS (4 g/l copper) will be treated in the low grade SX module.

The two SX modules will each consist of four conventional Mixer-Settlers configured as two extract and two strip stages operating in series. A standard ketoxime-aldoxime organic copper extractant will be used in a high flash point kerosene diluent. The SX plants will chemically purify and concentrate the PLS copper solution, while rejecting other anions and cations such as iron, manganese, calcium and chloride.

The individual raffinate streams discharged from the two SX modules will remain separate allowing the more acidic high grade raffinate to be recycled to dilute incoming leach feed while the less acidic and lower copper tenor low grade raffinate will be recycled as the CCD wash stream. This routing of the two raffinate streams will ensure that the amount of copper and acid passing out as underflow from CCD thickener 5 to the tailings storage facility will be minimised.

The combined strong electrolyte streams from the two SX modules will be pumped through multi-media filters ahead of electrowinning. The filters will contain garnet, sand and anthracite, to remove any solids and entrained organic which would adversely affect the copper cathode quality.

The electrowinning tankhouse has been designed with two identical wings, each with their own electrolyte circuit, with a centrally located semi-automatic stripping machine serviced by an overhead crane that harvests the cathodes manually for stripping. Two 48,000 A rectifiers will supply a direct current to both sides of the tankhouse.

In the tankhouse, electrowinning will take place at a Current Density of 290 Amps/m² and a Current Efficiency of 90%. A bleed stream of electrolyte will be returned to the leach circuit, as necessary, to maintain the iron level in electrolyte below 2 g/l to maintain an adequate current efficiency. The stainless steel cathode blanks will be stripped of their deposited copper on a seven day cycle. The harvested copper cathodes will be sampled and bundled prior to export.

The process plant will be operated from a single control room using a Supervisory Control and Data Acquisition (SCADA) System, a standard feature of the modern Zambian leach/SX plants. The process plant will also include the following facilities:

- Flocculant & coagulant preparation and dosing
- Sulphuric acid storage and dosing
- Diluent and extractant storage for the SX plant
- Fire water system
- SX and slurry event ponds.

The design of reagent storage facilities has taken cognisance of the logistical difficulties involved in maintaining a reliable supply to the DRC. Consequently on-site storage facilities have been designed to handle between two and three months supply of all reagents.

SX Fire Protection

The SX plant will be equipped with a comprehensive fire monitoring, detection and alarm system.

The plant fire water system is not designed to fight fires in the two solvent extraction circuits. Following a risk review, and consideration of major fires that have occurred in similar facilities, if an ignition incident occurs in either of the SX circuits and manual foam extinguishers cannot quickly extinguish the fire, then personnel will be evacuated and the fire will be allowed to continue until the available fuel is consumed.

In the event of a fire and unplanned drainage of one or more mixer settlers, the organic and aqueous liquors will initially pool inside the SX bund and then drain via a fire trap to the SX event pond. The fire trap is designed to starve the liquor of oxygen, and extinguish any fire, as it passes through.

If an SX module is lost due to fire, then the remaining module can be reconfigured into Series/Parallel operation to enable plant operations to continue at approximately 75% of capacity.

25.4 Recoverability

25.4.1 Kinsevere Stage I

Items 8.3 and 20.1 describe the removal of the EAF from the Stage I flowsheet and the impact on Stage I recoverability.

25.4.2 Stage II Metallurgy

The metallurgical testwork for the Stage II Project was predominantly performed at Independent Metallurgical Laboratories (IML) in Perth, Western Australia with specialised testwork and analyses organised at other leading facilities elsewhere in Australia.

The testwork was performed between July and December 2006 based on a programme originally developed by Internet Engineering. Anvil Mining, Miller Metallurgical Services and Lycopodium Engineering provided direction as the various testwork phases progressed.

The programme was divided into two sections. Initial testwork was performed on master composites representing each of the three deposits. This work included optimisation of the operating parameters affecting leaching; grind size, pH and leach duration. Subsequently various tests were repeated on a series of full diamond core samples from the three deposits to investigate variability through the orebodies. This full diamond core was also used to provide suitable samples for SAG milling testwork.

Mineralogy

Samples of the Master Composites (refer to Item 18) were screened at 0.1 mm and the fractions subjected to Heavy Liquid separation (HLS) using tetra-bromo-ethane

(SG – 3.0). Polished sections of the individual sinks fractions were then produced prior to examination by Scanning Electron Microscopy (SEM).

The results of the SEM analysis is summarised in Table 25.14.

Table 25.14 Summary of SEM Analysis

	Tshifufia		Tshifufiamashi		Kinsevere Hill	
	+0.1mm Sinks	-0.1mm Sinks	+0.1mm Sinks	-0.1mm Sinks	+0.1mm Sinks	-0.1mm Sinks
Goethite FeO(OH)	Maj	Maj	Maj	Maj	Min	Maj
Malachite Cu ₂ (OH) ₂ (CO ₃)	Maj	Maj	Maj	Maj	Dom	Maj
Azurite Cu ₃ (OH) ₂ (CO ₃) ₂	Tr					
Cuprite Cu ₂ O	Acc	Tr	Tr	Tr		
Tenorite CuO	Acc		Tr			
Bornite Cu ₅ FeS ₄	Tr					
Chalcocite Cu ₂ S	Acc	Min	Acc	Acc	Tr	
Chalcopyrite CuFeS ₂	Acc	Min	Tr	Tr		
Covellite CuS		Acc		Tr		Tr
Digenite Cu ₉ S ₅	Tr	Acc	Acc	Acc	Tr	Tr
CopperCu				Tr	Tr	
Gold	Tr					
Hematite Fe ₂ O ₃			Tr	Acc		
Mn Oxides			Acc	Acc		Tr
Pyrite FeS ₂		Min	Acc	Acc	Tr	Acc
Ti Oxides				Acc	Acc	Acc

Key:

- Dominant >50%
- Major 20-50%
- Minor 10-20%
- Accessory 1- 10%
- Trace <1%

Malachite is the predominant copper mineral present in all composites.

Significant chalcopyrite was present as discrete grains up to 1 mm in the Tshifufia sample. Based on previous mineralogical examination, this is considered to be due to the inclusion of discrete zones of sulphide mineralisation in the composite which should be separable during the mining process.

A previous mineralogical examination of ore from the three deposits in 2005 reported that the main copper minerals present were malachite and pseudomalachite, Cu₅(PO₄)₂(OH)₄, which has a similar CuO content to malachite (~70%). The pseudomalachite, which is soluble in sulphuric acid, was finer than the malachite in the dominant goethite and micaceous silicate gangue.

Several examples of a copper bearing heterogenite (cobalt hydroxide), were detected in a Tshifufia sample. X-ray diffraction analysis of these earlier samples identified the major constituents of the samples listed in Table 25.15.

Table 25.15 Major Constituents of the Kinsevere Deposits (Wt%)

Constituent	Tshifufia	Tshifufiamashi	Kinsevere Hill
Quartz	40-45	45-50	40-45
Chlorite	15-20	10-15	10-20
Biotite	10-15	10-15	10-20
Muscovite	20-25	15-20	15-20
Talc	<1	<1	<1
Malachite	4-6	6-7	3-4
Pseudomalachite	1-3	2-3	1-2

Master Composite Analysis

A comprehensive Inductively Coupled Plasma analysis was performed on each Master Composite. The key constituents are summarised in Table 25.16.

Table 25.16 Summary of Master Composite Analyses

	Units	Tshifufia	Tshifufiamashi	Kinsevere Hill
T Cu - Ultratrace	%	4.02	4.09	2.55
T Cu - SGS	%	3.94	3.97	2.41
Cobalt	%	0.19	0.15	0.01
Iron	%	3.23	4.64	3.94
Sulphur _{Total}	%	0.450	0.105	0.056
Sulphide Sulphur	%	0.20	0.09	0.05
Calcium	%	0.03	0.14	0.04
Magnesium	%	5.00	3.34	2.21
CO ₃ ²⁻	%	3.0	1.7	1.2
Cl	ppm	300	300	300
Silica	%	29.3	29.8	29.9
Manganese	ppm	185	804	143
Phosphorous	ppm	1200	1130	500

The relatively high level of phosphorous in the samples supports the presence of pseudomalachite which was reported in the preliminary mineralogical examination.

A Sequential Leach Analysis was also performed on the Master Composite samples to quantify the amount of each type of copper species present. Table 25.17 illustrates the results.

Table 25.17 Sequential Leach Analysis of Master Composite Samples

	Units	Tshifufia	Tshifufiamashi	Kinsevere Hill
Cu _{Acid Soluble}	%	3.30	3.80	2.37
Cu _{Cyanide Soluble}	%	0.420	0.165	0.030
Cu _{Residue}	%	0.220	0.133	0.116
Cu _{Total}	%	3.94	4.10	2.52
Cu _{Acid Soluble}	% of T Cu	83.8	92.7	94.2
Cu _{Cyanide Soluble}	% of T Cu	10.7	4.0	1.2
Cu _{Residue}	% of T Cu	5.6	3.2	4.6

The analyses indicated less Acid Soluble Copper in the Tshifufia composite than was expected. The nearly 11% of the copper extracted in cyanide suggests that Chalcocite (100% dissolution expected), Bornite (80%) or Covellite (100%) may be present. Approximately 20% of any Chalcopyrite that is present may also dissolve.

The higher Sulphide Sulphur levels of 0.2% detected in the detailed Tshifufia Master Composite analysis (Table 25.16) also indicates elevated proportions of Sulphide minerals being present. This is thought to be due to some transitional ore from around 100 metres depth being included in the Tshifufia Master composite.

The mineralogical examination detected accessory levels of both Chalcocite and Chalcopyrite in the Tshifufia composite. Significant Chalcocite was also identified in the Tshifufiamashi composite but not in the Kinsevere Hill composite.

The residue percentages, which often relates to the presence of refractory Chalcopyrite or impervious silicates, are similar for all composites (3.2 – 5.6%).

Size by assay analyses of the Master Composites indicated that there was a gradual decrease in copper grade with particle size. The Tshifufia sample assays varied from 1.53% copper for the minus 38 micron fraction up to 5.45% copper for the plus 11.2mm material. However economic upgrading of the sample by desliming did not appear feasible.

Physical Parameters

Determinations of the dry Specific Gravity of the various master composites were performed. The values reported are detailed in Table 25.18.

Table 25.18 Dry Solids SG of Master Composite Samples

Sample	Units	Tshifufia	Tshifufiamashi	Kinsevere Hill
1	-	2.84	2.58	2.69
2	-	2.70	2.70	2.65
Average	-	2.77	2.64	2.67

25.4.3 Comminution

SAG Milling Testwork

There are two main mechanisms of breakage in a SAG mill - Impact (high energy) and Abrasion (low energy). The Impact Breakage Parameters, A and b, are normally determined using the JK Drop weight test in which rock particles are broken under impact at various energy levels by dropping a weight from a pre-determined height.

For a wide range of energy inputs, particle sizes and ore types, the resulting various product size distributions remain similar in shape and can be identified by a single point on the distribution. The convention is to use the percentage passing one-tenth of the original particle size (t_{10}).

The Drop Weight test requires 120kg of drill core. However the SMC test has been developed to provide similar information from a much smaller quantity of drill core based on the Drop Weight index (Dwi). The Dwi is directly related to the JK breakage parameters, A and b, which are used to predict SAG mill performance. It is also related to the Point Load Index and the Unconfined Compressive Strength (UCS). Ideally a full JK Drop weight test should be performed for each batch of SMC tests to provide the most accurate correlation between the Dwi and A and b values.

The fresh drill core was examined by a representative of Orway Mineral Consultants to assist in selecting the samples for the JK Drop Weight and SMC tests. It was concluded that the core was extremely soft and that insufficient competent material was available to perform a full Drop Weight test.

Six separate samples of the most competent ore were selected from each of the four drill holes for SMC tests. The precise locations of these samples are detailed in Table 25.19. However three of these samples subsequently proven too soft and friable to test at JK Tech and are identified by shading in the table.

Table 25.19 Source of SMC Test Samples

	From (m)	To (m)		From (m)	To (m)		From (m)	To (m)		From (m)	To (m)
TF1-a	14.3	16.4	TF2-a	27.0	32.0	TFM-a	7.8	23.0	KS-a	15.4	30.5
TF1-b	31.0	33.2	TF2-b	43.0	52.0	TFM-b	24.6	26.5	KS-b	36.0	41.3
TF1-c	44.5	48.6	TF2-c	59.8	67.0	TFM-c	27.0	44.0	KS-c	41.3	48.7
TF1-d	57.0	60.6	TF2-d	73.8	78.6	TFM-d	44.0	49.0	KS-d	48.7	56.0
TF1-e	76.0	85.0	TF2-e	79.0	81.0	TFM-e	49.0	63.0	KS-e	69.0	77.5
TF1-f	86.2	88.5	TF2-f	87.6	90.0	TFM-f	83.5	91.0	KS-f	78.5	85.5

The SMC test results indicated that all three deposits were generally characterised by very soft ore. See Table 25.20.

Table 25.20 Summary of SMC Test Results

	Dwi kWh/m ³	A	B	A*b	t ₁₀ @1kWh/t
Average	2.0	61.4	3.33	237.9	55.7
Maximum	4.5	88	11.87	1016.4	88.0
Minimum	0.2	44.5	0.96	55.1	32.3

The lower the value of A*b, the harder is the ore and the more resistant it is to impact breakage. The lowest A*b value recorded, 55.1, is rated in the medium range.

Lower t₁₀ values indicate that less material passes one-tenth of the original particle size and that there is a higher resistance to abrasion breakage.

The average DWi, A*b and t₁₀ values are both rated as being “very soft” based on the JK Tech database.

The summary of JK Tech ratings for the individual A*b and t₁₀ values, based on historic testwork results, are summarised below in Table 25.21.

Table 25.21 Summary of SMC Test Ratings

Rating	A*b	t ₁₀
Medium	4.8%	9.5%
Moderately Soft	4.8%	-
Soft	38.0%	33.3%
Very Soft	52.4%	57.2%

Impact Crushing Work Index and UCS Determinations

Bond Impact Crushing Work Indices and Unconfined Compressive Strength (UCS) determinations were performed at Amdel on the most competent pieces of fresh core remaining after the SMC test samples had been extracted. The results are summarised in Table 25.22.

Table 25.22 Summary of Impact Crushing Work Index and UCS Results

Test		Tshifufia	Tshifufiamashi	Kinsevere Hill
ICWi – kWh/t	Average	2.7	1.4	3.5
	Maximum	8.0	4.1	6.7
	Minimum	0.7	0.8	0.7
UCS - MPa	Average	7.6	16.4	8.9
	Maximum	13.9	36.2	16.2
	Minimum	0.1	0.3	3.8

Rod & Ball Mill Work Indices

Various Rod Mill (RMWi) and Ball Mill Work Index (BMWi) determinations were performed on the Master and Variability composites. This included debris from the SMC tests which was used to determine BMWi values for these samples. Table 25.23 details the RMWi determinations and Table 25.24 the BMWi values.

Table 25.23 Summary of Rod Mill Work Indices (kWh/t)

	Tshifufia	Tshifufiamashi	Kinsevere Hill
Master Composite	13.1	11.7	10.3
Variability Samples			
- Average	12.8	12.5	10.1
- Maximum	13.4	13.5	10.6
- Minimum	12.0	11.7	9.3
- No. of Tests	4	3	3

Table 25.24 Summary of Ball Mill Work Indices (kWh/t)

	Tshifufia	Tshifufiamashi	Kinsevere Hill
Master Composite	9.1	8.9	6.6
Variability Samples			
- Average	10.0	11.7	7.0
- Maximum	13.9	16.1	8.5
- Minimum	8.0	8.9	5.9
- No. of Tests	14	4	6

The Ball Mill work Indices are generally typical of soft oxidised ore bodies.

Grindability of Fine Samples

Some of the samples were too fine for conventional BMWi testing. Those samples with F_{80} values less than 2 mm were subjected to Levin testwork, a technique that was developed to determine the grindability of fine sands. Even so one SMC sample had a F_{80} of only 0.3mm, which was considered too fine even for the Levin procedure. The Levin Test results are detailed in Table 25.25.

Table 25.25 Summary of Levin Tests

	Feed Sizing F80 - µm	Product Sizing P80 (µm) at Various Applied Energy Levels			
		2 kWh/t	4 kWh/t	6 kWh/t	8 kWh/t
Average	1542	263	127	91	73
Maximum	1967	493	183	116	91
Minimum	553	111	79	63	52

Abrasion Indices

Single determinations of the Abrasion Index were performed on the Master Composites (MC). The results are summarised in Table 25.26. Earlier determinations of this parameter for the Options Study (OS) are also included for comparison.

Table 25.26 Summary of Abrasion Index Results

Sample	Units	Tshifufia	Tshifufiamashi	Kinsevere Hill
MC	-	0.0293	0.0092	0.0094
OS	-	0.0203	0.0231	0.0098

All the Abrasion indices are extremely low by normal standards.

Site Water

A water sample from the Muombe River near the Kinsevere Project was analysed in South Africa and the results are compared with a typical Perth tap water in Table 25.27.

Table 25.27 Comparison of Site Water and Perth Tap Water

	Units	Muombe River	Perth Tap Water
Total Dissolved solids	mg/L	227	270
pH		8.26	7
Alkalinity - Bicarbonate	mg/L	239	20
Calcium	mg/L	47.5	4.3
Magnesium	mg/L	23.9	4.9
Potassium	mg/L	2.3	2.4
Sodium	mg/L	7.3	42
Iron	mg/L	<0.03	<0.05
Chloride	mg/L	5.19	83
Silicate	mg/L	18.4	6
Sulphate	mg/L	6.22	13

The site water was shown to be relatively good quality with no indication of dissolved elements or anions that were likely to be deleterious to the process.

As there were problems associated with importing site water into Australia, it was decided to utilise Perth tap water for the metallurgical testwork programme.

Knight Piesold completed a groundwater analysis study as part of the Project hydrogeological assessment. Groundwater samples from the aquifer characterisation boreholes were collected during the pump testing of the boreholes. Groundwater samples from the monitoring bores around the proposed TSF were also collected. The samples were delivered to a South African National Accreditation System (SANAS) accredited laboratory in South Africa where they were analysed for the following chemical constituents:

- pH, conductivity, total dissolved solids, total hardness, total alkalinity, total suspended solids, turbidity, dissolved oxygen, chemical oxygen demand, Ca, Mg, Na, K, CaCO_3 , HCO_3 , CO_2 , Cl, SO_4 , NO_3 , N, F, Hg, As, Se, Al, Ni, Mn, Fe, Zn, Pb, Co, Cu, Cr, Zr, Bi, Cd, Sr, P, Mo, Ba, Ag, Li

The results indicated that the groundwater in the Project area is a good quality calcium-magnesium-bicarbonate water type which is characteristic of dolomitic aquifers. A comparison against the World Health Organisation standards for drinking-water quality indicated a few elevated concentrations of aluminium, iron and barium.

25.4.4 Acid Leaching

Optimising Leach Parameters

To investigate the effect of Grind Size, leach pH and Leach Duration a series of leach tests were performed on each Master Composite.

The tests were performed at Grind P_{80} 's of 150 and 212 microns and the Leach pH was maintained at either pH 1.5, 1.8 or 2.0 for a period of 12 hours with sulphuric acid.

The leach results appear relatively consistent with recoveries for Total Copper after 12 hours, ranging as follows:

- Tshifufia: 76.8% - 79.6%
- Tshifufiamashi: 91.2% - 93.6%
- Kinsevere Hill: 90.5% - 92.3%

The Total Copper recoveries generally increased as both grind P_{80} and leach pH were reduced, although the trend was less marked in the Tshifufia tests.

The leach extractions for TF, based on total copper, were lower than expected but this appears to be due to a higher proportion of sulphide in the composite, as confirmed by the Sequential Analyses described earlier. A mineralogical examination of the Tshifufia leach tailings indicated that the predominant copper minerals present were chalcopyrite and digenite.

Table 25.28 summarises the leach results, based on Acid Soluble Copper recoveries after 12 hours.

Table 25.28 Summary of Acid Soluble Recoveries (% AS Cu)

Deposit	Leach Time (hours)	pH 1.5		pH 1.8		pH 2.0	
		P ₈₀ 150µ	P ₈₀ 212µ	P ₈₀ 150µ	P ₈₀ 212µ	P ₈₀ 150µ	P ₈₀ 212µ
Tshifufia	2	91.1	92.6	90.8	90.9	88.2	87.3
	4	93.8	94.8	93.0	92.4	91.6	93.2
	8	95.6	97.0	94.4	94.6	94.1	94.3
	12	96.8	97.3	95.9	96.7	95.6	96.4
Tshifufiamashi	2	97.1	90.8	95.4	85.8	92.9	84.5
	4	98.3	95.0	97.1	92.5	97.1	92.7
	8	99.3	97.0	97.7	96.8	96.9	95.9
	12	98.3	97.6	98.0	96.6	97.8	96.7
Kinsevere	2	88.4	87.6	87.1	88.2	84.3	82.9
	4	89.2	95.3	90.4	93.7	89.7	91.2
	8	94.3	95.4	95.0	95.1	95.4	96.2
	12	96.9	96.8	96.4	96.7	96.8	95.8

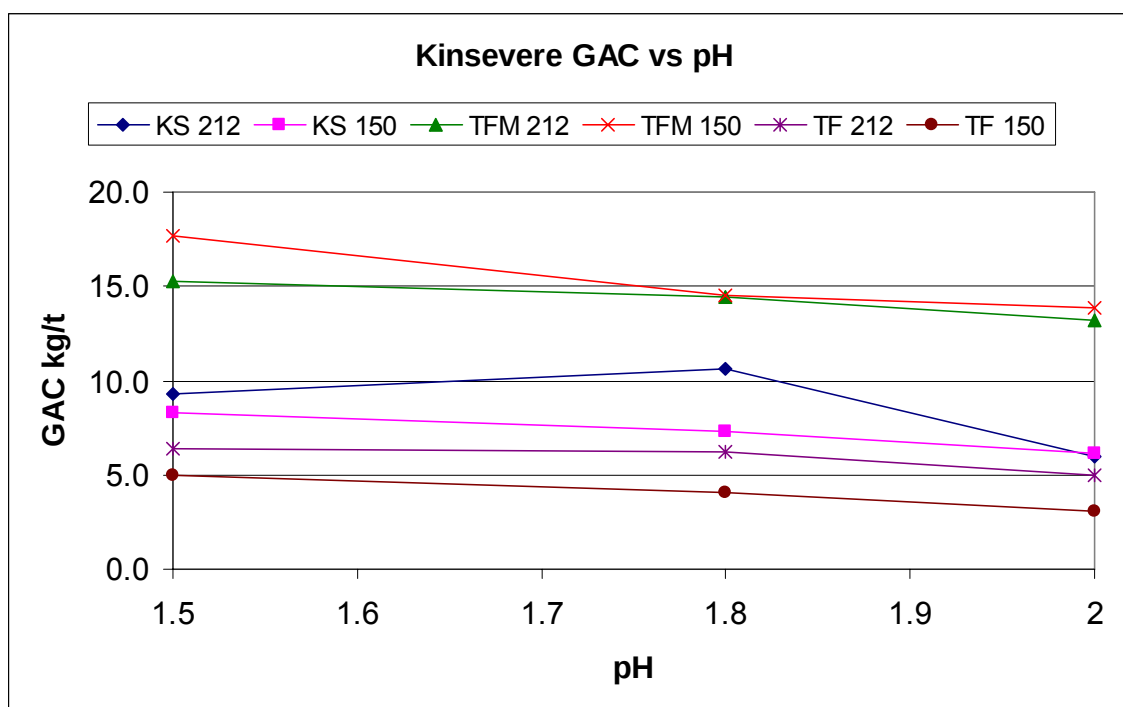
The Acid Soluble Copper results indicate the following:

- Leaching is recovering a very high proportion of the Acid Soluble Copper from all the Master Composites.
- Increasing the leach time from 4 hours to 8 hours raises the AS Cu recovery by around 2.4%.
- Further increasing the leach time from 8 to 12 hours improves extractions by an additional 1%.
- Lowering the pH from 2.0 to 1.8, on average, increases AS Cu recovery by 0.1% over an 8 hour leach. Lowering it further to 1.5 improves it another 0.8%, on average. However there is significant variation in the individual results.
- The benefit of the finer 150 micron grind is patchy. However Tshifufiamashi ore appears to respond to a finer grind, improving AS Cu recoveries between 0.7% and 1.4% over an 8 hour leach.

Gangue acid consumption (GAC) increases only slightly as the leach pH is reduced. The GAC figures are all very low, ranging as follows:

- Tshifufia 3.1 – 6.4 kg/t
- Tshifufiamashi 13.2 – 17.7 kg/t
- Kinsevere Hill 6.0 – 10.6 kg/t

Figure 25.15 Variation of GAC and Leach pH



Based on the above conclusions, a confirmatory test was performed on each Master Composite using a Grind P_{80} of 150 microns and a Leach pH of 1.5. The results were generally similar to the previous tests performed under the same conditions. However the reported GAC for the Tshifufia composite was significantly higher at 13.1 kg/t than was previously recorded (Table 25.29).

Table 25.29 Comparison of Initial and Confirmatory Leach Tests Performed at a Grind P_{80} of 150 microns, pH 1.5 for 8 Hours

		Tshifufia		Tshifufiamashi		Kinsevere	
		Initial	Confirm.	Initial	Confirm.	Initial	Confirm.
T Cu Rec.	%	77.5	79.2	94.6	94.1	89.9	93.8
AS Cu Rec.	%	95.6	96.6	99.3	99.0	94.3	98.5
Co Rec	%	49.9	49.8	50.9	51.4	23.8	18.7
GAC	Kg/t	6.8	11.9	15.7	19.2	8.8	7.7

The variability samples were leached based on the optimised conditions. ie. Grind P_{80} of 150 microns and a Leach pH of 1.5 for eight hours. Table 25.30 illustrates the results.

Table 25.30 Results for Variability Samples Leached at Selected Optimum Conditions (Grind P₈₀ of 150 microns, pH 1.5 for 8 Hours)

Sample	Head Grade			Recoveries			GAC kg/t
	TCu %	ASCu %	Co %	TCu %	ASCu %	Co %	
TF1	10.46	10.29	0.29	98.2	99.8	8.7	-0.8
TF2-a	4.31	4.22	0.25	97.5	99.5	16.6	3.6
TF2-b	6.80	3.80	0.36	51.1	91.3	1.4	6.2
TFM-a	3.29	3.11	0.26	93.4	98.9	3.9	4.1
TFM-b	7.07	6.90	0.40	98.2	100.6	17.2	7.4
KS-a	4.06	4.01	0.01	98.6	99.7	18.8	0.2
KS-b	1.92	1.82	0.01	95.4	100.7	24.9	5.9

The variability samples provided a spread of head grades ranging from 1.92% to 10.46% T Cu. The acid soluble copper content, with one exception, fell in the range of 94.5% to 98.8% of total copper levels. One sample, TF2-b had an acid soluble copper content of only 55.9%.

Excluding sample TF2-b, with the low proportion of Acid Soluble Copper, the acid soluble copper recoveries from all samples were extremely high and consistent, ranging from 98.9% to 100%. However the cobalt recoveries were generally low ranging from 1.4% up to 24.9%.

The reported Gangue Acid Consumption rates were very low with a maximum value of only 6.2 kg/t determined for sample TF2-b.

Mineralogical Examination of Leach Residues

The Master Composite leach residues were examined to identify the residual copper minerals.

In the Tshifufia residue chalcopyrite and digenite occurred as discrete grains up to 0.25 mm in diameter.

Covellite was the main copper mineral present in the Tshifiamashi residue, either as discrete particles or locked with pyrite.

Digenite and covellite was identified in the Kinsevere Hill residue, together with some discrete native copper particles up to 0.2 mm in diameter.

Leach Solutions

The resulting leach solutions from the suite of leach tests were analysed to determine if any deleterious elements or ions were present.

A summary of the results for the tests performed at the selected optimum leach conditions is included in Table 25.31. Only the most significant elements are listed.

Table 25.31 Summary of Leach Solution Analyses

Species	Units	Tshifufia	Tshifufiamashi	Kinsevere Hill
Copper	mg/l	7,330	8,440	6,070
Cobalt	mg/l	202	194	8.5
Iron	mg/l	73	84	38
Calcium	mg/l	61	273	53
Magnesium	mg/l	124	151	50
Silica	mg/l	50	100	60
Manganese	mg/l	124	121	17
Molybdenum	mg/l	<0.05	<0.05	<0.05
Chloride	mg/l	152	164	196

Iron can transfer to the tankhouse either chemically or by physical entrainment. When the concentration in the tankhouse exceeds 2 g/l, Current Efficiency is adversely affected. However the iron levels in all solutions are very low.

Calcium can cement out as gypsum in pipelines and can also generate crud in the SX plant and reduce the phase disengagement times. The levels detected in the Project solutions were far below the 600 ppm at which precipitation normally begins.

Manganese can be transferred to the tankhouse by entrainment where it can become oxidised to permanganate. The permanganate ion can severely degrade both the organic extractant and diluent. The manganese levels present in the three solutions are relatively low.

Molybdenum can be chemically extracted and lead to phase disengagement problems. However the element is below detectable limits in all leach solutions.

Soluble silica can lead to solid crud formation. However the concentrations detected are much lower than the levels at which problems generally occur (500 mg/l).

Chloride ions can be transferred to the tankhouse by entrainment where they can cause severe pitting of the stainless steel cathodes. The levels reported in solution are significantly below those at which problems may take place (500 mg/l).

In summary, all the leach solutions generated were very clean and visually resemble pure blue copper sulphate solution.

The comments above take no account of possible contaminant build-up due to recirculation of solution from the tailings dam. Potentially chloride could attain levels in solution that influenced the selection of stainless steel for construction. Consequently leach tests were performed on Tshifufia and Tshifufiamashi master composite samples, using demineralised water at 20% pulp density, to confirm the extent of chloride dissolution from the ore. The resulting chloride tenors of 15 mg/l and 7 mg/l respectively, indicated that chloride was unlikely to rise to significant levels in plant solutions.

Tailings Leach Tests

“Bucket” leach tests were performed on non-neutralised leach tailings from each deposit. These tests were intended to simulate the conditions in the tailings dam if acidified tailings disposal was adopted. This disposal technique is used by First Quantum Ltd at their Bwana Mkubwa project, and the reported benefits include additional copper recovery in the dam as well as reduced operating costs for acid, limestone and lime.

The samples were leached for a period of 16 weeks and the results tabulated in Table 25.32.

Table 25.32 Tailings Leach Test Results (Sequential Copper assays)

Deposit		Units	T Cu	AS Cu	CN Sol Cu	Leach Residue
Tshifufia	Head Grade	ppm Cu	8300	900	5000	2300
	Recovery	%	75.5	69.2	92.2	47.8
Tshifufiamashi	Head Grade	ppm Cu	2753	600	1000	1200
	Recovery	%	65.2	84.1	82.8	52.4
Kinsevere Hill	Head Grade	ppm Cu	1760	500	200	1100
	Recovery	%	41.5	51.3	53.2	48.9

The head grades reported in the table relate to the tailings grades of the conventional leach tests.

The cyanide soluble copper recovery indicates the dissolution of secondary copper sulphide minerals (chalcocite and covellite). The leach recovery of the secondary sulphides ranges from 53% to 92%. Notably the relatively high levels of secondary sulphides present in the Tshifufia conventional leach tailings are nearly all recovered during the tailings leach. In addition the results suggest that at least 50% of the residual acid soluble copper can be dissolved in the tailings dam.

The copper analyses for the final sequential leach residue indicate the levels of the most refractory copper minerals present such as chalcopyrite, bornite and native copper. The results suggest that approximately half of these minerals will dissolve in the dam.

To put these indicative results into perspective, the results of the overall leach recoveries for these three samples are reproduced in Table 25.33.

Table 25.33 Overall Leach Results – Conventional (12 hours) & Tailings Leach (16 weeks)

Deposit	Conventional Leach Recovery		Total Leach Recovery	
	T Cu %	AS Cu %	T Cu %	AS Cu %
Tshifufia	79.8	97.3	95.1	99.2
Tshifufiamashi	93.7	98.5	97.8	99.8
Kinsevere Hill	93.3	98.0	96.1	99.0

Not all of the additional copper that dissolves in the tails dam would be recovered. A portion would remain locked in the solution filling the interstitial spaces between the tailings particles.

Thickening Testwork

Prior to the thickening testwork samples of flocculant were obtained from various manufacturers who market their product in Southern Africa. These samples were evaluated by IML to discover the most effective polymers for both pre- and post-leach thickening.

The most suitable products identified were:

Pre-leach

- SAX 839 (Improchem)
- A 100 (Cytec)
- Magnafloc 155 (Ciba)

Post-leach

- N 100 (Cytec)
- Magnafloc 10 (Ciba)

The initial batch of tests was performed on the three Master composite samples. A second set of tests were carried out on the variability samples. To reduce the potential number of tests to be performed, IML performed a preliminary screening of the variability pre-leach samples to identify the sample from each deposit with the poorest settling characteristics.

As a result of these preliminary tests the following variability samples were selected for dynamic settling testwork:

- TF 2A
- TFM 6.9 – 23.8 m
- KS 15.4 – 35.9 m

A summary of the most promising results obtained during the tests are illustrated in Table 25.34 and Table 25.35.

The testwork indicated that highly diluted flocculant is required to obtain good overflow clarities. Two point flocculant dosing is standard procedure for thickener testing and usually provides adequate results. However in most of the pre- and post-leach testing this technique was not effective and occasionally significantly higher overflow suspended solid levels were recorded. Most of the tests were performed with the flocculant added to the dilution water prior to slurry dilution.

Table 25.34 Summary of Pre-leach Hi-Rate Thickening Testwork

Run No.	Solids (t/m2h)	Diluted Feed		Flocculant			Underflow		Overflow
		% w/w	Liquor Rise Rate	Type	Dose Point	Dose (g/t)	%w/w	Yield Stress (Pa)	Clarity ppm
TF3	0.84	11.7	6.6	SAX839	Diluent	20	58.9	81	140
TF4	0.84	11.7	6.6	SAX839	2pt	20	59.1	88	100
TF2A-2	0.94	11.2	7.7	SAX839	Diluent	20	60.7	70	40
TFM3	0.84	11.9	6.5	SAX839	Diluent	26	54.8		140
TFM4	0.60	11.4	4.9	SAX839	Diluent	26	56.8		120
TFM1A-2	0.95	11.6	7.5	SAX839	Diluent	23	59.6	62	50
KS3	0.84	9.5	8.3	SAX839	Diluent	43	57.7	97	390
KS5	0.86	12.7	6.2	SAX839	Diluent	36	58.2	105	340
KS1A-1	1.00	12.6	7.3	SAX839	2pt	32	55.0	50	290
KS1A-4	0.55	12.3	4.1	SAX839	Diluent	35	54.8	64	300

Table 25.35 Summary of Post-leach Hi-Rate Thickening Testwork

Run No.	Solids (t/m2h)	Diluted feed		Flocculant			Underflow		Overflow
		% w/w	Liquor Rise Rate	Type	Dose Point	Dose (g/t)	%w/w	Yield Stress (Pa)	Clarity ppm
TF2	0.95	11.8	7.4	Mag10	Diluent	28	57.1	30	70
TF3	0.77	14.7	4.7	Mag10	Diluent	33	58.7		40
TF2A-1	1.01	12.4	7.4	Mag10	Diluent	29	63.1	46	60
TFM1	0.71	11.2	5.9	Mag10	Diluent	34	58.1	33	80
TFM2	1.08	12.8	7.7	Mag10	Diluent	28	57.6		70
TFM1A-1	1.01	12.0	7.7	Mag10	2pt	29	61.1	56	110
KS1	0.80	11.8	6.2	Mag10	Diluent	41	57.3	100	70
KS3	1.04	15.4	6.0	Mag10	Diluent	38	56.3	75	70
KS1A-3	0.82	12.8	5.8	Mag10	2pt	39	56.0	51	250
KS1A-5	0.81	12.4	5.9	Mag10	Diluent	40	52.2	56	70

Clarification

Overflow samples from the post-leach thickening tests performed on the Master Composite samples by Outokumpu were sent to ConSep in Perth for testwork aimed at providing sizing data for a pinned bed clarifier (PBC).

The clarities of the samples provided were not ideal, as the suspended solids present were too low. The Total Suspended Solid Levels (TSS) ranged from only 61 ppm to 86 ppm, whereas PBC performance is optimal at TSS levels around 5,000 ppm. In normal operation fine solids need to accumulate under the pinned bed to provide a filtration media.

Each sample was tested at two different rise rates and two different flocculant dosage rates. Results are listed in Table 25.36.

Table 25.36 Summary of Clarification Testwork

Sample	Parameter	Units	Test No.			
			1	2	3	4
Tshifufia						
	Rise Rate	m/h	10		14	
	Flocc. Dose	g/m3	1	2	1	2
	Feed TSS	ppm	86			
	TSS Removed	%	57	55	50	48
Tshifufiamashi						
	Rise Rate	m/h	10		14	
	Flocc. Dose	g/m3	1	2	1	2
	Feed TSS	ppm	62			
	TSS Removed	%	64	40	26	15
Kinsevere Hill						
	Rise Rate	m/h	10		14	
	Flocc. Dose	g/m3	1	2	1	2
	Feed TSS	ppm	61			
	TSS Removed	%	58	40	29	23

Clarification performance was enhanced at both the lower rise rate and lower flocculant dosage. The suspended solids appeared well flocculated prior to testing which may account for the latter observation.

25.4.5 Tailings Neutralisation

Limestone & Lime Characterisation

Samples of limestone and burnt lime were obtained from Gécamines Likasi operation.

Head analyses of the samples were performed. A summary of the results are included in Table 25.37. A comparison with the products available from Ndola Lime in Zambia is illustrated (based on data from the Ndola Lime website).

Table 25.37 Comparison of Gécamines Limestone and Quicklime Quality with Ndola Lime Product Specifications

Species	Units	Gécamines Limestone	Ndola Limestone	Gécamines Quicklime	Ndola Quicklime
Available CaO	%	0.1		38.2	84.0
CaO		50.4	54.0 min	61.63	
CaCO ₃		98.0	96.0		
MgO		3.5	<1.0	1.5	<2.0
Reactivity	Seconds (to reach 40°C)	NA		Failed to reach 40°C	<300
MnO	%	0.08	<0.01	0.05	<0.01
LOI	%	NA		32.3	4.0
SiO ₂	%			1.4	<2.0
+45 mic grit (after slaking)	%			24.2	

The Gécamines limestone quality is generally acceptable but the low *Available CaO* and high *Loss on Ignition* values suggests that the lime burning process is very inefficient resulting in poor quality quicklime.

A size analysis of the Gécamines quicklime indicated that the P₈₀ size was approximately 2.1 mm, but 6.9% of the sample was greater than 12.5 mm.

The Gécamines limestone was found to have the following comminution characteristics:

- Ball Mill Work Index – 7.2 kWh/t
- Abrasion Index – 0.009

Neutralisation Testwork

A set of neutralisation tests were performed on a Tshifufia raffinate sample using Gécamines limestone ground to various P₈₀ values. The reaction time was 3 hours. See results in Table 25.38.

Table 25.38 Comparison of Neutralisation Performance of Gécamines Limestone at Different Grind Sizes

Limestone P ₈₀ (mic)	Limestone Addition (kg/t solids)	Final pH
106	69.0	6.66
75	73.1	6.48
45	73.6	7.01

The tests indicated that a grind P_{80} of 106 microns produced as effective neutralisation as the finer grind sizes. However the effectiveness of lime addition decreases as the pH rises and the addition rates reported were extremely high.

A series of tests were performed using a combination of limestone (P_{80} – 106 microns) and lime addition to neutralise Tshifufia raffinate. See Table 25.39. Good quality Perth hydrated lime was used for the latter two tests.

Table 25.39 Effect of Combined Limestone and Lime for Neutralisation

Limestone Addition (kg/t solids)	Lime Addition (kg/t solids)	Lime Source	pH – 1st Lime Addition	Time to Reach pH8 (mins)	Final pH
75.0	1.6	DRC	5.4	64	7.9
16.4	10.6	Perth	4.3	182	12.4
14.6	5.2	Perth	4.5	81	11.6

The tests indicated that the overall amount of neutralisation reagents could be markedly decreased by commencing lime addition at a lower pH.

Solvent Extraction

Two Major SX extractant manufacturers were invited to IML to test their products with PLS generated from the individual Kinsevere deposits.

25.4.6 Cognis

The copper tenors of the three PLS samples tested were:

- Tshifufia – 7.1 g/l
- Tshifufiamashi – 8.3 g/l
- Kinsevere – 5.4 g/l

All the PLS solutions were very clear and blue and Cognis noted that they resembled pure copper sulphate liquor in appearance.

Standard Cognis Phase Disengagement Time (PDT) tests using 10%v/v Lix® in Shellsol 2046 diluent were performed. The Phase Disengagement Times recorded in both Organic Continuous (OC) and Aqueous Continuous (AC) operating modes are detailed in Table 25.40.

Table 25.40 Cognis Phase Disengagement Testwork Results

Sample	Phase Disengagement Time (Seconds)	
	Organic Continuous	Aqueous Continuous
Tshifufia	76	74
Tshifufiamashi	79	69
Kinsevere Hill	68	62

The PDT's recorded are all within the normal range recommended by Cognis of 60 to 90 seconds.

Three organic test solutions were prepared containing varying proportions of Lix984N, a C9 Aldoxime, based on the PLS copper grade. Each organic sample was contacted at various O/A ratios with the corresponding PLS samples and the resulting copper concentrations in the Organic and Aqueous phases measured. The data generated was used to simulate the performance of various SX circuit configurations using the Cognis ISOCALC™ software. See Table 25.41.

Table 25.41 Summary of Copper Equilibrium Extraction Data and ISOCALC™ Modelling

Parameter	Units	Tshifufia		Tshifufiamashi		Kinsevere Hill	
Circuit Configuration		2E/1S	3E/1S	2E/1S	3E/1S	2E/1S	3E/1S
PLS	Cu g/L	7.09	7.09	8.25	8.25	5.36	5.36
Adv. O/A		1.00	1.00	1.00	1.00	1.00	1.00
LIX984N	Volume %	25	25	28	28	18	18
Cu Recovery	%	91.1%	96.0%	89.7%	95.2%	88.9%	95.1%
Raffinate	Cu g/L	0.628	0.285	0.846	0.400	0.594	0.262
Load Org	Cu g/L	11.051	11.400	12.495	12.951	8.161	8.500
Strip Org	Cu g/L	4.589	4.596	5.091	5.100	3.395	3.402
Net transfer	gpl Cu/%LIX	0.258	0.272	0.264	0.280	0.265	0.283
Max Load	Cu g/L	12.786	12.786	14.386	14.386	9.087	9.087
% Max Load	%	86.4%	89.2%	86.9%	90.0%	89.8%	93.5%

Cognis concluded that that all the PLS samples tested were readily amenable to processing using their extractant Lix 984N.

25.4.7 Cytec

Extract and Strip Isotherms were generated using both Acorga® M5774, an ester modified C9 Aldoxime and Acorga® MAK5510, an ester modified C9 Aldoxime/Ketoxime blend.

The mass balances in the isotherm data was not as reliable as expected, even though all solutions were re-assayed, and Cytec recommended that the results

should not be used for design purposes. Iron tenors measured at all the isotherm points were <1 ppm in all cases.

Load/Strip cycle testing was performed with both Tshifufia and Tshifufiamashi leach solutions. No crud accumulation or interfacial rag³ was apparent in any tests after fifteen contacts.

Phase disengagement times recorded with MAK5510 were significantly higher than those achieved with M5774, although Cytec did not consider them excessive considering the reagent concentration (25%) and cool ambient temperature.

Neither extractant exhibited an increase in phase disengagement time after multiple contacts suggesting that the organic phase is not affected by contaminants transferring from the aqueous phase.

Cytec concluded:

- The leach liquors were very clean and did not interfere with the physical aspects of solvent extraction performance.
- With high copper and low iron levels in solution, the chemical transfer of iron was minimal.

25.4.8 Rheology

The rheology properties of the Kinsevere Hill slurries were investigated in two testwork sessions, A and B. The tests investigated various pulp densities, as illustrated in Table 25.42.

Table 25.42 Range of Rheology Testwork

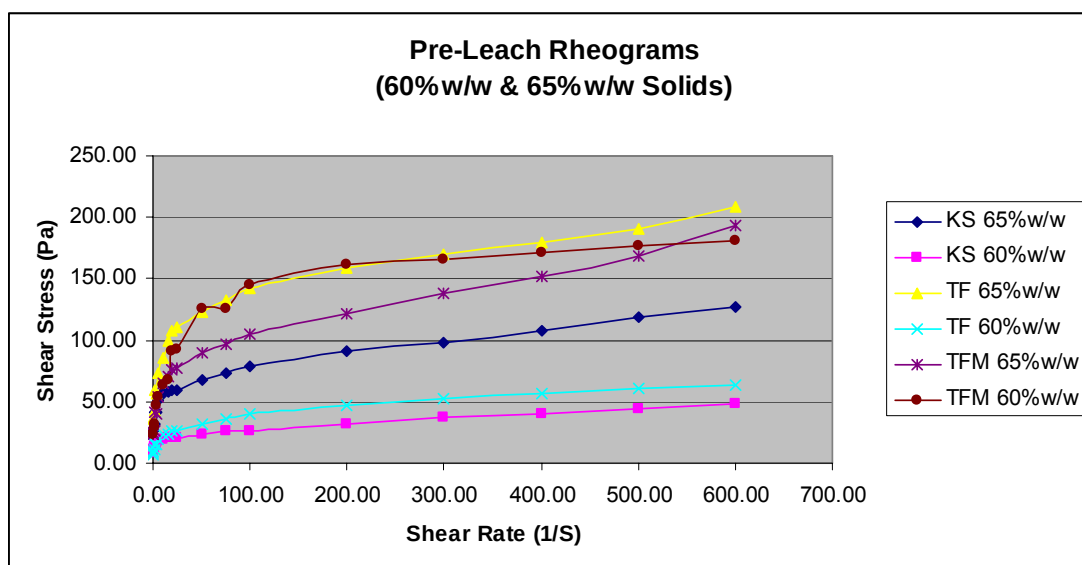
Sample	Pulp Density (%w/w)								Deposits
	25	30	40	45	50	55	60	65	
Pre-leach (A)						X	X	X	TF,TFM,KS
Post-leach (A)						X	X	X	TF,TFM,KS
Post-leach (B)	X	X	X		X				TF,TFM
Neutralised Post-leach (B)				X	X	X	X		TF,TFM

All the pre-leach samples demonstrated higher shear stress than the post-leach samples at similar pulp densities in the first set of tests. The only sample that exhibited settling of solids was the Kinsevere Hill post-leach sample at 55% solids.

Figure 25.16 demonstrates the rheograms for the higher pulp density slurries of the various deposits.

³ Rag is a relatively stable network of organic film that is suspended from the interface in the aqueous phase. If organic matter, such as 'rag', builds up excessively, it will be swept out of the settler with the aqueous phase and be lost from the solvent extraction unit.

Figure 25.16 Pre-leach Rheogram



All samples exhibited shear thinning characteristics with viscosity decreasing with increased shear rate.

The Tshifufia post-leach sample was retested at a pulp density of 65% solids 7 weeks after the original testwork. The results indicated that both the shear stress and viscosity readings had increased significantly at each shear rate which suggests the effects of ageing. Consequently for the second set of tests fresh samples were used.

The Tshifufia post-leach sample in Test 2 demonstrated low shear stress values at all pulp densities with the 25, 30 and 40% pulp densities showing zero values at the lower shear rates.

Similarly the Tshifufiamashi sample also showed low shear stress values for all pulp densities with the 50% pulp density sample settling at the lower shear rates.

Tshifufia and Tshifufiamashi post-leach slurry samples both demonstrated shear thinning behaviour when the pulp density reached 50%w/w solids. Plotting viscosity curves for more dilute slurries was not possible as the viscosity values were too low for detection by the viscometer.

The neutralised post-leach Tshifufia and Tshifufiamashi samples exhibited low shear stress values at the lower pulp densities with zero results recorded at the lower shear rates.

Viscosity curves for both samples demonstrated shear thinning behaviour at the higher pulp densities.

25.4.9 Effluent Treatment

As a consequence of the 'direct tailings disposal' design, testwork was performed to investigate potential effluent treatment options for excess solution in the tailings dam.

A sample of Tshifufia raffinate was treated with 10% lime solution to raise the pH up to 10.6 over 2.75 hours (Figure 25.17 and Table 25.43). The elevated pH was selected in order to precipitate heavy metals such as cadmium and nickel as well as iron, zinc and copper which are usually removed at pH 9.5.

Although raising the pH to 10.5 was effective at precipitating heavy metals, this degree of alkalinity is too high for most effluent discharge regulations. Consequently a test was performed to determine the amount of acid required to bring the pH back to 7. The resulting solution was analysed to determine if there had been significant re-dissolution of the hydroxide precipitate.

Figure 25.17 Variation of Effluent pH with Lime Addition

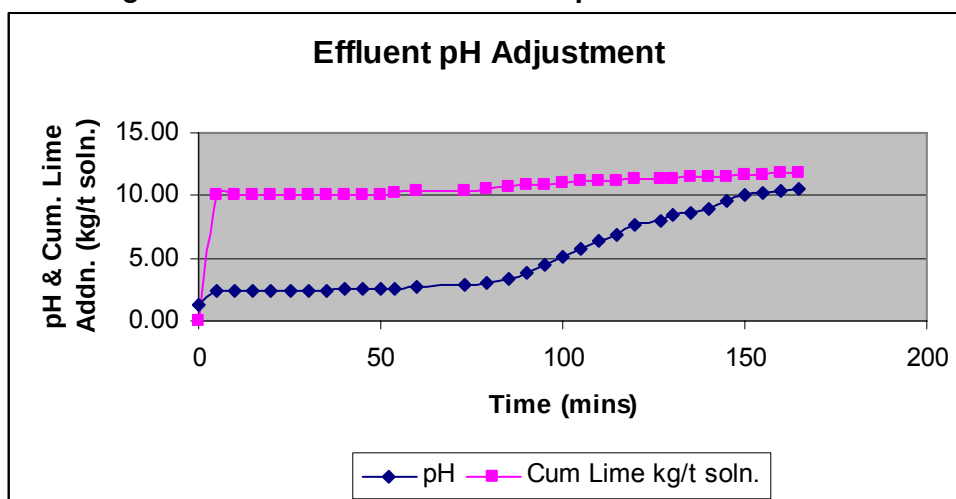


Table 25.43 Acid Consumption for Effluent pH Adjustment

Time (seconds)	pH	Cum Acid (g/m ³ soln.)
0	9.84	0
499	8.25	23.5
971	7.02	33.5

Table 25.44 indicates that there was minimal indication that the hydroxide precipitates redissolved as the pH was lowered down to 7.

Table 25.44 Effect of pH Adjustment on Effluent Quality

Element	Units	Raffinate @ Various pH Levels					
		1.23	6.93	8.49	9.53	10.61	7.02
Copper	mg/l	72.9	0.95	0.15	0.10	0.05	0.10
Cobalt	mg/l	168	129	57.4	1.10	0.10	0.21
Chromium	mg/l	0.95	<0.05	<0.05	<0.05	<0.05	<0.02
Iron	mg/l	103	0.1	<0.1	<0.1	<0.1	0.10
Manganese	mg/l	24.8	17.9	9.6	0.55	0.05	
Nickel	mg/l	3.50	2.88	1.90	0.15	<0.05	<0.02

A review of the effectiveness of a range of flocculants to settle the precipitated solids indicated that Magnafloc 10, Magnafloc 336, Magnafloc 800HP, Magnafloc 340 or Nalco 83376 were able to produce reasonable settling rates (Table 25.45).

All of the flocculants, bar Magnafloc 340, produced clarites of 46 on the clarity wedge (indicating low levels of suspended solids). One of the supernatant samples treated with Magnafloc 10 was able to achieve a TSS value of 15 ppm.

Table 25.45 Flocculant Effectiveness

Parameter	Units	Magnafloc 10	Magnafloc 336	Magnafloc 800 HP	Magnafloc 340	Nalco 83376
Dose Rate	ppm	2.5	2.5	2.5	2.5	2.5
Settling Speed	m/h	4.9	17.0	7.2	5.3	13.4
Clarity @ 10 minutes		46 (Max)	46 (Max)	46 (Max)	39	46 (Max)
Compaction (% original soln. volume)	%	54	52	40	56	56

25.4.10 Production Ramp-up

The projected recovery and processing rate for the proposed ramp-up period is listed in Table 25.46.

Table 25.46 Production Ramp-Up Limits

Ramp-up Period	Recovery	Capacity
Month 1	74.6%	30%
Month 2	78.1%	60%
Month 3	80.9%	75%
Month 4	85.4%	85%
Month 5	89.4%	95%
Month 6	91.4%	99%
Month 7	92.14%	100%

25.5 Tailings Management

The Stage II tailings storage facility (TSF) is located at the south east corner of the mining lease. The site was cleared and the starter embankment completed in 2008 to a height of 15 m (1,267.8 mRL). The final embankment will be built to 1,288 mRL and covers an area of approximately 1 km².

25.5.1 Design Concept

In 2008, Anvil proceeded with the detailed engineering design for the Stage II TSF to suit direct tailings disposal. Relative to the concept reported in the 2008 Technical Report, a number of changes were made to the design, specifically to the footprint dimensions, the embankment construction method and the lining specifications.

All four embankments will now be built by the downstream raising method, using waste rock from all pits. The embankment construction schedule is based on providing waste from the pits as it is available rather than to suit the rate of tailings deposition (ie, a longer timeframe). The facility is designed for a plant throughput of 1.6 Mtpa and will have the capacity to store a 1 in 1,000 wet year storm event lasting 72 hours. In addition, the TSF will be furnished with an emergency spillway for the sake of operating safety. Other design attributes are as follows:

- volume capacity = 20.5 million m³ of tailings, based on a slurry concentration of 40% to 50% by weight;
- catchment area = 92 ha;
- minimum/maximum pond volume = 15,000 m³ / 280,000 m³ for average conditions;
- three construction zones within the embankments:
 - o zone A consisting of weathered saprolite with at least 30% fines;
 - o zone B consisting of a mixture of weathered saprolite (less than 50% saprolite) and waste rock; and
 - o zone C consisting of waste rock (no fines).
- the base of the TSF will be scarified and recompacted to create a low permeability clay/silt layer covered with a 1.5 mm HDPE liner;
- the upstream slopes of the embankment will be built of zone A materials covered with 1.5 mm HDPE liner;
- a single HDPE geomembrane liner is installed on the bottom and side walls of the facility and is overlain by drainage layer which serves to reduce the pressure head acting on the liner; and
- a seepage detection and underdrainage/collection system is incorporated, comprising 160 mm diameter HDPE pipes connecting to a collection sump from which return water will be pumped using submersible pumps.

The Environmental Impact Assessment prepared for Kinsevere Stage I included information on Kinsevere Stage II but did not address direct tailings disposal nor the classification of the tailings material.

The DRC mining regulations classify tailings according to their degree of hazardousness. Analysis carried out by KP as part of the detailed TSF design process, confirmed that under the regulations classification structure, the tailings from Stage II are classified as high risk. The classification of the tailings has implications for the liner requirements in the design of the TSF, to the extent that the DRC regulations require that where tailings are classified at the degree of hazardousness confirmed for Stage II, a double liner is required. The original design specifications for the TSF had provided for a single liner only.

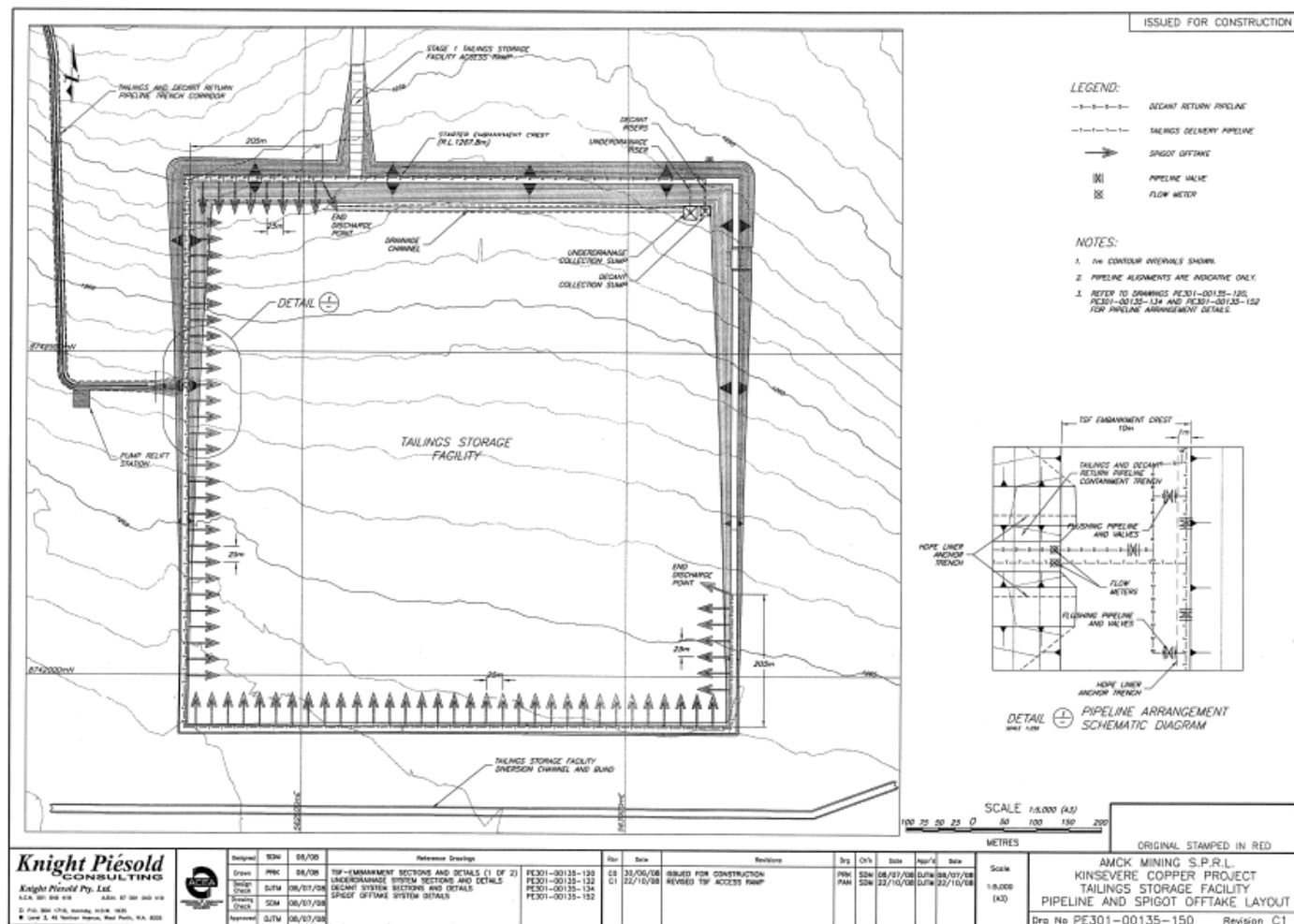
Figure 25.18 shows the revised TSF design.

25.5.2 Tailings Storage Facility Design Assessments

KP consultants undertook laboratory analyses and computer modelling to determine the expected geochemical properties of the tailings in accordance with requirements of Annexes IX and XI of the DRC Mining Regulations (2003). The conclusion from this work is that the Stage II tailings material is classified as high risk.

The proposed lining design, as described above, is comparable to the Level B management measures described in Article 62 of Annex IX, but does have slightly different technical specifications. The most notable difference is the incorporation of the drainage layer. KP conducted a seepage modelling exercise (KP, September 2008) that demonstrated that the performance of the proposed design, in terms of permeability, will be better than two of the three Level B management measures described in Article 62 of Annex IX.

Figure 25.18 Revised TSF Design



25.5.3 Potential Contaminant Leakage

The construction activities in the TSF will take place above the groundwater table (refer to Table 25.47 and will not have any influence on the groundwater. Therefore, no environmental impacts are expected on the groundwater quantity and quality due to the TSF construction.

Table 25.47 Boreholes Drilled for Hydrogeological Investigations

Borehole Name	Location	X Coord	Y Coord	Borehole depth (m)	Water Level (m)	Blow Yield (l/s)	Final Diameter
TSFMON01	TSF	563,030	8,741,620	80	69.98	<1.0	5.5" (0-80 m)
TSFMON02	TSF	563,400	8,743,080	81	46.30	<1.0	5.5" (0-81 m)
TSFMON03	TSF	562,400	8,743,220	80	41.84	<1.0	5.5" (0-80 m)

As a worst case scenario, KP (October 2007) simulated contaminant leakage from the TSF to the groundwater horizon due to a failed liner system. The modelled leakage equated to 4,000 mg/l of sulphate (SO₄). Three scenarios were compiled in order to predict the movement of possible contaminants 15 years into operation, 100 years later and then 150 years later.

The simulated migration of possible contaminants is mainly in a northerly direction. After 15 years, the simulated contaminant plume migrated a distance of approximately 600 m and after 100 years and 150 years, approximately 1 km in the same direction. Under this worst case scenario, the size of the contaminant plume is significant but will not extend beyond the Project area.

The KP (September 2008) analysis report addressed the proposed TSF basin lining system with respect to seepage potential and a comparison with the DRC Mining Regulations. The analysis results demonstrated that the proposed design is equivalent to the prescribed standards in the regulations. Based on the results of laboratory and field test work completed in August 2008, to confirm the permeability parameters assumed in previous conservative seepage modelling, the KP analysis concluded that:

- The compacted soil liner permeability at 98% standard maximum dry density is 8.5×10^{-7} cm/sec, which is better than the value assumed in previous modelling.
- Constant head permeability tests in six boreholes indicated an insitu permeability in the upper soil stratum of 4.3×10^{-6} cm/sec, which is also better than the value assumed in previous modelling.

All boreholes in the Project area will be monitored for both groundwater quantity and quality during the life of the operation. There are currently three monitoring bores around the Stage II TSF site.

25.6 Stage II Implementation Plan

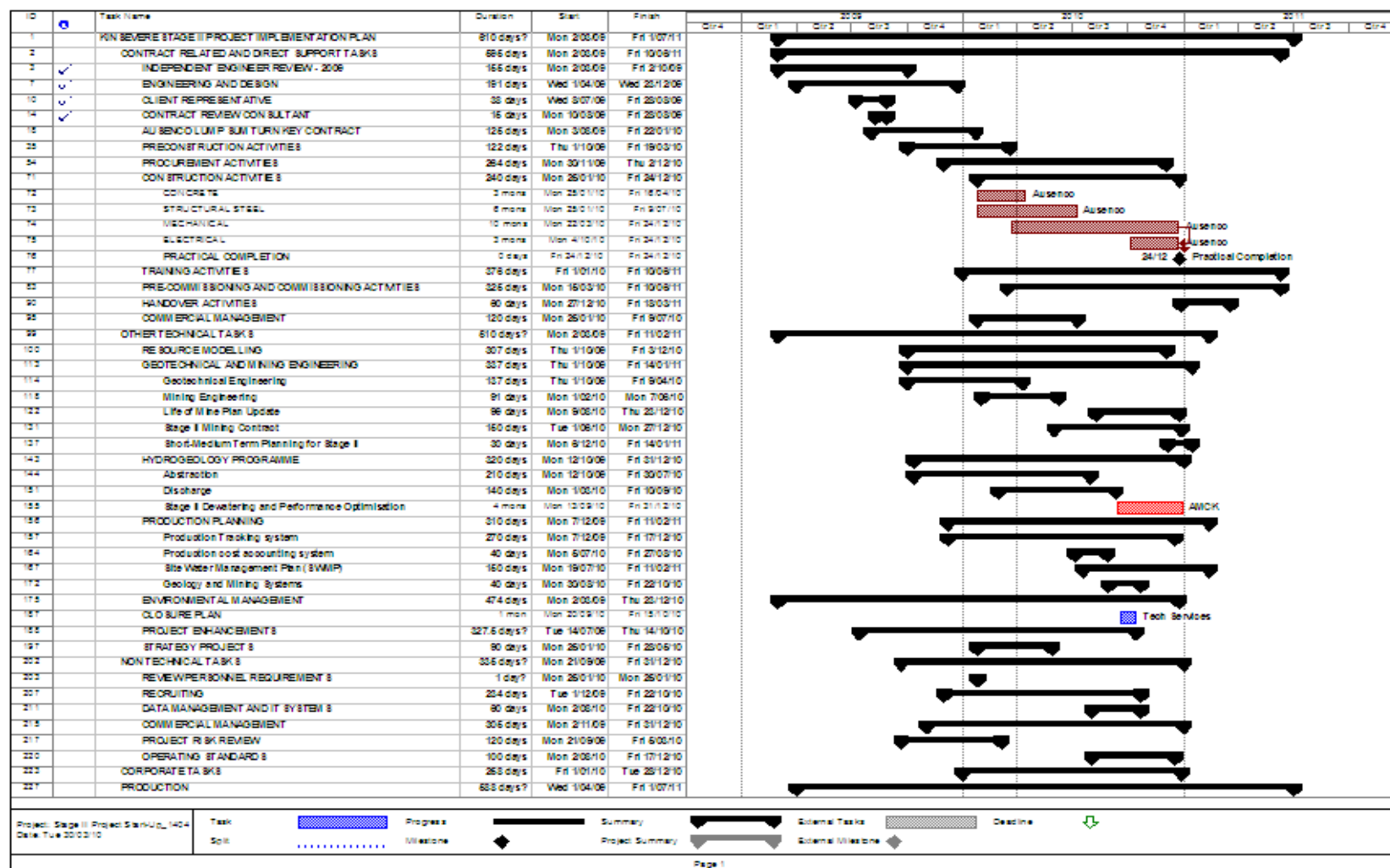
The Company has prepared a detailed Implementation Plan (Anvil, February 2010) to describe the principal management functions and tasks that will be used to execute the Stage II construction and commissioning scopes of work. In addition, there are certain management functions required at a corporate level to strategically position the Company during the Stage II construction and commissioning phases, specifically in the development of growth opportunities and maintaining position as a leader in regional corporate social responsibility.

The Implementation Plan defines the tasks, responsibilities and timeframes, in addition to formulating the procedures for managing the Stage II implementation process, developing “operational readiness” and maintaining a corporate strategic focus through the period to the end of 2010.

The Total Project Schedule which is included in the Implementation Plan can be summarised into the high level components shown in Figure 25.19. The Project tasks for the 2010 construction period can be broken down into:

- specific tasks related to the LSTK contract for construction of the SX-EW facility;
- general tasks required to be completed by the Company in parallel with, and in support of the SX-EW plant construction;
- other technically related tasks required to be completed by the Company and specialist consultants or contractors, in Perth or elsewhere;
- a host of non-technical tasks to be completed by the Company in diverse areas such as administration, human resources (ie, recruiting) and IT, all of which are required for successful project delivery; and
- corporate strategic tasks including social investment strategy, public affairs management and investor/stakeholder communication management.

Figure 25.19 Stage II Project Schedule



25.7 Markets

The proposed 60,000 tpa SX-EW plant will produce grade A cathode copper. Trafigura has an off-take agreement for Stage I HMS concentrate and Stage II cathode.

25.8 Contracts

There are a number of contracts (or agreements) in place with various parties, as follows:

- Offtake Agreement between the Company and Trafigura dated November 16, 2009 for the sale of product from the Company's Kinsevere mine.
- Technical Services Agreement between the Company and Trafigura dated November 16, 2009 for Trafigura's provision of consulting and technical support.
- Project Loan Facility Agreement between the Company and Trafigura dated December 16, 2009.
- Agreement with Ausenco Projects Limited dated January 21, 2010 in connection with the provision of engineering procurement and construction management services for Kinsevere Stage II.
- Agreement with Ausenco Solutions Pty Ltd dated January 21, 2010 in connection with the engineering procurement and construction management services for Kinsevere Stage II.
- Agreement with *Société Nationale d'Électricité* (SNEL) for the supply of high voltage electricity from the DRC national hydroelectric grid to Kinsevere.

These agreements adhere to normal commercial terms. A number of other shorter-term agreements are also in place for Stage I mining, explosives supply, grade control drilling and groundwater bore drilling.

The mining contract agreement will need to be re-tendered for Stage II operations whilst the grade control and groundwater bore drilling agreements may need to be extended for services beyond 2010.

Agreements to be negotiated in 2010 in relation to services to the SX-EW plant include:

- laboratory services; and
- supply of consumables, such as acid, diluent, extractant, grinding media and stores.

25.9 Environmental Considerations

DRC Government approval of the EIA for Stage I and Stage II was issued in writing by CAMI on October 15 2007. Approval of a variance to the design and operation of the Stage II TSF was issued by the Directorate for the Protection of the Mining Environment (DPEM) on 28 October 2008.

An additional Environmental and Social Impact Assessment (ESIA) was completed by KP in October 2009 (KP, October 2009). The ESIA provides a more comprehensive review and management plan for the undertaking of Stage II of the Project. It is a document required by Project financiers to ascertain the Company's compliance with the requirements of the Equator Principles and the Performance Standards of the IFC.

The ESIA presents significance ratings for potential environmental and social impacts, both before and after mitigation.

For the Pre-construction Phase of the Project, the identified Highly Significant impacts are:

- permanent overlying change in land use with compaction of the subsoil being confined to the footprint areas and erosion of the more clay rich sandy clays and slightly hydromorphic soils;
- destruction and fragmentation of vegetation;
- loss of important flora, eg *cissus scmitzii* and six other Congo Basin endemics to the miombo woodland; and
- increased harvesting of vegetation resources.

For the Construction Phase of the Project, the identified Highly Significant impacts are:

- permanent overlying change in land use with compaction of the subsoil being confined to the TSF footprint area;
- increased harvesting of vegetation resources;
- stream flow reduction (due to water abstraction);
- artisanal miners' loss of direct access to Mineral Resources resulting in the loss of higher income;
- loss of access to land at the mine site due to the construction of facilities;
- villagers' access to communications due to establishment of mobile phone network;
- job opportunities for people made available, and opportunity to learn new skills;
- increase in road traffic and hence potential increased road accidents;
- increasing mining players coming onto the scene;
- skills building for local personnel; and
- international confidence in DRC as an investment destination.

For the Operational Phase of the Project, the identified Highly Significant impacts are:

- extraction of a non-renewable resource;
- deterioration in the groundwater quality;
- loss of important flora due to the continuous operation of the WRD and TSF and influx of people;
- destruction and fragmentation of vegetation. Mining of Kinsevere Hill is of relevance as this will result in a threat to the rare *Gladiolus robiliartianus* plant;
- increase in harvesting vegetation resources;
- streamflow reduction due to pit dewatering programme;
- deterioration in surface water quality;
- increased employment opportunities in the community, resulting in more access to money;
- employment opportunities reduced due to eventual scaling down of activities;
- increasing mining players coming onto the scene;
- skills building for local personnel;
- community benefit from royalties;
- revenue from taxes; and
- international confidence in DRC as an investment destination.

For the Closure Phase of the Project, the identified Highly Significant impacts are:

- open pits will be allowed to flood;
- compaction and erosion of soils during rehabilitation activities;
- contamination of surface water if groundwater discharges into the Kifumashi River – increase in salts and possibly trace metals;
- contamination of surface water;
- restoration of the Kifumashi River levels;
- boost to vegetation growth arising from rising water table;
- reduction in groundwater quality due to TSF seepage;
- loss of maintenance to the site access road;
- loss of employment at closure;
- social distress due to loss of employment and business;
- loss of medical support;
- loss of benefit from royalties; and
- loss of businesses.

The ESIA includes the Environmental and Social Management Plan (ESMP) which is focussed on objectives and measures to mitigate all of the identified environmental and social impacts, including those listed above.

The Environmental and Social Action Plan (ESAP) (Anvil, November 2009):

- a) commits AMCK to comply with the Equator Principles except in certain specified respects; and
- b) sets out a plan for AMCK working towards full compliance with the Equator Principles.

The ESAP is an Action List which is used to track the implementation of recommendations made by the Independent Engineer (SRK, September 2009) in their review of the ESIA. SRK's report (September 2009) makes specific recommendations to better align the ESIA with the requirements of the Equator Principles and the IFC Performance Standards. SRK specifically:

- noted that the Company has demonstrated substantial progress towards full EP and IFC compliance;
- concludes the Company does not comply fully with five of the nine Equator Principles, with further work required on biodiversity studies, the access road corridor, community management and social management plans, and stakeholder engagement plans;
- concludes that in relation to the eight IFC Performance Standards, further work is required for full compliance on each of these, and principally on aspects of construction to operations transition, labour and working conditions, surface and groundwater impacts to the Kifumashi River, greenhouse gas emissions, community health and safety, traffic management, and compensation for land acquisition; and
- suggests that deficiencies should be addressed as Action Items in the ESAP.

Depending on the hydrogeological modelling to be carried out in 2010 (Item 25.1.7), it may be necessary to prepare an addendum to the EIA to address the discharge of increased volumes of abstracted groundwater.

25.10 Operations

Details of the operations plan including information on manning numbers and the operations organisation chart are reproduced below from the 2008 Technical Report. Where appropriate, updated information is provided.

25.10.1 Organisation Charts

The organisation charts proposed for the Stage II operations have changed since the 2008 Technical Report. Updated Figures 25.20 to 25.23 refer to the Administration, Technical Services, Processing and Maintenance organisation charts.

25.10.2 Management

Stage II operations will be managed by the General Manager who will report directly to Anvil's Chief Operating Officer. The General Manager will be responsible for the management of the mine, and specifically for the following functions:

- compliance with all relevant DRC legislation, including environmental and safety procedures;
- recruitment of Departmental Heads;
- overall management of operations at Kinsevere;
- ensuring agreed throughput and cost budgets for the operations are met;
- liaising with local NGO's to ensure harmony with the local population; and
- liaison with local authorities.

The following Senior Managers will report to the General Manager:

- Administration Manager (administration, Human Resources)
- Senior Accountant (payroll)
- Health Safety and Environment Manager (camp, health, safety and training)
- Security Manager (site security, security contract supervision)
- Logistics/Supply Manager (warehousing, logistics/transport)
- Mining Manager (drill/blast, mining contract supervision)
- Plant Manager (Plant and tailings storage facility operations, and laboratory)
- Maintenance Manager (Maintenance planning, plant maintenance, light vehicle maintenance and, minor site construction projects)

25.10.3 Administration and Finance

AMCK uses a centralised accounting system (Pronto) to control it's financial accounting and reporting. This system is centred in Anvil's office in Perth, Western Australia with site access for payroll, creditor control and sales. A Senior Accountant has been appointed at the mine to ensure that necessary controls and staff are in-place such that accurate financial records can be kept and reported in an orderly manner.

The Senior Accountant is supported by the Perth based Financial Controller. The Administration Department is headed by an Administration Manager who is responsible for all Human Resource, Government and Community Liaison functions.

Other departments sitting directly alongside the Administration function, include Logistics and Supply, Health/Safety/Environment and Security.

Anvil has learnt that for its operations to work effectively in the DRC, a co-ordinated approach to Transport and Logistics needs to be taken to ensure that operating

supplies are delivered to the mine in a timely manner and product is shipped out effectively.

A site based Logistics and Supply Manager will be appointed to co-ordinate the delivery of construction materials and supplies to the Project. Under DRC law it is a requirement that officers from the Departments of Customs (OFIDA) and Inspection (OCC) are based at site to control the inflow and export of goods from the mine. The Logistics and Supply Manager will co-ordinate with these officers to ensure that goods entering and exiting the mine are cleared in a timely manner.

The Health, Safety and Environment (HSE) Department will be headed by a HSE Manager who is directly responsible to ensure that the mine operates in accordance with its safety policy, and also conforms to the environmental commitments outlined by the DRC regulations. The HSE manager will also be responsible for co-ordination of relevant training programmes for staff.

In addition, the HSE Manager will supervise a contracted paramedic who is employed under AMCK's HSE policy to treat trauma patients from accidents at the mine or surrounding districts.

The Security Department will be headed by an expatriate Security Manager who reports to the General Manager. Assisting the Security Manager will be assistants who cover for the Security Manager during his rostered period off site. The Security Manager and his assistants must be fluent in French. Security guards and their required supervisors will continue to be outsourced from a security firm, Group 4. The security company is responsible for the supply of Mines Police required by DRC law to be present on the mine. The Mines Police officers are armed.

25.10.4 Technical Services

The proposed Site Technical Services department incorporates geology, mining and surveying. Additional personnel to be recruited include a Mining Manager, mining engineers and another geologist. Prior to the recruitment process, the Manager Group Technical Services will be responsible for developing job descriptions for all mining and technical services personnel.

Detailed Life of Mine planning and strategic mining engineering and Mineral Resource studies will continue to be managed in Perth by the Group Technical Services Manager.

25.10.5 Processing

The Processing Department will be headed by an expatriate Plant Manager who will be assisted by an expatriate Process Superintendent, two expatriate HMS trainers, and a Chief Metallurgist.

Process Plant staff will work a 12 hour shift on a 2 by 2 by 2 roster, and where possible local villagers will be employed and trained to fill plant operator and assistant plant operator roles. It is envisaged that the staff trained on the HMS plant will be transferred to the SX-EW plant once it is commissioned.

25.10.6 Maintenance

The Maintenance Department will be headed by a Maintenance Manager who will be assisted by a Maintenance Superintendent, a Mechanical Engineer and an Electrical Engineer. Reporting to the Superintendent will be at least three Foremen and two Maintenance Planners, all of whom will be expatriate.

It has been assumed that the Maintenance Department will require the services of an Electrical Engineer because of the 220 kVA transmission line which will connect the mine to the local SNEL grid and its associated switch gear. Several Instrumentation Engineers may also be appointed for an initial two year contract period.

The Maintenance Department will initially be based on shift, working the same roster as the Process Department. Once operations have settled down the staff members will be transferred back to day shift though key staff will be accommodated on site for call outs.

25.10.7 Staff Resourcing

The majority of skilled personnel will be sourced from Lubumbashi, situated an estimated 40 minutes bus ride to the south of the Project. These staff will be bused to site on a daily basis with only call-out staff and rostered-on supervision remaining at the mine after hours. Travel time to and from site will not be included in the normal eight hour shift and staff travelling to and from site will travel in their own time.

Much of the semi and non skilled workforce will be recruited from the local villages. However, care will need to be taken when hiring these staff to ensure that they are from the local Balamba Tribe and not from migrant artisanal miners who are predominately Tchokwe.

25.10.8 Rosters and Accommodation

All expatriate staff will be employed on a 'fly in fly out' roster and be accommodated within the operations camp on the mining lease.

The majority of operators at Kinsevere will be residentially based and not employed on a 'fly in fly out' roster due to the proximity of Kinsevere to Lubumbashi. Though AMCK does not intend to construct a mine village as such, infrastructure in the surrounding villages will be improved such that the living standards of the local employees, their families and the surrounding population will improve. Specifically schools, clinics and potable water supplies will be targeted for upgrading during the development phase of the Project.

The Project will continue to operate on a 7 day a week, 24 hours a day basis, with production only being suspended for annual maintenance shutdowns and public holidays.

Where expatriate staff are based in the DRC on a single status basis, they will work an eight weeks on and six weeks off roster. During their rostered period in-country they will be expected to work a 13 day fortnight.

DRC national staff who are not on a shift roster, will work a 48 hour week made up of six shifts of eight hours per week. Those staff not on a shift roster but employed in the maintenance, transport and warehouse departments will be contracted in such a manner that seven day a week coverage of these departments is maintained.

National staff working on a shift roster in the process, mining, maintenance and laboratory departments work twelve hour shifts based on a 2 by 2 by 2 roster. This roster has worked well at the Dikulushi Mine and consists of 2 day shifts followed by a 24 hour break, then 2 night shifts followed by 2 days off. Maintenance staff will initially work this roster until the processing facilities are bedded down.

25.10.9 Staffing Summary

The manning numbers in the cash flow model are summarised in Table 25.48. Although these have been revised since the 2008 Technical Report, further amendments will be made to the Stage II organisation chart and manning numbers during 2010.

Table 25.48 Manning Numbers

Department	Expat #	Cadre #	National #	Total #
Mining	2	12	34	48
Technical Services	1	5	3	9
SX-EW Processing	10	6	81	97
Maintenance	19	0	79	98
Administration	6	18	28	52
Total	38	41	225	304

Note: Expat# : Total# = 12.5% which is less than the allowable 15% limit

Figure 25.20 Stage II Operations, Organisation Structure – Administration

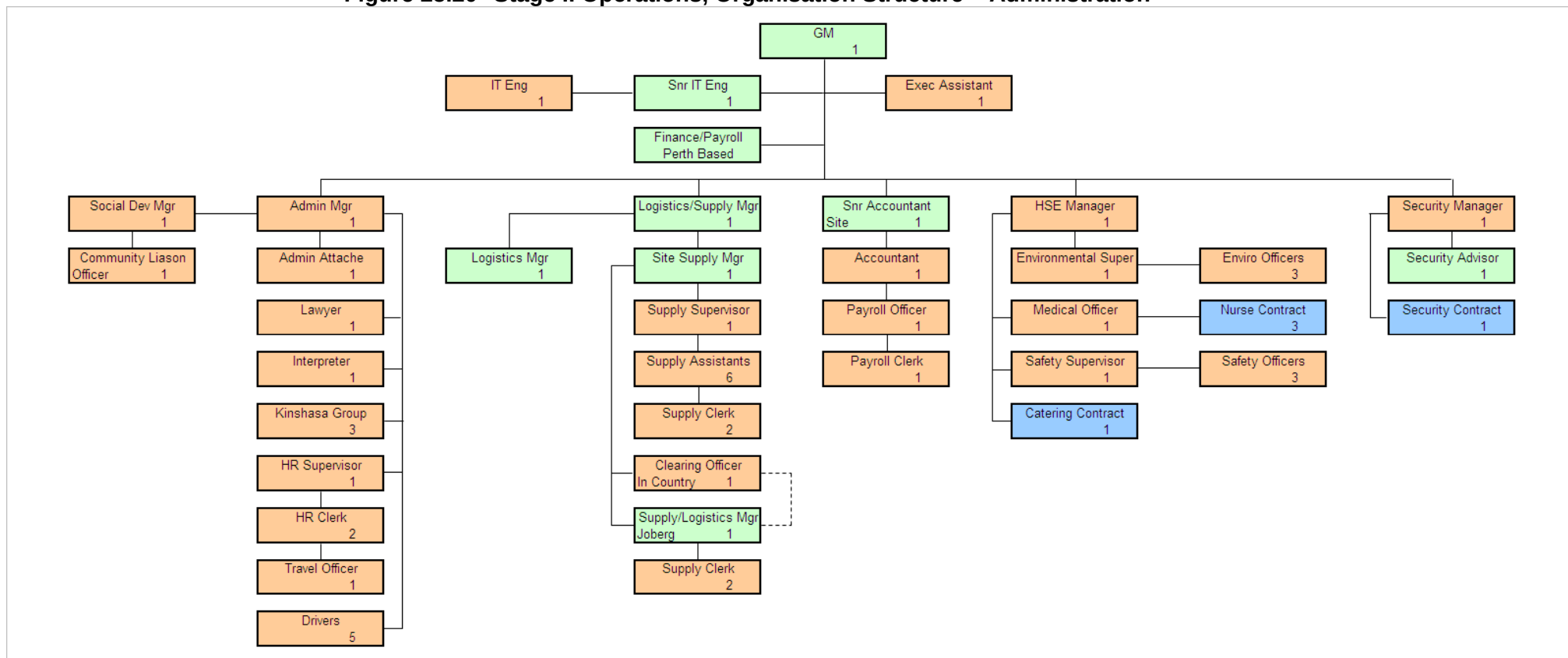


Figure 25.21 Stage II Operations, Organisation Structure – Technical Services

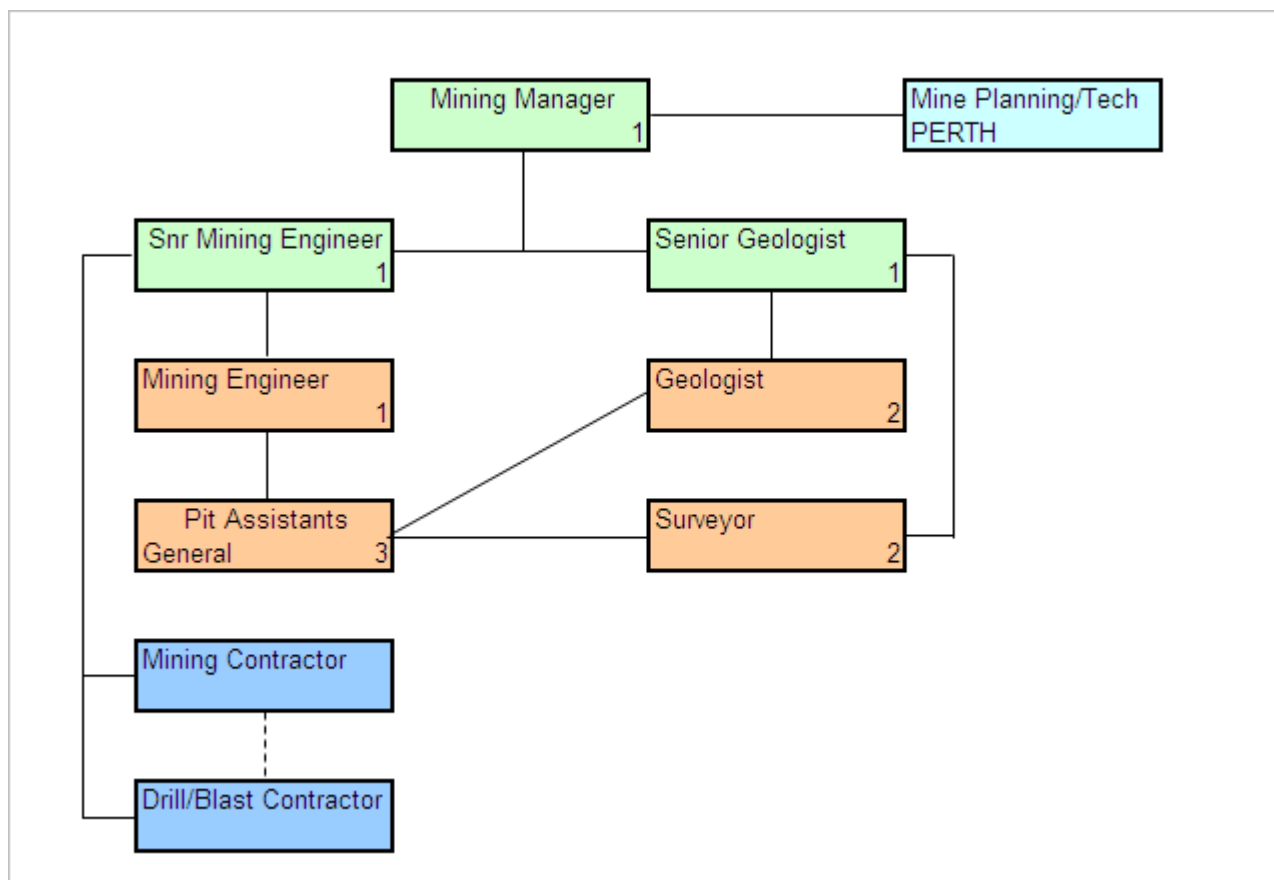


Figure 25.22 Stage II Operations, Organisation Structure – Processing

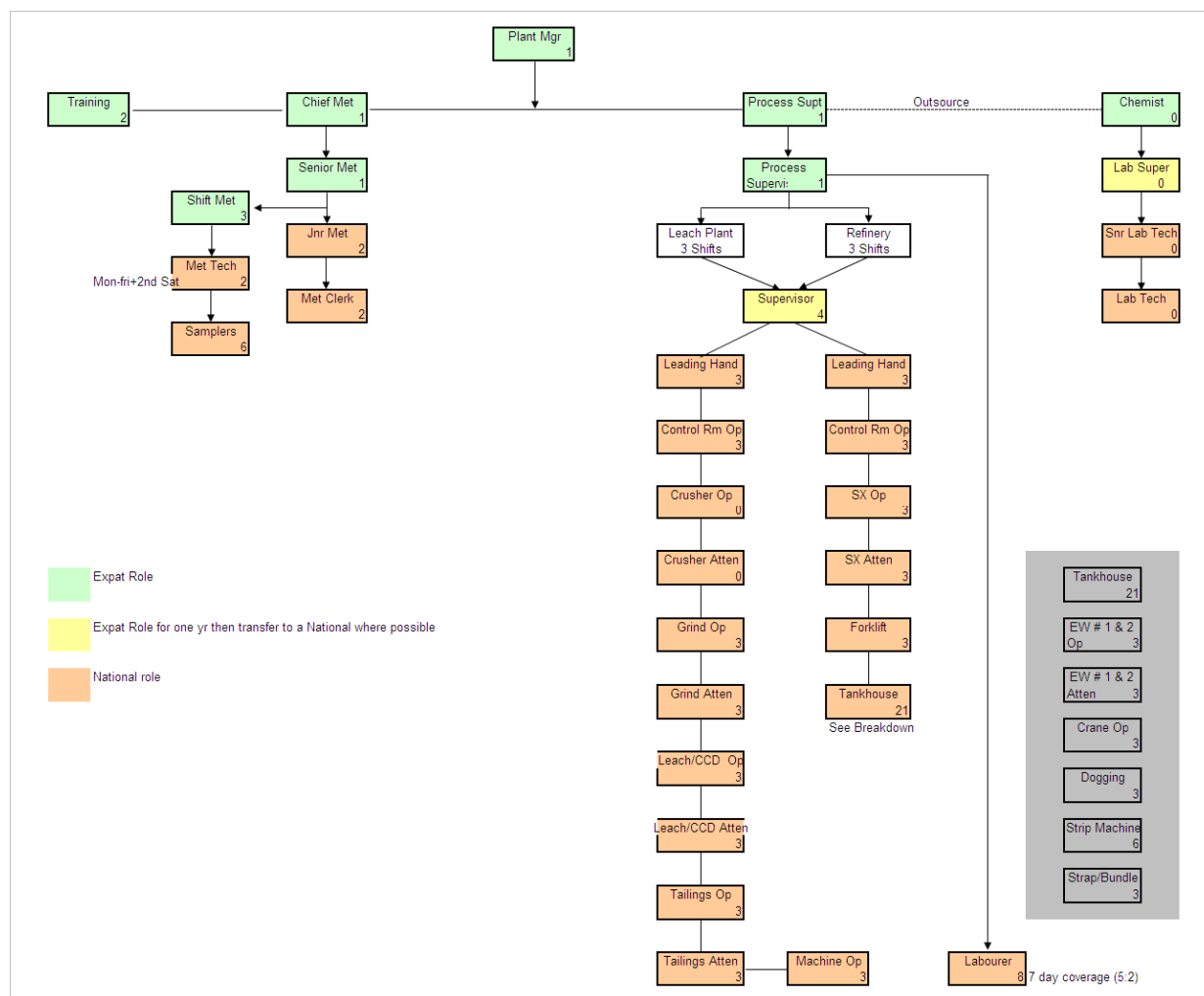
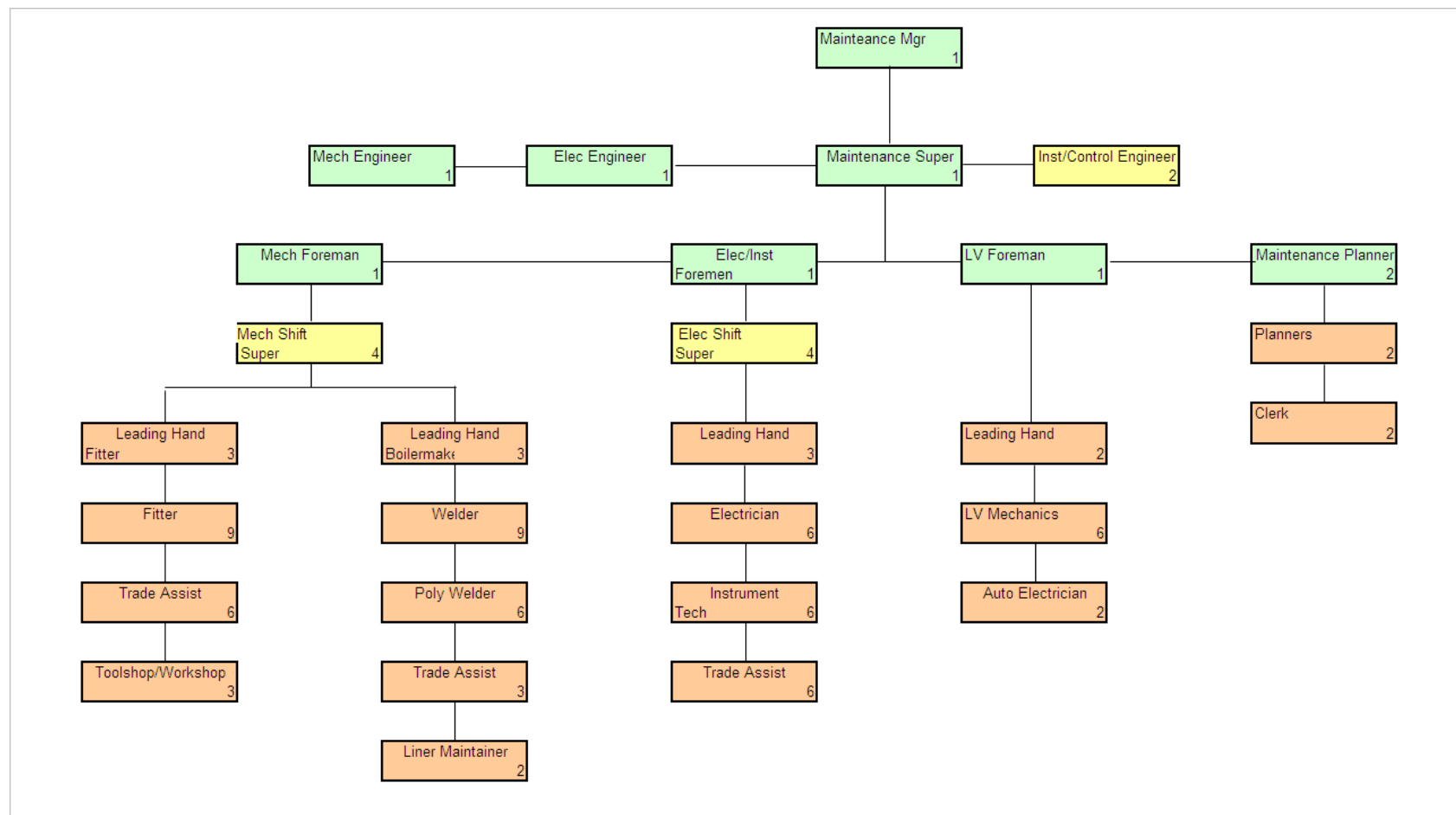


Figure 25.23 Stage II Operations, Organisation Structure – Maintenance



25.10.10 Operational Procedures

The operational procedures described in the March 2008 Technical Report are reiterated in the following commentary. Work remains in progress on developing these procedures.

Safety Policy

It is AMCK's policy that each employee will be provided with a safe and healthy place in which to work. To achieve this policy, AMCK has set the following objectives:

- Ensure compliance with statutory regulations, and maintain constant awareness of new and changing regulations.
- Aim to eliminate or control safety and health hazards in the working environment to achieve the highest possible standards for occupational health and safety in the mining industry.
- Ensure prospective employee's suitability for appropriate employment through a structured recruitment procedure.
- Provide relevant occupational health and safety information and training to all personnel.
- Develop and constantly review safe working practices and job training.
- Conduct regular departmental safety meetings and provide an open forum for input from employees.
- Provide effective emergency arrangements for the protection of employees and the public.
- Maintain good morale and safety awareness through regular employee assessment and counselling.
- Ensure contracting companies while engaged in work on site adopt the above policies and objectives and maintain the safety standards of the operation for their employees and subcontractors.
- Develop public awareness of the safety standards and objectives at all its operations.

In conjunction with this policy, a series of policy statements on specific individual safety and health matters are to be maintained.

Pre-Employment Procedures

A structured recruitment procedure has been adopted to ensure prospective employees are suitable for appropriate employment. It is AMCK's stated policy that local residents from the districts surrounding the operations namely the Kinsevere and Muombe-Bushinda villages will be employed in preference to non-local labour in all positions to which they hold suitable qualifications.

Prior to employment, all staff will be required to undergo a medical examination as required by DRC law. This will be carried out by a registered medical practitioner for local staff and by a registered travel clinic in their country of origin for all expatriate staff.

Training

AMCK recognises the beneficial impact that a well-trained operating workforce can have on the safety and economics of a project and as such has structured a training programme which addresses the following areas for both the mining and processing areas:

- General Principles for Stages and Unit Processes
- Standard Operating Procedures
- Safe Work Procedures
- Emergency Procedures
- First Aid Training
- Environmental Procedures

Training will commence prior to SX-EW commissioning and will form part of ongoing operations. Progress through the training programme will be locked in with wage rates in recognition of the higher skill level achieved through training.

Training will be initially undertaken at the Kinsevere training centre. Provision will be made for operations personnel to be exposed to a similar operating facilities (where possible) in order to provide hands-on experience prior to commissioning.

Procedure Manual and Operating Systems

Specialised contractors and consultants are to be engaged for the development of all manuals and operating systems required for the smooth operation of the Project.

It is expected that a significant amount of the operating systems and procedures will be adapted from Anvil's Dikulushi and Kulu operations. The development of manuals and operating procedures by contractors and consultants will allow Departmental Managers to concentrate on getting the operating systems running.

As a minimum, the following manuals will be developed for the various operations:

- Safety manual
- Environmental management plan
- Human resources manual
- Accounting procedures manuals
- Disciplinary code
- Training manuals for geology, mining, process plants
- Maintenance manual and procedures
- Mining operations manual and procedures
- Plant operations manual and procedures
- Laboratory procedures manual (by contractor)

25.11 Cost Estimates

25.11.1 Operating Costs

Operating cost estimates have been revised since the 2008 Technical Report, as follows:

- Stage I operating costs, based on 2009 estimates by Anvil, are as listed in Table 25.49, and include contractor mining costs derived from the MCK Contract rates;
- Stage I processing costs include a supervision charge of \$10.00/tonne of concentrate sold and a 1% weight franchise charge (on gross revenue);
- other Stage I budgeted operating cost allowances include non-cash items such as stock based compensation and stock drawdown provisions;
- Anvil estimated Stage II variable operating costs are listed in Table 25.50, whilst Stage II fixed operating costs are listed in Table 25.51;
- the Stage II process operating costs include the cost of sulphuric acid at a supply rate of \$280/tonne (although Trafigura has since advised a price of \$210/tonne supplied to Kinsevere); and
- import duties are levied on fuel and reagent costs at a rate of 40%, whilst all other consumables are subject to a rate of 30%.

Table 25.49 Stage I Operating Cost Inputs (Real)

Item	2009 Average/month	2010 Average/month
Mining	\$518,455	\$458,769
Technical Services	\$78,015	\$75,796
Crushing	\$20,112	\$19,734
Processing	\$212,040	\$195,709
Mobile Plant Maintenance	\$81,346	\$71,329
Grid Power	\$171,575	\$171,575
Power Supply	\$7,536	\$9,036
Maintenance	\$46,097	\$45,927
Civils	\$54,632	\$0
Transport	\$44,917	\$49,053
<i>Site Support Services:</i>		
Security	\$158,081	\$157,003
Health & Safety	\$38,169	\$33,305
Camp	\$36,868	\$36,501
Supply	\$28,640	\$26,390
Human Resource	\$66,782	\$35,594
IT	\$49,879	\$48,090
Administration (inc Kins & Joburg)	\$239,565	\$237,662
Logistics	\$43,259	\$41,152
Finance	\$80,902	\$75,450
Social Development	\$31,683	\$28,441
Total Site Operating Cash Costs	\$2,008,553	\$1,816,515

Table 25.50 Stage II Variable Operating Cost Inputs (Real)

Mining		
Mining Contractor (incl Rise & Fall)	\$/bcm	\$5.88
Drill and Blast – Oxide	\$/t total material	\$0.06
Drill and Blast – Saprolite	\$/t total material	\$0.12
Drill and Blast – Bedrock	\$/t total material	\$0.24
Grade Control	\$/t ore	\$0.12
Mining Contractor Dayworks	% Cost	3.20%
Processing		
Reclamation Costs - LT s/p to crusher (incl Rise & Fall)	\$/t ore	\$2.24
SXEW ROM Feed - ROM s/p to crusher (incl Rise & Fall)	\$/t ore	\$0.95
Power Crush and Mill @ \$0.038/kWhr	\$/t ore	\$1.40
Power Cost SXEW – Contract @ \$0.038/kWhr	\$/t ore	\$4.30
Chemical Reagents: Crush and Mill (excl import charges)	\$/t ore	\$5.87
Chemical Reagents: SXEW Variable Cost (excl import charges)	\$/t ore	\$5.32
Tailings Dam Variable Cost	\$/t ore	\$0.66
Process Water (Filtered)	\$/t ore	\$0.01
Plant Lab Variable Cost	\$/t ore	\$0.08
Maintenance		
Treatment Plant Slurry Section	\$/t ore	\$1.39
SX Section	\$/t ore	\$0.19
EW Section	\$/t ore	\$0.23
Reagents	\$/t ore	\$0.50
Tailings Dam Variable Cost	\$/t ore	\$0.05
Transport and Marketing		
% Shipped to Johannesburg	%	60%
% Shipped to Dar Es Salaam	%	40%
Transport Rate to Johannesburg	\$/t Cathode	\$240
Transport Rate to Dar Es Salaam	\$/t Cathode	\$180
Infrastructure Levy	\$/t Cathode	\$30
Margin over Libor	% p.a.	3%
Days to Bill of Landing Johannesburg	Days	0
Interest to B/L + 7 Johannesburg provisional	%	90%
Days to Bill of Landing Dar Es Salaam	Days	0
Interest to B/L + 7 Dar Es Salaam provisional	%	90%
Export Charges (per 32T truck)	\$/32T Load	2,800
OFIDA / Export Duty Payable	%	1%
OCC (per 32T truck)	\$/32T Load	300
Containerisation (Rail to FOB Johannesburg)	\$/t Cathode to J'burg	25
FOB Johannesburg	\$/t Cathode	30
FOB Dar Es Salaam	\$/t Cathode	28
Ocean Freight	\$/t Cathode	50
Insurance	%	0.025%

Table 25.51 Stage II Fixed Operating Cost Inputs (Real)

Mining	\$ pa
Total Labour Costs	\$1,600,130
Drilling and Sampling of Stockpiles	\$100,000
Mining Contractor Fixed Costs - Mobilisation	\$306,015
Mining Contractor Fixed Costs – Management (incl. Rise & Fall)	\$988,649
Stage II - Dewatering - (In-pit, Perimeter, Consult))	\$598,000
Stage II - Owner Fuel - (Drill & Blast, Vehicles & Other)	\$837,900
Stage II - Travel (R&R)	\$41,000
Stage II - Other Fixed Costs (inclusive of import)	\$462,792
Subtotal Fixed Costs (before import duty)	\$4,934,485
Technical Services	\$ pa
Total Labour Costs	\$663,854
Light Vehicles	\$105,368
Others subject to import	\$151,540
Others not subject to import	\$208,400
Fuel (inclusive of import duty)	\$45,000
Travel R&R	\$20,500
Subtotal Fixed Costs (before import duty)	\$1,194,661
Processing	\$ pa
Total Labour Costs	\$3,173,775
SX-EW – Vehicles	\$52,920
SX-EW - Subj to import	\$110,500
SX-EW - Subj to import (Headcount)	\$882
SX-EW - Not subj to import	\$675,283
SX-EW - Not subj to import (Headcount) - Travel	\$20,500
Fuel (inclusive of import duty)	\$22,268
Subtotal Fixed Costs (before import duty)	\$4,056,128
Maintenance	\$ pa
Total Labour Costs	\$3,115,987
Subject to import	\$110,500
Subject to import (Headcount)	\$69,678
Vehicles	\$105,840
Not subj to import	\$290,900
Not subj to import (Headcount)	\$161,500
Fuel (inclusive of import duty)	\$44,536
Subtotal Fixed Costs (before import duty)	\$3,898,941
Transport and Marketing	\$ pa
Total Labour Costs	\$122,840
Transport and Marketing (not subject to import)	\$51,000
Subtotal Fixed Costs (before import duty)	\$173,840
Administration	\$ pa
Total Labour Costs	\$4,397,883
Admin - Start Up Training	\$299,328
Vehicles	\$264,600
Subject to Import	\$244,250
Subj to import (Headcount)	\$24,696
Not Subject to Import	\$3,565,700
Not subj to import (Headcount) - Travel R&R	\$123,000
Not subj to import - Contract Bus Transport	\$2,093,992
Fuel (inclusive of import duty)	\$111,340
Subtotal Fixed Costs (before import duty)	\$11,124,789
TOTAL FIXED COSTS (before import duty)	\$25,382,844

Table 25.52 lists the Stage II unit cash operating costs, in real terms averaged over the life of the Project. The cash costs shown bracketed in red are those averaged over the period of steady state metal production. Under these circumstances, the mining cost is higher because it is spread over the life of mining, rather than processing (ie, plant feed is drawn from stockpiles after mining has ceased). The processing costs are lower because total annual costs are not included for those periods at the beginning and end of the Project life, when metal production is ramping up and then decreasing with declining head grade, respectively.

Table 25.52 Stage II Unit Operating Cost Summary Averaged Over the Life of Project

Average SX-EW Operating Costs	Units	LOM Real Cash Costs
Mining Costs	\$/lb	0.14 (0.18)
Fixed Technical Services Costs	\$/lb	0.01 (0.01)
Processing Costs	\$/lb	0.34 (0.26)
Maintenance Costs	\$/lb	0.07 (0.06)
Administrative Costs	\$/lb	0.11 (0.08)
Cash Cost at Mine Gate	\$/lb	0.67 (0.59)
Transport and Marketing Costs	\$/lb	0.20 (0.20)
Export Duties	\$/lb	0.02 (0.02)
Total C1 Cash Costs	\$/lb	0.89 (0.81)
Royalties Paid	\$/lb	0.11 (0.11)
Depreciation and Amortisation	\$/lb	0.33 (0.26)
Total C3 Operating Costs	\$/lb	1.33 (1.17)

25.11.2 Capital Costs

As at Q4 2009 approximately \$199 million (\$179.5 million spent, \$19.5 million committed) of the revised budgeted cost of \$395 million had been invested. The increase in capital cost from the estimate reported in the March 2008 Technical Report is a result of several factors including: revised Ausenco estimates; redundancy payments to Kinsevere Stage II personnel; payment of demobilisation costs to contractors; charges associated with the cancellation of the engineering and design works; penalties in connection with the cancellation and suspension of orders; provision for re-establishment of contracts and remobilisation and some escalation primarily due to the contract now being a Lump Sum Turn Key (LSTK) contract rather than an engineering, procurement and construction management (EPCM) contract.

The breakdown of Stage II capital costs is provided in Table 25.53 for the LSTK contract and in Table 25.54 for the LSTK plus other costs. Before payment allowances to Gecamines and Stage II creditors, the subtotal capital cost (ie, remaining cost) is \$195.86 million. When added to the \$199 million of funds already spent, the total cost in the Project cash flow model is approximately \$395 million.

Table 25.54 notes the "Owners" costs as being approximately \$52 million and the Owners contingency as being \$15 million. These figures were revised for the January 2010 News Release (Anvil, January 2010) and reported as \$58 million and \$12 million, respectively. Table 25.53 shows the LSTK cost as \$128.9 million, whereas in the News Release the value of the LSTK contract was noted as \$130 million, bringing the reported subtotal capital cost (ie, remaining cost) to \$200 million.

Table 25.53 Stage II LSTK Costs

LSTK COSTS	USD
PROCESS PLANT FACILITY	
Earthworks	\$8,754,204
Civil	\$8,892,103
Structural	\$8,845,307
Platework	\$9,088,759
Mechanical	\$7,822,431
Procurement	
Piping	\$14,812,102
Electrical	\$12,231,717
Architectural	\$1,942,076
	\$72,388,699
SITE SUPPORT	
Site Accommodation	\$2,621,223
Transport	\$500,000
Construction Equipment	\$144,250
Temporary Construction Facilities	\$3,953,798
	\$7,219,271
COMMISSIONING	\$1,500,000
SPARES - COMMISSIONING	\$500,000
OTHER	
Construction Camp Provision	\$260,000
Earthworks Provision	\$650,000
Concrete Works Provision	\$322,500
Stock Steel	\$115,500
Potable Water - camps etc.	\$180,000
Fees - Ausenco for LSTK quotation	\$272,100
	\$1,800,100
EPCM LABOUR	\$12,134,240
EPCM EXPENSES	\$2,433,781
ACCURACY PROVISION	\$6,635,955
ESCALATION ALLOWANCE	\$2,605,000
AUSENCO CONTINGENCY	\$9,670,000
DEFECTS LIABILITY / WARRANTY	\$1,563,000
AUSENCO PROFIT FEE	\$10,420,000
AUSENCO TOTAL	\$128,870,046

Table 25.54 Stage II Capital Cost Summary

CAPITAL COSTS	USD
LSTK COSTS	\$128,870,046
OWNERS COSTS	
Capital purchases including transport and duties	\$21,791,812
Insurance & commissioning spares	\$13,408,377
Commissioning	\$10,060,328
During construction expenses	\$3,983,004
Subcontractors	\$207,686
Services contract - complete design	\$2,534,143
	\$51,985,350
OWNERS CONTINGENCY	
Contingency on Owners Cost	\$5,000,000
Contingency on LSTK Contract	\$10,000,000
	\$15,000,000
ALLOWANCES	
payment to Gecamines	\$5,000,000
Stage II creditor payments	\$6,400,000
	\$11,400,000
TOTAL IN CASH FLOW MODEL	\$207,255,396

The capital spend profile has also been revised since the March 2008 Technical Report, as follows:

- the capital expenditure for Stage II construction commences in December 2009 and is scheduled as shown in Table 25.55, incorporating a spend profile as advised by Ausenco on 25 November 2009, but with a flat profile for the Company's capital expenditure and contingency provisions (included in Table 25.58);
- additional capitalised expenditure is included in the Stage II construction figures in Table 25.55 to cover a final payment of \$5 million to Gecamines in January 2010, plus Stage II creditor payments of \$6.4 million paid in equal instalments in January, February and March 2010;
- over the period that mined waste is available to be hauled to the site, the expenditure for the TSF as shown in Table 25.55 relates to construction earthworks for the embankments, contractor fixed charges, plus supply and extension of the HDPE liner up the TSF internal walls;
- **additional sustaining capital costs** are assumed at 2.5% of operating costs; and
- **additional Project closure costs** of \$23.45 million (real) are assumed to be spent in the last year of operations (ie, 2024).

Table 25.55 Capital Expenditure Schedule (\$'000) (Real)

2009	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	SubTotal
SII Const												\$22,644	\$22,644
SII TSF													\$0
2010	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	SubTotal
SII Const	\$18,567	\$16,772	\$15,896	\$16,580	\$17,471	\$17,840	\$17,192	\$15,363	\$13,299	\$13,980	\$11,515	\$10,088	\$184,563
SII TSF													\$0
2011	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	SubTotal
SII Const													\$0
SII TSF	\$730	\$730	\$730	\$730	\$730	\$730	\$730	\$730	\$730	\$730	\$730	\$730	\$8,764
2012	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	SubTotal
SII Const													\$0
SII TSF	\$686	\$686	\$686	\$686	\$686	\$686	\$686	\$686	\$686	\$686	\$686	\$686	\$8,229
2013	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	SubTotal
SII Const													\$0
SII TSF	\$682	\$682	\$682	\$682	\$682	\$682	\$682	\$682	\$682	\$682	\$682	\$682	\$8,180
2014	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	SubTotal
SII Const													\$0
SII TSF	\$578	\$578	\$578	\$578	\$578	\$578	\$578	\$578	\$578	\$578	\$578	\$578	\$6,940
2015	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	SubTotal
SII Const													\$0
SII TSF	\$491	\$491	\$491	\$491	\$491	\$491	\$491	\$491	\$491	\$491	\$491	\$491	\$5,893
2016	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	SubTotal
SII Const													\$0
SII TSF	\$750	\$750	\$750	\$750	\$750	\$750	\$750	\$750	\$750	\$750	\$750	\$750	\$9,004
2017	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	SubTotal
SII Const													\$0
SII TSF	\$799	\$799	\$799	\$799	\$799	\$799	\$799	\$799	\$799	\$799	\$799	\$799	\$9,588
2018	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	SubTotal
SII Const													\$0
SII TSF	\$241	\$241	\$241	\$241	\$241	\$241	\$241	\$241	\$241	\$241	\$241	\$241	\$2,890
TOTAL													TOTAL
SII Const													\$207,207
SII TSF													\$59,489

25.11.3 Corporate and Other Charges

Corporate and other charges have been revised since the 2008 Technical Report, as follows:

- operating cost allowances for SMK are budgeted at an average of \$70,781/month for the duration of Stage I and Stage II;
- operating cost allowances for Anvil's Perth office are budgeted at an average of \$351,356/month for the duration of Stage I and Stage II;
- a total provision of \$783,430 is made for Anvil Dikulushi costs, through to March 2010;
- a total provision of \$1.51 million is made for Stage II costs incurred during 2009 and 2010, and relating to expatriate staff, travel and insurance;
- a total provision of \$3.9 million is made for Anvil's Canadian office costs incurred during 2009 and 2010;
- budgeted cash outflows include payment plan and creditor payments in the period up to December 2010, as listed in Table 25.56;
- budgeted cash inflows include debtor payments, as listed in Table 25.56; and
- *pas de porte* payments relating to SMK of \$7,198,334 were paid in January 2010 (with the balance of \$7,125,000 budgeted for January 2011).

Table 25.56 Corporate Cost Provisions (Real)

September 2009 to December 2010 Provisions	Amount
Stage II creditors	\$12,666,941
Distribution to Dikulushi Trust	\$1,750,000
Operating Creditors & Commitments	\$15,859,921
Kapulo Joint Venture	\$1,289,177
Stage II Electrical Sub Station	\$1,017,187
Rehabilitation payments - AMCK	\$500,000
Mutoshi <i>Pas de Porte</i>	\$7,198,334
Capital Budget Stage I - dewatering & grade control drilling	\$1,360,900
Corporate and holding costs	\$15,820,200
Less sundry revenue – debtors	(\$9,819,068)
TOTAL	\$47,643,591

25.12 Economic Analysis

25.12.1 Base Case Cash Flow

The Base Case Cash Flow Model has the following fundamental inputs and assumptions:

Physicals:

- the mining and processing physicals are based on LOM Plan Scenario H, as described in Item 25.2, yielding average Stage II mining and production figures as shown in Table 25.57; and
- the mining and processing start dates are 1 January 2010 for Stage I and 1 January 2011 for Stage II.

Table 25.57 Production Assumptions for Stage II

Annual Mining and Production	Units	Average Annual Rate
Material mined	bcm	2,960,000
Material mined	tonnes	5,020,000
Ore mined	tonnes	2,230,000
Tonnes Processed	tonnes	1,600,000
Payable Copper	tonnes	46,220 ^{1,2}

1. Includes reprocessing of Stage I floats and effluent.

2. The average is less than 60,000 tonnes due to the lower head grade from 2019.

Copper Pricing:

- a Base Case copper pricing scenario was used based on LME forward pricing trends as listed in annual average terms in Table 25.58; and
- a US CPI rate of 2% has been used to convert to nominal terms from the real terms shown in Table 25.59.

Table 25.58 Copper Pricing Scenarios (Nominal)

Copper Price	2009	2010	2011	2012	2013	2014	2015	2016	2017
LME forward price (Nov 2009)	\$2.32	\$2.94	\$2.93	\$2.91	\$2.88	\$2.85	\$2.82	\$2.80	\$2.78
Copper Price	2018	2019	2020	2021	2022	2023	2024	2025	2026
LME forward price (Nov 2009)	\$2.77	\$2.77	\$2.83	\$2.89	\$2.95	\$3.01	\$3.08	\$3.15	\$3.22

Table 25.59 Copper Pricing Scenarios (Real)

Copper Price	2009	2010	2011	2012	2013	2014	2015	2016	2017
LME forward price (Nov 2009)	\$2.29	\$2.85	\$2.79	\$2.71	\$2.63	\$2.55	\$2.48	\$2.41	\$2.35
Copper Price	2018	2019	2020	2021	2022	2023	2024	2025	2026
LME forward price (Nov 2009)	\$2.29	\$2.23	\$2.23	\$2.23	\$2.24	\$2.24	\$2.24	\$2.25	\$2.25

Cash flows:

- the cash flows are calculated from 1 January 2010 (except where sunk capital costs are included);
- royalties to Gecamines are paid at a flat 2.5% rate of gross revenue, whilst royalties paid to the DRC Government are at a rate of 2% of net smelter revenue;
- Stage I cashflows, between 1 September 2009 and 31 December 2010, are included for valuation purposes and consolidated cash balances (and Stage I revenues are based on the sale of HMS plant concentrate at 40% LME price payable);
- corporate expenditure is included for the same reasons;
- DRC Government Mining Review payments made in July 2009 (\$10 million) and January 2010 (\$5 million) are amortised over the life of mine;
- DRC company tax rate is 30%;
- Stage I depreciation is incurred at \$1 million per month, for the 21 month period between April 2009 (ie, the recommencement of HMS processing) and December 2010, whilst Stage II depreciation is calculated at 60% of declining balance commencing in February 2011 (ie, the month following the commencement of Stage II processing);
- Stage I amortisation amounts to \$3.1 million over financial years 2009 and 2010; and
- Working capital for Stage II (ie, in relation to sales revenue) is calculated assuming 90% receivable in 7 days and the balance in 90 days.

Tables 25.60 and 25.61 summarise the Project cash flows for circumstances where the sunk costs (ie, before January 2010) are excluded and included, respectively.

Table 25.60 Cash Flow Summary, Excluding Sunk Costs

Year		Sum	Pre 2009	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Ore Processed	Mtpa	21.82		-	-	1.39	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.21	-
Cash Flows		-																		
Revenue	USD m	3,475.26		-	-	256	360	348	337	328	320	310	302.9	270	200	153	115	99.5	74.7	3.08
Operating Costs excl. Royalties	USD m	-		-0.15	-1.21	-87.3	-104	-108	-108	-108	-108	-108	-108	-84	-77	-72	-68.5	-66.9	-51.8	-4.13
Royalties Paid	USD m	-153.49		-	-	-9.63	-16	-15	-15	-14.5	-14.2	-13.7	-13.4	-12	-9.11	-6.9	-5.21	-4.44	-3.4	-0.46
Post Financing Tax Paid	USD m	-442.83		-	-	-0.21	-0.36	-31	-53.5	-56.9	-56.6	-53.4	-51.6	-60	-32.3	-21	-11.7	-8.24	-5.95	-
Capital Expenditure	USD m	-315.28		-22.6	-173	-11.5	-11.3	-11	-10.1	-9.04	-12.2	-12.7	-5.98	-2.4	-2.19	-2	-1.88	-1.83	-24.8	-
Kinsevere Other Corporate Costs	USD m	-5		-	-5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other Stage II Costs Incurred during Construction	USD m	-6.35		-	-6.35	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Anvil Corporate Costs (Perth and Canada only)	USD m	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Operating and Investing Cash Flows	USD m	1,287.07		-22.8	-186	148	228	181	150	139	129	122	124.2	111	79	50.6	27.4	18.2	-11.2	-1.52

Table 25.61 Cash Flow Summary, Including Sunk Costs

Year		Sum	Pre 2009	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Ore Processed	Mtpa	21.82		-	-	1.39	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.21	-
Cash Flows																				
Revenue	USD m	3,475.26		-	-	256.5	359.9	348	336.7	327.6	319.7	310	303	270	199.6	153	115	99.5	74.7	3.08
Operating Costs excl. Royalties	USD m	-		-0.15	-1.21	-87.3	-104.1	-108.4	-108	-108	-108	-108	-108	-83.8	-77	-72	-68.5	-67	-51.8	-4.13
Royalties Paid	USD m	-153.49		-	-	-9.63	-16	-15.46	-15	-14.5	-14.2	-13.7	-13.4	-12.1	-9.11	-6.9	-5.21	-4.4	-3.4	-0.46
Post Financing Tax Paid	USD m	-442.83		-	-	-0.21	-0.36	-31.31	-53.5	-56.9	-56.6	-53.4	-51.6	-60	-32.3	-21	-11.7	-8.2	-5.95	-
Capital Expenditure	USD m	-537.03	-221.74	-22.64	-173	-11.5	-11.32	-11.37	-10.1	-9.04	-12.2	-12.7	-5.98	-2.44	-2.19	-2	-1.88	-1.8	-24.8	-
Kinsevere Other Corporate Costs	USD m	-5		-	-5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Other Stage II Costs Incurred during Construction	USD m	-6.35		-	-6.35	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Anvil Corporate Costs (Perth and Canada only)	USD m			-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Operating and Investing Cash Flows	USD m	1,065.32	-221.74	-22.8	-186	147.8	228.1	181.49	149.8	139.3	128.8	122	124	111	78.97	50.6	27.4	18.2	-11.2	-1.52

25.12.2 Base Case Project Value

A summary of the key Project outcomes for the Base Case Cash Flow Model is provided in Table 25.62, in real terms. These outcomes **exclude corporate costs**.

Note that this outcome represents the **value of the Project as at 1 January 2010** (the date coincident with the Mineral Reserve statement) and ignores sunk capital costs.

In comparing between the economic results of the Feasibility Study and the current Base Case, points to note are as follows:

- the Stage I NPV from the Feasibility Study reflected production of “black copper” from the EAF during 2008 and 2009;
- the current Stage I NPV only accounts for production of HMS concentrate (and hence lower revenue) for the period from January to December 2010;
- the duration of the Stage II Project is now shorter than envisaged in the Feasibility Study, due to the revised Mineral Resource and Mineral Reserve estimates described in Item 19, and due to the changed methodology for selecting the ultimate mining pits;
- the estimated total capital cost for the Stage II plant has increased by over 60% since the Feasibility Study, whilst the TSF cost has increased as a result of the change in design as described in Item 25.5;
- sustaining capital costs are now calculated at 2.5% of operating costs, instead of 2.5% of the Stage II plant capital expenditure, as adopted in the Feasibility Study; and
- the adopted current long term copper price, however, is over \$0.80/lb higher than the price used in the Feasibility Study, which has a significant impact on the Project cashflow (ie, price and ore grade are the most sensitive Project variables).

The January 2010 News Release (Anvil, January 2010) advised the following economic results, based on the exclusion of sunk costs incurred before 1 January 2010:

- Stage II NPV at 95% equity = \$683 million;
- Stage II Internal Rate of Return = 79%; and
- Payback period of less than 2.5 years.

These results differ slightly from the key outcomes listed in Table 25.62 due to production schedule changes made between the date of the News Release and this Technical Report (refer to Item 25.1.9):

- Stage II NPV at 95% equity = \$668 million;
- Stage II Internal Rate of Return = 73%; and
- Payback period of less than 2.5 years.

The economic results when sunk costs incurred before 1 January 2010 are included are also shown below and in Table 25.63:

- Stage II NPV at 95% equity = \$457 million;
- Stage II Internal Rate of Return = 34%; and
- Payback period of less than 3.6 years.

Table 25.62 Summary of Key Project Outcomes (Real), Excluding Sunk Costs

Description	Unit	2007 Feasibility Study Base Case	2010 Base Case
Economic Return, Excluding Sunk Costs			
Discount Rate	%	10%	10%
Stage I+II NPV	\$ million	\$281.22	\$720.47
Stage I NPV	\$ million	\$83.32	\$17.49
Stage II NPV	\$ million	\$197.91	\$702.97
Stage II IRR	%	25%	73%
Stage I+II NPV , 95% Equity	\$ million	\$267.16	\$684.44
Stage I NPV, 95% Equity	\$ million	\$79.15	\$16.62
Stage II NPV, 95% Equity	\$ million	\$188.01	\$667.82
Operations Life			
Mining duration: Stage I, Stage II	Years	2.25, 10.75	1.0, 8.0
SX-EW duration (incl floats/effluent processing)	Years	16.5	14.00
Mining			
Ore mined	million t	25.74	18.41
Waste mined	million t	43.68	22.68
Total material mined	million t	69.42	41.10
Processing			
Stage I HMS - ore processed	million t	1.00	0.73
Stage I HMS – Cu recovered	Tonne	41,937	16,799
Stage II SX-EW - ore processed (inc floats/effluent)	million t	26.23	21.82
Stage II SX-EW – Cu recovered	tonne	842,487	647,075
Total Tonnes of Cu metal produced	tonne	882,484	663,874
Operating Costs			
Mining + Technical Services	\$/lb	\$0.14	\$0.19
Processing - Stage I	\$/lb	\$0.015	-
Maintenance - Stage I	\$/lb	\$0.002	-
Transport and Marketing - Stage I	\$/lb	\$0.01	-
Processing - Stage II	\$/lb	\$0.26	\$0.26
Maintenance - Stage II	\$/lb	\$0.06	\$0.06
Transport and Marketing - Stage II	\$/lb	\$0.18	\$0.22
Administration	\$/lb	\$0.08	\$0.08
Depreciation and Royalties	\$/lb	\$0.24	\$0.37
Total Operating Cost, Stage I and Stage II	\$/lb	\$0.98	\$1.19
Capital Costs			
Capital cost - Stage I	\$ million	\$4.22	-
Capital cost - Stage II – Expended (Sunk)	\$ million	-	\$199.09
Capital cost - Stage II - Remaining	\$ million	-	\$195.86 ¹
Capital cost - Stage II - Total	\$ million	\$241.50	\$394.95
Tailings Storage Facility	\$ million	\$39.83	\$59.49
Drilling and sampling of stockpiles	\$ million	\$1.70	\$0.80
Sustaining capital	\$ million	\$6.93	\$36.49
Closure costs	\$ million	\$8.19	\$23.45
Total Capital Cost	\$ million	\$302.37	\$515.27
Other Costs			
Corporate charges	\$ million	\$12.74	\$15.00
Revenue			
Long term Copper Price, Average Copper Price	\$/lb	\$1.43, \$1.53	\$2.24, \$2.44
Total copper revenue	\$ million	\$2,984.58	\$3,475.26
Charges			
Royalty - Gecamines	\$ million	\$55.49	\$86.88
Royalty - DRC	\$ million	\$54.79	\$63.83
Cash flow			
Cash flow after capital and tax	\$ million	\$783.24	\$1,287.07

1. Excludes final payment to Gecamines and Stage II creditor payments (cf Table 25.56)

Table 25.63 Summary of Key Project Outcomes (Real), Including Sunk Costs

Description	Unit	2007 Feasibility Study Base Case	2010 Base Case
Economic Return, Including Sunk Costs			
Discount Rate	%	10%	10%
Stage I+II NPV	\$ million	\$281.22	\$498.72
Stage I NPV	\$ million	\$83.32	\$17.49
Stage II NPV	\$ million	\$197.91	\$481.23
Stage II IRR	%	25%	34%
Stage I+II NPV , 95% Equity	\$ million	\$267.16	\$473.79
Stage I NPV, 95% Equity	\$ million	\$79.15	\$16.62
Stage II NPV, 95% Equity	\$ million	\$188.01	\$457.17

25.13 Sensitivity Analysis

Due to the relatively high average oxide Mineral Resource grade at Kinsevere, the key economic outcomes listed in Tables 25.60 and 25.61 are highly sensitive to variations in copper price, with every 10% change in the copper price resulting in a change of \$120 million in the NPV (Figure 25.24). Movements in other key variables have a less significant impact (Figure 25.25). For example, a 10% change in capital expenditure results in only a 2% change in the NPV, whilst a 10% change in the operating costs results in a 7% change in the NPV.

Figure 25.24 Sensitivity to Copper Price (\$/lb) - Based on Project NPV Including Sunk Costs to Date

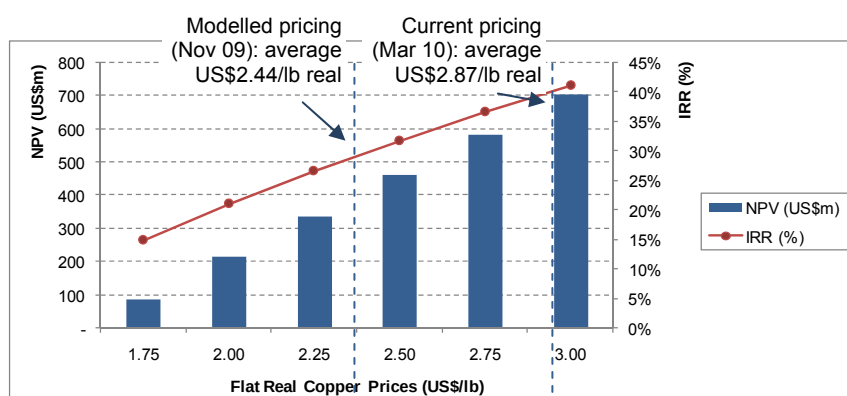
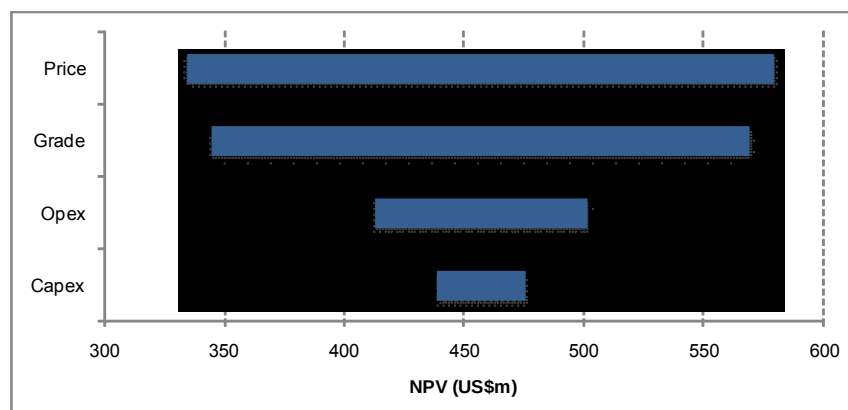


Figure 25.25 NPV Sensitivity to 10% Changes in Capex, Opex, Grade and Copper Price



Further sensitivity analyses were undertaken to address the concerns of the Independent Engineer (SRK, 2009) as noted in Item 21.3. The results are summarised in Table 25.64.

Table 25.64 NPV Sensitivity Analysis Summary (Real), SRK Scenarios

Test Scenario	Price Scenario	Stage I NPV (\$ million)	Stage II NPV (\$ million) ¹	Corporate NPV (\$ million)	Combined Asset Value (\$ million)
1	Base Case at current LME Forward Copper Price, averaging \$2.44/lb (real)	17	457	(83)	391
1a	Scenario 1 without HMS floats and effluent processing in Stage II	17	439	(81)	374
2	Scenario 1 with allowance for work-in-progress, inclusion of environmental liability provisioning as operating expenditure and inclusion of terminal benefits liabilities on closure	17	449	(83)	383
3	Scenario 2 with production 3 month commissioning delay, increased capital expenditure of +15%, increased dewatering costs and capital expenditure for additional bores	17	396	(83)	330
4	Scenario 3 with downward adjustment of metal content (grade) by - 10%	17	287	(83)	221
5	Scenario 3 with incorporation of additional other corporate costs within the Kinsevere Project	17	317	-	334
6	Scenario 5 with retrofit of the SX-EW plant to replace MIR/DTD flowsheet and a long term increase to SX-EW operating expenditure of \$9.00/t	17	235	-	251

1. Including sunk costs prior to 1 January 2010.

In relation to the Stage II Project, and including sunk costs prior to January 2010, these results show that:

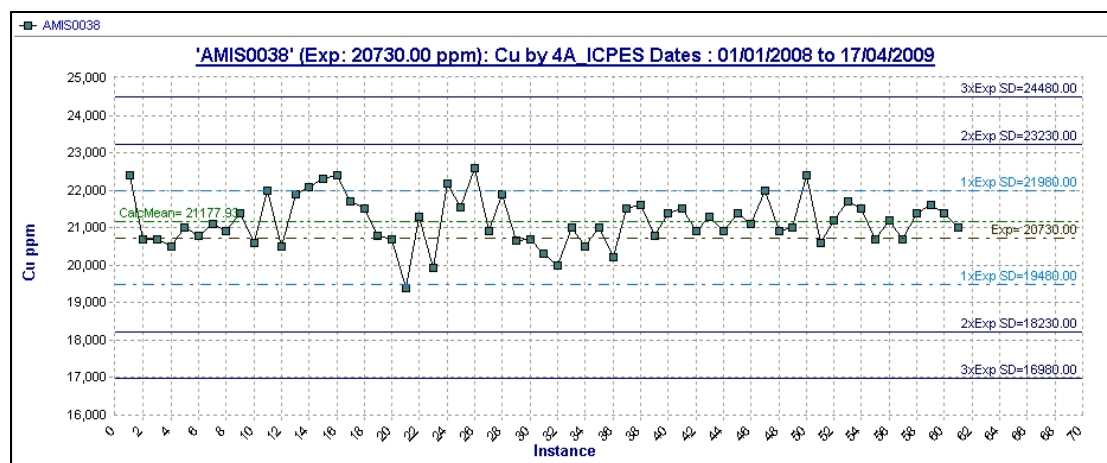
- excluding the floats and effluent (as opportunistic feed) from the Stage II production profile reduces the Base Case NPV by only 4%;
- including adjustments as per Scenario 3 (reflecting a three month commissioning delay and increased capital expenditure) reduces the Base Case NPV by 13%; and
- as a worst case, including additional corporate costs, further adjustments for plant retro-fitting and increased SX-EW operating costs reduces the Base Case NPV by 49%.

Appendix A CRM Batch Control Charts

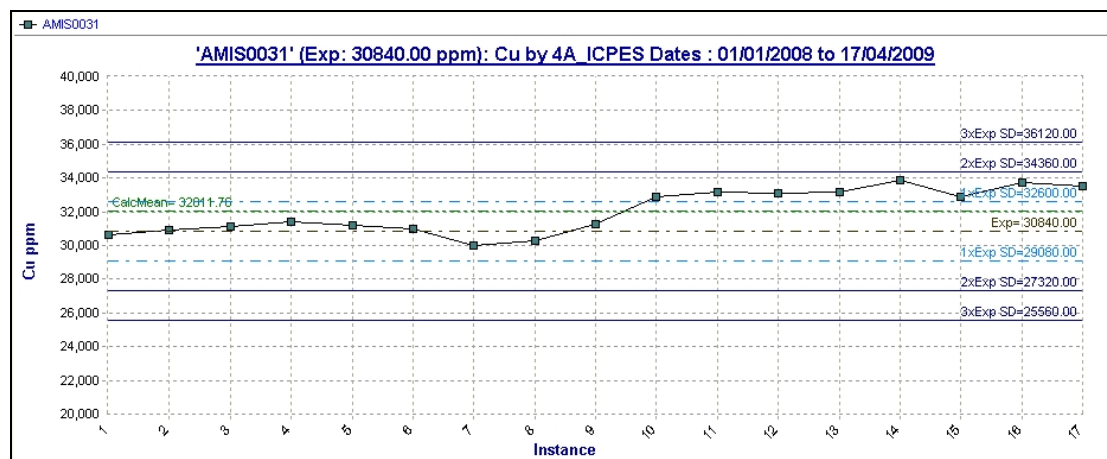
QAQC monitoring charts in the form of CRM, MAPD and duplicate control charts are appended herewith as reference checks in support of commentary provided in the previous memorandum.

Phase 4 Drilling: Sample Batch Control Charts:

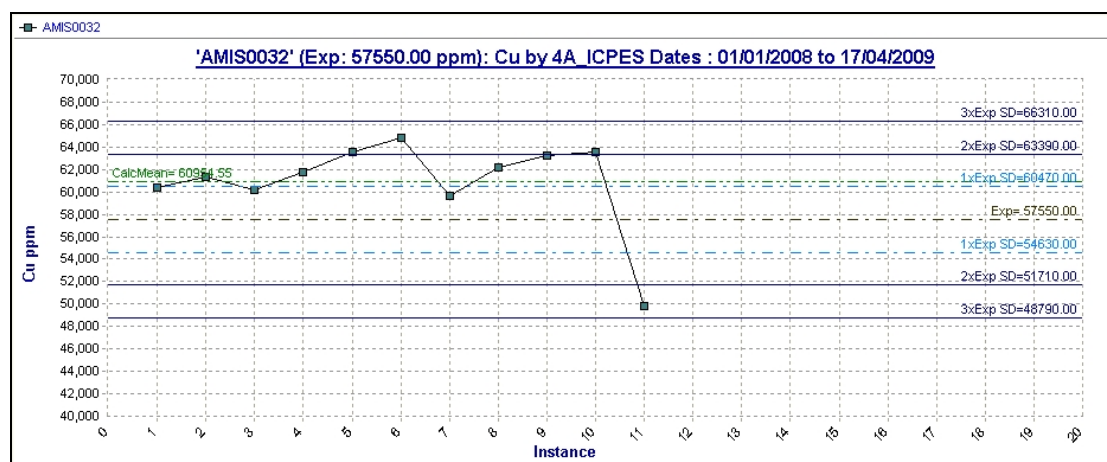
a) CRM = AMIS0038



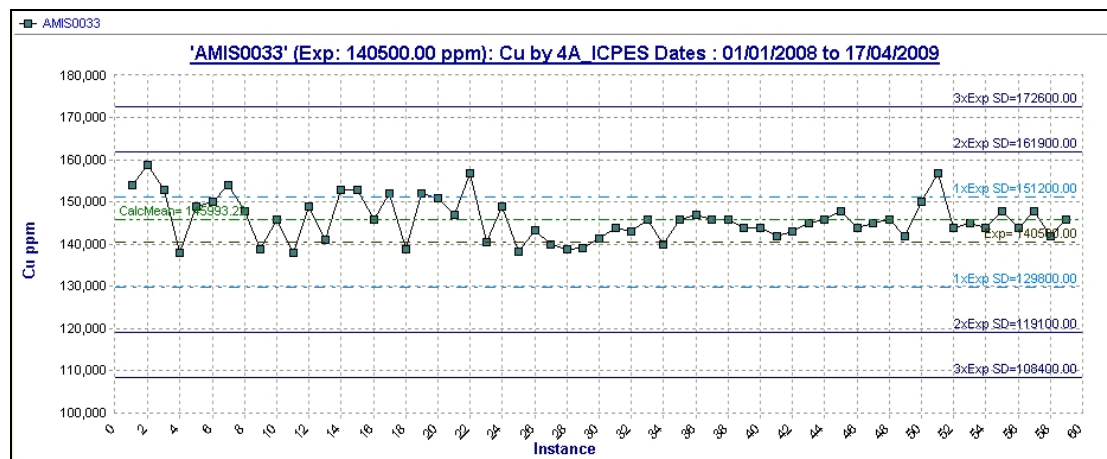
b) CRM = AMIS0031



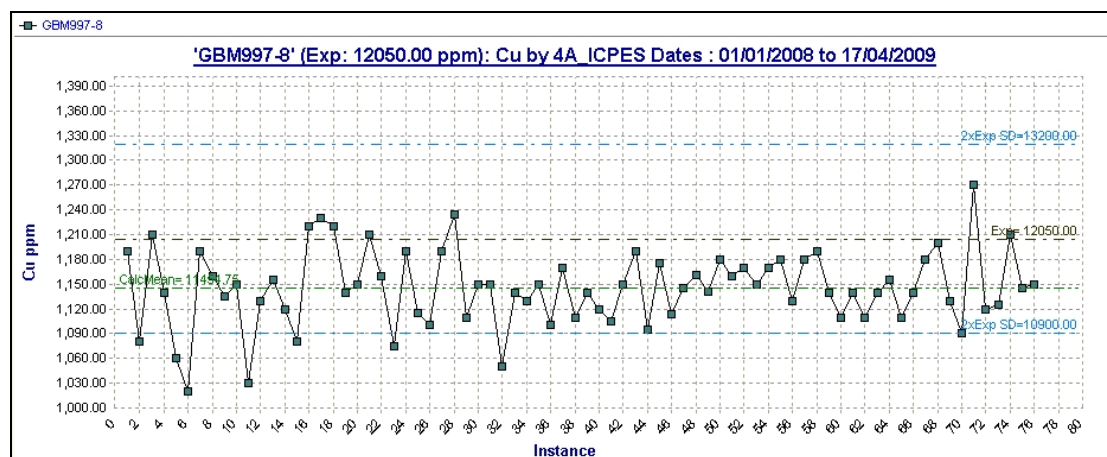
c) CRM = AMIS0032



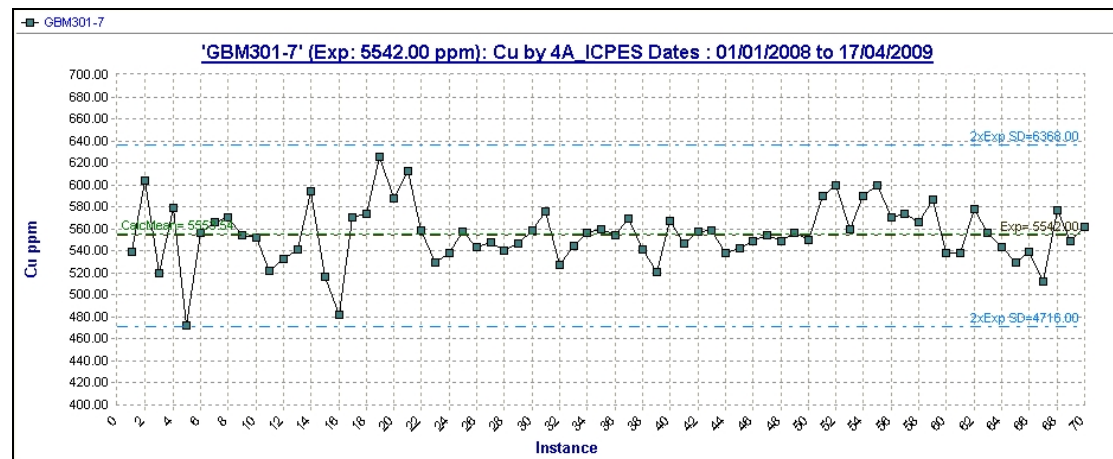
d) CRM = AMIS0033



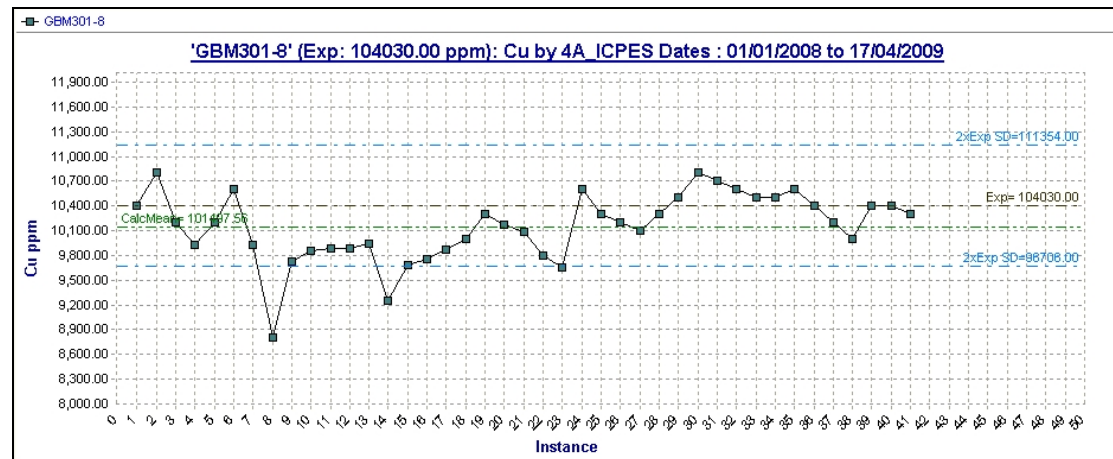
e) CRM = GBM997-8



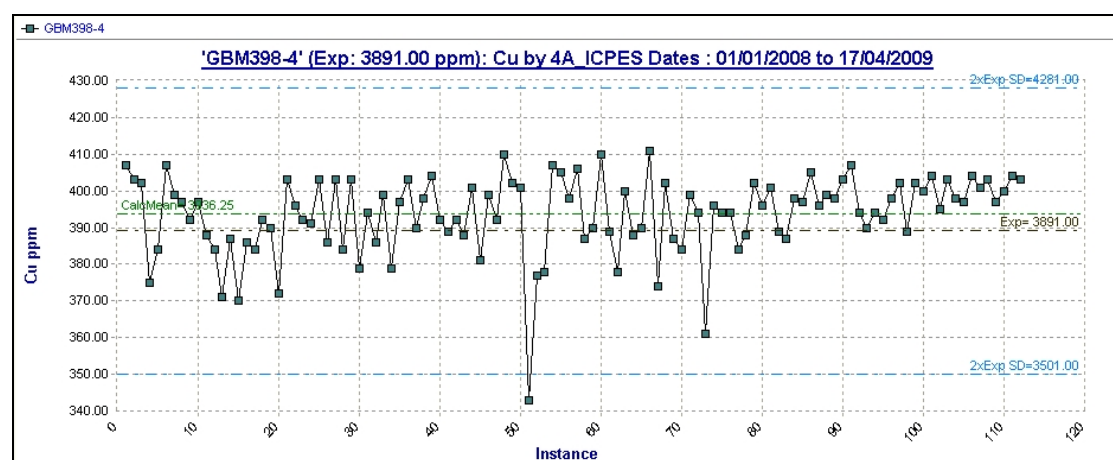
f) CRM = AMIS0033



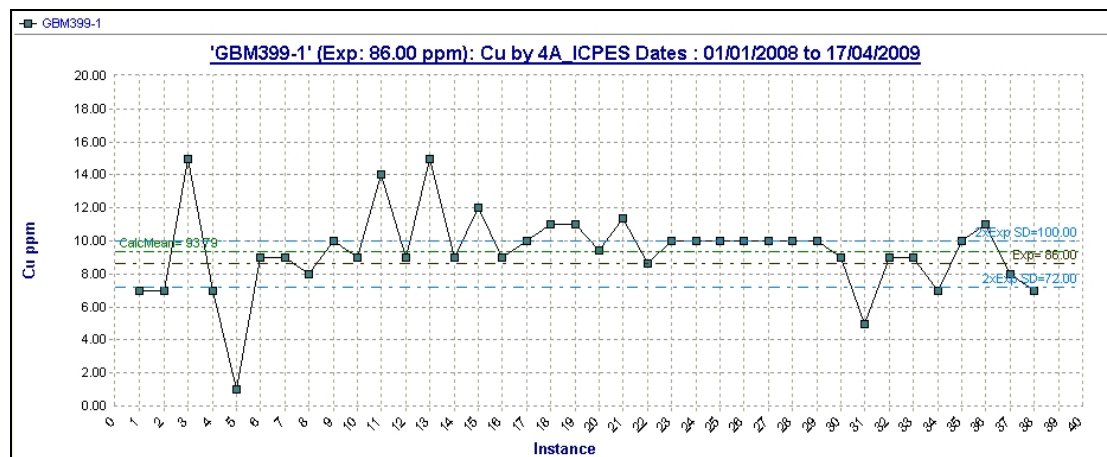
g) CRM = GBM301-8



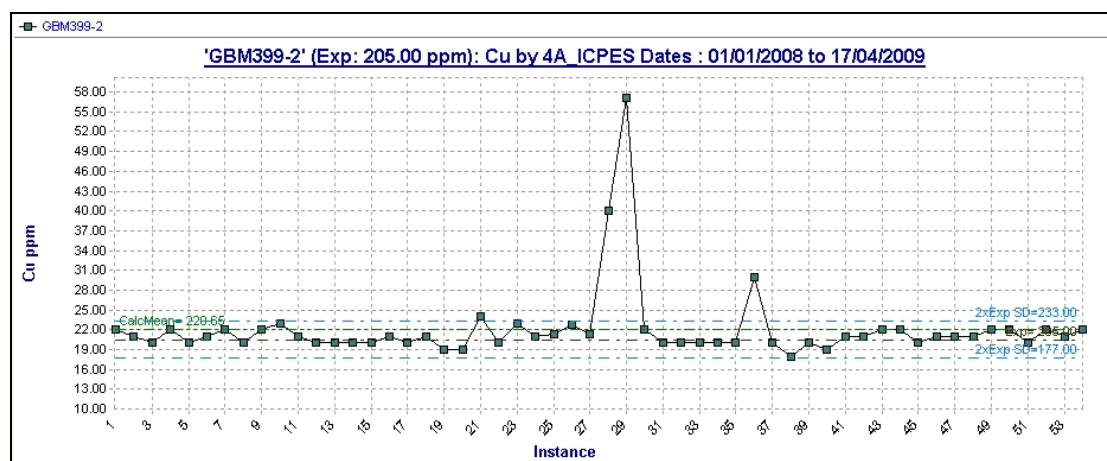
h) CRM = GBM398-4



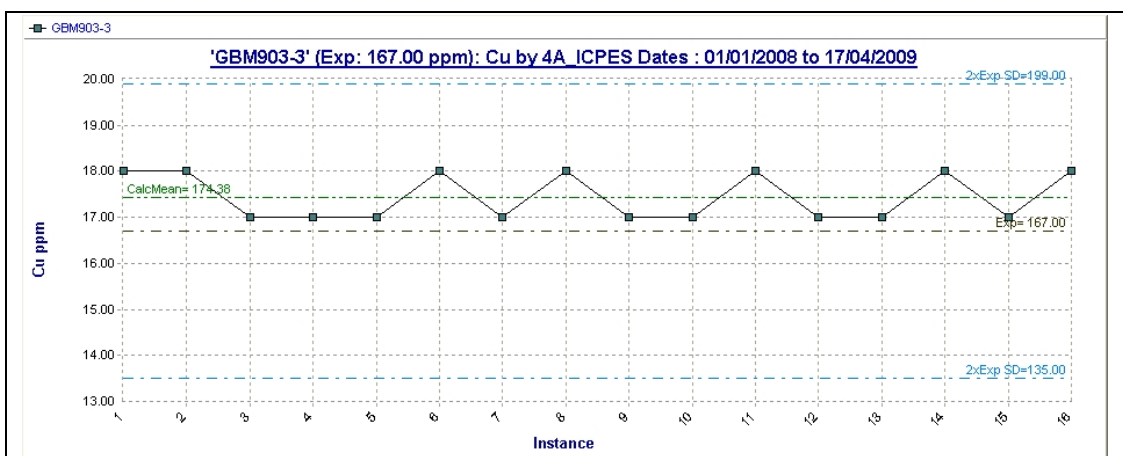
i) CRM = GBM399-1



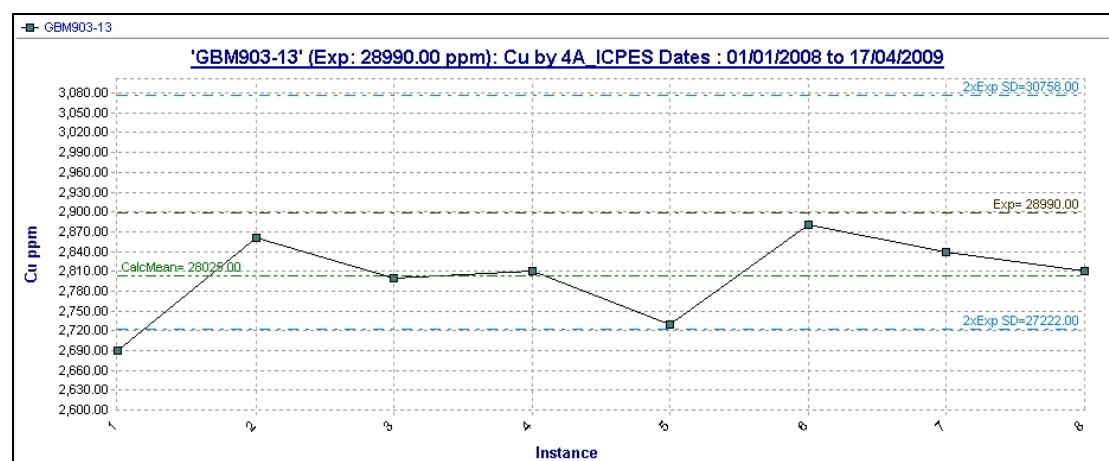
j) CRM = GBM399-2



k) CRM = GBM399-2

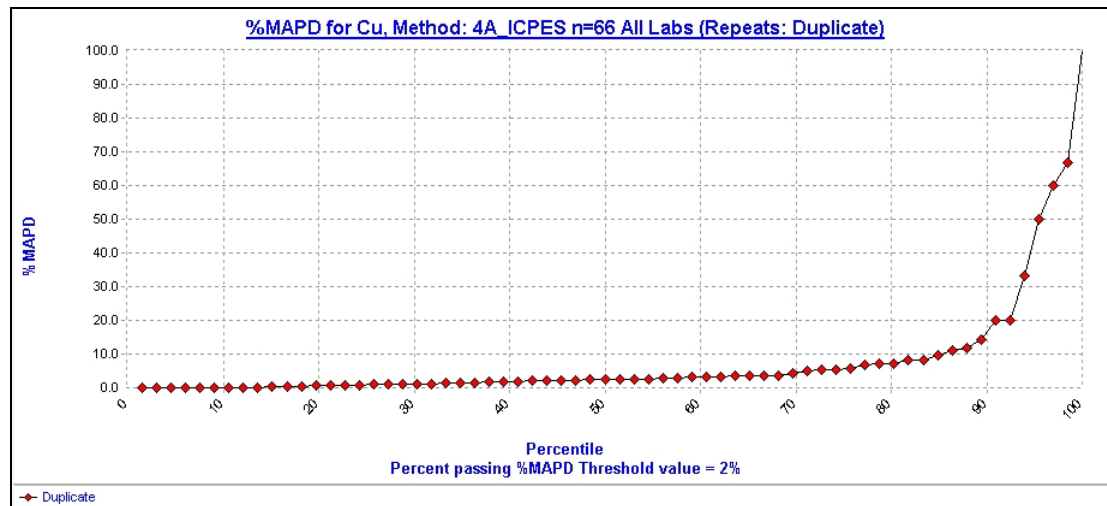


k) CRM = GBM903-13

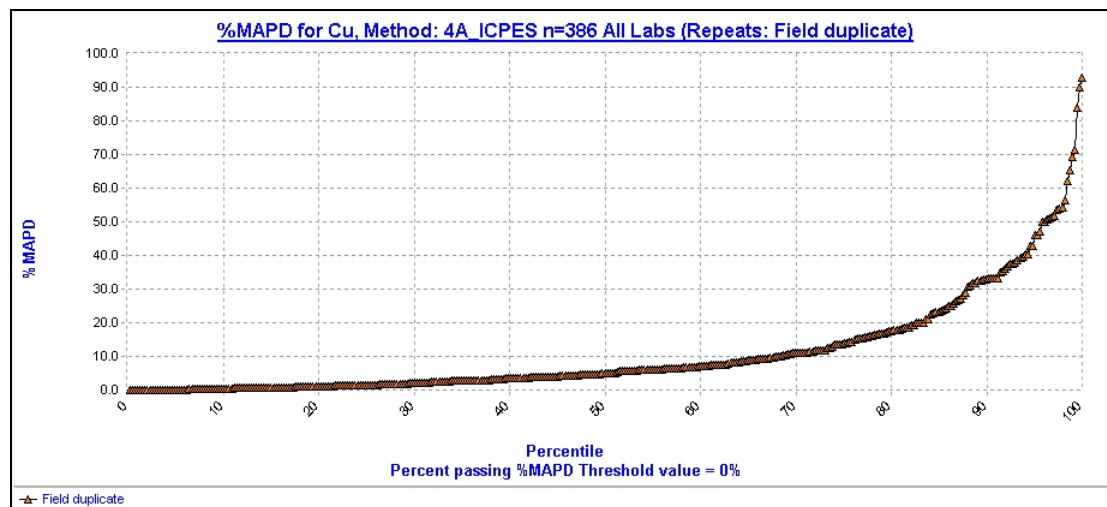


Phase 4 Drilling: Mean average paired differences – Percent passing 2%

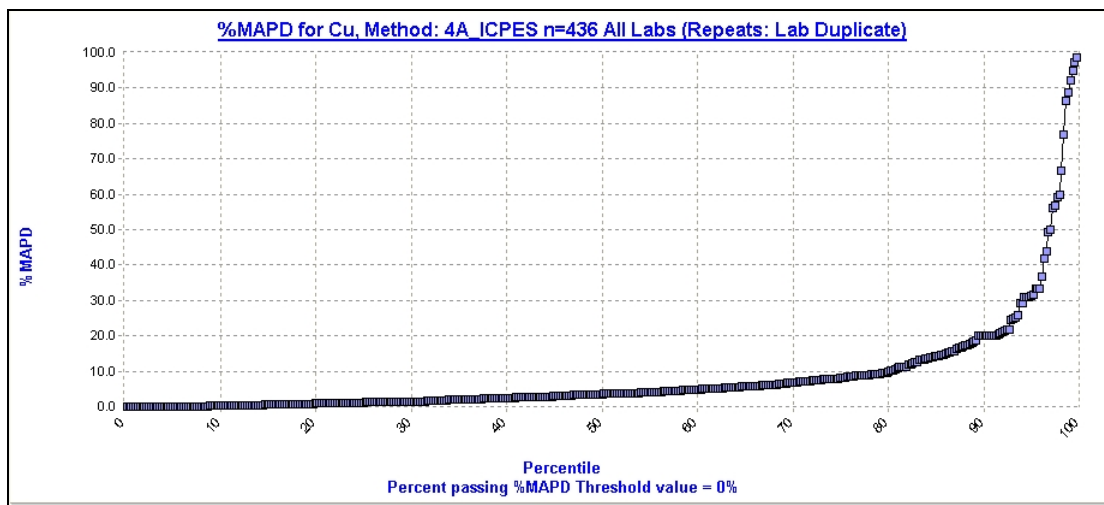
a) Lab splits



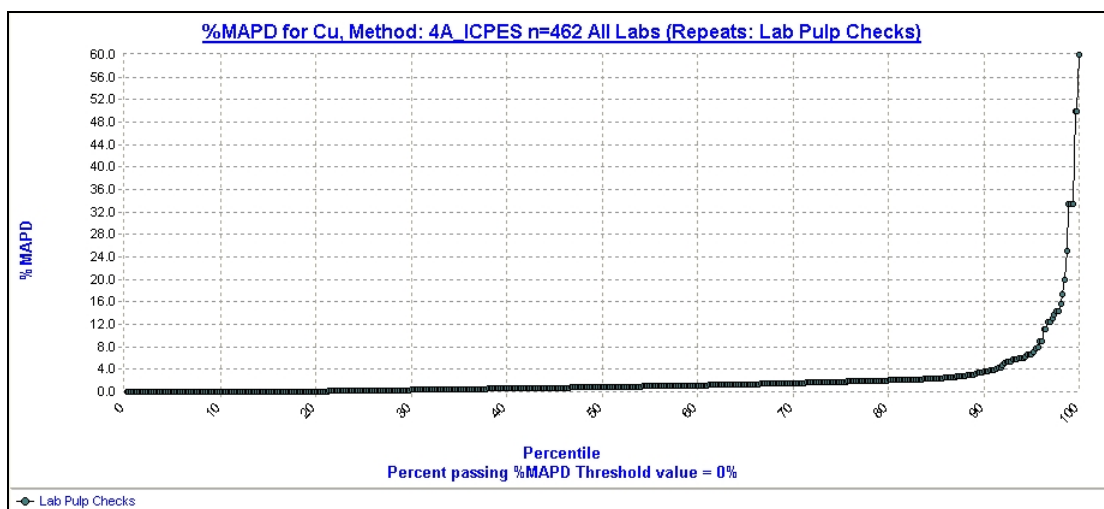
b) Field Duplicates



c) Analytical Duplicates

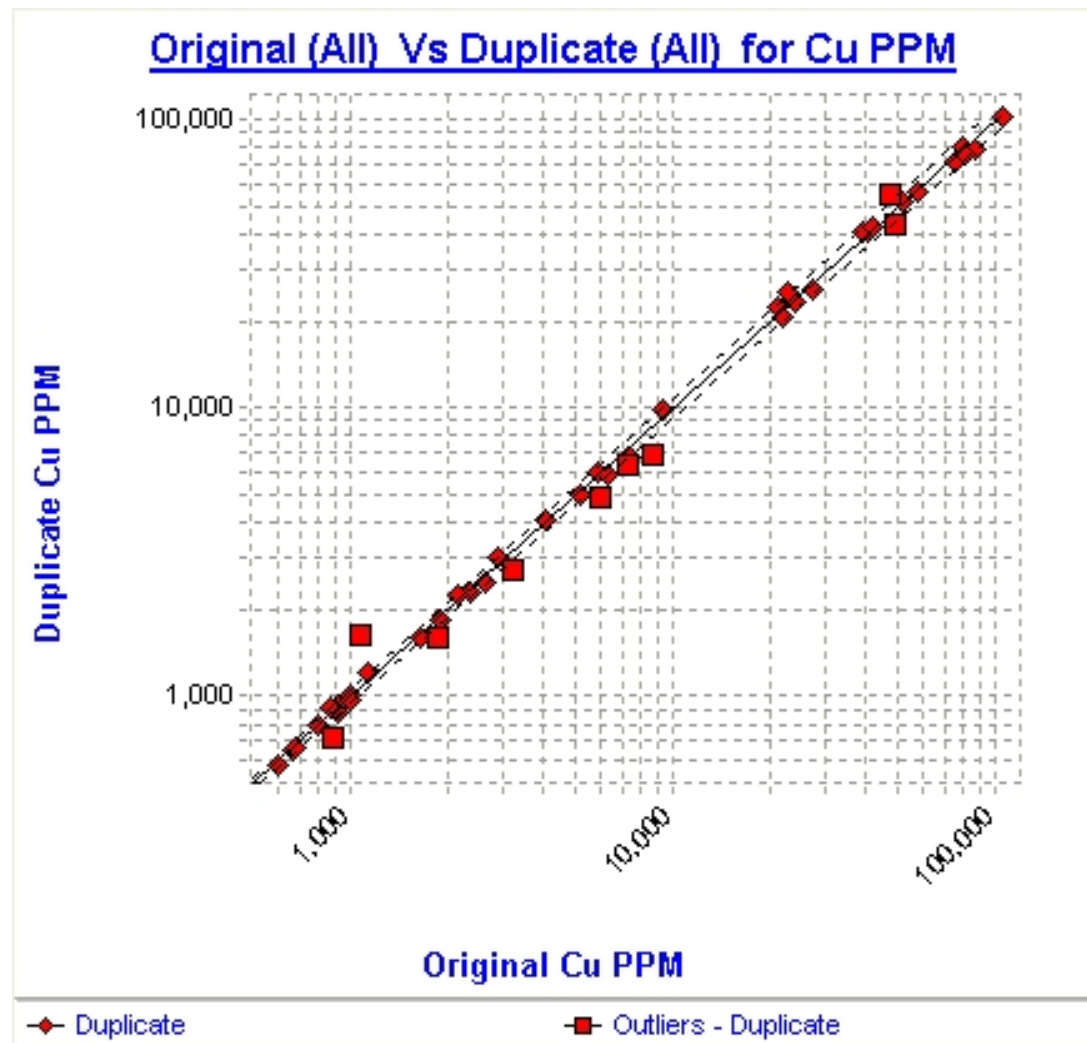


d) Sample Replicate

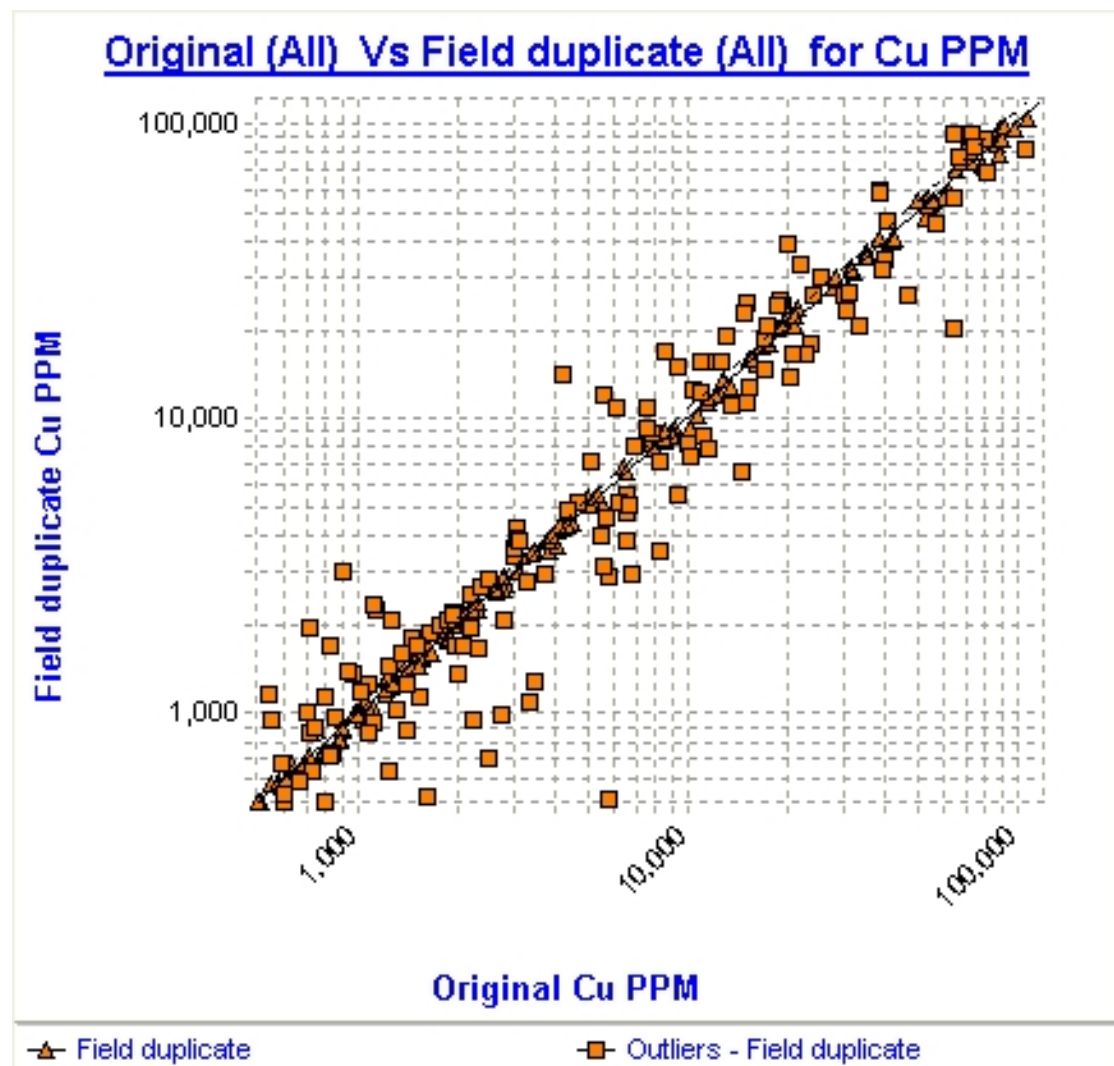


Phase 4 Drilling: Scattergrams

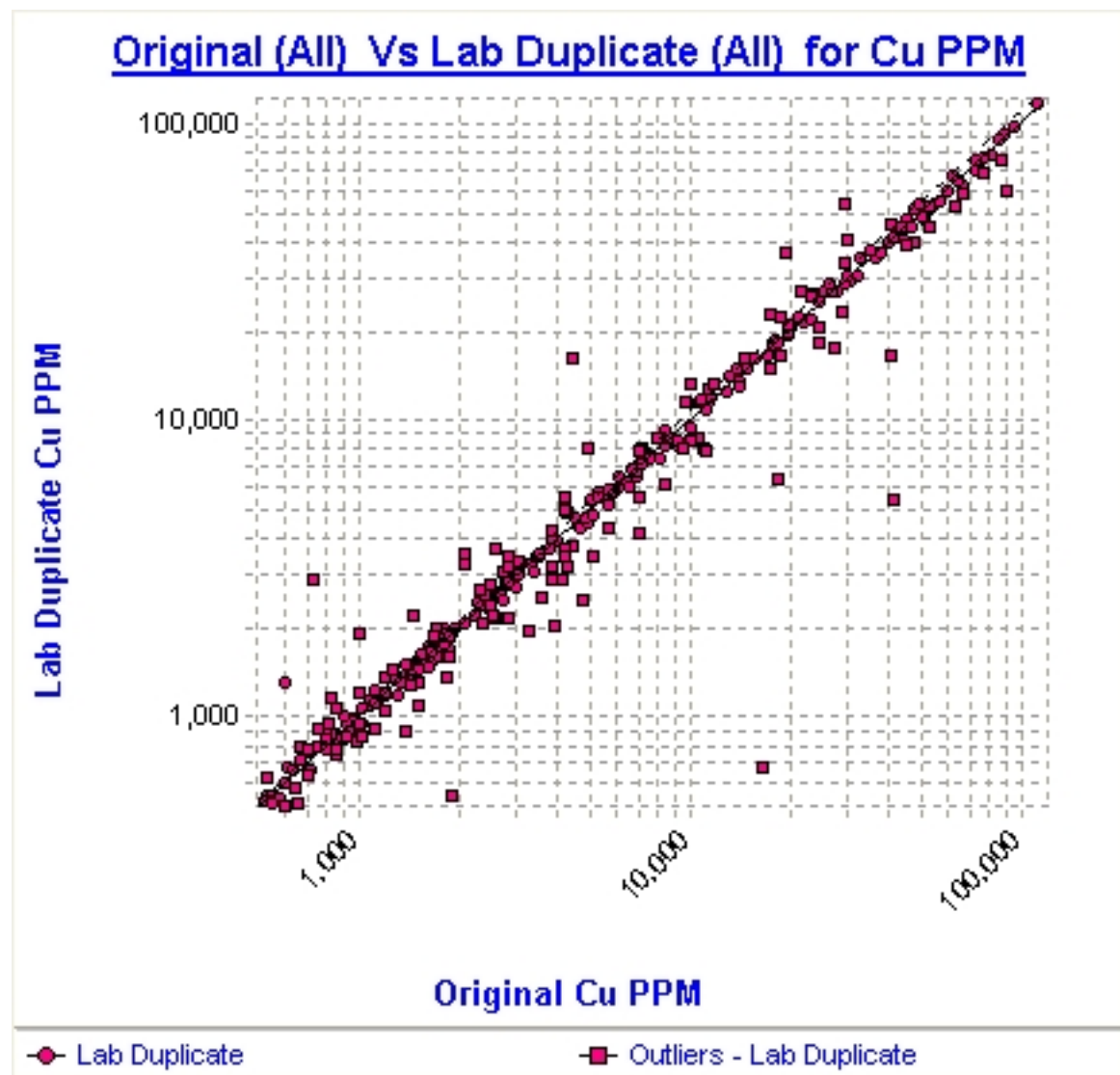
a) Lab Splits



b) Field Duplicates



c) Analytical Duplicates



Anvil Mining Cu Standards Submitted with Original Assays

Cu Standards			ALS					Mc Phar					ACT Labs				
			No. of Samples	Calculated Values				No. of Samples	Calculated Values				No. of Samples	Calculated Values			
Standard Code	Value	SD		Mean Cu	SD	CV	Mean Bias		Mean Cu	SD	CV	Mean Bias		Mean Cu	SD	CV	Mean Bias
AMIS0031	30840	0.176											17	32012	1318.561	0.041	1037895
AMIS0032	58280	0.28											11	60955	4041.624	0.066	1045791
AMIS0033	140500	10700						8	141325	3531.39	0.025	0.5872	51	146725	4915.601	0.034	4.431
AMIS0038	20730	0.125	5	20480	719.027	0.035	987840	10	20785	695.417	0.033	1002572	46	21339	570.956	0.027	1029284
Blank	6	-	125	773	6246.79	8.079	12786.7	32	-51	64.086	0	950.521	94	214	1875.476	8.775	3462.06
BLANK_GW	-	-	5	4	13.416	3.354	-										
BLANK_RAW	-	-	86	35	83.351	2.35	-	51	-39	74.586	0	-	128	12	33.608	2.74	-
GBM301-8	104030	3662	14	97179	3219.216	0.033	-6.586	3	101867	1059.874	0.01	-2.0795	24	103971	2341.215	0.023	-0.0569
GBM399-1	86	7	28	92	28.681	0.312	6.7276	10	99	6.867	0.069	15.5814					
GBM399-2	205	14	20	207	14.903	0.072	0.9756	12	262	113.982	0.435	27.8455	22	210	8.985	0.043	2.6608
GBM903-13	28990	884	8	28025	640.87	0.023	-3.3287										
GBM997-8	12050	575	60	11433	504.289	0.044	-5.1245	16	11538	240.653	0.021	-4.2474					

